EIDO International Doctoral School

DOCTORAL DISSERTATION

Causes of color variation in the populations of the marine gastropod Littorina saxatilis *from the Rías Baixas*

Juan Gefaell Borrás

2024
International Mention

Universidade Vigo

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Supervised by:

Emilio Rolán Álvarez

Juan Galindo Dasilva

2024

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Emilio Rolán Álvarez and Juan Galindo Dasilva,

DECLARE that the present work, entitled "Causes of color variation in the populations of the marine gastropod *Littorina saxatilis* from the Rías Baixas", submitted by Juan Gefaell Borrás for the attainment of a Doctoral degree, was carried out under their supervision in the PhD program "Methodology and Applications in the Life Sciences [*Metodoloxía e Aplicacións en Ciencias da Vida*]"

Vigo, September 3, 2024

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Emilio Rolán Álvarez e Juan Galindo Dasilva,

FAN CONSTAR que o presente traballo, titulado "Causas da variación na cor en poboacións do gasterópodo mariño *Littorina saxatilis* das Rías Baixas", que presenta Juan Gefaell Borrás para a obtención do título de Doutor, foi elaborado baixo a súa dirección no programa de doutoramento "Metodoloxía e Aplicacións en Ciencias da Vida".

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By Violeta Caballero-Lopez

"*We therefore must not recoil with childish aversion from the examination of the humbler animals. Every realm of nature is marvelous, (…) so we should venture on the study of every kind of animal without distaste; for each and all will reveal to us something natural and something beautiful*."

Aristotle, *Parts of Animals*, 645a-1.

"The sea is not made of water. (…) Look down as you crouch over the shallows and you will find a periwinkle or a prawn, a claw-displaying crab or a cluster of anemones ready to meet you. No need for binoculars or special stalking skills: go to the rocks and the living will say hello."

Adam Nicolson, *Life Between The Tides*, Introduction.

"Many of us are fascinated by the biological world around us. We marvel at the diversity of color, pattern, form, shape, size, ferocity, tameness, speed, and ingenious things that animals and plants do to find food and mates and avoid being eaten. (…) All of this is so enthralling that some of us not only want to know why the world is the way it is; we also want to explore, examine, and test ideas about it in order to make our own discoveries. We are evolutionary biologists."

Peter R. and B. Rosemary Grant, *40 Years of Evolution*, 3–4.

"*My success as a man of science, whatever this may have amounted to, has been determined, as far as I can judge, by complex and diversified mental qualities and conditions. Of these, the most important have been–the love of science–unbounded patience in long reflecting over any subject–industry in observing and collecting facts–and a fair share of invention as well as of common sense*."

Charles Darwin, *Autobiography*, 79.

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Gefaell J, Vigo R, González-Vázquez AH, Galindo J & Rolán-Alvarez E. 2024. Temporal stability and directional change in a color cline of a marine snail from NW Spain. *Current Zoology* 70: 163–173. [Open Access]

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Additionally, I have formally or informally co-supervised a series of undergraduate and master dissertations directly related to the topic of this dissertation:

González AH. 2022. Frecuencia de colores de la concha en poblaciones de la Ría de Vigo en el caracol marino *Littorina saxatilis*: estabilidad temporal desde el siglo pasado. Undergraduate Dissertation.

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Mosqueda MC. 2022. Evaluación de las posibles causas de un polimorfismo de coloración en ultravioleta en conchas de *Littorina saxatilis*. Undergraduate Dissertation.

Camesella R. 2023. Evaluación de la construcción de nicho como explicación del mantenimiento de un polimorfismo de color de concha de *L. saxatilis*. Undergraduate Dissertation. **[Formal co-supervision]**

Vigo R. 2023. Evolutionary causes of a shell color cline in *Littorina saxatilis*: Temperature and predation experiences. Master Dissertation.

Sánchez-Cantón O. 2024. Evaluación de una hipótesis selectiva de mantenimiento de un polimorfismo de color de concha en *Littorina saxatilis*. Undergraduate Dissertation.

Outomouro I. 2024. Explicación del patrón clinal de coloración de la concha en *Littorina saxatilis* en la Ría de Vigo: estimas de fecundidad. Undergraduate Dissertation.

Domínguez P. 2024. Tolerancia a la salinidad y su contribución al mantenimiento de un polimorfismo de color de concha en *Littorina saxatilis* de la Ría de Vigo. Undergraduate Dissertation. **[Formal co-supervision]**

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Articles:

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Gefaell J, Prieto T, Abdelaziz M, Álvarez I, Antón J, Arroyo J, Bella JL, Botella M, Bugallo A, Claramonte V, Gijón J, Lizarte E, Maroto RM, Megías M, Milá B, Ramón C, Vila M & Rolán-Alvarez E. 2020. Acceptance and knowledge of evolutionary theory among third-year university students in Spain. *PLoS ONE* 15: e0238345.

Gefaell J, Varela N & Rolán-Alvarez E. 2020. Comparing shape along growth trajectories in two marine snail ecotypes of *Littorina saxatilis*: A test of evolution by paedomorphosis. *Journal of Molluscan Studies* 86: 382–388.

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Torres L, Sánchez-Robles J, Moscoso B, Gijón J, Linarer M, Linares D, Pimentel A, Botella M, Bugallo A, **Gefaell J** & Rolán-Alvarez E. Submitted. Acceptance of evolutionary theory in students and educational service teachers from Ecuador.

Book chapters:

Gefaell J & Saborido C. 2023. Incommensurability in Evolutionary Biology: The Extended Evolutionary Synthesis controversy. In: Viejo JM, Sanjuán M, eds. *Life and Mind: New Directions in the Philosophy of Biology and Cognitive Sciences* (Interdisciplinary Evolution Research, Vol. 8). Cham: Springer, 165–183. https://doi.org/10.1007/978-3-031-30304-3_8

Gefaell J & Rolán-Alvarez E. 2023. Nuevas tendencias en biología evolutiva: el caso de la síntesis evolutiva extendida. In: Botella López MC, Gijón MK, Gijón Puerta J,eds. *Después del origen de las especies: la teoría darwiniana, paradigma de hoy*. Granada: Editorial Universidad de Granada, 165–193. [In Spanish]

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Book reviews:

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Gefaell J. 2022. The psychology of science denialism. Review of *Science Denial: Why It Happens and What to Do About It,* by Gale M. Sinatra and Barbara K. Hofer (OUP, 2021). *Metascience* 31: 453–456.

Gefaell J. 2022. A taxonomy of pseudoscience. Review of *On the Fringe: Where Science Meets Pseudoscience*, by Michael D. Gordin (OUP, 2021). *Metascience* 32: 169–172.

Edited book:

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Summary

Color variation is a classic topic in evolutionary ecology, encompassing color polymorphism, where populations exhibit multiple discrete morphs, and color clines, characterized by a gradual progression in morph frequencies along a geographic transect. These phenomena provide insights into how evolutionary processes maintain within-species diversity and the functional role of coloration. *Littorina saxatilis*, a marine gastropod found in the intertidal ecosystems of the northern Atlantic coasts, is a widely used organism in evolutionary ecology due to its remarkable phenotypic diversity, particularly in shell color. Despite this, shell color variation in *L. saxatilis* has been relatively understudied. The Rías Baixas, a series of four inundated river valleys located in the northwest Iberian Peninsula (Galicia, Spain), stands out for its shell color diversity in *L. saxatilis*, featuring both color polymorphic populations and a clinal variation from wave-sheltered to wave-exposed localities. This thesis aims to advance the understanding of the evolutionary and ecological mechanisms underlying shell color variation in *L. saxatilis* from this region. Chapter 1 reviews the current knowledge of color variation in marine gastropods, highlighting natural selection as the primary maintenance mechanism, though the evidence for this is in most cases indirect. Chapters 2 and 3 investigate mate choice-driven negative frequency-dependent selection as a potential mechanism for shell color polymorphism in the wave-sheltered localities of the Rías Baixas. Chapter 2 presents a field study that measures negative assortative mating for color in various *Littorina* species, suggesting that this pattern exists and may be ancestral in the *Littorina* clade. Chapter 3 further explores the link between negative assortative mating and mate choice through a laboratory experiment, providing overall

supportive evidence. Chapters 4, 5, and 6 focus on the *L. saxatilis* color cline of the Rías Baixas. By comparing two temporal series of morph frequencies, Chapter 4 concludes that the cline has remained relatively stable over the last forty years and that morph frequencies are influenced by ecological factors. Chapter 5 describes a field tethering experiment testing selection for crypsis as a maintenance mechanism for the cline, finding suggestive but inconclusive evidence. Finally, Chapter 6 examines various shell traits and ecological factors across populations from representative localities in the four Rías Baixas, revealing an analogous clinal variation in shell color that might have evolved repeatedly in each ria, along with substantial co-variation between different shell traits. Taken together, these studies suggest that mate choice-driven negative frequency-dependent selection may help explain intrapopulation shell color variation in the wave-sheltered localities of *L. saxatilis* from the Rías Baixas, while spatially varying selection, probably related to crypsis, may drive the recurrent evolution and stability of the cline. The dissertation concludes with a discussion of relevant hypotheses and a unified account of color diversity evolution in *L. saxatilis* from the Rías Baixas.

Resumo estendido

É un feito coñecido que as especies mostran diversidade xenética e fenotípica nas súas poboacións. Esta diversidade é crucial para a evolución, na medida en que o cambio evolutivo depende da súa existencia para que a selección e outros mecanismos poidan operar. Entre a variación que amosan as especies está a diversidade de cor. Se nos centramos nos animais, por exemplo, as distintas estruturas tegumentarias (pelaxe, escamas, cunchas, cutículas, plumas) poden variar nas súas propiedades cromáticas. A este fenómeno coñéceselle como variación na cor ou variación cromática. En tanto en canto a cor dun organismo cumpre varias funcións, como a termorregulación, a comunicación con conespecíficos ou a evitación da depredación, a variación neste carácter presenta un gran interese para os biólogos evolutivos: Que patróns pode seguir? Que forzas son responsables do seu mantemento?

En termos xerais, a variación na cor pode dividirse en variación intrapoboacional, tamén coñecida como polimorfismo de cor, e variación xeográfica, dentro da cal destacan as clinas. O polimorfismo de cor é un dos principais temas de estudo na ecoloxía evolutiva. Varias hipóteses teñen sido propostas para explicar por que certas poboacións de animais son diversas a nivel cromático. Entre as posibles respostas destacan as distintas modalidades de selección natural, tales como a selección negativa dependente das frecuencias, a selección sexual, ou unha combinación destes e outros mecanismos. Na selección negativa dependente das frecuencias, por exemplo, os individuos que presentan unha cor máis rara son aqueles que máis probabilidades teñen de sobrevivir e reproducirse. Pola súa banda, as clinas de cor defínense como cambios graduais neste carácter ao longo dunha área xeográfica. Estes cambios na cor

poden tomar a forma de progresións nalgún parámetro cromático cuantitativo (por exemplo, a claridade do pelaxe) ou dun cambio paulatino na frecuencia dunha serie de morfos discretos, o que se coñece como unha clina na ratio de morfos. En canto as causas evolutivas das clinas de cor, destacan de novo a selección natural, neste caso a que varía ao longo do espazo, e o fluxo xénico.

Unha especie particularmente popular no eido da ecoloxía evolutiva é o caracol mariño *Littorina saxatilis*, tamén coñecido como bígaro bravo. *L. saxatilis* habita nos ecosistemas intemareais rochosos das costas europeas e americanas do hemisferio norte, ocupando o nicho ecolóxico de ramoneador das microalgas que crecen sobre as rochas. Alomenos dous feitos relacionados con *L. saxatilis* fano un organismo especialmente valioso para entender o modo no que funciona a evolución. En primeiro lugar, *L. saxatilis* ten un sistema de reprodución ovovivíparo no cal os embrións desenvólvense no interior dun saco de cría ata alcanzar o estadio de xuvenís maduros, momento no cal son liberados ás rochas pola nai. Isto contrasta co modo reprodutivo da maioría de gasterópodos mariños, cuxas femias liberan os ovos fecundados á columna de auga do océano, onde estes permanecen durante boa parte do seu desenvolvemento ata que, tras unha serie de metamorfoses, retornan ás costas ás que lles leven as correntes. En termos evolutivos, o sistema reprodutivo de *L. saxatilis* favorece o illamento das súas poboacións, de xeito que entre elas non se produza intercambio de xenes. A consecuencia máis palpable deste fenómeno é a evolución de adaptacións locais. En segundo lugar, o ecosistema onde vive *L. saxatilis* (isto é, o intermareal) é enormemente diverso tanto en termos xeolóxicos e fisicoquímicos como biolóxicos. Así, por exemplo, as distintas costas varían no seu grao de exposición ás ondas, o que xera diferencias progresivas na composición de especies habitan nelas. Isto pode facer que a cor de fondo sobre a que se establece *L. saxatilis* sexa variable ao longo do espazo.

L. saxatilis é especialmente dada a mostrar gran variedade de cores de cuncha. Así, esta especie presenta dende cunchas brancas puras ata negras, pasando por toda unha gamma intermedia de tonalidades, se ben estruturada en torno a morfos máis ou menos discretos: branco-crema ou gris, amarelo, laranxa, marrón clariño, e marrón escuro. A maiores, en moitos destes morfos poden engadirse distintos patróns (liñas, bandas, mosaicos), que a súa vez poden ter distintas cores. Isto xera unha enorme diversidade cromática, pouco vista noutras especies de caracois mariños. A pesares da enorme diversidade cromática que presenta *L. saxatilis*, o modo en que esta variedade se mantén nas poboacións naturais (isto é, a súa ecoloxía evolutiva) ten sido comparativamente pouco estudado.

Dende o punto de vista das cores de *L. saxatilis*, unha área ao noroeste da Península Ibérica, as Rías Baixas (Galicia, España), asemella especialmente apta para o seu estudo. As Rías Baixas constitúen una serie consecutiva de catro vales fluviais inundados cuxa orixe data da transgresión do Holoceno (ao redor de 10000 anos a. p.), momento no cal o nivel do mar subiu substancialmente e anegou as concas fluviais que ata o momento dábanse na zona actualmente ocupada polas rías. De norte a sur, as rías de Muros e Noia, Arousa, Pontevedra e Vigo caracterízanse todas elas por unha forma semellante a un funil que alberga alomenos un gran río cas baña situado na súa zona máis interior. Estas rias mostran unha substancial variación físico-química e xeolóxica ao longo das súas costas. Concretamente, as distintas zonas das Rías Baixas presentan claras variacións no grao de exposición ás ondas.

Toda esta variación físico-química, unida á de carácter propiamente ecolóxica, fan do litoral das Rías Baixas unha contorna enormemente diversa, algo que se deixa relucir en *L. saxatilis*. Así, o bígaro bravo das Rías Baixas mostra variación xeográfica gradual (isto é, unha clina) na cor da súa cuncha dende as zonas máis protexidas das ondas ata as máis expostas, ao que se suma unha rechamante variación intrapoboacional (é dicir, un polimorfismo de cor) en moitas localidades puntuais desta latitude. Malia que a variedade de cor do bígaro bravo das Rías Baixas ten sido recoñecida por algúns malacólogos clásicos, ata o momento apenas tense estudado dunha maneira sistemática e centrada en entender a súa evolución. A presente tese doutoral ten como obxectivo principal comezar a botar luz sobre este fenómeno, o cal pasa tanto por realizar unha descrición o máis axeitada posible da súa ecoloxía, así como poñer a proba algúns dos mecanismos evolutivos que poden explicar a súa existencia sostida no tempo. Para iso, esta tese estuda por unha banda o polimorfismo de cor de cuncha en certas poboacións de *L. saxatilis* das Rías Baixas, e por outra, a variación clinal que en termos xerais amosan todas elas cando se estudan en conxunto, con especial énfase na Ría de Vigo. A maiores, e cara a contextualizar a variación de cor de *L. saxatilis* no eido máis amplo da variación de cor nos gasterópodos mariños, realízase tamén un traballo teórico de revisión bibliográfica. En consecuencia, a tese divídese en seis capítulos, dentro dos cales o primeiro ocúpase da revisión bibliográfica, os dous seguintes da variación intrapoboacional, e os tres últimos da variación clinal. Pasamos agora a facer unha breve descrición do contido e principais achados de cada un deles.

Capítulo 1: Polimorfismo de cor en gasterópodos mariños

O primeiro capítulo o conforma a revisión bibliográfica sobre os polimorfismos de cor nos caracois mariños, entendendo “polimorfismo” en sentido amplo (abranguendo tanto variación

intrapoboacional como xeográfica). A revisión comeza estudando a cor dende a perspectiva máis básica (a da súa física e bioquímica), rematando coa máis integradora (a ecolóxica e evolutiva). Dende un punto de vista molecular, a cor dos gasterópodos é o resultado combinado das propiedades estruturais e pigmentos biolóxicos da súa cuncha. Dende o punto de vista xenético, aínda cos xenes concretos responsables da cor dos caracois mariños apenas se coñecen, múltiples evidencias directas e indirectas suxiren que, en efecto, ditos xenes existen. Outra perspectiva necesaria no estudo do polimorfismo de cor é a da ecoloxía sensorial. Na medida en que a cor dun animal pode ter funcións de comunicación cos conespecíficos e a evitación dos depredadores, é preciso ter en conta os sistemas sensoriais destes á hora de entender o seu posible valor adaptativo. Nese sentido, o máis probable e que, mentres cos propios caracois mariños non teñen visión cromática, a maioría dos seus depredadores si a posúen.

Unha vez revisados brevemente todos estes enfoques, a revisión focalízase nos patróns de diversidade que poden amosar os polimorfismos de cor. Nos gasterópodos mariños téñense descrito tanto casos de variación intrapoboacional como xeográfica. En cambio, moita menos literatura ten analizado os cambios temporais na frecuencia de morfos de cor ao longo do tempo. Finalmente, neste apartado descríbese a distribución do polimorfismo de cor ao longo dos distintos grupos taxonómicos de caracois mariños, facendo notar que aínda que se trata dun fenómeno xeralmente raro, é mais abundante nas especies con desenvolvemento directo, como *L. saxatilis*.

Tralos patróns de diversidade, a revisión aborda os seus mecanismos evolutivos, tema ao cal adícaselle a maior extensión. En termos xerais, a literatura apunta ao rol predominante da selección natural no mantemento da variación de cor nos gasterópodos mariños. Agora ben, dependendo do contexto ecolóxico co que se rechea este mecanismo, o resultado pode ser ben distinto. Por exemplo, non é o mesmo que a selección actúe directamente sobre a cor a que opere sobre un carácter correlacionado con esta. De xeito análogo, non é o mesmo que a selección sexa dependente das frecuencias a que sexa independente. Aínda que existen exemplos da maioría de modalidades de selección nos caracois mariños, a evidencia a favor deles é máis ben indirecta. Isto entronca co seguinte apartado da revisión, no que se presentan as distintas metodoloxías empregadas para estudo do polimorfismo de cor neste grupo de animais. O achado máis importante é que a maioría de investigacións sobre este fenómeno teñen empregado metodoloxías frecuentistas, nas que se analizan as frecuencias de distintos morfos en relación ao seu hábitat como forma de obter pistas acerca do valor adaptativo da cor. Tal e como se salienta na última sección da revisión, adicada ao futuro do campo, é desexable que estes estudos –valiosos de por si– se complementen con outros focalizados en revelar de

maneira máis directa os mecanismos que contribúen ao seu mantemento (como experimentos de manipulación, espectrometría ou técnicas -*ómicas*).

Capítulo 2: Apareamento asociativo negativo e mantemento do polimorfismo de cor en especies do xénero *Littorina*

O mantemento da variación de cor intrapoboacional pode producirse por mecanismos de carácter reprodutivo, coma a selección sexual. Dentro desta, unha modalidade especialmente interesante é a selección sexual negativa dependente das frecuencias (SSNDF), na que a eficacia biolóxica de natureza sexual dun morfo varía de forma inversa á súa frecuencia. Unha forma indirecta de estudar este mecanismo e a través das cópulas, xa que unha predición da SSNDF é que ditas cópulas serán maioritarias entre individuos de cores distintas, algo coñecido como apareamento asociativo negativo (AAN). Normalmente, o proceso condutual que media entre a SSNDF e o AAN é a elección de parella, na cal os espécimes do sexo elixidor (no caso dos litorínidos, o macho) escollen activamente aparearse cos membros do sexo oposto que presentan unha cor distinta á súa.

Unha vantaxe dos caracois do xénero *Littorina* –incluído *L. saxatilis*– é a relativa abundancia de cópulas que poden atoparse na súa contorna natural. Aproveitando este feito, no Capítulo 2 estudouse a posible existencia de AAN nunha serie de poboacións polimórficas de tres especies do xénero *Littorina* das Rías Baixas: as especies irmás *L. fabalis* e *L. obtusata*, máis *L. saxatilis* (en conxunto, o subxénero *Neritrema*). Para iso, recolléronse cópulas destas tres especies no campo, ás que se lle engadiron os resultados previamente publicados dunha delas (*L. fabalis*). Ditas cópulas foron tomadas seguindo un protocolo validado, polo cal colleitáronse tamén datos da cor dos individuos máis próximos á cópula para dar conta da escala da elección dos caracois. O obxectivo do estudo foi dobre: por unha banda, avaliar se a SSNDF mediada por elección de parella pode constituír un mecanismo de mantemento do polimorfismo de cor nas poboacións de *L. saxatilis* das Rías Baixas, especialmente daquelas situadas nas localidades más protexidas das ondas. E por outra, establecer ata que punto dito mecanismo pode estar operando simultaneamente noutras especies do xénero, é dicir, ata que punto é un carácter biolóxico ancestral.

Empregando o índice de apareamento asociativo I_{PSI}, o estudo de cópulas no campo revelou que, cando se agrupan os morfos de cor das distintas especies en cores claras e escuras, o cómputo global das distintas poboacións mostra unha clara tendencia cara ao AAN. Por si mesmo, isto constitúe evidencia a favor da hipótese de que o polimorfismo de cor de certas poboacións de *L. saxatilis* está a ser mantido por SSNDF mediada por elección de parella, se ben

dita evidencia é indirecta. De xeito análogo, o resultado tamén suxire o carácter ancestral do AAN nas especies do subxénero *Neritrema*. Por outra banda, as excepcións ao patrón de AAN de dúas poboacións de *L. fabalis* das Rías Baixas da pistas sobre o posible valor selectivo da cor neste grupo de caracois mariños. Dado que as especies de *Littorina* carecen con toda probabilidade de visión cromática, é probable que a cor de cuncha sexa seleccionada porque está vencellada a nivel xenético con outros caracteres con función adaptativa, quizais a través dunha inversión cromosómica. De probarse o vínculo entre o AAN e a SSNDF mediada por elección de parella, isto abriría novas vías de investigación sobre os mecanismos xenéticos do polimorfismo de cor en *Littorina*.

Capítulo 3: Evidencia experimental da elección de parella como mecanismo detrás do apareamento asociativo negativo para a cor da cuncha observado en *L. saxatilis*

O estudo correspondente ao terceiro capítulo pode lerse como unha continuación do anterior, adicado a revelar os mecanismos de mantemento evolutivo do polimorfismo intrapoboacional observado en certas localidades das Rías Baixas. De cara a dito obxectivo, no Capítulo 3 descríbese un experimento de laboratorio no que se empregou un paradigma experimental poucas veces usado antes en *L. saxatilis* para vencellar o patrón de AAN coa elección de parella. O paradigma experimental consistiu no uso de cámaras ou bolas transparentes de cópula nas que se introduciron espécimes de *L. saxatilis* de diferente sexos e cores. Dado que nesta especie os machos perseguen e seleccionan ás súas parellas seguindo o rastro de moco que estas últimas deixan ao desprazarse, é posible concluír que as cópulas observadas nas bolas xorden dunha decisión explícita de apareamento por parte dos machos (é dicir, dunha elección de parella). Se dito nexo fose corroborado, isto constituiría unha proba máis robusta do mantemento do polimorfismo de cor por SSNDF.

Para o experimento descrito, recolléronse exemplares de *L. saxatilis* de tres localidades do interior da Ría de Vigo que son representativas da zona protexida das ondas de todas as Rías Baixas. Dous desas tres localidades foron as mesmas que se analizaron no estudo correspondente ao Capítulo 2. Os exemplares empregados pertencían aos tres morfos de cor máis comúns nas devanditas localidades: branco-crema ou agrisado, amarelo e marrón. En termos xerais (en dúas de tres localidades), o experimento corroborou o patrón de AAN, que esta vez puido atribuírse dunha forma máis fiable á elección de parella. Isto analizouse non só a través do índice I_{PSI}, senón tamén por medio de modelos matemáticos. Non obstante, certos problemas

permanecen na medida en que a poboación discordante é unha das cales nas que o AAN si foi detectado no Capítulo 2. Ademais, a cuestión relativa a como os machos de *L. saxatilis* detectan as femias dunha cor distinta á súa, así como os posibles beneficios derivados das cópulas entre diferentes, seguen sen ser resoltos. Polo tanto, aínda que os estudos correspondentes aos Capítulos 2 e 3 apoian o rol da SSNDF mediada por elección de parella no mantemento do polimorfismo de cor das poboacións protexidas de *L. saxatilis* das Rías Baixas, algunhas incógnitas permanecen.

Capítulo 4: Estabilidade temporal e cambio direccional nunha clina de cor en *L. saxatilis* da Ría de Vigo

Se ben os anteriores capítulos centrábanse no polimorfismo intrapoboacional, os seguintes teñen como obxectivo describir e dar conta da variación clinal. Dende tempo atrás, ven sendo apreciado que as Rías Baixas presentan unha variación gradual na cor das cunchas de *L. saxatilis*. Así, diversos investigadores na segunda metade do século pasado describiron un patrón polo cal un morfo de cor branco-crema ou agrisado (*fulva*) sería especialmente abundante na zona interior das rías, mentres que un morfo con liñas (*lineata*) predominaría nas zonas máis exteriores. Entre ambas zonas existiría unha zona intermedia con gran diversidade de cores. A pesares da valiosa contribución destes investigadores, os seus obxectivos eran maioritariamente taxonómicos, polo que apenas profundaron en describir e entender o mantemento evolutivo desta clina. O Capítulo 4 describe un estudo no cal mostrexáronse un total de 36 localidades ao longo da Ría de Vigo que xa foran previamente estudadas na década de 1970 por Cesare Sacchi, de forma que, a través da comparativa entre as frecuencias dos distintos morfos en cada unha destas series temporais, puidérase avaliar o grao de estabilidade ou cambio experimentado pola clina. Ademais, en ditos mostrexos recolléronse tamén datos acerca da fauna e flora das localidades co fin de empregar dita información para a formulación de hipóteses acerca do mantemento evolutivo da clina.

Os resultados de dita comparativa mostran claramente como, en termos xerais, esta clina tense mantido estable nos últimos corenta anos. Certo é que tamén detéctanse certos cambios aleatorios e direccionais na forma da clina, mais ditos cambios non ensombrecen o feito global de que a clina segue lucindo moi semellante a coma fora descrita xa nos anos 70 do século pasado. Isto, por si mesmo, constitúe un forte indicio de que a clina estase a manter por selección que varía no espazo, conxectura que ven reforzada tamén polo carácter preditivo de certas variables ecolóxicas. Así, por exemplo, as regresións múltiples descritas no capítulo aquí referido mostran que a frecuencia do morfo *lineata*, abundante no exterior, ven explicada pola

presenza ou abundancia de especies propias das zonas máis expostas ás ondas, como o caracol mariño *Monodonta lineata* e o lique *Lichina pygmaea* na costa norte ou *Melarhaphe neritoides* na sur. Pola contra, a frecuencia de *fulva*, maioritario no interior, é predita por distintas especies e parámetros físico-químicos asociados aos ambientes protexidos das ondas. Aínda que ditas regresións non permitan inferir mecanismos causais (de modo análogo a coma acontecía co estudo de cópulas no campo no Capítulo 2) polo menos proporcionan indicios acerca dos mesmos.

Capítulo 5: Contribúe a depredación diferencial de distintos morfos de cor de *Littorina saxatilis* ao mantemento da súa clina nas Rías Baixas?

O quinto capítulo da presente tese describe un experimento de campo destinado a avaliar a verosimilitude dun dos posibles mecanismos responsables do mantemento da clina de cor en *L. saxatilis*: a selección por cripsis. Esta hipótese sostén que se ao longo dun transepto xeográfico existen diferenzas progresivas na composición de especies (coma é o caso nas Rías Baixas), isto pode xerar cores de fondo distintas nos ecosistemas intermareais. Se as especies depredadoras en ditos ecosistemas son animais principalmente visuais, isto pode xerar a evolución dunha clina de cor nas súas presas por medio de selección por cripsis.

L. saxatilis é depredada por unha multitude de depredadores visuais. Entre eles inclúense, cun alto grao de probabilidade, os caranguexos, entre os que destaca o queimacasas (*Pachygrapsus marmoratus*). Esta especie é especialmente abundante nas zonas exteriores das Rías Baixas, onde tamén xoga un rol clave na diferenciación dos ecotipos coñecidos como *Crab* e *Wave*. No quinto capítulo da tese detállase un experimento de *tethering* no que atáronse distintos morfos de cor dunha poboación polimórfica ás rochas dunha localidade representativa da zona exposta que é monomórfica para *lineata* co fin de determinar se os morfos menos crípticos son maioritariamente depredados polos queimacasas. Desta forma, poderíase explicar por que nesas localidades só un morfo, o *lineata*, prolifera (a saber, porque é o máis críptico nas rochas alcalino-graníticas que caracterizan a parte exterior das Rías, cubertas ou non por arneirións), e por conseguinte botar luz acerca dos mecanismos evolutivos da clina de cor en *L. saxatilis* das Rías Baixas.

Ademais, e de cara a complementar a información proporcionada polo tethering, en dito estudo fixéronse unha serie de medicións das cunchas dunha mostra de morfos representativa daqueles empregados no experimento: amarelos, negros, e *lineata*. Particularmente, o grao de

cripsis dos distintos morfos avaliouse obxectivamente empregando espectrometría de reflectancia. Dista maneira púidose corroborar, por exemplo, a asunción da que partiu o experimento, é dicir, que o morfo amarelo é o menos críptico nas poboacións expostas e que o *lineata* camúflase moi satisfactoriamente. Mais alá destas medición suplementarias, o experimento de tethering proporcionou evidencia suxerinte de que os queimacasas depredan preferentemente aos caracois amarelos. Así, atopouse unha maior taxa de individuos amarelos con marcas de depredación nunha das dúas sesións experimentais levadas a cabo, e sempre e cando se entendan as fracturas pequenas na cuncha coma casos de depredación. É por isto que as probas a favor da selección por cripsis aportadas por dito experimento son caracterizadas só como suxerintes, en contraposición a concluíntes. Neste sentido, no capítulo discútense as dificultades propias dos experimentos de campo, as distintas posibles explicacións da discrepancia obtida entre sesións do experimento, e posibles fórmulas metodolóxicas para atallar a cuestión. Entre elas, destaca a necesidade dun experimento futuro de laboratorio ou mesocosmos no que se constate que as fracturas pequenas nas cunchas dos caracois constitúen, en efecto, intentos infrutuosos de depredación por parte dos queimacasas.

Capítulo 6: Análises preliminares sobre un caso potencial de evolución paralela dunha clina de cor nas poboacións de *Littorina saxatilis* das Rías Baixas

Finalmente, o sexto capítulo estende o estudo da clina de cor en *L. saxatilis* da Ría de Vigo ao resto das Rías Baixas (Pontevedra, Arousa, e Muros e Noia). Ata o momento, a totalidade de estudos que compoñen a presente tese doutoral estivo realizada na Ría de Vigo. A escasa literatura existente sobre as clinas e o polimofismo de cor das poboacións do bígaro bravo das Rías Baixas destaca a existencia de patróns análogos de variación nas catro rías, motivo polo cal en todo momento asumiuse que a Ría de Vigo é representativa do patrón xeral. Sen embargo, esta hipótese non ten sido nunca avaliada de maneira sistemática. Por ese motivo, o sexto e último capítulo da tese tivo como obxectivo contrastar dita hipótese.

Dende fai tempo, un dos temas máis transcendentais na bioloxía evolutiva é se a evolución é repetible: evolucionan os organismos as mesmas adaptacións ante condicións ecolóxicas similares, ou pola contra, a evolución das distintas liñaxes incorpora elementos exclusivos non presentes noutros grupos ou poboacións? A primeira hipótese subliña o papel da selección natural como xeradora de homoxeneidade evolutiva, mentres que a segunda salienta o rol do azar, en gran medida causado por deriva xénica e outros mecanismos aleatorios. Para

estudar esta clase de cuestións, os biólogos evolutivos a miúdo fan uso de *experimentos naturais*, isto é, de poboacións, enclaves e procesos evolutivos que recrean unha serie de condicións análogas de maneira independente. Comparando ditas poboacións e os seus hábitats, estes investigadores poden chegar a conclusións acerca de en que medida a evolución é repetible.

Pola súa xeomorfoloxía histórica, as Rías Baixas asemellan un bo candidato a experimento natural: todas elas teñen unha orixe recente, unha forma semellante, e, supostamente, unhas condicións ecolóxicas homologables. Do mesmo xeito, *L. saxatilis* tamén parece un organismo axeitado para estudar procesos de evolución repetida ou paralela. En primeiro lugar, como xa se ten dito, é unha especie que presenta unha gran facilidade para evolucionar adaptación locais. E segundo lugar, e máis importante, xa téñense descrito casos de presumible evolución paralela no bígaro bravo. Así, os ecotipos *Crab* e *Wave* asemellan ter evolucionado de maneira independente en distintas rexións das costas europeas. Por todo isto, a motivación última de estudar se o patrón clinal en *L. saxatilis* é homologable nas catro rías é comezar a botar luz na cuestión de se a evolución é repetible.

O sexto e último capítulo da tese está formado por unha serie de análises preliminares que constitúen os primeiros pasos nesta tarefa. Máis en concreto, estes análises buscaban acadar tres obxectivos: en primeiro lugar, confirmar que, en efecto, as Rías Baixas son xeomorfoloxica e ecoloxicamente homoxéneas. En segundo lugar, establecer en que grao existe unha clina de cor homologable nas poboacións de *L. saxatilis* das catro rías. E en terceiro lugar, avaliar se existe co-variación entre distintos caracteres de cuncha para poder formular hipóteses acerca das presións selectivas que manteñen a clina e a base xenética da cor. Para isto, analizáronse máis de 3000 caracois recollidos en 55 localidades das catro Rías Baixas. En cada unha destas localidades tomáronse datos dalgunhas das súas propiedades físico-químicas, así como da presencia e abundancia de varias especies do intermareal. Nos exemplares de *L. saxatilis* colleitados, pola súa banda, medíronse diversos caracteres de cuncha, coma súa cor, forma, grosor, tamaño ou escultura.

En termos xerais, atopouse que as Rías Baixas son moi similares en termos xeolóxicos e, particularmente, en termos ecolóxicos. Todas elas mostran un gradiente de exposición ás ondas dende as partes interiores ata as exteriores, se ben a linearidade desta relación varía entre rías, sendo maior en Vigo en Muros e Noia que en Pontevedra e Arousa. Ademais, en todas elas confirmouse unha clara relación entre as cores de *L. saxatilis* e o grao de exposición ás ondas: as localidades protexidas mostran poboacións con unha maioría de individuos branco-crema, acompañados de amarelos e marróns, mentres que nas áreas expostas abunda o morfotipo

lineata, con fondo branco e liñas escuras. Non obstante, a forza da asociación entre cores e ecosistemas es, de novo, variable entre rías (maior en Vigo e Muros e Noia, e menor en Arousa e Pontevedra). En terceiro lugar, atopouse que a maioría de caracteres de cuncha analizados tamén varían no gradiente de exposición ás ondas, e que existe covariación entre estes e a cor dos caracois. Así, os caracois branco-crema das zonas protexidas tenden a ser máis grandes, e a ter unha cuncha máis grosa, lisa, aguzada e cunha abertura menor. Por contra, os *L. saxatilis* da variedade *lineata* das rexións expostas tenden a ser máis pequenos e cunha cuncha máis fina, con costelas máis marcadas, máis redonda e cunha abertura de maior tamaño. Sen embargo, e a pesares desta covariación, os análises suxiren que existe variación na cor que é independente da variación no resto de caracteres fenotípicos. Todas estas descubrimentos confirman as Rías Baixas coma un potencial experimento natural, axeitado para o estudo da evolución paralela. Ademais, suxiren diversas hipóteses acerca do mantemento da variación clinal nas *L. saxatilis* desta contorna. En concreto, en base aos datos proporcionados, proponse que as cores do bígaro bravo mantéñense por selección por cripsis, sendo os probables depredadores visuais peixes percomorfos. O resto de rasgos de cuncha correlacionados coas cores constituirían adaptacións á depredación por caranguexos da especie *Carcinus maenas* nas localidades protexidas, e ás ondas nas localidades expostas.

En resumo, a presente tese doutoral constitúe un primeiro paso na comprensión da variación de cor nas poboacións de *L. saxatilis* das Rías Baixas. Mentres que a variación intrapoboacional parece estar sendo mantida alomenos en parte por SSNDF, a variación clinal pode estar séndoo por selección por cripsis. A tese remata cunha posta en conxunto dos resultados obtidos e unha discusión de posibles vías futuras de investigación. Aínda que queda moito traballo relacionado coa ecoloxía evolutiva e xenómica deste fenómeno por realizar, a labor levada a cabo durante estes anos senta as bases para futuras investigacións neste eido.

General introduction

This dissertation is about color variation, using *Littorina saxatilis* and the Rías Baixas as the study organism and area, respectively. Due to practical considerations, I will distinguish between intrapopulation color variation, also known as color polymorphism, and geographic color variation, within which I'll focus on color clines. In the real world, both phenomena are intermingled, so their conceptual separation might seem somewhat artificial. But as some philosophers of science have argued (e.g., Potochnik, 2017), oversimplified (or even outright wrong) models of phenomena are unavoidable in science, as otherwise, the complexity of reality would knock us down before we even start to study it. I hope, nonetheless, that with this simplification I can shed light on the nature of color variation and provide an adequate context for the ensuing chapters of the dissertation.

Before starting, I should also note that references to *Littorina saxatilis* and the Rías Baixas have been intentionally reduced in the first two sections of this General introduction. These topics are covered in detail in the third section and the main chapters of this dissertation. This approach allows for a deeper exploration of intrapopulation and clinal color variation in general, regardless of the specific organism and area studied. Additionally, several subsidiary topics of significant interest to the evolutionary study of color polymorphism and color clines, though not directly related to this dissertation, are included in Appendices 1 and 2, respectively.

1.- Intrapopulation color variation: Color polymorphism

1.1.- Definition of color polymorphism

One way a species can exhibit color variation is through color polymorphism (Bond, 2007; White & Kemp, 2016). Color polymorphism is a type of polymorphism that involves the coloration of an organism, so before discussing its mechanisms, it's important to clarify the precise meaning of *polymorphism*. According to traditional definitions (e.g., Ford, 1940, 1945; Huxley, 1955), polymorphism (from Ancient Greek *polús* –many–, and *morphé* –form, shape–) refers to the temporary or permanent co-occurrence of two or more genetically dependent discrete variants of a trait in the members of a species. More specifically, this variation must affect individuals of the same sex and developmental stage that coexist in an interbreeding population inhabiting a particular area. In addition, according to traditional accounts, the different morphs of a polymorphic species must occur at such frequencies that the rarest among them is not solely maintained by recurrent mutation (Ford, 1945). Instead, some evolutionary force must act within the population to sustain the polymorphic condition over multiple generations, even if it eventually disappears.

So, putting all these requirements together, we have that a polymorphism represents the state of a population in which there is variation in a trait that:

- is discrete;
- has a genetic basis;
- occurs in members of the same sex and developmental stage that coexist in the same spatial region and whose members can interbreed;
- is not maintained by recurrent mutation.

Consequently, the previous definition excludes all cases where the variation in a trait is continuous, plastic, or not directly dependent on genes (*polyphenism*; Mayr, 1963), seasonally induced (*seasonal color molting*; Zimova *et al*., 2018), affects only individuals of different sexes (*sexual dimorphism*; Mesnick & Ralls, 2018), occurs in separate populations or geographically distant regions (*geographical variation*; Olvido & Mousseau, 2012), or is present at such a low frequency that is maintained by a repeated mutational process (i.e., *recurrent mutation;* Futuyma & Kirkpatrick, 2017). Unsurprisingly, such a stringent definition has led many researchers to abandon it in favor of a more liberal stance regarding the criteria that a polymorphism must meet (e.g., Roulin, 2004; Gefaell *et al*., 2023).

In line with the previous paragraphs, a color polymorphism is a type of genetic polymorphism that involves the color of a given integumentary structure. This could include the skin of an anuran (Hoffman & Blouin, 2000), the cuticle of an arthropod (Nokelainen, Stevens, & Caro, 2018), the shell of a mollusk (Cook, 2017), the scales of a lizard (Abalos, 2021; Madsen *et al.*, 2022), the coat of a mammal (Ramey & Nash, 1976), or the beak and the feathers of a bird (Galeotti *et al.*, 2003; Roulin, 2004; Enbody *et al.*, 2021) (Figure 1). Since the color of the integument depends on both pigments and nanostructures (Cuthill *et al.*, 2017), any trait that shows a discrete and genetically dependent variation in these components could, in principle, exhibit a color polymorphism.

1.2.- Maintenance mechanisms of color polymorphism

How color polymorphism –or any other kind of genetic polymorphism– is maintained in populations has historically been, and continues to be, a central question in evolutionary biology (for a historical recount see Appendix 1, section A1.1). In a way, color polymorphism exemplifies the 'breaking' of evolutionary rules, as natural selection and genetic drift, the main evolutionary processes driving population evolution, generally tend to erode genetic diversity and consequently reduce polymorphisms (Lewontin, 1974; Nielsen, 2005). Thus, it is unsurprising that the evolutionary transition from color polymorphism to monomorphism is much more common than the reverse (Hugall & Stuart-Fox, 2012). Interestingly, however, the very same mechanisms that typically eliminate polymorphisms can also help maintain them, provided certain ecological and genetic conditions hold. In other words, both natural selection and, to a lesser extent, genetic drift have been invoked –and sometimes corroborated– as the primary processes behind the persistence of color polymorphism in natural populations.

There are various ways to classify the different maintenance mechanisms of color polymorphism, both in terms of form and content. Elsewhere (Gefaell *et al*. 2023, corresponding to Chapter 1 of this dissertation), we have adopted an algorithm-like approach to sort out these mechanisms based on the direction in which selection operates. Here, I will follow a slightly different approach to avoid overlapping the contents between both texts. I will start off with what most evolutionary biologists consider the chief maintenance mechanism of color polymorphism –negative frequency-dependent selection– and proceed to discuss the remaining mechanisms in order of importance.

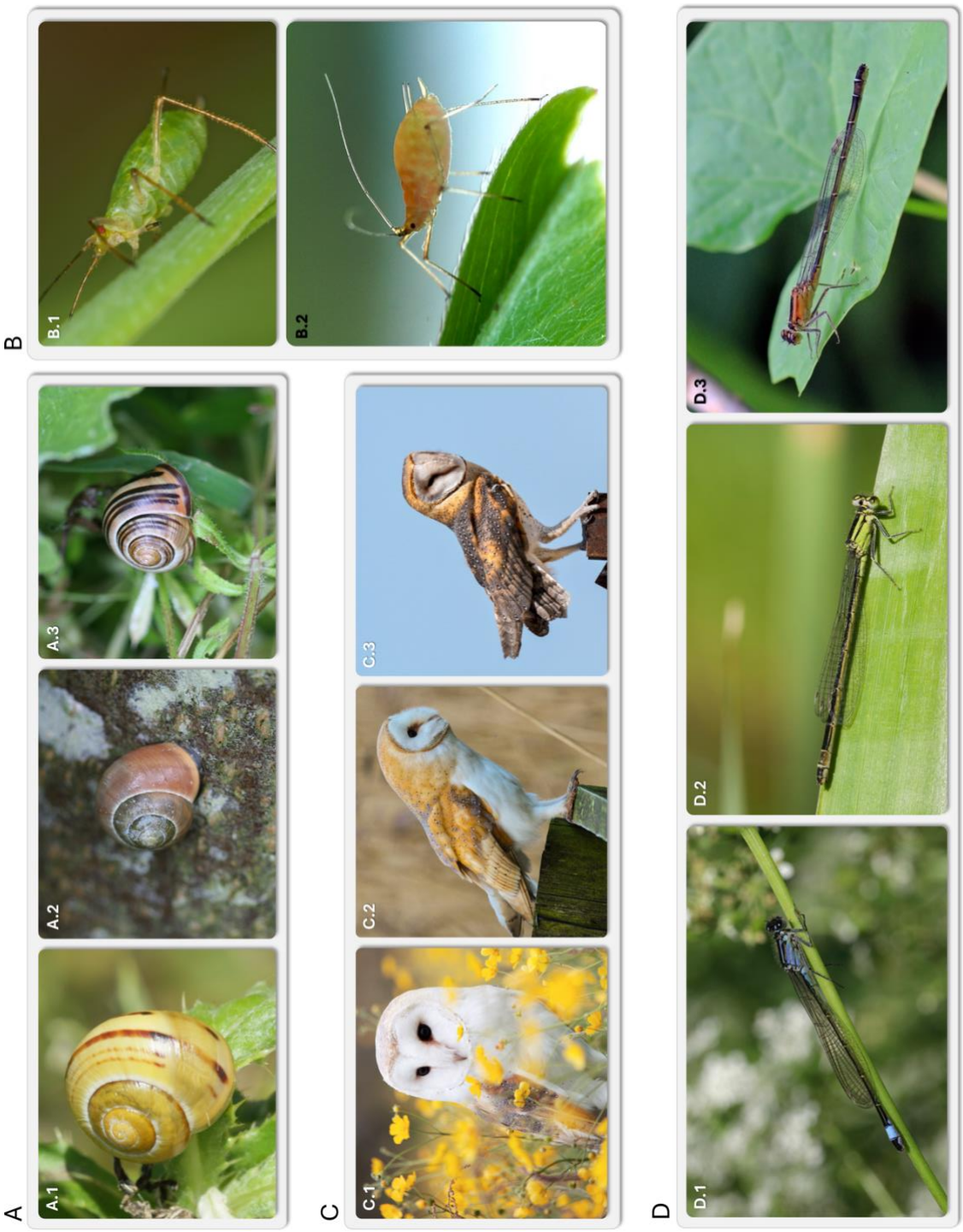

Figure 1. Various examples of color polymorphism that are discussed in the present General introduction. A: Some color morphs of *Cepaea nemoralis*. These morphs vary depending on their background color, as well as the number, width, and intensity of their bands. A.1 displays a yellow individual with faded bands, A.2 a brown bandless snail, and A.3 a banded specimen with fawn-brownish background. B: Pea aphid (*Acyrthosiphon pisum*) color morphs. B.1 shows the green morph, and B.2 the red morph. C: Different (continuous) color morphs of the barn owl (*Tyto alba*). C.1 shows the whitish morph, and a more brownish coloration gradually increasing in C.2 and C.3, respectively. D: *Ischnura elegans* female color morphs; D.1 shows the andromorph morph (resembling a male), D.2 the *infuscans* morph, and D.3 the *infuscans obsoleta* morph. All photos were collected from Wikimedia Commons (CC copyright license). Photographs A.1, A.2, and A.3 were taken by S. Rae; B.1 by Andy Murray, and B.2 by Jpeccoud; C.1 by Andy Chilton, C.2. by Alun Williams, and C.3 by Jens Oldenburg; D.1 and D.3 were taken by Charles J. Sharp, and D.2 by L. B. Tettenborn. None of the photos are at scale.

a) Negative frequency-dependent selection

The most important mechanism for explaining the maintenance of color polymorphism is negative frequency-dependent selection (henceforth NFDS; Ayala & Campbell, 1974; Brisson, 2018). In NFDS, the fitness of a morph varies depending on how rare it is: the rarer the morph, the higher its fitness. This mechanism prevents infrequent morphs from becoming extinct by increasing their reproductive output just as they are getting close to disappearance. Since the fitness of a morph under NFDS is inversely related to its frequency, this mechanism can create balancing situations where the position as the minority morph alternates over time among the different coexisting morphs. Imagine two snail color morphs co-occurring in a population, m_1 and m_2, with m_1 being the commonest and the latter the rarest at a given time, t_1 (see Figure 2). At this point, m_2 will have higher fitness due to its lower frequency, causing its numbers to increase. In contrast, at that same temporal interval (t_1), m_1 will exhibit lower fitness, causing its frequency to decline. This shift can eventually result in m_2 becoming the new commonest morph and m_1 the rarest at a hypothetical time t_2, thereby causing the tide to turn and giving m_1 a higher fitness than m_2, and so on. In some cases, though, populations might reach an equilibrium of frequencies from which it is difficult to deviate.

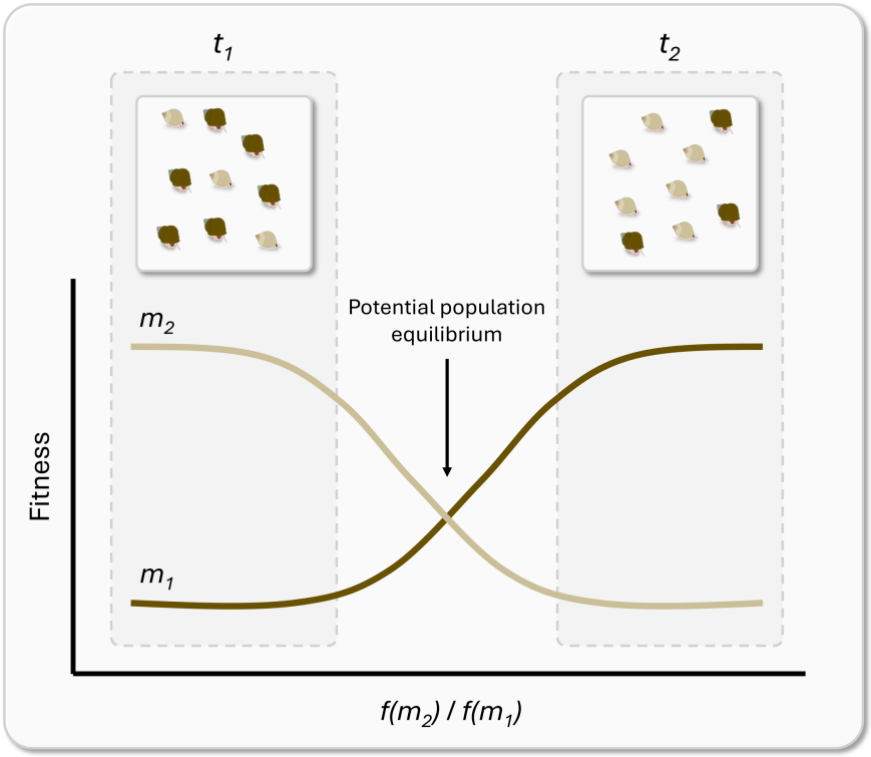

Figure 2. Schematic representation of how NFDS works at a formal level. Here two morphs (m_1 and m_2, corresponding to brown and fawn snails, respectively) show varying fitnesses depending on how rare they are, expressed as the relative frequency of m_2 [$f(m_2)$] divided by the relative frequency of m_1 [$f(m_1)$]. This can generate two different temporal intervals (t_1 and t_2) with contrasting finesses, separated by a potential population equilibrium.

This is how NFDS works at a formal level, but the evolutionary scenarios it generates can vary significantly depending on the ecological and sexual contexts with which you fill it. NFDS can involve interactions between predators and prey, parasites and hosts, or sexes (see two examples in Figure 3). One ecological context that can generate NFDS is apostatic selection. In apostatic selection, a predator forms a search image of its most frequent prey, thereby ignoring the morphs that deviate from this image. This confers the less frequent morphs a survival advantage[1]. A paradigmatic example of apostatic selection might occur in the color polymorphic terrestrial snails from the genus *Cepaea* (Clarke, 1962; Cook, 2017; Figures 1A and 3A; see also Appendix 1, section A1.1c). In this case, rare-colored individuals would have a higher fitness because they would go unnoticed by their main predator, the song thrush (*Turdus philomenos*). These birds would prey by forming images of their most common prey, thus tending to ignore those snails that deviate from the prevalent pattern. Beyond the song thrush, evidence suggests that other predator species also prey by forming search images (Ajuria Ibarra & Reader, 2013; Bond, 2007; Pietrewicz & Kamil, 1979). Putative cases of apostatic selection maintaining a color polymorphism have been described in guppies (*Poecilia reticulata*) and adders (*Vipera berus*) (Olendorf *et al.*, 2006; Madsen *et al.*, 2022; for an extended discussion of apostatic selection, see Appendix 1, section A1.2d).

Besides apostatic selection, other situations can lead to NFDS. One such situation is the rare partner mating advantage hypothesis (Knoppien, 1985), where –typically, but not always– females prefer to mate with males bearing the rarest colors. This process has also been described in guppies (Hughes *et al.*, 2013), although it remains unclear why females of this species prefer rare-colored males. A plausible interpretation is that females might gain indirect fitness effects from mating with the rarest individuals through the mating success of their sons, a possibility known as the *sexy sons*' effect (Potter *et al.*, 2023). A related phenomenon is the mating preference for dissimilar individuals, which can also result in NFDS. In certain polymorphic populations of *Littorina saxatilis* from the Rías Baixas, color polymorphism might be maintained through a male mate choice-driven negative assortative mating, which would in turn result in NFDS (see Chapters 2 and 3 of this dissertation, corresponding to Gefaell *et al*. 2021, 2024a). Lastly, NFDS can also result from non-transitive competitive mating strategies, such as those described in the side-blotched lizard (*Uta stansburiana*) (Sinervo & Lively, 1996).

[1] Although some researchers (e.g., Abalos, 2021; Madsen *et al*., 2022) use NFDS and apostatic selection interchangeably, I distinguish between the two here. As used in the text, NFDS refers to the formal evolutionary process expressed in terms of frequencies and fitness consequences, while apostatic selection corresponds to a particular proximal mechanism by which NFDS can occur (i.e., the formation of images of the target organism). However, I acknowledge that both are closely related, so in many contexts it would not be problematic to treat them as synonymous.

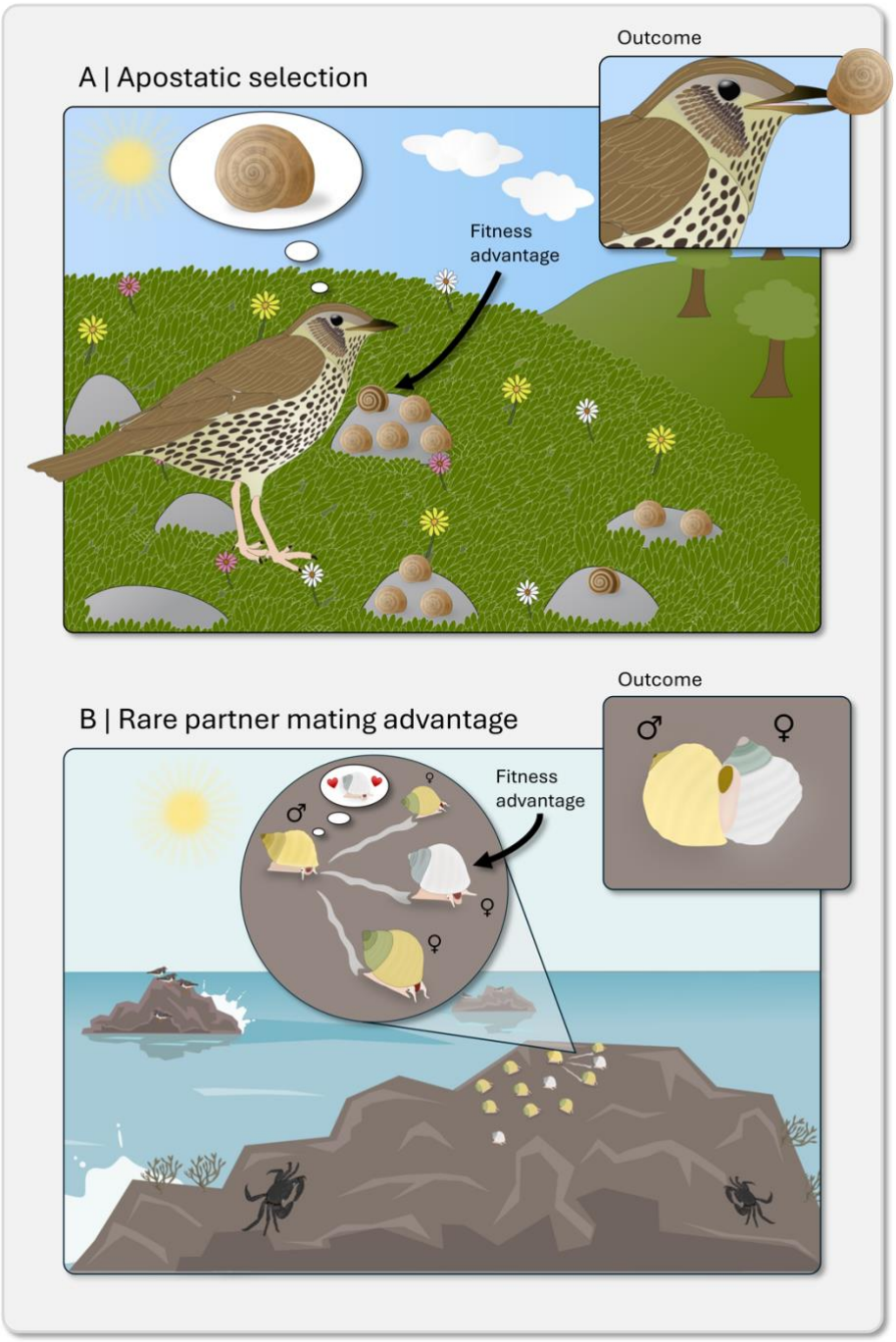

Figure 3. One ecological and one sexual context that can eventually result in NFDS. A: Apostatic selection. A predator bird (in this case a song thrush, *Turdus philomenos*) preys upon the most frequent morph (without bands) of a *Cepaea* sp. population because the predator has formed a search image of their commonest prey. This indirectly confers the rare morph (banded) a fitness advantage. B: A potential case of rare partner advantage in *Littorina saxatilis*, by which the least frequent morph (white) is preferentially chosen by the picking sex (in this case, the male). This example is hypothetical, although a relatively similar process, by which male *Littorina saxatilis* prefer to mate with dissimilar color morphs, has been invoked as an explanation for the maintenance of color polymorphism in certain populations of this species from the Rías Baixas (see Chapters 2 and 3).

b) Disruptive selection: Environmental heterogeneity

Another mechanism that can theoretically promote the maintenance of color polymorphism is disruptive selection in spatially, temporally, or ecologically heterogeneous habitats. The main functions of animal color patterns are thermoregulation, communication with conspecifics, and defense against predators (Endler, 1978; Endler & Greenwood, 1988). Thus, spatial or temporal variation in the thermal or illumination properties of a habitat, or in the types of predators it faces, can lead to the evolution of color polymorphism.

For instance, consider crypsis, or the concealment of the organism by resembling a random sample of the background color pattern of its habitat (Endler, 1978). This process can favor the evolution of color polymorphism if (1) different morphs resemble the different patches of a spatially or temporally diverse background, as perceived by a given predator; or if (2) each morph is adapted to concealment from different predator species with dissimilar sensory ecologies operating over the same background. In either case, each color morph would represent an alternative phenotypic optimum in a bimodal (or multimodal) distribution brought about by disruptive selection.

There are several examples illustrating how spatially heterogeneous environments can drive the evolution of color polymorphism (see Bond, 2007; Nilsson & Ripa, 2010 and references therein). Indirect evidence for this can be seen in the uneven distribution of color morphs among different habitats, as demonstrated in species like the orthopteran *Acrida ungarica* (Pellissier *et al.*, 2011) and several many marine gastropod species discussed in Gefaell *et al*. (2023). Stronger support for this hypothesis comes from experimental studies. For instance, experiments with the stick insect *Timema cristinae* have shown that different color morphs exhibit lower vulnerability to predation by lizards and birds in their preferred microhabitats, where they appear to be more cryptic compared to competing morphs (Sandoval, 1994).

Different predators with presumably distinct sensory ecologies acting in opposite directions can also drive the evolution and maintenance of color polymorphism. A potential illustration of this phenomenon is observed in the pea aphid (*Acyrthosiphon pisum*), which exhibits color-dimorphic populations of green and red morphs (Figure 1B). This aphid species faces parasitoidism by the wasp *Aphidius ervi* and predation by the ladybird beetle *Coccinella septempunctata*. Despite being able to learn from experience (Langley *et al.*, 2006; see Appendix 1, section A1.2d), *A. ervi* wasps tend to infect the green morph (the more common color), while *C. septempunctata* prefers to prey on red individuals (Losey *et al.*, 1997). Consequently, under conditions of high parasitoidism rates relative to predation (influenced by *A. ervi* density), the frequency of red morph individuals increases. Conversely, higher predation rates relative to

parasitoidism (due to higher *C. septempunctata* densities) lead to an increase in the green morph (Losey *et al.*, 1997). Although the role of crypsis in this interaction is not entirely clear, the density dynamics of these two predator species seem to contribute to the pea aphid color dimorphism through disruptive selection.

c) Selection on correlated characters and correlational selection

So far, we've assumed that color is the primary target of selection, directly contributing to fitness. However, this isn't always the case. Color polymorphism tends to go hand in hand with variation in a suit of behavioral, physiological, anatomical, or life history traits, meaning different color morphs may differ in one or more of these features, if not in all (Mckinnon & Pierotti, 2010). This entails that color polymorphism can persist in a population because color variation, which may be neutral or quasi-neutral for a particular selective pressure, is genetically linked to variation in more adaptive traits. This can be called *selection on correlated characters*, as selection does not act directly on color but on its correlated traits. Alternatively, color variation can contribute directly to fitness, but only when it occurs alongside specific trait combinations. This is termed *correlational selection* (Lande & Arnold, 1983; Sinervo & Svensson, 2002; Svensson *et al.*, 2021).

Correlations between color and other traits abound in color polymorphic species. For instance, the barn owl (*Tyto alba*; see Figure 1C) exhibits associations between color and tail length, among other traits (Roulin, 2006). Similarly, female color morphs of the blue-tailed damselfly, *Ischnura elegans* (Figure 1D), display covariation with morphology, growth rate, and overall size. The androchrome morph, which resembles male conspecifics, tends to grow faster, be larger compared to at least one other coexisting morph in the species, and exhibit a more male-like morphology (Abbott & Svensson, 2008). A comprehensive compilation of correlations between color polymorphism and other phenotypic characteristics can be found in Table 1 of Mckinnon & Pierotti (2010), highlighting the widespread nature of these associations.

As mentioned, the role of color in these correlations can either be neutral or quasi-neutral, or pivotal for survival –a distinction that allows us to differentiate between selection on correlated characters (when color is neutral for a specific selective pressure) and correlational selection (when color directly contributes to fitness). While these categories often overlap, this distinction helps to make sense of certain scenarios. For example, different color morphs of *Littorina saxatilis* from the White Sea appear to show varying physiological tolerances to saline stress (Sokolova & Berger, 2000). In this case, shell color is likely neutral under saline stress selection, and color polymorphism might have evolved due to its genetic association with stress

tolerance mechanisms. Conversely, in some insect species, integument coloration correlates with habitat selection (Jaenike & Holt, 1991). Here color seems to play an active role in adaptation: morphs preferring microhabitats in which they are conspicuous instead of cryptic would be penalized by natural selection. Therefore, the optimal fitness strategy would typically involve a combination of a specific color and a behavioral preference for cryptic habitats.

Box 1. Genetic mechanisms of selection on correlated characters and correlational selection

At least two different and non-exclusive genetic mechanisms can contribute to the association of color with other traits: linkage disequilibrium and pleiotropy (Mckinnon & Pierotti, 2010). Linkage disequilibrium refers to the non-random association of two alleles at two or more loci (Futuyma & Kirkpatrick, 2017; Sætre & Ravinet, 2019). Linkage disequilibrium may in turn be caused by non-random mating, certain mutations (such as chromosomal inversions; Mckinnon & Pierotti, 2010), genetic drift, or correlational selection itself, in a potential example of reciprocal causation (e.g., Svensson, 2018). However, unless such correlational selection is persistent, recombination may erode linkage disequilibrium over time (Svensson *et al.*, 2021). Putative evidence of a link between correlational selection involving a color polymorphism and a chromosomal inversion causing linkage disequilibrium has been suggested in *Heliconius* butterflies (Kronforst *et al.*, 2006). Pleiotropy, on the other hand, refers to a single gene directly or indirectly affecting multiple phenotypic traits. For example, it has been posited that genes responsible for producing brown to black eumelanin in vertebrates also influence various physiological and behavioral traits (Ducrest, Keller, & Roulin, 2008).

Although not much is known yet about the genetic underpinnings of color polymorphism and its association with other phenotypic traits, this area constitutes a cutting-edge research program in contemporary evolutionary biology (San-Jose & Roulin, 2017; Svensson *et al.*, 2021), so presumably much more will be learned about this phenomenon in the upcoming years.

d) Heterozygote advantage

A further mechanism that can contribute, at least in theory, to the maintenance of color polymorphism is heterozygote advantage, where heterozygotes for a specific locus show higher fitness compared to their homozygotic counterparts (Futuyma & Kirkpatrick, 2017). This phenomenon can prevent recessive alleles from disappearing, thereby maintaining a polymorphic condition in the population. Heterozygote advantage has been well-documented in genetic polymorphisms like sickle cell anemia (Futuyma & Kirkpatrick, 2017; Ridley, 2004). However, few examples have so far been described where this mechanism contributes to the maintenance of color polymorphism. A recent exception might be the aposematic polymorphic

moth *Arctia plantaginis* (De Pasqual *et al.*, 2022), which exhibits two color morphs (white and yellow) that depend on a single locus with two alleles, *W* and *y*. Genotypes with *WW* and *Wy* produce the white morph, while *yy* genotypes result in the yellow morph. Interestingly, this locus also influences several life history traits, and white heterozygotes (*Wy*) have been shown to have higher fertility, hatching success, and offspring survival than the white and yellow homozygotes (*WW* and *yy*), in a case of entangled heterozygote advantage and pleiotropy (De Pasqual *et al.*, 2022; see Box 1). Heterozygote advantage has also been associated with negative assortative mating for color morphs due to potential immune benefits of heterozygotes (Maisonneuve *et al.*, 2021 and references therein).

e) Non-selective mechanisms

Finally, the last category of mechanisms that can bring about color polymorphism encompasses non-selective evolutionary forces. As recurrent mutation is explicitly excluded from color polymorphism by definition, the remaining non-selective mechanisms include, essentially, genetic drift and migration. Genetic drift refers to the stochastic changes in allele frequencies, which can lead to the fixation or loss of specific genetic variants despite their neutral or quasi-neutral contribution to fitness (Futuyma & Kirkpatrick, 2017). Darwin himself suggested genetic drift as a potential explanation for some cases of polymorphism in *On the Origin of Species*: 'Variations neither useful nor injurious would not be affected by natural selection, and would be left a fluctuating element, as perhaps we see in the species called polymorphic' (Darwin, 1859: 63–64). Population genetics theory, however, emphasizes that genetic drift typically erodes genetic diversity over time, and therefore, polymorphism (Lewontin, 1974).

In the specific context of color polymorphism, few studies have explicitly assessed the role of drift. It is generally acknowledged that genetic drift operates in populations, particularly those with low effective population size (N_e) (Caballero, 2020), suggesting that stochastic processes could contribute to fluctuations in color morph frequencies over time. One example of the role of genetic drift in color polymorphism is observed in the northern leopard frog, *Rana pipiens* (Hoffman *et al.*, 2006). Using molecular tools, it has been shown that the polymorphism of this species is maintained solely by genetic drift. Similarly, a molecular approach revealed that genetic drift plays a significant role in shaping population divergence in morph frequencies of the Skyros wall lizard (*Podarcis gaigeae*) from the mainland island and its associated islets (Runemark *et al.*, 2010). Fluctuations in gene frequencies related to color polymorphism over a 60-year period in the scarlet tiger moth *Callimorpha dominula* -have also been attributed to

genetic drift (O'Hara, 2005). Random changes have also been detected in the frequencies of *Littorina saxatilis* color morphs from the Ría de Vigo (see Gefaell *et al*. 2024b, corresponding to Chapter 4 of this dissertation).

Box 2. Adaptationism and the study of color polymorphism

Whether selection or non-selective forces are the ultimate causes of evolution has probably been one of the most enduring debates in evolutionary biology. In population genetics, the neutralist/selectionist controversy dominated most discussions in the field during the second half of the twentieth century. As Richard Lewontin (1929–2021) put it in a Population Genetics Group meeting held in Bangor in 1971, paraphrasing several texts from the Christian tradition, by that time population genetics was 'divided into two warring sects. These are the adherents to the Epistle of St. Sewall [Wright] to the Japanese [Motoo Kimura], who believe that *the race is not to the swiftest nor the battle to the strong … but time and chance happeneth to them all* [i.e., that most variation is due to neutral causes], and the followers of St Ronald [Fisher], who believe that *many are called but few are chosen*.' (Charlesworth & Charlesworth, 2017: 2) A similar debate swept through evolutionary ecology, which prompted Stephen J. Gould (1941–2002) and, again, Richard Lewontin, to denounce adaptationism and what they called the *Panglossian paradigm*, or the tendency of many biologists to assume that most traits are adaptations evolved by natural selection without sufficient evidence (Gould & Lewontin, 1979). Gould and Lewontin's paper, in turn, elicited responses from prominent biologists (e.g., Lande & Arnold, 1983; Mayr, 1983). For example, Ernst Mayr (1904–2005) conceded that Gould and Lewontin were partly right in their complaint, but argued that if we do away with adaptationism, we might be throwing the baby out with the bath water. Properly applied, adaptationism constitutes a powerful research program that has helped to solve many puzzling features of organisms and populations.

Regarding color polymorphism, many theoretical and empirical arguments suggest that this condition is in most cases maintained by selective forces (see Gefaell *et al.*, 2023; Chapter 1 of this dissertation). Accordingly, the relative significance (sensu Beatty, 1997) of non-selective mechanisms might be lower than that of selection. However, we must not forget Gould and Lewontin's admonition. Non-selective forces can play a role as well in the maintenance of color polymorphism. Therefore, commitment to the adaptationist program, although warranted due to its explanatory power, should be accompanied by an awareness of its limitations. In practice, this means that a fair shot should be given to non-selective forces when testing different maintenance mechanisms for color polymorphism.

As for gene flow (i.e., the exchange of genetic material between populations; Futuyma & Kirkpatrick, 2017), it can theoretically contribute to the maintenance of color polymorphism if there is constant migration of morphs between monomorphic adjacent populations (a situation that sometimes can also result in clines; see the next section). Consider two morphs adapted to specific habitats in nearby areas, each representing a phenotypic optimum in its respective

habitat. Despite potential selective disadvantages in arrival populations, migrating individuals can maintain a transient color polymorphism if there is enough migration, especially if the selective disadvantage is not severe. This scenario has been invoked as an explanation for why color polymorphism exists in aposematic species, where strong positive frequency-dependent selection normally favors monomorphism (Briolat *et al.*, 2019; see also Abalos, 2021). On other occasions, gene flow can interact with selective forces to generate color polymorphic populations, as observed in the isopod *Idotea baltica* (Merilaita, 2001).

2.- Geographic color variation: Color clines

2.1.- Definition of color clines

In addition to color polymorphism, species can exhibit geographic color variation, where populations show diverse colorations across areas (Endler, 1977; McLean & Stuart-Fox, 2014). A specially interesting case of geographic variation is clines (from Ancient Greek *klinein* –to lean–), in which there is a measurable gradient or a regular progression in the species' coloration along a geographic transect (Ridley, 2004; Futuyma & Kirkpatrick, 2017; Gefaell *et al.*, 2023). Typically, this gradual color change goes hand in hand with steady variations in one or more environmental parameters (Mayr, 1963), such that the latter is the ultimate cause of the former, although this doesn't always need to be the case.

A measurable gradient in coloration can manifest as either a continuous change in the average value of a quantitative chromatic parameter (e.g., hue or brightness) or a steady variation in the frequencies of a series of discrete morphs, sometimes referred to as a *morph ratio cline* (Wunderle, 1981; Smith & Smith, 1998; Gefaell *et al.*, 2024b). The latter case illustrates how color polymorphism and geographic color variation can intermingle in natural populations, as the relative frequencies of coexisting color morphs change over a geographic range. As happens with color polymorphism, whose classic definitions include several stringent conditions, orthodox accounts of clines (i.e., Huxley, 1938; Mayr, 1963) also impose some strict criteria. Among these, the most important is that clines must allegedly affect, or be defined by, a single trait, which means that, for a given set of populations of a species, there could be as many clines, each exhibiting its own pattern, as traits showing geographic variation.

Box 3. Concepts related to clines

Discussions of color clines (or clines in general) in the literature heavily rely on several related terms. While these terms purportedly refer to different phenomena, they are not always easily distinguishable from clines (Endler, 1977; Barton & Hewitt, 1985). Among these concepts, *ecotype* and *hybrid zone* stand out:

-*Ecotype*: An ecotype is a population or set of populations of a species that represents a local genetic adaptation to a particular environment (Futuyma & Kirkpatrick, 2017; Smith & Smith, 1998; Turesson, 1922). Ecotypes would differ from clines in that (1) they involve multiple traits rather than just one (Lowry, 2012); (2) they do not necessarily show a gradual progression with other populations of the species; and (3) they are usually defined in terms of a series of selective pressures that would have driven their evolution. Nevertheless, distinguishing between ecotypes and clines is often challenging, prompting some authors to argue for the dissolution of the dichotomy (Linhart & Grant, 1996; Stebbins, 1950; but see Lowry, 2012). After all, as noted in Box 1, genetic mechanisms such as linkage disequilibrium and pleiotropy tend to create co-variations between traits. This means that a putative (color) cline might end up being better described as an ecotypic variation upon closer inspection, as the focal trait (i.e., color) can be found to strongly covary with other phenotypic characters. Moreover, the distinction between both phenomena is blurred because the extremes of a cline can consist of ecotypes genetically adapted to contrasting local environments (as seen with the *Crab* and *Wave* ecotypes of *L. saxatilis*; Westram *et al*., 2018), or because the selective pressures maintaining a presumptive ecotypic variation are not yet well understood.

-*Hybrid zone*: According to Barton and Hewitt's influential paper (1985), hybrid zones are 'narrow regions in which genetically distinct populations meet, mate, and produce hybrids' (113; see also Barton, 1979; Hewitt, 1988). As these authors note, many interrelated but distinct meanings have been conferred to the concept of hybrid zone in the literature, some of them overlapping with both *clines* and *ecotypes*. For example, Barton and Hewitt (1985) observe that some authors have used hybrid zone to refer to narrow clines, where there is an abrupt change in the focal trait over a short geographic range (see section 2.2), while others have applied it to clines involving more than one character (that is, to what has just been characterized as an ecotype). Although Barton and Hewitt express a preference for the term to have been used exclusively for clines maintained by both selection against hybrids and dispersal, they comply with what up to that moment was the prevalent usage of the term, which simply equated it with clines (Barton & Hewitt, 1985). The former definition is assigned by Barton and Hewitt to the notion of *tension zone* (Barton & Hewitt, 1985).

The differences between clines on the one hand, and ecotype, hybrid zone, and related concepts on the other shouldn't be overlooked when studying color clines and their mechanisms, as these distinctions can shed light on the unique aspects of each case. However, these differences should be seen as set in stone, or as if they were 'carving nature at its joints' (to use Plato's famous quote), given the fuzzy boundaries that tear them apart biological phenomena on most occasions (Dupré, 1993). Actually, it is not surprising that the close connection between all these concepts, coupled with the rigidity of the traditional definition of a cline, have led some authors to use these terms rather loosely.

So, wrapping up, color clines are defined as intraspecific variation in coloration that:

- shows a gradient (gradual change or progression) over a geographic transect;
- tends to vary along environmental parameters (but not necessarily);
- can either involve a quantitative chromatic parameter or the frequency of discrete color morphs;
- affects a single trait (in this case color), with different clinal patterns for other characters.

Additionally, it should be noted that clines can arise from either genetic or plastic differences (e.g., Schilthuizen & Kellermann, 2014; Trussell, 2000). Building on the previous section, here I'll focus only on genetically based clinal variation, which would turn the resulting phenotypic cline into a cline in gene frequencies (Futuyma & Kirkpatrick, 2017).

2.2.- Shape and temporal dynamics of color clines

Two properties of clines can help to shed light on their underlying causes: shape and temporal dynamics (Figure 4; for a discussion of other properties, see Appendix 2, section A2.2b).

a) Shape of color clines

Clines can take several forms, but among them, linear and sigmoid are the most common (Figure 4A). Linear or sigmoid clines can be distinguished based on whether the rate of change ($\frac{dy}{dx}$) in the mean value of the trait involved (*y*) per unit of change in a geographic transect or an environmental parameter (*x*) is constant (linear), or whether it changes along the cline (sigmoid). Linear clines can be described using a simple linear equation:

$$y = bx + a$$

where *b* stands for the slope or rate of change in the value of the trait or the frequency of a given color morph (*y*) per unit of transect or environmental parameter (*x*), and *a* for the intercept or value of the trait at the left edge of the cline. Linear clines can in turn vary in their slope, being either shallow (with low *b* values) or steep (with high *b* values), which could be interpreted as how dramatic the change is in the involved trait along the geographic transect.

By contrast, in sigmoid clines, the rate of change in the value of the trait varies along the gradient. These clines can be described with a logistic function:

$$y = \frac{L}{1 + e^{-k(x-x_i)}}$$

where *L* represents the maximum value of the trait (normalized) or the maximum frequency of the focal morph (in case of a morph ratio cline), *k* stands for the steepness of the function (i.e., how quickly goes from the minimum to the maximum value of the trait), and x_i denotes the value of *x* (the unit of transect) at the function's midpoint (i.e., the point of steepest slope). Again, sigmoid clines can vary in their slope.

Closely related to the slope of a cline is its clinal width (sometimes denoted as *w*), which represents the inverse of the maximum slope (Endler, 1977; Mallet *et al.*, 1990). Steep clines tend to have narrow clinal widths, while shallow ones have wide clinal widths. Different factors can cause a cline to have a narrower or wider width (i.e., be steep or shallow). For example, if the environmental parameter changes smoothly along the geographic range, the cline will likely have a wide width, while the reverse would cause the cline to have a narrow width. The latter can happen, for instance, in ecotones (Smith *et al*., 1997). If selection is involved in the maintenance of the cline, then the slope of a steep cline will also depend on the fitness difference of the morphs along the cline (including that of intermediate phenotypes or hybrids) (Endler, 1977; Barton & Hewitt, 1985). Lastly, the intensity of gene flow along the populations of a cline can also exert a substantial influence on its shape (Barton, 1979; Ridley, 2004).

b) Temporal dynamics of color clines

The temporal dynamics that a cline might experience over time –measured as fluctuations in its clinal width, for example– can also provide important insights as to how it is maintained. Clines can either be stable or subjected to change. Changes in dynamic clines can, in turn, be stochastic, cyclical, or directional (see Figure 4B).

Certain clines remain stable over more or less prolonged periods. For instance, by comparing present-day samples to historical data from the 1960s, Ramos-Gonzalez & Davison (2021) found an overall stability in the color clines of *Cepaea nemoralis* from the Pyrenees. Similarly, Gefaell *et al*. (2024b), corresponding to Chapter 4 of the present dissertation, found that, in general, the clinal pattern shown by the different color morphs of *Littorina saxatilis* from the Ría de Vigo (the southernmost ria of the Rías Baixas; see section 3.2) has kept relatively stable over the last forty years. The temporal stability of a cline is a powerful suggestion of natural selection, as no other evolutionary mechanism can account for the absence of change in the shape of a cline over time (Endler, 1986).

In other cases, though, clines do change. Sometimes, this change is stochastic, meaning that no discernible direction of change is detected when the temporal dynamics of the cline are analyzed. This random variation can, in turn, derive from non-selective forces, particularly genetic drift, or from unsystematic variations in the strength of a selective factor. As for genetic drift, studies in *C. nemoralis* over long periods have found idiosyncratic patterns of change in certain populations that are most likely random, in addition to other, more directional changes (Ożgo & Schilthuizen, 2012; Cameron, Cook, & Greenwood, 2013). As for selection, it is well known that the strength of selective pressures can fluctuate from year to year (Siepielski, DiBattista, & Carlson, 2009), which can produce changes in the shape of a cline that are not due to random forces (i.e., genetic drift), yet do not necessarily persist for long periods or show a consistent pattern. For instance, it has been found that the frequency of unbanded (light-colored) snails of *Theba pisana* tends to be higher in years with hotter, sunnier summers (Johnson, 2011). Even though this trend conforms to the thermal melanism hypothesis (see Appendix 2, section A2.2b), by which lighter snails have higher fitness than darker specimens under conditions of high thermal stress, the fact that yearly variation in weather is to some degree stochastic (regardless of the overall directional trend imposed by anthropogenic climate change) makes some of these changes in morph frequencies transient.

Changes in clines can also follow cyclical patterns that presumably recur over time. These include seasonal fluctuations in morph frequencies. For instance, in the color polymorphic damselfly species *Nehalennia*, seasonal variation in morph frequencies has been described in populations from Central Canada (Van Gossum *et al.*, 2007).

Finally, clines can exhibit directional change, such that their shape moves systematically along a certain course. This has been detected, for instance, in the peppered moth, *Biston betularia* (see Appendix 1, section A1.1c). As governments in the UK and the US started to pass legislation aimed at improving air quality, the once populous melanic morph of *B. betularia* began a steady decline that is correlated with decreases in atmospheric sulfur dioxide (Grant *et al.*, 1998; Saccheri *et al.*, 2008). However, nowhere is clinal directional change more tangible than in species affected by climate change. A consistent change in the thermal profiles along a geographic transect due to climate change can alter the frequencies of melanic and non-melanic morphs, which are often maintained by thermal selection regimes. This phenomenon has been described in several taxa, such as gastropods (Ożgo & Schilthuizen, 2012; Cameron *et al.*, 2013; but see Silvertown *et al.*, 2011), insects (de Jong & Brakefield, 1998; Brakefield & de Jong, 2011; Zeuss *et al.*, 2014; Clusella-Trullas & Nielsen, 2020), and amphibians (Hantak *et al.*, 2021). Perhaps the most dramatic case involves the two-spot ladybird (*Adalia bipunctata*). A cline in the

frequency of the melanic morph of this species along a thermal gradient in the Netherlands has gradually become shallower over the last decades, up to the point of virtually disappearing (de Jong & Brakefield, 1998; Brakefield & de Jong, 2011). The rise in temperatures in the coolest parts of the cline, where the melanic form was once fitter due to its ability to absorb more solar radiation, has dissolved this fitness advantage, resulting in uniform morph frequencies across the entire clinal range.

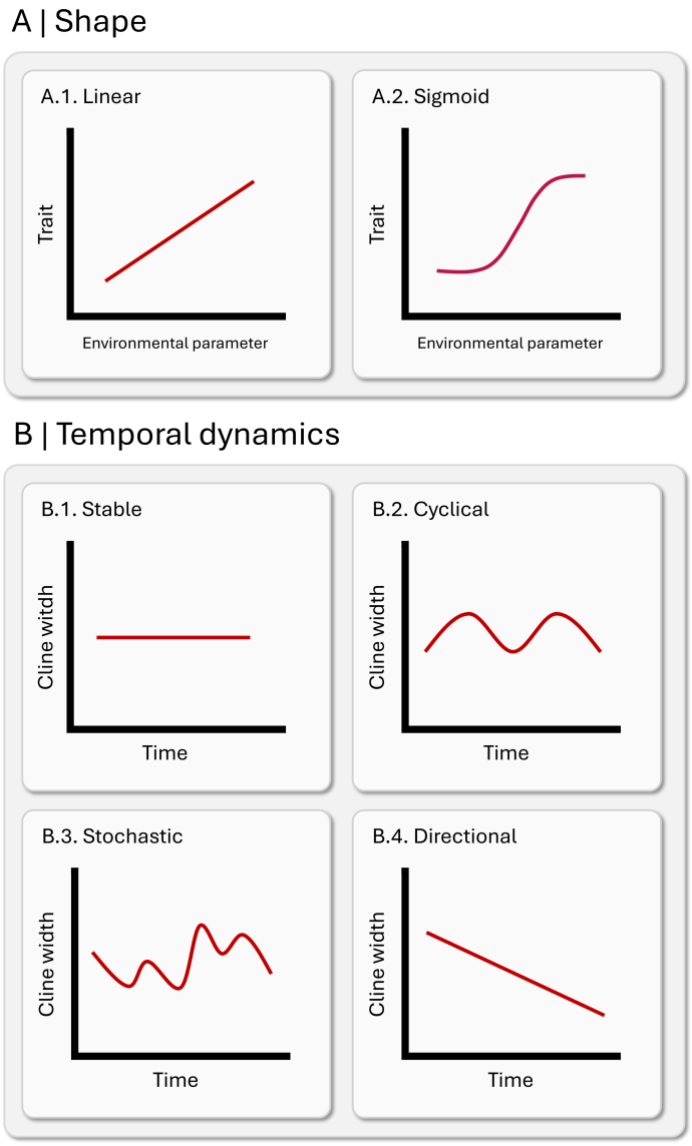

Figure 4. Different clines according to their shape (A) and temporal dynamics (B). Based on shape, clines can either be linear (A.1) or sigmoid (A.2), while based on temporal dynamics, they can either be stable (B.1) or dynamic (B.2 to B.4). Temporal dynamics can be assessed by plotting the clinal width or any other clinal shape parameter against time. Dynamic clines can experience cyclical (B.2), stochastic (B.3), or directional (B.4) change. Directional change (B.4) can also occur in the opposite direction than the one depicted in the graph (with positive slope, or $b > 0$).

2.3.- Maintenance mechanisms and origins of color clines

a) Maintenance mechanisms of color clines

Clines are the result of two contrasting sets of forces: those that contribute to the differentiation of populations (selection and genetic drift), and those that promote their homogenization (gene flow) (Barton & Hewitt, 1985; McLean & Stuart-Fox, 2014; Futuyma & Kirkpatrick, 2017). On the one hand, spatially varying selection promotes the evolution of local adaptations, while genetic drift can favor the fixation of different alleles in different populations. On the other hand, migration prompts the exchange of alleles between populations, which can cause the differences between populations to wither away.

To illustrate how the tug-of-war between these contrasting forces brings about the maintenance of color clines, let's focus on the interplay between selection and gene flow –that is, let's leave genetic drift aside for a moment. The ratio between gene flow (measured either as the migration rate –*m*– or the migration variance –σ_m^2–, depending on whether the cline is patchy or continuous, respectively) and selection (assessed through the selection coefficient –*s*–) is associated with the approximate shape of a cline, as determined by its width (*w*) (Futuyma & Kirkpatrick, 2017). If gene flow is much higher than selection, then it can end up eroding all locally adapted alleles, a process known as gene swamping (Futuyma & Kirkpatrick, 2017). This results in very shallow transitions in allele frequencies. Conversely, if selection significantly outweighs gene flow, locally adapted alleles might become fixed over time, establishing steep clines that could potentially facilitate a speciation process (provided that such balance between gene flow and selection persists; see Appendix 2, section A2.2c).

Typically, though, genetic drift must also be considered, as continuous populations living along a transect often diverge following an isolation-by-distance model (Wright, 1943; Slatkin, 1993). Normally, genetic drift is expected to affect small populations living at the edges of a species range, or populations that are relatively isolated from other demes, as both are normally less influenced by gene flow (McLean & Stuart-Fox, 2014). Genetic drift has been shown to play a role in some color clines. Brakefield (1990), for instance, obtained evidence for genetic drift and founder effects in the populations of the color polymorphic spittlebug *Philaenus spumarius* from the islands of the Scilly archipelago. Similarly, Oxford (2005) proposed a model to explain morph ratio clines in the candy stripe spider (*Enoplognata ovata*), which incorporates genetic drift as a key mechanism. According to this model, drift accounts for fluctuations in a morph when it is present at frequencies between 0.05 and 0.3, beyond which selection starts to operate.

Examples such as these, where genetic drift contributes to explaining a cline, are likely to abound in nature. However, this should not be interpreted as if genetic drift and selection

would have the same explanatory weight. Population genetics theory indicates that even minimal migration between populations is likely to counteract the diverging effects of genetic drift (Futuyma & Kirkpatrick, 2017). Therefore, for a cline to be maintained solely by genetic drift, genetic flow would need to be negligible –an unplausible scenario. The only feasible alternative for a cline to be preserved in the face of gene flow is, then, that spatially varying selection acts alongside genetic drift to accentuate the divergence process.

Given the greater significance of selection compared to genetic drift, from now on I'll focus on the former mechanism. When it comes to clines, a crucial distinction is between endogenous (or intrinsic) and exogenous (extrinsic) selection (Mallet & Barton, 1989; Barton, 2001; Wang *et al.*, 2022). Endogenous selection involves genetic incompatibilities that reduce the fitness of hybrids independently of ecological (i.e., exogenous) conditions, so that a cline can be maintained by a balance between selection against hybrids and random dispersal (Wang *et al.*, 2022). Endogenous selection amounts to a tension zone as defined by Barton & Hewitt (1985). Alternatively, in exogenous selection clinal divergence is driven by environment-coupled fitness differences between genotypes, such that each morph (or trait value) constitutes a local adaptation to a given selective pressure that varies gradually along a transect.

Instances of endogenous selection have been described for chromosomal rearrangements, such as those affecting *Podisma pedestris* grasshopper populations in the Alps. Here, a tension zone has been formed due to reduced fertility of heterozygotes for a Robertsonian translocation (Barton & Hewitt, 1989; see also Futuyma & Kirkpatrick, 2017). As for exogenous selection, various environmental factors that vary along gradients –temperature, substrate, humidity, urbanization, pollution, ecological communities– can directly or indirectly result in the maintenance of a color cline (Burla & Gosteli, 1993; Mullen & Hoekstra, 2008; Johnson, 2011; Wittkopp et al., 2011; McLean *et al.*, 2015; Cosentino *et al.*, 2023; see Appendix 2, section A2.2d for illustrations of each factor). Essentially, any abiotic or biotic parameter exhibiting a continuous progression along a geographic transect can potentially exert selective pressures that lead to the formation of a cline. For instance, the degree of wave exposure, which varies smoothly across the Rías Baixas– explains a substantial portion of the variance in morph frequencies of *Littorina saxatilis*, thus offering insights into the selective pressures at work (Gefaell *et al.*, 2024b; Chapter 6).

In sum, selection pressures can be categorized as endogenous or exogenous, each influencing the shape of a cline. As a final note, though, it is worth mentioning that both endogenous and exogenous selection may act simultaneously within a cline (Wang *et al.*, 2022), and each may interact with mechanisms maintaining intrapopulation color polymorphism. Such

interactions can, in fact, offer comprehensive explanations for certain color clines (McLean & Stuart-Fox, 2014). For example, NFDS (a mechanism for color polymorphism) has been suggested to play a role in the morph ratio cline of the *Ischnura senegalensis* populations from Japan (Takahashi *et al.*, 2011). In this species, the andromorph's fitness is lower than the gynomorph in the southern range of the cline but higher in the north. Despite this variation, both morphs seem to be protected from extinction in localities where they are less frequent due to NFDS (Takahashi *et al.*, 2011). The interplay between spatially varying selection and NFDS could similarly provide the most plausible explanation for the color variation observed in the populations of *Littorina saxatilis* from the Rías Baixas.

Box 4. Tools for the study of color clines

The relative weights of selection and genetic drift in maintaining clines cannot only be deduced from theoretical models but can also be empirically assessed using molecular tools. Among these, population pairwise F_{ST} comparisons between neutral loci and those presumably under selection are among the most popular (e.g., Inomata *et al.*, 2015; Takahashi, 2015). These comparisons quantify differences in the widths of clines between the two sets of loci: if selection is acting on a locus, its clinal width should be narrower than that of neutrally evolving loci. For example, Takahashi *et al.* (2014) compared F_{ST} values for neutral loci and a locus putatively associated with color variation in the damselfly *Ischnura senegalensis*, revealing that selective factors alone maintain the color cline in Japanese populations of this species. Given that selection often takes a leading role in the maintenance of color clines, yet genetic drift also contributes, a useful approach to the study of gradual progressions in color is to consider the isolation-by-distance model as a null hypothesis against which to test for natural selection, much like when studying other evolutionary phenomena (Duret, 2008; but see Pigliucci & Kaplan, 2006).

The exogenous selective pressures contributing to the maintenance of a color cline can be, in turn, studied through various methods, out of which frequentist-observational are probably the most common (Gefaell *et al.*, 2023a). In this approach, the frequencies of color morphs or trait values along a gradient are correlated with more environmental factors using regression analyses, such that the latter is interpreted as a potential cause for the evolution of the former (Endler, 1986; Takahashi, 2015). The inferential power of this approach –its ability to provide conclusive evidence about causal factors– though, is limited. Therefore, manipulative experiments are typically necessary to corroborate whether a specific environmental parameter plays a causal role in maintaining the cline.

b) Origins of color clines

Closely tied to discussions on the maintenance mechanisms of color clines is the question of how these clines are formed in the first place (Endler, 1977). Simply put, clines generally originate either through primary differentiation or secondary contact (Figure 5). In

primary differentiation, environmental variation along a species' geographical transect can lead to population differentiation, primarily driven by spatially varying selection. Alternatively, primary differentiation can occur during a range expansion, in which successive populations colonize previously unoccupied areas. If these newly colonized areas exhibit distinct environmental conditions compared to the species' original range and if gene flow between these populations is not curtailed, a cline can evolve through spatially varying –exogenous– selection. A putative case of color cline developed through primary differentiation is described in Gefaell *et al*. (2024b), corresponding to Chapter 4 of this dissertation, as well as in Chapter 6.

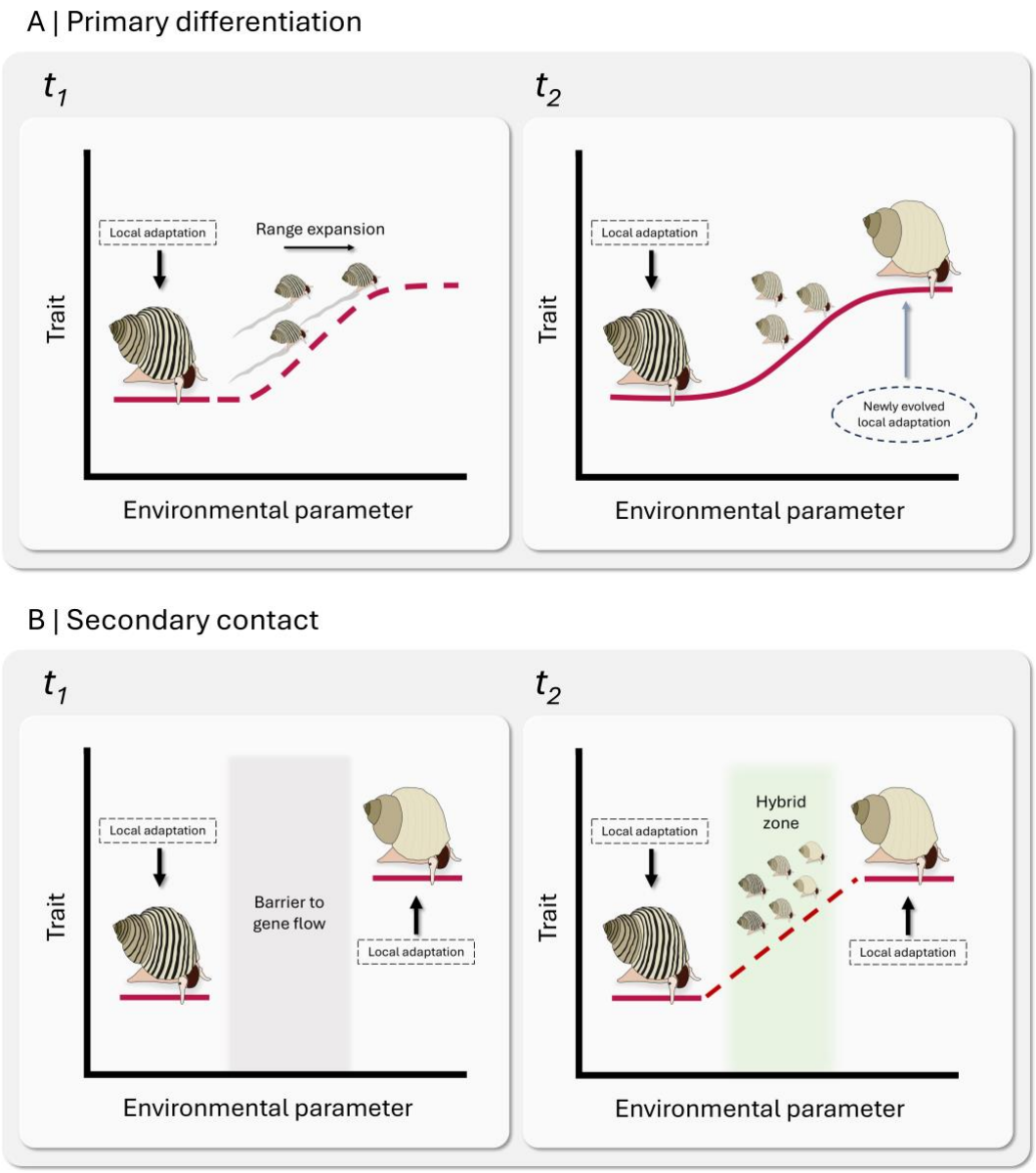

Figure 5. The two main mechanisms for cline formation. A: In primary differentiation, a cline can arise from environmental heterogeneity within a species' range or from colonization of new areas with distinct characteristics (as depicted in the figure). B: Alternatively, clines may form through the contact of previously isolated populations, known as secondary contact.

Conversely, in secondary contact, a cline forms when two isolated populations of a species that had been hitherto evolving in allopatry become connected. A potential case of a color cline resulting from secondary contact has been described in the orthopteran *Hemideina maori* from southern New Zealand (King *et al.*, 1996). This species exhibits two distinct color morphs: a black morph in the southernmost regions and a yellow morph in the northern areas, with intermediate colorations between them. Based on several molecular markers and the distribution of color morphs, King *et al*. (1996) have proposed that this cline might have arisen from the secondary contact of two previously isolated populations with diverging colorations. Normally, secondary contact leads to the formation of a hybrid zone, or more specifically, of a tension zone, in which hybrids display lower fitness than their parental phenotypes. Anyway, clines resulting from secondary contact are also expected to be maintained by exogenous selection; otherwise, the newly established gene flow between the parental populations would result in the loss of their respective local adaptations (i.e., gene swamping).

Determining whether a cline has formed through primary differentiation or secondary contact is of high interest, as it can provide clues on its shape and maintenance mechanisms. Theoretically, clines originated through secondary contact would have steeper clines in gene frequencies (or narrower clinal widths), as allopatry tends to favor population divergence. However, the gene flow initiated after the contact of the parental populations would tend to lessen the genetic differences between populations, thereby making it more similar to clines brought about by primary differentiation. In practice, this means that without historical data it's normally quite difficult to establish whether a given cline has formed through primary differentiation or secondary contact (Barton & Hewitt, 1985). It is for this reason that information on the historical geomorphology of the area in which a color cline occurs is of uttermost importance for its understanding.

3.- Study organism and area

Now that we have addressed the main processes to which this dissertation is devoted – color polymorphism and color clines–, it's time to introduce the organism and area chosen to study them: *Littorina saxatilis* and the Rías Baixas, respectively.

3.1.- Study organism: *Littorina saxatilis*

a) Basic biology of Littorina saxatilis

Littorina saxatilis (Olivi, 1792; Figure 6), also known as the rough periwinkle, is a marine gastropod living in the rocky intertidal ecosystems of the northern Atlantic coasts. In Europe, this

species ranges from the shores of the Iberian Peninsula or even further south, up to those of Iceland (Reid, 1996; Heller, 2015). *L. saxatilis* belongs to the genus *Littorina* (from new Latin *littor-*, seashore), or the winkles, which comprises eighteen species inhabiting intertidal ecosystems worldwide (Rolán-Alvarez *et al.*, 2015a). The winkles are quite abundant in these ecosystems, comprising up to around 85% of their biomass in some instances (Rolán-Alvarez *et al.*, 2015a). *L. saxatilis* is a medium-sized snail within the winkles, measuring around one centimeter in shell height. In the wild, it can live up to nine years in northern latitudes and around two to four in the south (Hughes, 1995; Sergievsky, Granovitch, & Sokolova, 1997; Gefaell *et al.*, 2024b); however, most individuals fall prey to predators before reaching their maximum longevity. The rough periwinkle reproduces throughout the year, with gravid females found in all seasons (Reid, 1996). As for its trophic position, *L. saxatilis* gently grazes the microalgae (particularly diatoms) and other microorganisms that grow on the surface of the rocks (displayed in Figure 6A) and is preyed upon by a variety of seashore predators, including shorebirds, crabs, fishes, and even other predatory gastropods, such as *Nucella lapillus* (Pettitt, 1975). Two especially important predators in the populations of the Rías Baixas are the marbled rock crab, *Pachygrapsus marmoratus*, and the green European crab, *Carcinus maenas*.

Detailed activity budgets for *L. saxatilis* are not yet available (although they exist for other winkles, e.g., Ng *et al.*, 2021), but informal observations repeatedly show that these snails tend to be most active when the tides are close to the supratidal zone –the specific intertidal zone in which they live. This is when the rocks on which they graze and copulate (Figure 6B) have been recently splashed by the waves and are thereby moist. The mobility range of this species is quite limited, moving between one and four meters in three months (Janson, 1983). Furthermore, this movement is not always cumulative and directional, as *L. saxatilis* tends to exhibit homing behavior (Rolán-Alvarez, pers. obs.). During the central hours of the day or at low tides, most individuals avoid desiccation by standing on their mucus (Chapperon *et al.*, 2017; see Figure 6C), thus minimizing the surface contact with the rocks, as other marine gastropods do (e.g., Miller & Denny, 2011; Ng et al., 2017). Additionally, the rough periwinkle avoids heat stress by hiding in crevices with conspecifics, or even with members of other gastropod species (Figure 6D).

b) The role of Littorina saxatilis *in evolutionary ecology*

L. saxatilis has long been a model organism in evolutionary ecology (Reid, 1996; Johannesson, 2003, 2016; Rolán-Alvarez *et al.*, 2015a; Johannesson *et al.*, 2024). This is due to several aspects of its biology and environment. For a start, the rough periwinkle is present at high densities in many intertidal localities (Reid, 1996), and its passive way of life and short mobility

range make it relatively straightforward to sample. Additionally, shells are both important for the organisms bearing them (that is, they are probably under selection; Vermeij, 1993), and easily quantifiable; indeed, several potentially relevant shell features can be measured or scored with relative ease, from shell thickness to shell shape (e.g., Johannesson, 1986; Carvajal-Rodriguez *et al.*, 2005; Larsson *et al.*, 2020). Furthermore, mating events in winkles are quite common, and especially at southern latitudes, they occur throughout the year, making them ideal organisms to study sexual selection (Rolán-Alvarez & Ekendahl, 1996; Rolán-Alvarez & Caballero, 2000; Rolán-Alvarez *et al.*, 2012).

Figure 6. Main behaviors displayed by *L. saxatilis* in the wild. A: Grazing and feeding on microalgae that grow on the surface of the rocks. B: Copulating (male on the left, female on the right). C: Standing on their mucus attached to the rock to avoid desiccation. D: Hiding from sun exposure at a crevice, in this case, bathed by seawater forming a small pond. Photos by J. Gefaell.

So far, so good. But these features are not only shared by virtually all winkles but by many snail species. In that sense, they do not make the rough periwinkle special. What, then, has ignited the interest of many evolutionary ecologists in this humble animal? Part of the answer lies in *L. saxatilis*' remarkable tendency to evolve local adaptations (Behrens Yamada, 1987; Reid, 1996; Rolán-Alvarez, 2007; Johannesson, 2016). This species exhibits various ecotypes or morphs adapted to different parts (or types) of the seashore. This, in turn, is largely due to its

reproductive biology. Instead of releasing their embryos into the water column, as most marine gastropods do, the females of this species carry their fertilized eggs in a brood pouch until these are released as fully-grown juveniles (Reid, 1996). Given that, as we've just seen, the mobility range of the rough periwinkle is remarkably limited, this feature dramatically curbs the dispersal capabilities of their populations. This limitation, in turn prevents gene flow from eroding the potential local adaptations that they can evolve.

In addition, the intertidal ecosystem, far from homogeneous, constitutes a highly diverse environment in several respects (Raffaelli & Hawkins, 1999; Little, Williams, & Trowbridge, 2009). Not only are there different kinds of intertidal ecosystems depending on a suite of abiotic and biotic parameters (such as wave exposure, temperature, salinity, and so on), but also within a given intertidal ecosystem, there are various utterly different microhabitats. Some of these microhabitats follow clinal trends –going, for instance, from the more wave-exposed to the most wave-sheltered areas– while others exhibit a more patchy distribution, like the different lichens growing on the rocks. This astounding diversity creates a myriad of selective pressures, many of them intense, that contribute to the acceleration of local adaptation processes. Moreover, the relatively high number of chromosomal inversions in the *L. saxatilis* genome favors linkage disequilibrium and trait correlations among many fitness-bearing characters, which in turn can tighten the adaptive coupling of certain phenotypes to their microhabitats (Faria *et al.*, 2019; Koch *et al.*, 2021, 2022; Reeve *et al.*, 2023).

Within the *L. saxatilis* system, the most studied variation has been, by far, that related to the *Crab* and *Wave* ecotypes (Rolán-Alvarez, 2007; Johannesson, 2016; Koch *et al*., 2021). These ecotypes represent local adaptations to the different zones of the intertidal ecosystem and have evolved in parallel at least on the Swedish, British, and Spanish coasts (Butlin *et al.*, 2014). The ecology of the *Crab* and *Wave* ecotypes has been extensively characterized during the last three decades, with a genomic perspective added in recent years (Koch *et al.*, 2022). The study of the genomes of these ecotypes has opened up exciting new avenues of research, which is why the *Crab* and *Wave* ecotypes continue to be a flourishing topic of investigation today. However, the genomic characterization of both ecotypes would not have been as valuable as it is without the extensive ecological knowledge gathered in the previous decades.

c) Color variation in Littorina saxatilis

Other traits in *L. saxatilis* have, however, been much less studied. Among these, shell color, a putatively genetically dependent character (Gefaell *et al*., 2023), probably stands out. Even a casual look at the rough periwinkles from many localities reveals the staggering diversity

of shell colorations within this species (see Figure 7). *L. saxatilis* shells can range from the palest white to the darkest black, passing through many intermediate colorations. Alongside this background chromatic diversity, several patterns can be superimposed, including bands, lines, and tessellations. In fact, Reid (1996) documents 35 named color varieties described in the taxonomic literature. Some of these variations suggest a clear functional significance, such as instances where a significant majority of individuals in a population exhibit a shell color that matches the chromatic properties of their habitat background. Other cases of shell color variation, while not initially suggesting an obvious adaptive hypothesis, show some rhythm or pattern, such as color clines involving multiple contiguous populations across geographic or ecological transects. Yet, in a third scenario, neither functional significance nor any discernible pattern can be immediately elucidated just by examining the snails' coloration, thus requiring hands-on field and lab research to uncover their evolutionary maintenance mechanisms.

An area where research in *L. saxatilis*' color variation has been particularly understudied is the Rías Baixas, a series of four coastal inlets in the northwestern part of the Iberian Peninsula (Galicia, Spain; see the next section). Since the 1960s, several studies have been conducted on the populations of this region by early pioneers in *L. saxatilis* research, such as Edouard Fischer-Piette, Jean M. Gaillard, Cesare Sacchi, or Alberto Torelli (Fischer-Piette & Gaillard, 1960, 1961, 1971; Sacchi, 1979; Sacchi & Malcevschi, 1982; Torelli, 1983). Many of these studies are echoed in David Reid's classic textbook (1996), where he notes, among other things, that a '(...) striking trend is found in Galicia, where on exposed coasts shells are mostly a lineated morph and in shelter mostly yellow and fawn, with high colour diversity and less predictable occurrence of morphs at intermediate localities' (321-322). However, beyond the valuable work of these early pioneers –who were primarily interested in taxonomy– no further research has been conducted on the color morphs of these populations. This gap has left many potentially interesting questions unanswered, ranging from how this 'striking trend' looks nowadays to the potential mechanisms maintaining it.

Since the early days of Fischer-Piette and colleagues, many ecological tools have been developed that can help us begin to understand both the intrapopulation and clinal color variations of *L. saxatilis* from the Rías Baixas. Additionally, much knowledge has been accumulated on the evolutionary causes of shell color polymorphism in marine gastropods and many other organisms. Given the paucity of research on this phenomenon –shell color polymorphism in *L. saxatilis* from the Rías Baixas– a crucial first step is to place this phenomenon within the broader context of marine gastropods, the taxonomic group to which it belongs (see Chapter 1 of the present dissertation).

Figure 7. Various color morphs of *L. saxatilis* from the Rías Baixas both in their natural habitats (A, B, and D to H) and in a lab setting (C). Photos by J. Gefaell.

Many of the ecological tools mentioned above are aimed at assessing the contribution of sexual selection to NFDS as a mechanism for maintaining color polymorphism in natural populations (Rolán-Alvarez *et al*., 2012; Rolán-Alvarez *et al*., 2015b; Estévez *et al*., 2018, 2020). For instance, both theoretical models and empirical studies have shown that a mate choice-driven negative assortative mating –where males choose to mate with females of a different color morph– can result in NFDS in the closely related species *L. fabalis* and *L. obtusata* (Estévez *et al*., 2020). Could this process be taking place in *L. saxatilis* as well? This would not only explain the maintenance of color polymorphism in the populations of this species from the Rías Baixas, but also suggest that mate choice is an ancestral behavioral trait in the Neritrema clade (which groups *L. fabalis*, *L. obtusata*, and *L. saxatilis*). This is explored using field data in Gefaell *et al*. (2021), corresponding to Chapter 2. Moreover, in recent years, new experimental laboratory paradigms have been developed to quantitatively assess mate choice in marine gastropods (Lau *et al.*, 2021), making it easier to draw causal inferences regarding the mechanisms underlying the observed negative assortative mating patterns in *L. saxatilis*. Gefaell *et al*. (2024a) and its correlative Chapter 3, addresses that possibility.

Similarly, the rich tradition of cline analysis in evolutionary ecology offers insightful perspectives on how geographic color variation can persist in natural populations. Especially interesting is the analysis of long-term trends in clinal patterns (e.g., Brakefield & de Jong, 2011; Ramos-Gonzalez & Davison, 2021; see Gefaell *et al*. 2024, corresponding to Chapter 4). Furthermore, the literature on shell color polymorphism in marine gastropods, alongside examples from other taxa, suggests various maintenance mechanisms that contribute to the persistence of color variation, such as divergent cryptic selection (see Chapter 5). Finally, a longstanding problem in evolutionary biology is whether evolution is repeatable (Blount, Lenski & Losos, 2018). The putatively repeated color cline of *L. saxatilis* from the Rías Baixas might constitute a promising model to study this topic in the future (Chapter 6). These issues are analyzed in varying degrees of detail throughout this dissertation; however, before delving deeper into them, it is essential to provide further context on the study area (i.e., the Rías Baixas).

3.2.- Study area: The Rías Baixas

a) General overview of the Rías Baixas

In geology, a ria is defined as a type of landform formed by a drowned incised river valley (Evans & Prego, 2003; Méndez & Vilas, 2005; Goudie, 2018). The concept of ria derives from *Riasküsten*, a term first introduced by the German geographer Baron von Richthofen (1833–1905) in 1886, who took inspiration from the Galician rias (Galicia, Spain) to coin it (von Richthofen,

1886). Even though von Richthofen's original definition required the drowned valley to be settled on a terrain that cut transverse to the geological strike (a feature that does not properly apply to the Galician rias; Evans & Prego, 2003), the concept of a ria has ever since been inextricably linked to these characteristic landforms from the northwestern corner of the Iberian Peninsula (Goudie, 2018).

Within the Galician rias, the Rías Baixas, a set of four coastal inlets situated at the south of the region, stand out (Figure 8A). From north to south, the Rías Baixas are composed of the Ría de Muros-Noia, the Ría de Arousa, the Ría de Pontevedra, and the Ría de Vigo. Their names correspond to the major cities bathed by each inlet (with the Ría de Muros-Noia having two small towns, Muros and Noia). All four Rías Baixas, oriented southwest to northeast, are characterized by a funnel shape, with a wide mouth and central area that gradually narrows towards their heads, where their main rivers flow (Muros-Noia: Tambre; Arousa: Ulla; Pontevedra: Lérez; and Vigo: Verdugo). There are, however, slight shape differences between them, with the Ría de Muros-Noia having a wider opening and the Ría de Arousa exhibiting a more meandering coastline. As a matter of fact, all the rias display specific physiographic peculiarities, such as the Bay of Baiona and the San Simón inlet in the Ría de Vigo, the Ría de Aldán in the Ría de Pontevedra, and the Grove inlet in the Ría de Arousa (Méndez & Vilas, 2005). Additionally, three out of the four Rías Baixas –all except Muros-Noia– contain at least one island and a series of subsidiary islets in their mouths (Arousa: Sálvora; Pontevedra: Ons; Vigo: Cíes, in this case, being an archipelago of three islands), which generates two entrances to the rias (north and south) and moderates swell inside them.

In terms of size, the total surface area of the Rías Baixas is approximately 700-750 km^2, with each ria occupying an area of about 125-230 km^2. The Ría de Muros-Noia is the smallest (125 km^2), while the Ría de Arousa is the largest, covering almost twice the area at around 230 km^2 (Méndez & Vilas, 2005). The depth of the Rías Baixas ranges from approximately 40-60 meters along their central axis to about 5-10 meters at their heads, where freshwater from river mouths forms estuaries due to constant flow. Geologists divide the Rías Baixas into two zones based on their hydrodynamic and sedimentologic properties: an inner and an outer area (Méndez & Vilas, 2005). As we will explore further, in addition to gradual changes in salinity due to continuous freshwater inflow from rivers, these inner and outer areas show many other steady variations in physicochemical and ecological parameters that shape the outlook of the Rías Baixas. The main types of rock that make up the Rías Baixas are granite (particularly alkaline-granitoid and calc-alkaline granitoid rocks), and metamorphic rocks, particularly schist and gneiss. Of these, the latter rock types are much less frequent, being interspersed between granitic formations. The

seafloor, in turn, is primarily composed of sandy sediments resulting from rock erosion, along with muds and silts that accumulate in the quieter parts of the rías, normally the inner zones. Biogenic materials are also present, with occasional rock outcrops near the coast (Vilas, Bernabeu, & Méndez, 2005).

Another important topic regarding the oceanography of the Rías Baixas is their currents. Currents in this region are influenced by wind patterns, tides, river inputs, and the topography of the rias themselves, which influences wind direction (deCastro *et al.*, 2000). As for wind, the northerly winds that occur during summer move the surface waters offshore due to the Coriolis effect. This movement causes nutrient-rich bottom waters from the ocean platform to enter the rias and rise to the surface, a process known as coastal upwelling, which accounts for the high nutrient abundance (Prego, Barciela, & Varela, 1999) and, in turn, the elevated shellfish productivity of the Rías Baixas (Figueiras, Labarta, & Fernández Reiriz, 2002). Conversely, during the autumn and winter months, winds come mostly from the south, which pushes surface waters towards the coast and prevents these from getting out of the rias. Tides, which at this zone are semidiurnal (two high and low tides every lunar day) and exhibit a meso-tidal regime (with spring tidal ranges of around 3 meters), move the water masses in and out as the tidal cycle unfolds. The tides are the most energetic process driving water circulation (Alvarez *et al.*, 2005). Finally, river discharges, which are especially intense during the rainy Galician winters, can also affect local currents by intensifying the flow of the less dense surface freshwater layer that drifts over the denser ocean water in the estuaries. This flow can be either outward (positive circulation) or inward (negative circulation), depending on wind direction –northerly and southernly, respectively (Rosón, Cabanas, & Fiz, 2008).

The Rías Baixas have an oceanic climate that nonetheless becomes more arid in the summer (Alvarez *et al.*, 2005). There are seasonal changes in temperature and salinity within the rias, with surface water being colder in the summer (due to upwelling) and less salty in the winter (due to rainfall patterns) (Alvarez *et al.*, 2005). Salinity drops caused by intense rains often lead to high mortality rates in many bivalve species (Parada, Molares, & Otero, 2012), and possibly affect *L. saxatilis* as well. This phenomenon might become worse in the future due to climate change (Des *et al.*, 2021). The coasts of the Rías Baixas are heavily altered by human activities, including industrial complexes, harbors, buildings, and sewage emissions (Méndez & Vilas, 2005), although the intensity varies depending on the ria and location (Pérez-Fernández, Viñas, & Besada, 2020). These activities result in various types of pollution, which have been documented in all four rias (Santos-Echeandia *et al.*, 2008; Pérez-Fernández *et al.*, 2020; Carretero, Gago, & Viñas, 2021).

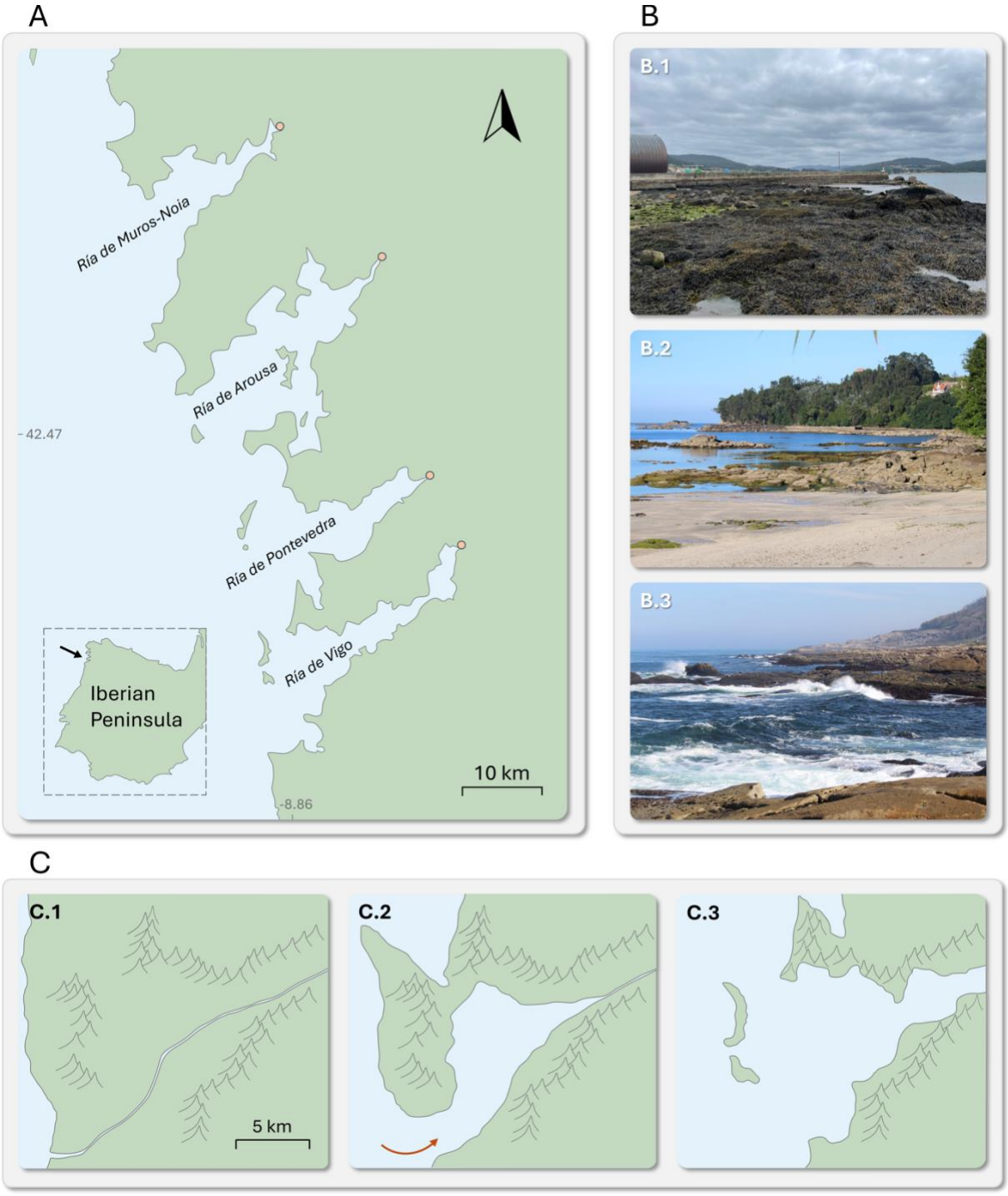

Figure 8. The Rías Baixas. A: Location of the four Rías Baixas within the Iberian Peninsula. Pink dots mark the mouths of the main rivers of each ria. B: Diversity of rocky intertidal ecosystems within the Rías Baixas. B.1 depicts a representative wave-sheltered ecosystem, abundant in brown algae; B.2 shows an intermediate wave-exposed ecosystem, and B.3 a fully wave-exposed ecosystem (photos by J. Gefaell). C: Historical evolution of the Rías Baixas, using the Ría de Vigo as a model. C.1 illustrates the area around 18,000 years ago, prior to the Holocene transgression, when the coastline extended much farther than the current mouth of the ria. C.2 depicts the ria around 11,000 years, marking the onset of the sea level rise and the inundation of the Rías Baixas (the red arrow indicates the direction of the incoming ocean water and the mountains the most elevated parts of the ria). Finally, C.3 portrays the present-day Ría de Vigo.

b) Diversity of intertidal ecosystems along the Rías Baixas

Among those not affected by human activities, there are three different types of intertidal ecosystems along the coasts of the Rías Baixas: rocky shores, sandy beaches, and estuarine environments. The latter is confined to river mouths, while the other two are scattered along the outer and inner littorals. Typically, the rocky shores are interspersed by long, recurrent sandy beaches, which in some areas dominate the coastal landscape. There are, however, segments, where the opposite pattern occurs, with most of the shoreline composed of rocks.

The Rías Baixas exhibit a wave exposure gradient from their inner to their outer parts that runs parallel to other physicochemical gradients (Gefaell *et al.*, 2024b; Chapter 6). Although there are exceptions, the inner zones of the rias are typically wave-sheltered, while the outer parts experience intense wave action. This creates an enormous diversity of intertidal ecosystems, which strongly influences species communities (Figure 8B). Wave-sheltered rocky ecosystems are characterized by high densities of brown algae (mostly *Ascophyllum nudosum*, *Fucus vesiculosus*, and *Pelvetia canaliculata*) and periwinkle species such as *Littorina fabalis*, *Littorina obtusata*, and *Littorina littorea*, among other animals. Conversely, highly wave-exposed coasts feature large patches of the lichen *Lichina pygmaea*, extensive barnacle coverage (*Semibalanus balanoides* and *Chthamalus* spp.), and different species of marine gastropods such as *Melarhaphe neritoides*, *Phorcus lineata*, *Nucella lapillus*, and *Gibbula umbilicalis*. Since wave exposure is a continuous parameter, this difference is more a matter of degree than kind, with intermediate wave-exposed localities sometimes exhibiting transitional characteristics between purely wave-sheltered and intensely wave-exposed environments (Chapter 6).

There are also differences in rocky intertidal ecosystems depending on the type of rock substrate. In many wave-sheltered localities of the Ría de Vigo, for instance, small rocks prevail, creating microhabitats beneath them where most organisms dwell. This contrasts with what is normally the case in wave-exposed zones, where large bedrock surfaces are more common. For many intertidal gastropods, including *L. saxatilis*, crevices serve as the equivalent of the below-the-rocks microhabitat in these heavily exposed areas providing protection against wave action and predators. Another distinction in rock substrates is visible when comparing schist and granitic surfaces along the Rías Baixas. Barnacle coverage appears to reach higher vertical limits on granite than on schist, where it seems to be confined to the sublittoral and littoral fringes. A plausible interpretation is that schist, being darker, reaches higher temperatures, thereby increasing the mortality rate of the uppermost settled barnacles. Although this has never been formally tested in the Rías Baixas, a similar situation has been observed in the northern Gulf of California, where barnacles' upper vertical averages 25 cm lower on basalt (also a dark stone,

but of volcanic origin) than on granite (Raimondi, 1988). In sum, the Rías Baixas exhibit a remarkable diversity of intertidal ecosystems that vary continuously and typologically.

c) Historical geomorphology of the Rías Baixas

Paraphrasing Grant & Grant (2008), to study the evolutionary causes of color variation in *L. saxatilis*, we need to know how the evolutionary play unfolded as the scenery of the Rías Baixas changed. The first step in this endeavor is to review what is currently known about the origins and coastal evolution of these landforms. The Rías Baixas, which are, as said, flooded river valleys, require an explanation of how these valleys were formed in the first place, and how they were subsequently inundated. The prevailing view suggests that the valleys of the Rías Baixas were formed by tectonic activity during the Miocene epoch, approximately 23 to 5 million years ago (Méndez & Vilas, 2005). This interpretation is supported by the observation that the width of the valleys largely exceeds that of the rivers they once contained, indicating that fluvial erosion alone could not have created these valleys in the first place. Instead, the valleys may have formed due to the falling or dislevelment of tectonic blocks, which would have caused part of the land to subside, thereby giving rise to the basins that were later filled by the rivers (Méndez & Vilas, 2005). This does not exclude, however, that fluvial and marine erosion could have also played a role in the genesis of the Rías Baixas, albeit this role would have only been a complementary one.

The next step in the formation of the Rías Baixas is the inundation of the river valleys with seawater (see Figure 8C). This phenomenon would have started after the last glacial maximum, around 12,000 years ago, in a process known as Holocene transgression (Arce-Chamorro, Vidal-Romaní, & Sanjurjo-Sánchez, 2022). The Holocene transgression was characterized by the retreat of large glaciers, which gradually raised the sea levels from approximately -120 m around 18,000 years ago (García-García, García-Gil, & Vilas, 2005) to their current levels about 6,000 to 3,000 years ago, thereby inundating the Rías Baixas. Until that moment, the coastline –mainly made of sand deposits– was well beyond the islands at the mouths of the rias (Vilas *et al.*, 2005; Arce-Chamorro *et al.*, 2022). As the sea level rose, the seawater gradually penetrated the rías, until eventually reaching the present moths of the rivers less than 3,000 years ago (Arce-Chamorro *et al.*, 2022). This suggests that the intertidal ecosystems of the Rías Baixas are relatively young.

Regarding recent coastal evolution, drawing largely on the pioneering work of Ramón Margalef (1919–2004) (Margalef, 1956, 1959), a model of the main stages in the Rías Baixas over the last 8,000 years –a period when sea levels were still 25 to 30 meters lower than today– has been proposed (Méndez & Vilas, 2005). According to this model, four main climatic periods might

have occurred: a Boreal period (8,500 to 7,000 years ago), characterized by a relatively warm climate with rapidly increasing temperatures; an Atlantic period (7,000 to 4,500 years ago), with a short cooling phase but still generally higher temperatures and more intense rainfall compared to the present; a Subboreal period (4500 to 2600 years ago), marked by dry and warm conditions; and a Subatlantic period (around 2600 years ago to the present), characterized by cool and rainy phases (Margalef, 1956; Méndez & Vilas, 2005). Within the Subatlantic period, different phases can be distinguished as well, with at least one spanning from 3,000 to 700 years ago and marked by intense estuarine circulation within the rias and a second one from around 500 years ago to the present in which water exchange between the ocean and the rias increased and seasonal upwelling began (Méndez & Vilas, 2005). These models offer valuable insights, which might eventually enable the development of plausible scenarios for the colonization of the Rías Baixas by *L. saxatilis*. As for more recent years, it is worth noting that no apparent temperature changes have occurred in the Ría de Vigo over the last 34 years, despite climate change (Otero *et al.*, 2023).

Objectives

As stated in its title, this thesis aims to shed light on the causes of color variation in the populations of *L. saxatilis* from the Rías Baixas. By *causes*, I refer to ultimate causes sensu Mayr (1961), that is, those addressing 'Why' questions related to evolution. And by *color variation*, intrapopulation color polymorphism and color clines. For this aim, this thesis relies heavily on the traditional methods of evolutionary ecology. Understanding evolution at a mechanistic or genetic level requires significant basic fieldwork to describe the evolutionary patterns and processes that an organism experiences. The specific goals of this research are the following:

a) Conduct an exhaustive review of the current knowledge on color variation in marine gastropods, the group to which *L. saxatilis* belongs. This will help familiarize with the relevant literature, including the different hypotheses, mechanisms, and methodologies used in studying color polymorphism. (Chapter 1)
b) Analyze whether there is negative assortative mating between color morphs in various species of *Littorina* from the wave-sheltered polymorphic populations of the Rías Baixas. This aims to test if mate choice-driven negative frequency-dependent selection maintains shell color polymorphism and whether this mechanism is ancestral in *L. saxatilis*. (Chapter 2)
c) Experimentally corroborate the link between mate choice and negative assortative mating to verify the role of mate choice-driven negative frequency-dependent

selection as a maintenance mechanism in the polymorphic populations of *L. saxatilis* from the wave-sheltered areas of the Rías Baixas. (Chapter 3)

d) Systematically study the clinal pattern of *L. saxatilis* from the Ría de Vigo and its predictors. Determine whether this cline has remained stable over the last forty years by comparing current and past data on morph frequencies. (Chapter 4)

e) Experimentally test one of the main hypotheses for the maintenance of the color cline of *L. saxatilis* from the Rías Baixas, specifically selection for crypsis, through a field tethering experiment. (Chapter 5)

f) Compare the patterns of clinal color variation across various localities in the four Rías Baixas to gain more detailed insights into its ecology and co-variation with other shell traits. (Chapter 6)

With these objectives, I aim to modestly contribute to a new research line in *L. saxatilis*, focused on understanding the patterns and causes of its color variation in the Rías Baixas. Chapter 1 has been published in the journal *Evolutionary Applications*, Chapter 2 in *Marine Biology*, Chapter 3 in the *Biological Journal of the Linnean Society*, Chapter 4 in *Current Zoology*, and Chapter 5 is currently under review. Chapter 6 summarizes some data and analyses from a larger project that is still in progress. Since each chapter was conceived as an independent article, there might be some redundancy in their contents and contextualization of the topics addressed.

Chapter 1

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REVIEW

Shell color polymorphism in marine gastropods

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Abstract

Marine gastropods are characterized by an incredible variation in shell color. In this review, we aim to introduce researchers to previous studies of shell color polymorphism in this group of animals, trying to provide an overview of the topic and highlighting some potential avenues for future research. For this, we tackle the different aspects of shell color polymorphism in marine gastropods: its biochemical and genetic basis, its patterns of spatial and temporal distribution, as well as its potential evolutionary causes. In particular, we put special emphasis on the evolutionary studies that have been conducted so far to reveal the evolutionary mechanisms responsible for the maintenance of shell color polymorphism in this group of animals, as it constitutes the least addressed aspect in existing literature reviews. Several general conclusions can be drawn from our review: First, natural selection is commonly involved in the maintenance of gastropod color polymorphism; second, although the contribution of neutral forces (gene flow-genetic drift equilibrium) to shell color polymorphism maintenance do not seem to be particularly important, it has rarely been studied systematically; third, a relationship between shell color polymorphism and mode of larval development (related to dispersal capability) may exist. As for future studies, we suggest that a combination of both classical laboratory crossing experiments and *-Omics* approaches may yield interesting results on the molecular basis of color polymorphism. We believe that understanding the various causes of shell color polymorphism in marine gastropods is of great importance not only to understand how biodiversity works, but also for protecting such biodiversity, as knowledge of its evolutionary causes may help implement conservation measures in those species or ecosystems that are threatened.

KEYWORDS
clines, crypsis, frequency-dependent selection, genetic drift, natural selection, snails

1 | INTRODUCTION

Color polymorphisms are widespread in nature (White & Kemp, 2016). Several taxa are widely known for their rich and beautiful color patterns (Cuthill et al., 2017), sometimes showing a striking intrapopulation diversity. The term "polymorphism" is often used to refer discrete, genetic, and intrapopulation variation in a trait (Mayr, 1963), but here we will adhere to a more relaxed definition,

Juan Galindo and Emilio Rolán-Alvarez contributed equally.

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as the intraspecific variation in a trait, whether it is geographically or temporally distributed, is more continuous than discrete, or if it is not directly dependent on genes. Color polymorphism and its maintenance have been a major topic of study in evolutionary biology since at least half a century (Clarke, 1962; Ford, 1965; Murray & Clarke, 1966). In the past decades, the number of studies of color polymorphism and its various aspects has grown exponentially (e.g., Bond, 2007; Gray & McKinnon, 2007; Jamie & Meier, 2020; McKinnon & Pierotti, 2010; McLean & Stuart-Fox, 2014; Svensson, 2017; Takahashi & Noriyuki, 2019; Wellenreuther et al., 2014; White & Kemp, 2016).

The terrestrial snail *Cepaea nemoralis* stands out as a classic model organism for color polymorphism, showing highly diverse shell colors (reviewed in Cook, 2017). However, marine gastropods, despite being as diverse in their shell colors as *Cepaea*, and probably much more conspicuous, have been much less studied to date. In these organisms, shells serve different purposes, most of them related to fitness, such as protection from various abiotic factors or predators (e.g., Reid, 1996), and color could also be involved in some of those purposes. This, along with the fact that marine habitats are highly heterogeneous, cause the shells to be subjected to different selective pressures that, presumably, account for the high variability in shape, pattern, and color shown by these species, and that leads to shell polymorphisms.

In the present review, we summarize what is known about shell color polymorphism in marine gastropods, including its biochemistry and genetic basis, the patterns of diversity for this trait (intrapopulation, spatial, and temporal variation) and the evolutionary mechanisms responsible for the polymorphism. Our target audience are those researchers interested in shell color polymorphism in this group of animals, and especially those who are new to this research topic and who wish to be introduced to its study. There have recently been published several reviews regarding shell color (Williams, 2017) or shell color polymorphisms (Cook, 2017) in mollusks. However, these reviews hardly refer to color polymorphisms in marine gastropods as a whole. Since most of marine gastropods form a monophyletic clade, and the marine environment is very different from the terrestrial one, we believe that a review focused only on this group of organisms is necessary. Furthermore, both reviews, while valuable, do not analyze in depth the evolutionary mechanisms responsible for color polymorphism. For this reason, in this review, we place special emphasis on studies that have investigated these evolutionary mechanisms linked to color polymorphism on marine gastropods.

1.1 | Search strategy

In January 2020, we searched for published papers on the topic "shell color polymorphism in marine gastropods" in the Web of Science (WOS). In a prospective search, we used various combinations of target terms, from which the three most effective combinations arose (see Figure 1). The first combination yielded 115 studies, the second 15 and the third 52. A total of 49 studies were kept after repeated or non-valid papers (i.e., those that do not deal with shell color polymorphism in marine gastropods) were excluded. To this list, we added some references found in these studies, as well as a few old studies published in non-English journals known by the authors and new studies published after January 2020, leading to a total of 61 studies. All of those studies were later classified on the basis of the intended structure of this review (Table 1).

2 | THE COLOR OF SHELLS

2.1 | A physical perspective

Colors stem from the pattern of reflection and absorption of electromagnetic radiation that a certain object shows. Reflected electromagnetic radiation can then be captured by the photoreceptors (usually cone cells) of the visual system of a given animal and transduced into nervous impulses that travel to the central nervous system, giving rise to color sensation. The different colors emerge from the different wavelengths (λ) of the reflected electromagnetic radiation. Every color-sensible species has a range of visible wavelength spectrum (Cuthill, 2006). For humans, this visible spectrum, which encompasses all visible colors, ranges from wavelengths of 400 nm (violet) to 700 nm (red). Between those wavelengths are the rest of primary colors (blue, 450 nm; green, 525 nm; yellow, 575 nm; orange, 625 nm) and their derivatives. Many species, however, can go beyond the human visible spectrum: certain fish, birds, and many groups of arthropods can reach 320 nm, what is called the ultraviolet range, and some butterflies and fish can detect wavelengths of up to 760 nm (Gerl & Morris, 2008; Land & Nilsson, 2012). This information should be taken into account when designing experiments on the contribution of predation to the evolutionary maintenance of shell color polymorphism.

2.2 | A biochemical perspective

At the molecular level, the color of a given organism is determined by the combination of the structural properties of their organs plus the effect of certain color biomolecules ("biological pigments") contained in it. Those pigments have a particular range of absorbed and reflected electromagnetic energy, resulting in a specific visible color (Prum, 2006). In most marine gastropods, observable shell color relies heavily on the pigments present in its outer layer or periostracum.

Although knowledge on the biochemical basis of shell color polymorphism is still scarce, in the last decade several discoveries have been made in this regard. Current research has shown that the shells of marine gastropods can contain different classes of pigments, such as tetrapyrroles (porphyrins, bilins), carotenoids, and melanins. Porphyrins may be responsible for red, brown, and purple, while bilins have been involved in the determination of blue, green, brown,

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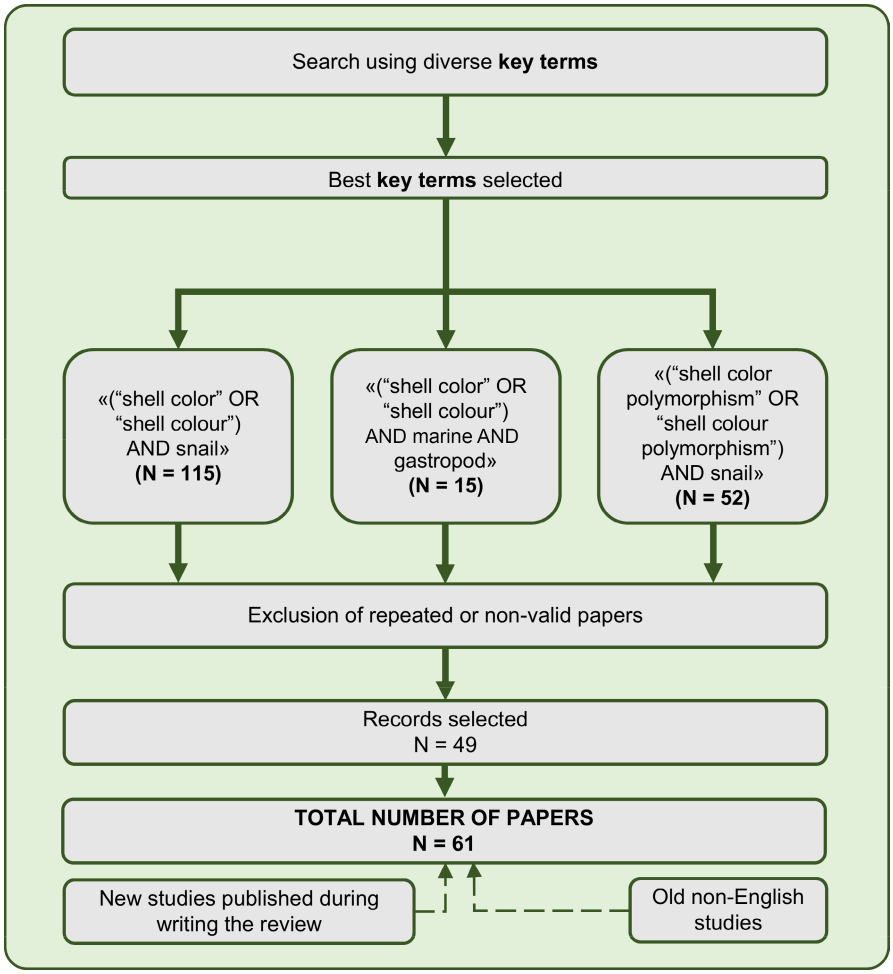

FIGURE 1 Flowchart of studies selected for the review

and red colors (Williams, 2017). For example, Williams et al. (2016) showed that uroporphyrins I and III are responsible for the pink-red color of strawberry topshells (*Clanculus pharaonius*), as well as the pink-red dots and lines of the early whorls and the yellow-brown color of later whorls in the beautiful clanculus (*C. margaritarius*). Nevertheless, porphyrins have not been identified in other marine snails that show a similar coloration (e.g., the European painted top shell, *Calliostoma zizyphinum*), suggesting that the same color may be caused by different pigments in different species of marine gastropods (Saenko & Schilthuizen, 2021; Williams et al., 2016).

Carotenoids have not been directly characterized in marine gastropods, but some polyene compounds found in various species have been interpreted as indirect evidence of them (Williams, 2017). Furthermore, based on their chemical properties, Kozminsky and Lezin (2007) have proposed that the yellow coloration in the shell of the flat periwinkle (*Littorina obtusata*) could be caused by carotenoids. Given that marine gastropods cannot synthesize carotenoids, these pigments may be acquired from algae ingestion (Williams, 2017), which constitutes the diet of many marine snail species (including *L. obtusata*). Melanins may be responsible for dark shell colors, as well as for red, brown, and yellow pigmentation. In a classical study, Comfort (1951) suggested that the dark shell colors of the common periwinkle (*Littorina littorea*) may be caused by melanoproteins analogous to those of bird feathers. Finally, other biological pigments have been suggested as potential causes of shell color, such as guanine, associated by Kozminsky and Lezin (2007) to the white coloration of *L. obtusata* (see also Kozminsky, 2014).

2.3 | A genetic perspective

Two hypotheses can account for the ontogenetic origin of shell color polymorphism in marine gastropods: (1) this trait may be caused by genetic differences in those genes that code for the biological pigments of the shell; (2) shell color is a plastic trait sensitive to different ecological factors, and therefore prone to showing polymorphic (i.e., polyphenic) variation. We will now consider the genetic hypothesis, leaving the phenotypic plasticity hypothesis for section 4.3.

Following Ruse's (2006) distinction between direct and indirect evidence for evolution, we can separate between direct and indirect evidence for the genetic basis of shell color. We shall call *direct evidence* all scientific data regarding the metabolic pathway from genes to (color) phenotype, and *indirect evidence* the inferred information about the purported allelic (or quantitative) basis of shell color, as well as its pattern of transmission, based on laboratory crosses and the phenotypic analyses of parents and offspring. Ideally, the genetic basis of a given trait should be supported by both direct and indirect evidence.

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TABLE 1 Summary of studies that have investigated shell color polymorphism in marine gastropods

Species	N° of morphs	Color trait[a]	Genetic candidate[b]	Pleiotropic effects	Biodiversity[c]	Evolutionary factor[d]	Ecological mechanism[e]	Methods[f]	Larval dispersion mode[g]	References
Littorina saxatilis	2–13	BC, BA	MGB, TGB, Linkage Group 6 and 17	Thickness, Salinity toleration, Weight, Aperture shape	SPA, SNL, IP	DS, NFDS, PS, BS	SC, APS, SHS	F, ME, CE, MP, -*Omics*	D	Atkinson and Warwick (1983), Byers (1990), Ekendahl (1998), Ekendahl and Johannesson (1997), Estévez et al. (2021), Fischer-Piette and Gaillard (1961), Fischer-Piette et al. (1961), Gefaell et al. (2021), Heller (1975), Hughes and Jones (1985), Johanneson and Ekendahl (2002), Johannesson and Butlin (2016), Koch et al. (2021), Kozminsky (2011), Raffaelli (1979), Sacchi (1979), Sacchi and Malcevschi (1983), Sokolova and Berger (2000), Torelli (1983)
Littorina fabalis	2–6	BC, BA	MG, DG	Mucus	SPA, IP, SNL, ONT	DS, NFDS, PS, HS, SS	DM, APS, SC	F, ME, MP, CS, RS	D	Ekendahl (1994, 1995), Estévez et al. (2020, 2021), Gefaell et al. (2021), Reimchen (1979, 1989), Rolán-Alvarez and Ekendahl (1996), Rolán-Alvarez et al. (2012), Rolán-Alvarez, Carvajal-Rodríguez, et al. (2015b)
Littorina obtusata	2–4	BC, BA	MGB, DG	Several behav. and morphol. traits (unspecified)	IP, SPAGR	DS	STS	ME, CE, MOC, -*Omics*	D	Kozminsky (2011), Kozminsky (2014), Kozminsky (2016), Kozminsky et al. (2010), Phifer-Rixey et al. (2008)
Littorina compressa, *Littorina arcana*	5, 10–13	BC/BA	ns	ns	SPA, IP	DS, NFDS	SC, APS	F	D	Atkinson and Warwick (1983), Byers (1990), Heller (1975)
Littorina (Littoraria) scabra, *Echinolittorina natalensis*	3–4	BC	ns	ns	IP	DS/NFDS	SC/APS, ev. against STS	ME, MP	P	Hughes and Jones (1985), Hughes and Mather (1986), Miller and Denny (2011)
Littoraria filosa	3	BC	ns	ns	SPA/IP	DS/NFDS	SC/APS	ME, F	P	McKillup et al. (2000), McKillup and McKillup (2002, 2008), Parsonage and Hughes (2002), Reid (1987)
Littoraria pallescens	2	BC	ns	Thickness	IP	DS, GD, PS	ns	F, ME, MP	P	Cook (1986, 1992), Cook and Kenyon (1993)
Littoraria luteola	3	BC	ns	ns	SPA/IP	DS	SC	ME	P	Parsonage and Hughes (2002)
Littoraria philippiana	3	BC	ns	ns	SPA/IP	DS	SC	ME	P	Parsonage and Hughes (2002)
Nucella lapillus	2–6	BC, BA	ns	ns	SPA/IP	DS	STS	F, ME	D	Etter (1988), Harris and Jones (1995)
Nucella emarginata	3	BC	MGT	ns	SPA	ns	ns	CE	D	Palmer (1984)
Nucella ostrina	ns	Color Brightness	ns	ns	ONT	PhP	ns	ME	D	de Bruyn and Gosselin (2014)
Urosalpinx cinerea	3	BC	MGT	ns	IP	ns	ns	CE	D	Cole (1975)
Oliva oliva	3	BC	ns	ns	IP	PhP	ns	F	D	Tursch et al. (1995)

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TABLE 1 (Continued)

Species	N° of morphs	Color trait[a]	Genetic candidate[b]	Pleiotropic effects	Biodiversity[c]	Evolutionary factor[d]	Ecological mechanism[e]	Methods[f]	Larval dispersion mode[g]	References
Asolene platae	2	BC/BA	MGB	ns	IP	ns	ns	CE	D	Tiecher et al. (2017)
Hastula cinerea	3	BC	ns	ns	IP	DS	STS	F	D	Molina et al. (2013)
Tegula xanthostigma	2	BC	ns	ns	SPA	ns	ns	F	P	Yamazaki et al. (2019)
Buccinum undatum	29	BC/BA	ns	ns	SPAGR	ns	ns	F	D	Magnúsdóttir et al. (2018)
Lottia asmi, *L. digitalis*, *L. pelta*	3, ns, 6	BC/BA	ns	ns	IP, SPA	DS/PhP	SC	F, ME	P	Lindberg and Pearse (1990), Sorensen and Lindberg (1991)
Batillaria attramentaria	6	BC/BA	ns	ns	SPA	DS/GD	STS	F	D	Miura et al. (2007)
Crepidula convexa	3	BC	MGB	ns	IP	DS	SC	F	D	Hoagland (1977)
Haliotis discus	3	BC	MGT	ns	IP	DS/PhP	SC	CE	P	Liu et al. (2009)
Concholepas concholepas	Cryptic color	BC	ns	ns	IP	DS/PhP	SC	ME	D	Manríquez et al. (2009)

Note: ns, not specified.

[a]BC, Background Color; BA, Bands, spots and tesselations.

[b]MG, Monogenetic; MGB, Monogenetic Biallelic; MGT, Monogenetic Triallelic; DG, Digenetic; TGB, Trigenetic Biallelic.

[c]IP, Intrapopulation; SPA, Spatial; SPAGR, Spatial Gradient; SNL, Seasonal; ONT, Ontogenetic change in color.

[d]DS, Direct Selection; GD, Genetic Drift / Neutralism; NFDS, Negative Frequency-Dependent Selection; BS, Balancing Selection; HS, Habitat Selection; PS, Pleiotropic Selection; SS, Sexual Selection; PhP, Phenotypic Plasticity.

[e]SC, Selection for Crypsis; APS, Apostatic Selection; SHS, Selection against Haline Stress; STS, Selection against Thermal Stress; DM, Dissortative Mating/Negative Assortative Mating.

[f]F, Frequentist; CE, Crossing Experiments; ME, Manipulative Experiments; CS, Cross-Sectional design; MP, Model Predictions; MOC, Mother-Offspring Correlation; *-Omics*, genomic, transcriptomic and/or proteomic techniques; RS, Reflectance Spectrometry.

[g]D, Direct development; P, pelagic larvae.

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Direct evidence for the genetic basis of shell color in marine gastropods is scarce: only two studies have been done so far on the gene-to-phenotype pathway for shell color (Koch et al., 2021; Williams et al., 2017). Williams et al. (2017) obtained direct evidence of the role of several enzyme-coding genes in the production of pigments that are ultimately responsible for shell color in two species of the genus *Clanculus*. In particular, these genes coded for several enzymes of the pathway responsible for the synthesis of uroporphirine I and III (see previous section above).

In the second study, Koch et al. (2021) used a QTL (quantitative trait loci) approach with F_2 families of the rough periwinkle (*Littorina saxatilis*), and they performed targeted sequencing to genotype the offspring for 25000 DNA fragments spread across the genome. The goal of the study was to map known adaptive traits between two divergent ecotypes of *L. saxatilis*, and although shell color was not shown yet to be adaptive, it varied between these ecotypes (beige vs black). Several QTLs for adaptive traits were associated with chromosomal inversions, similarly to previous studies (Westram et al., 2018), and QTLs for color were detected within two of these inversions. For example, a QTL for shell color was located in the same inversion as traits like weight, shell thickness, shell length, and aperture shape. Therefore, shell color might be evolving linked to traits under selection within a chromosomal inversion in these *L. saxatilis* populations, instead of being a direct target of selection. However, these results should be interpreted with caution, as QTLs represent large regions of DNA and no genes have been identified yet as responsible for this color polymorphism.

Indirect evidence for the genetic basis of color polymorphism in marine gastropods is more abundant. One of the first studies in this regard was carried out in the eastern oyster drill (*Uralpinx cinerea*; Cole, 1975). Results suggested that juvenile shell color is genetically inherited through a triallelic, single-locus genetic system, in which a dominance hierarchy between the different alleles exists. Palmer (1984) found a similar trend for shell color polymorphism in the emarginate dogwinkle (*Nucella emarginata*). Similarly, crossing experiments carried out by Ekendahl and Johannesson (1997) on populations of *Littorina saxatilis* from Iceland point to a relatively straightforward genetic basis for band shell color, which in this case would be a one-locus two-allele system.

During the last decade, a series of noteworthy cross-breeding studies on the various aspects of shell color polymorphism in the genus *Littorina* have been conducted by Kozminsky and collaborators (Kozminsky, 2011, 2014, 2016; Kozminsky & Lezin, 2007; Kozminsky et al., 2010). These researchers tested the different modes of inheritance of the various components of the shell color of *L. obtusata*: background color (Kozminsky, 2014), shell bands (Kozminsky, 2011, 2016; also *L. saxatilis* in Kozminsky, 2011) and the pattern of white spots (Kozminsky et al., 2010). Evidence suggested that simple Mendelian models successfully explained only some shell color components. For example, dark shell bands in *L. saxatilis*, and dark and white shell bands in *L. obtusata* were accounted for by a monogenetic biallelic system of inheritance (Kozminsky, 2011, 2016). However, background color in *L. obtusata* could not be explained by one or two Mendelian loci (Kozminsky, 2014). Given that background color in this species depends on several pigments, the author proposed that it can be caused by a gene system responsible for the separate inclusion of the different pigments in the shell.

Despite progress made in the last decade, to date, there are no marine gastropod species in which both direct and indirect inferences have allowed researchers to elucidate the proximate genetic mechanisms of color determination with as much detail as in other model organisms (see the peppered moth case reviewed in Cook & Saccheri, 2013).

2.4 | A sensory ecology perspective

To understand shell color polymorphism, it is important to take into account the role of sensory ecology (Stevens, 2013), or the way in which marine gastropods and their predators process color vision. This knowledge is needed for the success of experimental designs intended at revealing the potential fitness advantages of different shell colors in polymorphic species.

Only a few studies have dealt with the visual abilities of marine gastropods (Hamilton et al., 1983; Seyer, 1992), and in these studies, no clues are offered on whether they have color vision. In any case, it is highly implausible that (at least most) marine gastropods can see colors. Cephalopods, the mollusks with a more advanced visual system (Nilsson, 2013) typically have one kind of photoreceptor, therefore not being able to see colors (with some exceptions; Land & Nilsson, 2012). If cephalopods, whose behavioral ecology depends much more on vision than that of marine gastropods (e.g., Hanlon & Messenger, 2018), cannot see in color, it is highly unlikely that gastropods can. However, an alternative hypothesis is that even color-blind gastropods might be able to distinguish shell colors if these represent distinct levels of brightness/shading across the range from light to dark. Still, this hypothesis has not yet been tested in marine gastropods and would require further investigations.

Regarding the sensory ecology of predators, littorinid snails as well as other marine gastropods are usually preyed upon by intertidal crustaceans (such as crabs), shore birds (such as gulls and oystercatchers), pond fishes (such as blennies and gobies), and predator snails (such as the dog welk, *Nucella lapillus*; Pettitt, 1975). At least some species in these groups of predators (with the exception of the dog welk) have color vision, and evidence gathered in the past two decades suggests that some of them can even see beyond the human visible spectrum (see Marshall et al., 2003, 2019; Ödeen et al., 2010). This means predators may select their preys based on their visual properties, including those that go beyond the visible spectrum.

In sum, what we have seen throughout this section allows us to draw three conclusions: (1) the biochemical basis of shell color polymorphism depends on a limited number of pigments that vary between species; (2) simple Mendelian models can account for many shell color traits but molecular and physiological mechanisms are unknown; and (3) while marine gastropods cannot see color, most of their predators can.

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3 | PATTERNS OF DIVERSITY

3.1 | Intrapopulation variation

Shell color variation within populations is a common phenomenon in marine gastropods (see Figure 2). Perhaps, one of the most studied groups in this respect is the genus *Littorina*: within this genus at least five of 18 species show a clear and conspicuous intrapopulation shell color polymorphism, with the rest of the species showing only very subtle intrapopulation color variation (reviewed in Reid, 1996; Rolán-Alvarez et al., 2015a). *L. saxatilis* shows shell color variation within several populations of the Galician coasts (NW Spain), including orange, white and banded color morphs (Fischer-Piette & Gaillard, 1961; Fischer-Piette et al., 1961; Gefaell et al., 2021; Sacchi, 1979; Torelli, 1983). Northern populations of *L. saxatilis* in the British Isles, Sweden, Iceland, and Russia also show great intrapopulation polymorphism (e.g., Ekendahl & Johannesson, 1997; Raffaelli, 1979).

Regarding *L. obtusata* and *L. fabalis*, most studies have detected the existence of four different color morphs (brown, olive, yellow, and orange) that usually coexist in the same populations (Estévez et al., 2020; Gefaell et al., 2021; Reid, 1996; Rolán-Alvarez et al., 2012, 2015b). Hughes and Jones (1985) recognized three color morphs (yellow, red, and brown) coexisting in populations of *Littorina scabra* (now *Littoraria scabra*) from Queensland (Australia). In addition, patterns of intrapopulation shell color polymorphism have been described in species from genera other than *Littorina* (for instance, *Hastula cinerea*; Molina et al., 2013); colors beyond the visible spectrum have also been described, such as the near ultraviolet, which may be detected by some predators of snails and therefore subject to selection (Williams, 2017; see previous section). Based on current data, it is highly likely that most marine gastropods show at least a subtle degree of intrapopulation shell color variation. However, the prevalence of conspicuous intrapopulation shell color polymorphism in marine gastropods, as well as its potential correlation with developmental characteristics, genetic variation, and environmental variables, has not been yet systematically analyzed (an exception to this is the genus *Littorina*, Reid, 1996).

3.2 | Spatial variation

Shell color variation occurs not only within populations, but also between populations, showing spatial patterns. When this type of color variation is gradual, it is called *cline* (*sensu* Mayr, 1963). Clinal variation in shell color between populations has been observed in *L. saxatilis*, where yellow and fawn individuals are abundant in sheltered localities of estuaries, and gradually decrease in frequency as we approach the exposed shores, where they are almost completely replaced by a banded phenotype (Reid, 1996; Sacchi, 1979; Sacchi & Malcevschi, 1983 and references therein). A similar kind of spatial variation has been found in *Tegula xanthostigma*, in which light brown shells were found to be associated with sheltered environments, and black shells with exposed shores (Yamazaki et al., 2019).

In other occasions, clines may show gradual variation in the number of color morphs present at a given location. Several *Littorina* species (*L. obtusata*, *L. fabalis*, and *L. compressa*) have been reported to show higher numbers of color morphs in the southern parts of their ranges than in the northern parts (Reid, 1996). In a similar vein, Magnúsdóttir et al. (2018) found that in the north-Atlantic common whelk (*Buccinum undatum*), populations in the innermost parts of a bay located in western Iceland showed low color diversity than in the outer parts. This suggests that natural selection and other evolutionary mechanisms somehow promote the evolution of a higher degree of color variation (seen in the number of color morphs available) in certain locations compared to others, but the exact reasons for this pattern are not clear.

3.3 | Temporal variation

The most important temporal variation in color involves changes in the frequency of the different colors in the population over time. For example, in *L. fabalis* seasonal variation in shell color (yellow, brown) was studied in Swedish populations, showing that color frequencies remained similar between spring and autumn (Ekendahl, 1994, 1995). However, these studies revealed several microhabitat changes that could potentially affect survival. For instance, Ekendahl (1995) found that in autumn, yellow adult morphs were more frequent in the laminae of the seaweed *Fucus vesiculosus*, while brown adults were more frequent in stipes. On the contrary, in spring no differences were observed between adult morphs. These results suggest a seasonal pattern of habitat choice that could be linked to different selective pressures affecting the color morphs in the different seasons.

Besides seasonal variation, other studies have looked for monthly or annual variations in color frequencies (Estévez et al., 2020, 2021; Tursch et al., 1995). This is the case of *L. fabalis* (Estévez et al., 2020, 2021), which did show a reduced change in frequencies over a hypothetical equilibrium frequency in the course of 20 years. However, larger time spans have not yet been analyzed.

3.4 | Variation across taxa

It is not easy to find patterns in shell color polymorphism across gastropod species because the number of known polymorphic species is relatively low (27 species in Table 1). However, at least a few preliminary suggestions can be made. For example, it has been suggested that the pattern of larvae dispersal (direct *versus* pelagic) could influence the level of population genetic (and, therefore, color) variability (Bohonak, 1999). In species with direct development, eggs are laid in the immediate surroundings of the mother (or carried by the mother until they are mature, as in ovoviviparous species such

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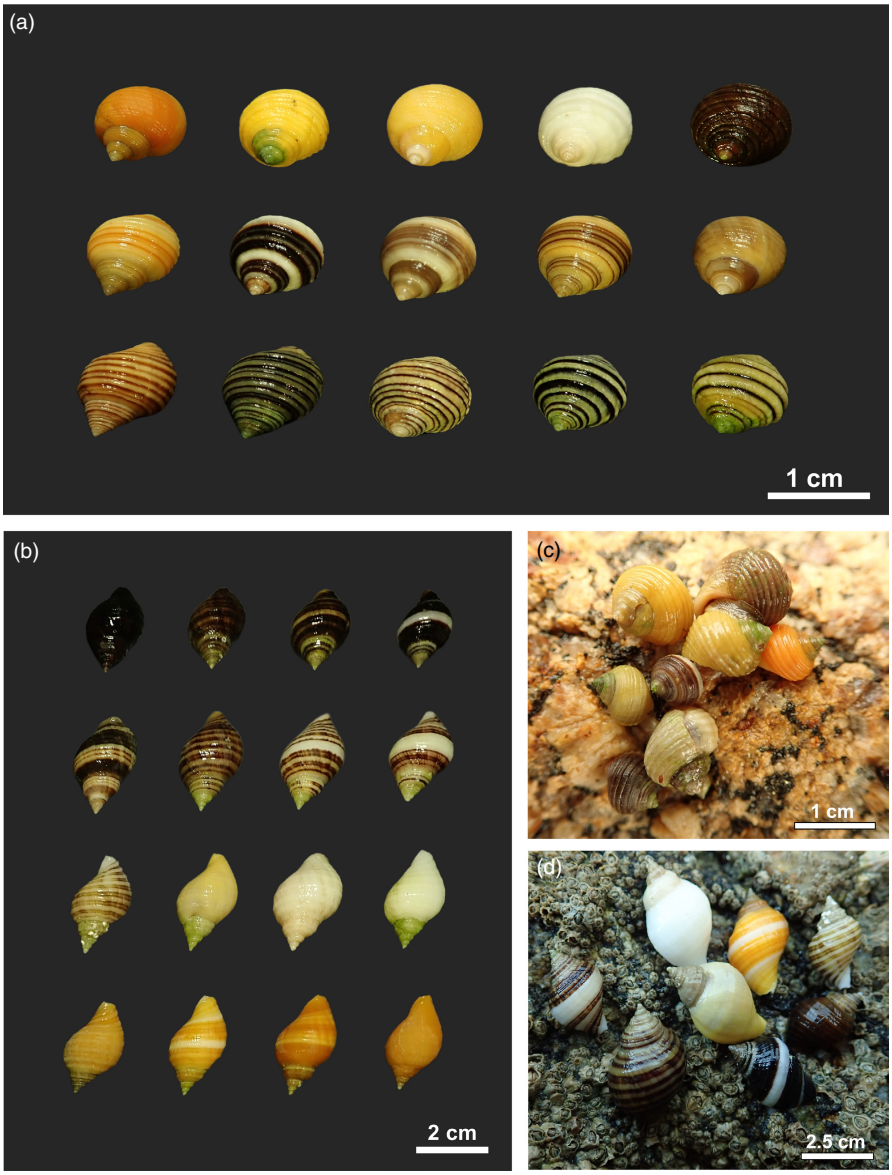

FIGURE 2 (a) Color polymorphism shown by *L. saxatilis*. (b) Color polymorphism shown by *Nucella lapillus*. (c) Some color morphs of *L. saxatilis* in their natural environment. (d) Some color morphs of *Nucella lapillus* in a natural setting

as *L. saxatilis*); pelagic species are those that go through a planktonic larval phase until metamorphosis, when they return to the shore (Hickman et al., 2007).

According to Table 1, the number of polymorphic species for shell color that are direct developers is higher than those with pelagic larvae (59.3% vs. 40.7%). This pattern seems to change if we broaden our focus to marine gastropods in general, whether polymorphic or not; in this group of animals, 45% undergo direct development, while about 55% of them show a pelagic larvae stage (Jablonski & Lutz, 1983). This suggests that direct development could somehow promote shell color polymorphism. This is further supported by the averaged maximum number of shell colors observed in direct developers (mean ±SD = 7.38 ± 7.41, N = 13) *versus* those with pelagic larvae (3.37 ± 1.92, N = 8). In addition, it has been suggested that phenotypic plasticity is more prevalent in organisms with high dispersal ability (Hollander, 2008). However, plasticity in shell color polymorphism is similarly distributed in direct (4 out of 16) and pelagic species (3 of 16) of marine gastropods.

4 | EVOLUTIONARY MECHANISMS

After reviewing the "How" and "What" of color polymorphism, that is, its physicochemical nature, genetic basis, and patterns of diversity, it is now time to review the "Why" of color polymorphism, or, in other words, the evolutionary mechanisms responsible for their evolution and maintenance within populations (Mayr, 1998).

Several different hypotheses have been proposed to account for color polymorphism (see White & Kemp, 2016). We have arranged these hypotheses following an algorithm-like scheme (see Figure 3). Once we have put aside the special case of phenotypic plasticity, the main distinction can be drawn between non-selective and selective hypotheses. In turn, selective hypotheses can be further subdivided into direct selection and selection on correlated traits. In addition, a distinction can also be drawn between directional and balancing selection.

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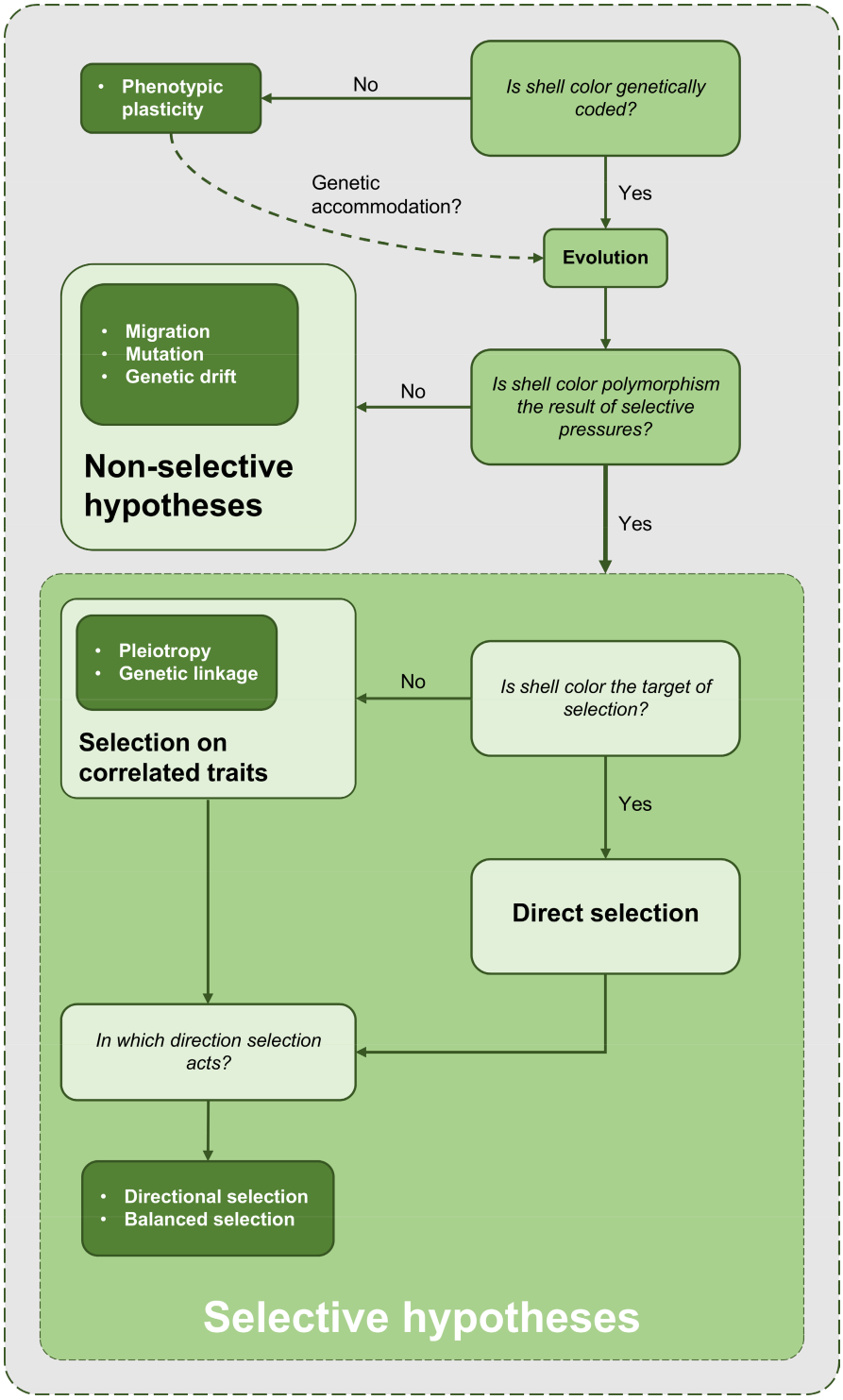

FIGURE 3 A summary of all proposed hypotheses to account for shell color polymorphism portrayed in an algorithm-like fashion. See text above for a detailed explanation

4.1 | Non-selective hypotheses

Three evolutionary mechanisms promote the evolution of populations besides selection: mutation, migration, and genetic drift (Freeman et al., 2017). Mutations introduce new (color) variations in a given population. Although it is reasonable to assume that several color polymorphisms are caused by mutations in genes involved in shell coloration, no particular mutation for shell color has yet been detected at the molecular level in marine gastropods.

Migration refers to the reproductive consequences of the transfer of one or many individuals from one population to another. From an evolutionary point of view, migration is important insofar as it contributes to gene flow, causing the homogenization of populations, thereby making it harder for specific adaptations to evolve (Freeman et al., 2017; Herron & Freeman, 2014). Evidence suggests that direct developers show a greater tendency toward evolving local adaptations than their planktonic counterparts (Behrens Yamada, 1989).

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Many stochastic changes in the genetic makeup of populations are due to genetic drift (see Freeman et al., 2017; Futuyma, 2009; Herron & Freeman, 2014). This mechanism explains the fixation of alleles that do not bear a reproductive advantage as a result of sampling error during mating. Among the non-selective mechanisms, genetic drift has classically stood out as a plausible explanation for the occurrence of phenotypic polymorphisms; for instance, Darwin himself noted in *On the Origin of Species* (1859/2006) that: "Variations neither useful nor injurious would not be affected by natural selection, and would be left a fluctuating element, as perhaps we see in the species called polymorphic." (1859/2006: 63–64).

The equilibrium between genetic drift and migration should be the null hypothesis when testing the role of other evolutionary mechanisms, particularly selection. Genetic drift assumes that the different color variations (whether intrapopulation or clinal) do not bear any fitness advantage, so they fluctuate in a random—but quantifiable—manner across locations and generations. In intrapopulation polymorphism, genetic drift and migration equilibrium can be rejected if the observed intergeneration allele variance is higher or lower than the one predicted based exclusively on effective population size (N_e; assuming absence of effective migration; see for example Estévez et al., 2020, 2021). In the case of clinal polymorphism, genetic drift and migration are expected to lead to unique and non-repeatable patterns of color variation at different locations (although the magnitude of variation across subdivided subpopulations can be to some extend predicted; see for example Johannesson & Butlin, 2017). This means that, if a given clinal polymorphism is observed in a similar ecological context in several (and at least partially isolated) locations, then the null hypothesis can be rejected (or at least considered highly unlikely).

Although most research tacitly assumes the equilibrium of genetic drift and migration as null hypothesis, very few attempts have followed this framework in the study of shell color polymorphism. Besides the examples cited above (Estévez et al., 2020, 2021; Johannesson & Butlin, 2017), another exception is Cook (1992), who indirectly tested genetic drift in the mangrove snail *Littoraria pallescens* by estimating the effective population size (N_e) either based on molecular markers or color variation. As the level of diversity for both genetic traits could be explained based on correspondingly estimated N_e, he concluded that color variation may be a trait as neutral as molecular variation, although he admitted some local exceptions putatively caused by selection (Cook, 1992). In any case, most previous studies (e.g., Cook, 1992; Estévez et al., 2020, 2021; Johannesson & Butlin, 2017) suggest that genetic drift and migration may not be a particularly important evolutionary mechanism for the explanation of shell color polymorphism in at least littorinid snails, but given that such mechanism is always working in nature (because $N_e \neq \infty$), it should be properly tested in each putative case before being formally ruled out.

4.2 | Selective hypotheses

Natural selection is considered the most important evolutionary mechanism by the majority of evolutionary biologists, and it is frequently invoked as an explanation to account for all complex traits that resemble an adaptation. Although selective explanations should be formulated with caution, given the widespread tendency to incur what Gould and Lewontin (1979) called the "Panglossian paradigm," its explanatory potential is undeniable (Mayr, 1983). Regarding shell color polymorphism in marine gastropods, an overwhelming majority of studies have assumed that natural selection is ultimately responsible for its evolution.

Natural selection can take many forms. A first distinction is based on whether a given trait (in our case, shell color) is the real target of selection. If this is so, then we can speak of direct selection; if, on the contrary, selection does not act directly on shell color, but on any other trait correlated with shell color, then we can speak of selection on correlated traits (Figure 3). Whether selection acts directly on shell color or on any other correlated trait, a further distinction can be drawn between directional and balancing selection. In directional selection, the direction of selection is constant in a given habitat during a significant number of generations, favoring those phenotypes at an initially lower frequency in the population; in balancing selection, selection acts in such a way that multiple variants in a population are maintained at larger frequencies than those expected by genetic drift. This can be ultimately achieved either by heterozygote advantage or frequency-dependent selection.

4.2.1 | Directional selection

Directional selection can promote shell color polymorphism if there exists habitat heterogeneity in which the different color morphs bear different fitnesses. When selection results from visual predators, then crypsis could evolve and shell color will tend to match its background (Bond, 2007; Endler, 1978); if this background is heterogeneous, then color polymorphism can be expected, maybe in conjunction with a behavioral mechanism of habitat choice (e.g., Edelaar et al., 2008). In other words, the main prediction of directional selection is that color polymorphism must reflect the chromatic heterogeneity of a given habitat.

In the past, many studies in marine gastropods have evaluated directional selection using frequentist-observational approaches that compare shell colors with their backgrounds (Byers, 1990; Ekendahl & Johannesson, 1997; Heller, 1975, 1979; Hoagland, 1977; Reimchen, 1979, 1989). For instance, Heller (1975) studied the frequencies of different colors in *Littorina saxatilis* from Wales. Among other things, he observed that reddish individuals were more prevalent in those localities where red sandstone was more abundant, suggesting that visual predation by rock pipits and crabs may be the main selective pressures there. Also in Welsh populations, Byers (1990) found that grey *L. saxatilis* individuals were more frequent in grey rocks, while brown individuals were more frequent in red rocks. One further example is that of *Littorina fabalis´* juveniles, which are cryptic with *Spirorbis* tubeworms, both found in the fronds of brown macroalgae, in order to avoid predation by blennies (Reimchen, 1989). These correlations between

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shell and background colors constitute powerful suggestions for crypsis, though not a definite corroboration, so manipulative experiments need to be carried out. For example, Ekendahl (1998) conducted tethering experiments in Swedish populations of *L. saxatilis* to test if crypsis was influenced by predation by the crab *Carcinus maenas*. For this, dark and white snails were glued to black and greyish stones, and then the number of preyed snails was counted. Predation results showed no differences in the number of preyed snails of both colors on each type of stones, suggesting that crabs hardly influence the color frequencies of these populations of *L. saxatilis*.

In another study, Hughes and Mather (1986) carried out mark-recapture and caging experiments on the various color morphs of the mangrove snail *Littorina scabra* (now *Littoraria scabra*), that lives in trees of the species *Avicennia marina*, which have a mosaic of colors in their trunks and leaves, ranging from brown to yellow. This species of mangrove snail presents three background colors (brown, yellow and red, in decreasing order) and a striped pattern that can superimpose on background color. The mark-recapture and caging experiments suggested that yellow individuals showed less predation and a higher survival. For instance, when the snails were placed at the cages, where predation by toadfishes (*Spheroides hamiltoni*) is restricted, yellow individuals decreased in frequency, whereas on control trees, with no cages preventing predation, the frequency of this morph increased. This suggests that yellow individuals are in one way or another protected against predators. Although Hughes and Mather did not rule out the existence of frequency-dependent selection, they suggest that selection for crypsis may play a role in explaining the frequencies of observed color morphs in this species of mangrove snail.

So far, we have assumed that selection for crypsis is caused by predators; however, a series of studies point to the possibility that parasitoid insects may be responsible for the crypticity of certain color morphs in *Littoraria filosa* mangrove snails (McKillup & McKillup, 2002; McKillup et al., 2000). *Sarcophaga megafilosia* and *S. meiofilosia* deposit larvae near the snails, so that they can infect the snails and feed on them (eventually killing them). Results by McKillup and colleagues showed that *S. megafilosia* infects more those snails that do not match their background color, while *S. meiofilosa* does not differentiate between colors. These results suggest that at least one of these parasitoid species (*S. megafilosia*) could be maintaining color polymorphism in *L. filosa* due to habitat color heterogeneity. Further studies have supported this interpretation (McKillup & McKillup, 2008).

In addition to crypsis, other selective pressure that can promote directional selection on shell color is thermal stress. Shell color is presumed to contribute to thermoregulation, with light colors preventing overheating, while darker shells attracting heat. Therefore, if there is a thermal cline at a given region, then a color polymorphism based on thermal tolerance can be expected: colors less prone to overheating should be more prevalent in the warmer regions. Several studies have observationally and experimentally explored this possibility in various intertidal gastropods (Cook, 1986; Etter, 1988; Harris & Jones, 1995; Miller & Denny, 2011; Phifer-Rixey et al., 2008). For example, in the Japanese mud snail (*Batillaria attramentaria*), darker morphs are predominant in colder regions, where almost no light-colored individuals exist, suggesting strong selection against them (Miura et al., 2007). In warmer regions, lighter individuals are more frequent, but dark ones are still found, suggesting weaker selection against dark individuals in these regions. Similarly, Phifer-Rixey et al. (2008) found that in *L. obtusata* from the Gulf of Maine the darkest individuals are found in colder regions, while the lighter ones are more frequent in warmer regions.

Experimental evidence also provides support for the thermal stress hypothesis. In a series of lab and field experiments conducted with dogwhelks (*Nucella lapillus*) from the northern coast of Massachusetts, Etter (1988) found that the brown morphs heated up faster and were less resistant to thermal stress than the white morphs in those locations with higher temperatures (the more protected ones).

The above results suggest that selection related to thermal stress may be a potential cause of the evolution of shell color polymorphism, also indicating a possible role for this trait in resilience to climate change. However, these results should be interpreted with caution as, for example, in the past decade Miller and Denny (2011) have shown that the snail's behavior makes a greater contribution to thermoregulation than shell color in the case of five littorinid species from California, based on field experiments and heat-budget models. That is, to prevent overheating, behavioral mechanisms tend to be much more efficient than shell colors.

4.2.2 | Balancing selection

Balancing selection can promote polymorphisms through heterozygote advantage and frequency-dependent selection. Heterozygote advantage occurs in those occasions in which heterozygote individuals show higher fitness than any of the two homozygotes (Futuyma, 2009). Frequency-dependent selection occurs when the frequency of a given trait in the population affects the fitness of its bearer organism. The most striking case of frequency-dependent selection is *negative* frequency-dependent selection (hereinafter NFDS), in which a trait or phenotype increases its fitness as its frequency decreases (Ayala & Campbell, 1974; Lewontin, 1974). Since the classic studies on shell color polymorphism in *Cepaea* (Clarke, 1962), NFDS has been invoked as a putative explanation for color polymorphisms in a wide variety of organisms (Gigord et al., 2001; Olendorf et al., 2006; Estévez et al., 2021; but see Ibarra & Reader, 2013). The most common case of NFDS is *apostatic selection*, where the rarer phenotypes of a prey species show greater fitness than the more frequent phenotype because certain predators look for prey by forming a "search image" of its most common phenotype, thus ignoring those phenotypes at a lower frequency (Ayala & Campbell, 1974; Clarke, 1962).

In the case of shell color polymorphism, apostatic selection predicts that the color morph or morphs at lower frequencies are less detected by predators because they do not match the predator's

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search image. Thus, though present at low frequencies, these rarer morphs are protected from extinction by NFDS. Several potential cases of apostatic selection have been described to account for shell color polymorphism (Atkinson & Warwick, 1983; Johannesson & Butlin, 2017; McKillup & McKillup, 2008; Reid, 1987). For example, correlations between the frequencies of different color morphs, as well as between the levels of phenotypic diversity and the presence/absence of color morphs between *Littorina saxatilis* and *L. arcana* across several British populations were proposed as an indication of NFDS (Atkinson & Warwick, 1983). In the case of *Littoraria filosa*, and in addition to selection for crypsis by parasitoid insects, as we previously mentioned, evidence was also obtained for apostatic selection on color driven by shell-crushing predators (most likely crabs) (McKillup & McKillup, 2008), suggesting that in some occasions different modes of natural selection may be operating simultaneously in opposing directions in a given species.

NFDS can also be driven by sexual selection, and among these scenarios, we find sexual conflict (Gosden & Svensson, 2007; Svensson et al., 2005), mate competition (Rios-Cardenas et al., 2018; Sinervo & Lively, 1996), and mate choice (Takashi & Hori, 2008). However, in organisms with a limited behavioral repertoire, such as most marine gastropods, only mate choice has been linked to NFDS for shell color (Estévez et al., 2020; Gefaell et al., 2021; Rolán-Alvarez & Caballero, 2000; Rolán-Alvarez & Ekendahl, 1996; Rolán-Alvarez et al., 2012, 2015b).

In the genus *Littorina*, negative assortative mating has been used as an indirect proxy for mate-choice-driven NFDS (see Gefaell et al., 2021 and references therein). Rolán-Alvarez et al. (2012, 2015b) collected mated and unmated individuals of *L. fabalis* from different microareas to determine whether a process of sexual selection or assortative mating could be playing a role in the evolutionary maintenance of shell color polymorphism. These researchers found a high level of negative assortative mating affecting the mating behavior of this species; a result that was consistent with that found 21 years earlier in the same population (Rolán-Alvarez & Ekendahl, 1996).

In a more recent study including seven consecutive years of analysis of a population of *L. fabalis*, Estévez et al. (2020) found both negative assortative mating and NFDS (via sexual selection) on shell color along the period studied, thus contributing to the maintenance of a polymorphism in this trait. Estévez and colleagues also used multimodel inference to determine whether mate choice, mate competition or both could explain the observed distribution of shell colors observed for mating pairs, showing that mate choice was always included in all potential explanatory models (Estévez et al., 2020). Recently, similar patterns of negative assortative mating have been found in other populations of *L. fabalis*, as well as in *L. obtusata* and *L. saxatilis*, suggesting that male mate choice may be an ancestral behavioral trait driving NFDS on shell color, and thereby promoting polymorphism in at least certain populations of these species (Gefaell et al., 2021). However, since the relationship between negative assortative mating, mate choice and NFDS is not straightforward and direct, experimental confirmations of this link must be made before the mate choice-driven NFDS hypothesis is considered definitively corroborated.

4.2.3 | Selection on correlated traits

The former selective hypotheses have assumed that natural selection acts directly on shell color; however, there could be cases in which the real target of selection may not be shell color itself, but any other correlated trait that may be the one which really bears a fitness advantage for the organism. Notice that in this case the mode of selection also occurs by either directional or balancing selection, as in the previous cases. We shall call this group of hypotheses *correlated traits' selection hypotheses* (CTSH).

Two genetic mechanisms have been proposed as explanations for CTSH: linkage disequilibrium and pleiotropy (see McKinnon & Pierotti, 2010). In linkage disequilibrium, two given alleles are nonrandomly associated; on the other hand, in pleiotropy one gene is directly or indirectly responsible for the expression of two phenotypically independent traits. Irrespectively of the genetic mechanisms involved, CTSH predicts a correlation between shell color and some other phenotypic trait, which has to demonstrate some fitness advantage for the carrying organism. If such correlation and fitness advantage are found, then we will be ready to study its genetic and molecular basis.

A few putative examples of CTSH involving shell color polymorphism have been proposed for marine gastropods. It has been suggested that shell color polymorphism may be associated with parasite resistance (Raffaelli, 1979), shell strength (Cook & Kenyon, 1993), and haline stress (Sokolova & Berger, 2000; see Sergievsky, 1992 for an early review of Russian studies). For instance, Cook and Kenyon (1993) tested the shell strength of different color morphs in *Littoraria pallescens*, and they found differences in shell resistance between these colors. More specifically, dark shells of two of three analyzed locations were found to be heavier and more resistant. This greater resistance was dependent on the structural properties of this morph's shell and not on the color itself. Cook and Kenyon (1993) associated this difference in shell thickness with crab predation, which it is probably more prevalent in bark and rock surfaces, where the dark morphs live.

Another putative example of CTSH is found in *L. saxatilis*, where a color gradient has been observed in the White Sea, with brown tessellated unbanded (BTU) shells being more frequent in the head of the estuaries, where there is low salinity. Laboratory experiments revealed that BTU morphs show higher survival rates under extremely low salinity conditions and under a combination of low salinity and low temperatures compared to typical morphs from marine environments (Sokolova & Berger, 2000). This points to the possibility that these physiological differences associated with the BTU morph may be the result of adaptation to haline stress. However, an important issue with this hypothesis is having color and population as confounded factors in the analysis, as both color morphs may differ in several other traits simultaneously. Therefore, future studies comparing colors from the same population are needed before the final confirmation of this trend.

A third presumed case of CTSH involves mate choice and NFDS. Estévez et al. (2020), as well as Gefaell et al. (2021), have suggested

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the possibility that in the abovementioned cases of negative assortative mating and mate choice for shell color in several species of the genus *Littorina*, the real target of (sexual) selection may not be the color itself, but another genetically correlated trait, tightly linked to it. An argument in favor of this hypothesis, in addition to the observed mating patterns in different *Littorina* populations (see Gefaell et al., 2021), is that it is highly unlikely that marine gastropods can see colors, a prerequisite for mate choice based on shell color. This has led to the hypothesis that littorinids can somehow recognize the different shell colors without any visual cue, or alternatively each of these colors may be associated with some other trait crucial to biological fitness. There is already evidence that at least certain littorinid species can detect with olfactory cues their conspecifics of the opposite sex (Seuront & Spilmont, 2015; Wyeth, 2019). However, it cannot be ruled out that color-blind snails could distinguish between colors based on the different brightness/shading effect produced by each color, so further studies are needed to test this possibility.

In cases such as the latter, the evolutionary mechanism at work may be heterozygote advantage. Although heterozygote advantage should be considered to affect to a minority of all loci in any species (Hedrick, 2012), this does not preclude that it may have a role in maintaining certain genes in some populations. A particular way in which this mechanism could have a role in maintaining shell color polymorphism could be by means of an association between shell color genes and chromosomal inversions. Chromosomal inversions can become overdominant under several scenarios (Kirkpatrick, 2010). Furthermore, chromosomal inversions have been recently detected in *L. saxatilis* associated with adaptative traits (Faria et al., 2019; Westram et al., 2018) and particularly to shell color (Koch et al., 2021). Therefore, heterozygote advantage should be further studied in relation to these findings.

4.3 | Phenotypic plasticity

Phenotypic plasticity is usually defined as the process by which one individual genotype is able to produce different phenotypes under different environmental conditions (see Fusco & Minelli, 2010). It has been suggested that, at least in some occasions, phenotypic plasticity can precede evolutionary change; that is, variations arising via phenotypic plasticity could later be fixed at the genetic level, thus being inherited and subject to evolutionary forces (Ehrenreich & Pfennig, 2016; Gilbert & Epel, 2009; West-Eberhard, 2003). This process, usually called genetic assimilation, has not yet been described in marine gastropods. However, some putative examples have been suggested in other taxons (e.g., Aubret & Shine, 2009; Badyaev et al., 2017), so the possibility of genetic assimilation in marine gastropods should not be ruled out. A first step toward investigating genetic assimilation is to corroborate the existence of phenotypic plasticity. In the present case, phenotypic plasticity predicts that shell color in a given population of snails is dependent on a given environmental parameter, thus changing if that parameter changes.

Most of the known cases of color polymorphism in marine gastropods caused by phenotypic plasticity are induced by changes in diet (Lindberg & Pearse, 1990; Liu et al., 2009; Manríquez et al., 2009; Palmer, 1984; Sorensen, 1984; Sorensen & Lindberg, 1991). This is the case of the black limpet (*Lottia asmi*) and the shield limpet (*L. pelta*), that show a large diversity of shell colors and patterns, securing cryptic coloration at the different types of substrates in which they live. It has been found that these colors depend on the type of diet available on each substratum, and that a change in the substrate (and therefore diet) triggers new shell growth that makes the limpet more cryptic in their new environment (Lindberg & Pearse, 1990; Sorensen, 1984; Sorensen & Lindberg, 1991). Another example is the so-called Chilean abalone, *Concholepas concholepas* (a muricid gastropod), which has a shell color that matches the background color of their habitat, made up of their most frequent prey: black individuals feed on mussels and light colored on barnacles (Manríquez et al., 2009). Juvenile individuals reared in a specific habitat in the laboratory developed a shell color that matched this habitat. Moreover, when the individuals were moved to a different habitat, they showed a color change that corresponded to that of the new habitat. In sum, shell color in marine gastropods is sometimes dependent on phenotypic plasticity. Plastic shell colors may contribute to camouflage in a given habitat, so phenotypic plasticity could potentially interact with selection by crypsis. It remains open whether these initially plastic traits can eventually be genetically assimilated (Table 1).

5 | METHODOLOGICAL OPTIONS

The methodology used to study a phenomenon may itself condition its understanding. For this reason, it is convenient to take into account the kinds of methodologies used when studying shell color polymorphism in order to detect some potential biases or limitations. Several methodologies exist, depending on the goals of the study (see Table 2). If our goal is the physical determination of color, or the precise characterization of color phenotypes, then these can either be scored visually by the researcher (e.g., against a color palette), analyzed based on digital images (e.g., RGB scale), or examined with reflectance spectrometry (RS; when the color is objectively estimated by a spectrometer through wave length reflectance). If we seek to analyze the molecular basis of shell color polymorphism, the pigments responsible for the different colors can be studied, for instance, through high-performance liquid chromatography (HPLC) or mass spectrophotometry (Saenko & Schiltuizen, 2021). If our goal is to understand the hereditary basis of shell color polymorphism, then we can employ crossing experiments (CE; when using breeding experiments in the laboratory to infer Mendelian-like proportions), "*-Omics*" approaches (OA; based on genomic, transcriptomic or proteomic data), or knockout experiments. Although knockout experiments or other gene-editing techniques have never been applied to the study of shell color polymorphism in marine gastropods, they have yielded interesting results when applied to other mollusk traits, such as biomineralization of the shell (Clark et al., 2020). In

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TABLE 2 Methods for the study of shell color polymorphism depending on different goals

Goals	Methodology
Physical determination	Visual scoring
	Digital imaging
	Reflectance spectrometry (RS)
Molecular characterization	High-performance liquid chromatography (HPLC)
	Mass spectrophotometry
Hereditary basis	Crossing experiments (CE)
	Knockout experiments
	Omic approaches
	Epigenetic approaches
Evolutionary mechanisms	Frequentist studies (F)
	Manipulative experiments (ME)
	Model predictions (MP)
	Cross-sectional designs (CS)

addition, and even though studies done so far strongly suggest that color is a genetically-codified trait, epigenetic approaches may help explain those shell color traits most difficult to explain by means of genetic mechanisms or that are related to plastic mechanisms. These possibilities have been discussed in other invertebrate taxons (e.g., Hiyama et al., 2012). Finally, if our goal is to ascertain the evolutionary mechanisms at play in the maintenance of shell color polymorphism, various approaches can be followed: frequentist (F; when the emphasis is made on the distribution of color frequencies within or across populations), manipulative experiments (ME; any field or laboratory experiment based on the deliberate control of potential causal factors), model predictions (MP; those approaches that develop computer-based explanatory models, especially when they are tested against empirical data) and cross-sectional or fitness estimation designs (CS; when color frequencies are compared at different life-cycle stages).

Each of these methods has its own strengths and limitations. For instance, ME are especially valuable to ascertain the causal role of the different evolutionary factors putatively involved in the maintenance of a polymorphism. However, ME are generally (though not always) carried out in the laboratory since there it is easier to control for intervening variables; but this raises several problems of its own. First, ME are only well-suited for organisms that can be bred and kept in laboratory conditions, thus limiting the spectrum of organisms amenable for this kind of studies. Second, it poses the problem of generalization, given that laboratory settings are highly artificial and controlled and it is not always easy to extrapolate those results to natural environments (Ruxton & Colegrave, 2006). On the other hand, when we want to avoid these limitations by carrying out a ME in the field, we are confronted with a lot of practical setbacks related to the difficulty of controlling for intervening causal variables from the natural environment. On the contrary, frequentist (F) studies, given their observational and/or correlational nature, can be easily done in the field. However, their main limitation has to do with a lack of causal control, which makes drawing causal inferences more difficult. Model Predictions (MP) allow the testing of a hypothesis in a straightforward way. Nevertheless, they sometimes made dubious assumptions that limit their explanatory scope.

In sum, there is no perfect method for studying shell color polymorphism. Researchers should think carefully about the available methodological options and weight them against each other in order to determine which is the best methodology for a specific goal or biological question.

6 | FUTURE OF THE FIELD

A considerable amount of knowledge has been gathered in the past decade regarding shell color polymorphism in marine gastropods (Table 1). However, many questions remain unanswered. In relation to the future of the field, and based on what we have previously exposed throughout this review, we propose new directions and goals to which new researchers could direct their efforts in order to take this field forward. These are centered around certain methodological issues and evolutionary mechanisms.

6.1 | Methodological Issues

According to Table 1, most studies done so far have followed a frequentist (F) approach (25 out of 61) and several of these show the existence of a correlation between shell colors and their various backgrounds. Looking ahead, frequentist studies can be still very useful if, for example, frequentist geographical data are combined with detailed geographic-scale environmental information, in order to detect correlations that could be interpreted in the light of potential causal relationships with environmental variables. Moreover, to date no study has yet analyzed the temporal consistency of geographic color morph frequencies in marine gastropods over decades, similarly to how Ramos-Gonzalez and Davison (2021) have recently done in *Cepaea nemoralis* populations from the Pyrenees. Although a few studies exist that focus on temporal variation in shell color

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frequency (see Estévez et al., 2020, 2021; Johannesson & Butlin, 2017), this approach could be of interest not only to determine how evolutionary forces operate over long periods of time, but also because it may give new insights on the impact of color polymorphism on the vulnerability of a species (Bolton et al., 2015, 2016; Forsman, 2016; Takahashi & Noriyuki, 2019). In this sense, given that some studies suggest that shell color may contribute to body thermoregulation, it would be very interesting to ascertain if there has been any kind of selective process due to environmental variation over long time scales, as this is directly linked to climate change research (e.g., Leung et al., 2019; Minuti et al., 2021; Oliveira et al., 2020). Such a process has already been described in some species of insects (Brakefield & de Jong, 2011; de Jong & Brakefield, 1998).

The second kind of most conducted studies as derived from Table 1 are manipulative experiments (ME) (21 ME vs 25 F). Despite all their problems, ME have proven to be very useful, allowing researchers to infer causal relationships between shell color polymorphism and the different evolutionary mechanisms. So we argue researchers must keep doing MEs. In order to avoid the problem of generalization, efforts should be directed toward manipulative experiments in the field, or at least mesocosms experiments, as only they guarantee the ecological validity of results. Tethering and caging experiments would be the most obvious based on previous studies, although adaptation of experimental protocols from other taxons should not be ruled out (e.g., Endler, 1980).

Another important methodological consideration for future studies refers to shell color characterization by researchers. To date, most studies in marine gastropods have analyzed shell color using a visual scoring method (i.e., the observational criterion of experts guided by color templates; e.g., Pettitt, 1973), that is, a qualitative method. Although valuable as a first approximation, this method is subjected to several potential biases, such as low reliability. Therefore, we recommend the use of reflectance spectrometry (RS), as well as other potential quantitative approaches, to study shell color polymorphism. RS has only been used a few times in marine gastropods (Dauphin & Denis, 2000; McKillup & Mckillup, 2008; Rolán-Alvarez et al., 2015b). In addition to render more robust and quantitative determinations of shell color, it can help to detect patterns of reflectance in the non-visible spectrum, like the ultraviolet range.

Finally, this review has highlighted the lack of studies using *-Omics* approaches (OA) (genomic, transcriptomic, and proteomic) to unravel the genetic basis of color polymorphism in marine gastropods (2 studies of 61). Crossing experiments (CE) have shown that shell color in these species could have a rather simple genetic basis (e.g., one to a few loci). This suggests that although the genotyping of families bred in the laboratory or genome-wide association studies (GWAS) represent very useful methods to determine the genetic basis of color, it still represents a daunting task, especially when a reference genome is not available in the species, and thousands of genetic markers (e.g., SNPs) are ideally needed (Koch et al., 2021). One of the reasons for the lack of such studies could be the relatively large size of the genome of these animals, with a lot of them being greater than 1.0 Gb and some having even larger genomes (e.g., dogwhelks 2.8 Gb; www.genomesize.com).

Another important reason is that until very recently whole-genome sequencing projects have been associated with model species or those with an economic interest, although nowadays these kinds of projects have also been carried out in species with an ecological or evolutionary interest (e.g., *Littorina saxatilis*, Westram et al., 2018). It is predicted that these *-Omics* studies will considerably increase in the next years with the availability of high-quality genome assemblies for a growing number of species, including marine gastropods, lower sequencing prices, and new knowledge about the relationship between shell color polymorphism and the different evolutionary mechanisms in marine gastropods. Certainly, a combination of *-Omics* studies with CE would be ideal to determine in the future the precise genetic basis of shell color and its polymorphism.

In recent years, GWAS type analyses have been widely used to study color polymorphism as a start point to depict the genomic regions associated with a specific color. Methodologically this has been done by WGS (Whole Genome Sequencing) of DNA pools of each color (e.g., Lopes et al., 2016, in canary birds; Gautier et al., 2018, in ladybirds; Andrade et al., 2019, in lizards), or by GBS (genotyping by sequencing) of individual samples and then combining the genotypes of each specific color (e.g., Nosil et al., 2020, in stick insects). The use of families from CE in the laboratory (e.g., Koch et al., 2021) is more powerful than GWAS, but it is also a lot more time consuming because of the breeding stage in the laboratory, which is highly dependent on the organism (marine gastropods are not ideal in that sense). In any case, what all the previous studies have in common is the use of a reference genome to map the short sequences obtained by NGS (Next Generation Sequencing), in some cases being an already existing genome in the databases, and in others a newly sequenced genome specifically for that study. If we focus on our phylum of interest, that of mollusks, transcriptomic studies mostly in bivalves of economic interest have been relatively common (e.g., Ding et al., 2015; Hu et al., 2019; Xu et al., 2019). These studies focus on the differential expression of genes across the whole transcriptome between different colors, assuming that a part of those transcripts would be candidates for being involved in the inheritance of color. Following these analyses, the functional annotation of the transcripts is necessary and the quality of this annotation is a key step in order to study the different functions of the candidates. Some studies (e.g., Xu et al., 2019) also include proteomic analyses (e.g., mass spectrometry-based proteomics) to support the transcriptomic results. It has been observed in these mollusk studies that the number of differentially expressed genes is large and that finding good gene candidates is difficult. Finally, although genomic studies might be more effective at the time of finding a candidate genomic region than candidate genes through transcriptomics, these genomic regions are usually rather large and study of gene expression of genes present in that region represents a common follow up of the GWAS analysis (Andrade et al., 2019; Gautier et al., 2018; Lopes et al., 2016). Therefore, even with the availability of NGS technologies, targeting a gene or a few causal genes for color polymorphism still represents a great challenge.

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6.2 | Evolutionary mechanisms

As we have already seen, there are plenty of studies that have analyzed the putative evolutionary mechanisms that may be maintaining shell color polymorphism in marine gastropods; however, several facts stand out. First, not all evolutionary mechanisms have been studied equally, direct selection has received much more attention than other selective and non-selective mechanisms (30 out of 61 studies). A conclusion that can be drawn from Table 1 is that the most frequent mechanism used to explain color polymorphism has been natural selection in its different guises (crypsis, apostatic, mate choice, etc.). This may be justified due to the heuristic potential of adaptive hypotheses (Mayr, 1983); however, we believe that each hypothesis deserves its fair chance. A very useful approach to solve this would be to explicitly employ the equilibrium between genetic drift and migration as null hypothesis of selective mechanisms when conducting evolutionary studies.

Second, the pleiotropic and genetic linkage hypotheses have a great but still unexplored explanatory potential, since suggesting evidence reviewed in previous sections points to the possibility that, at least in some marine gastropod species, shell color may not be the real target of selection. However, most of this research ultimately requires unraveling information about the genetic basis of shell color, and would demand more efforts in generating genomic data for those species that might show this kind of patterns. This explanatory recommendation is closely linked, therefore, to the abovementioned need of future studies to focus more on *-Omics* techniques.

Thirdly, in the past decade it has been suggested that the evolution and ecological performance of genetically polymorphic species (ecologically associated) may be greater than those of plastic (environmentally caused) polymorphic species, or than those species in which the association between the polymorphism and the environment is random (Forsman et al., 2008; but see Bolton et al., 2015, 2016). These studies suggest that more geographically variable species show broader niches, reduced intra-specific competition, decreased vulnerability to environmental changes (or genetic drift), and decreased extinction risk. Other authors have claimed that marine gastropods may have a special capability to resist at least some effects of climate change, such as temperature increases (Leung et al., 2019). We suggest that shell color polymorphism in marine gastropods may also be of great value for checking these hypotheses. For instance, as suggested in the previous section, if certain color morphs resist thermal stress better (as studies suggest; see section 4.2.1.), and if temperatures rise in certain populations due to climate change, then an increase in the frequency of such morphs in those populations can be expected.

Finally, we are currently witnessing a renewal of evolutionary theory, with promising new ideas and mechanisms, such as genetic assimilation, epigenetic inheritance, evo-devo, or niche construction, that are receiving increasing attention (Laland et al., 2015; Pigliucci & Müller, 2010). The study of shell color polymorphism in marine gastropods cannot leave behind these new trends in evolutionary biology. For example, phenotypic plasticity in gastropods is an established fact (Bourdeau et al., 2015), and here we have seen some putative cases of plastic shell color polymorphism in marine gastropods. However, its role as an evolutionary mechanism is not clear, because the possibility of genetic assimilation of plastic traits has not been yet evaluated in this group of organisms. Putative cases of genetic assimilation in other groups of animals (e.g., Aubret & Shine, 2009; Badyaev et al., 2017) may give us clues as how to test such hypothesis for shell color polymorphism in marine gastropods.

As a final commentary, we must say that, after reading this review, some may worry that, to date, "the" evolutionary mechanism behind shell color polymorphism in marine gastropods has not yet been resolved in many cases. However, as a philosopher of science John Beatty (1997) has argued, disputes in biology often revolve around the question of how many cases in a particular domain of phenomena does one theory or mechanism explains. Or, in other words, what is its *relative significance*. Applied to this case, the really pressing question is not what is "the" actual evolutionary mechanism responsible for shell color, but how many instances of shell color polymorphism does each evolutionary mechanism explain. Evolution has many ways of arriving at a particular scenario, so we should not expect to find a single answer to the causes of such a huge diversity of shell color polymorphisms that we observe in nature. Instead, we should adopt a more pluralist outlook and be open about the potential contribution of different evolutionary mechanisms to the explanation of shell color polymorphism.

7 | CONCLUSION

Shell color polymorphism in marine gastropods is a pervasive phenomenon. Throughout the last decade, great advances have been made in the understanding of the different aspects that comprise it. In this review, we have summarized such knowledge. In particular, we have focused our attention on the evolutionary forces that could be contributing to its maintenance in natural populations. We have also derived from such knowledge a series of recommendations regarding the most promising methodological options, as well as the more fertile evolutionary explanations, given current limitations. Although a quick glance at the explanatory mechanisms suggests that natural selection may have a prominent role in explaining shell color polymorphism in several marine gastropods, some gaps in our current knowledge of such phenomenon exist. Unraveling the evolutionary mechanisms responsible for the existence and maintenance of intraspecific variation—including color polymorphism– is a major topic of study in evolutionary biology, as they are the key to biodiversity. Shell color polymorphism in marine gastropods represents a well-suited model system to study those mechanisms, and the new *-Omics* techniques constitute a promising approach to unraveling those evolutionary mechanisms at the molecular level. For this reason, we urge evolutionary biologists and malacologists alike to focus their efforts on the study of the various aspects of shell color polymorphism in this diverse and beautiful group of organisms.

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CONFLICT OF INTEREST

The authors declare they have no conflict of interest.

DATA AVAILABILITY STATEMENT

Data sharing not applicable—no new data generated.

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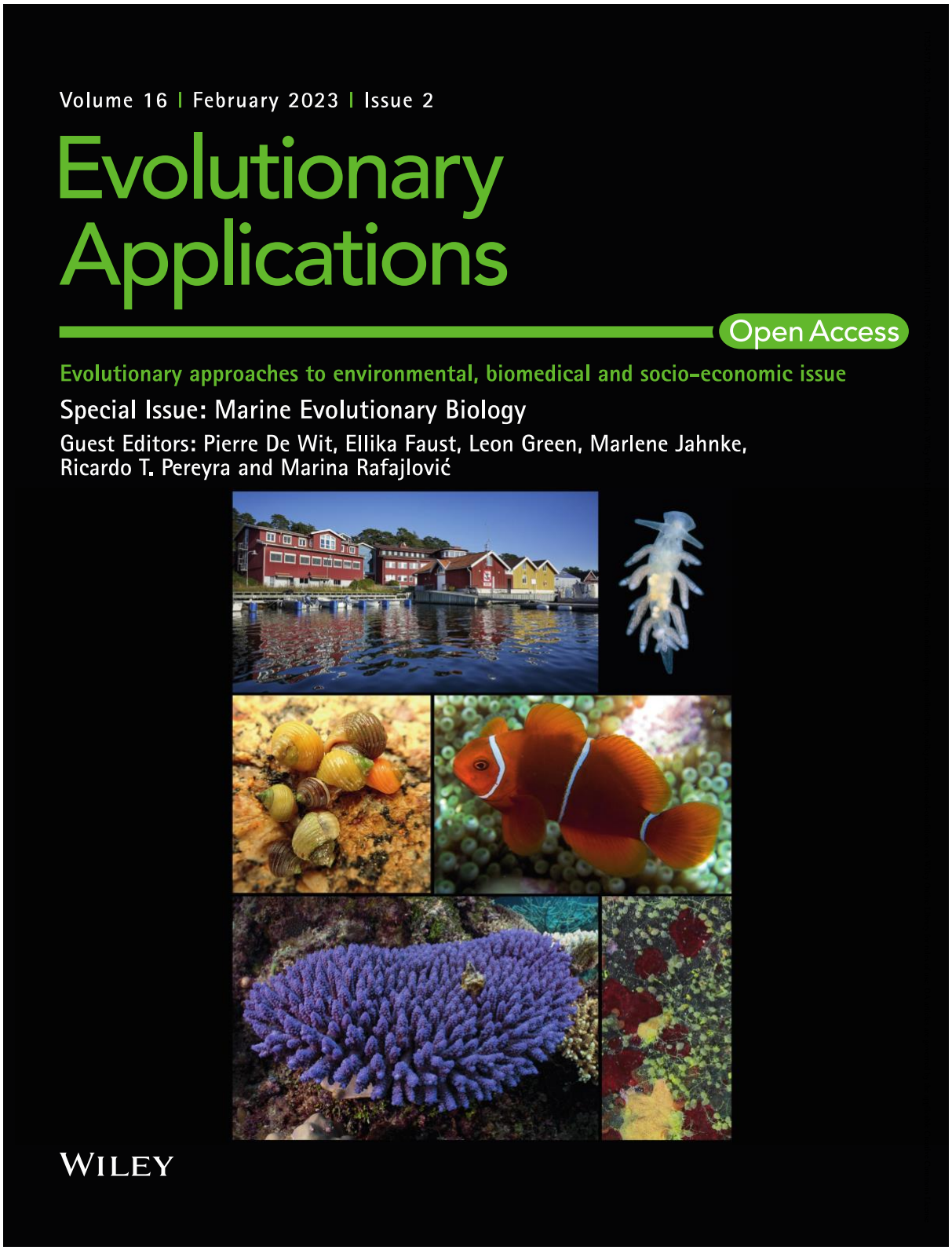

Figure S1. Journal issue cover displaying a photograph of color polymorphic *L. saxatilis* snails from the Rías Baixas (by J. Gefaell).

Chapter 2

Gefaell J, Galindo J, Malvido C, Nuñez V, Estévez D, Blanco S, González-Conde M, Martínez-Domínguez S, Novo I, Pérez-Fernández G, Rus B, Mosconi I & Rolán-Alvarez E. 2021. Negative assortative mating and maintenance of shell colour polymorphism in *Littorina* (*Neritrema*) Species. *Marine Biology* 168: 151. [CCC Reprint license number: 5766420904873]

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SHORT NOTES

Negative assortative mating and maintenance of shell colour polymorphism in *Littorina* (*Neritrema*) species

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Abstract
Colour polymorphism is a widespread phenomenon in natural populations of several species. In particular, it is especially common on marine gastropod species from the genus *Littorina*. Recently, it has been argued that intrapopulation shell colour polymorphism in *Littorina fabalis* could be caused by negative frequency-dependent sexual selection via a mechanism of mate choice (indirectly estimated through negative assortative mating). Here we try to determine the existence of negative assortative mating in three species of the subgenus *Neritrema* (*L. fabalis*, *L. obtusata, L. saxatilis*) that share a similar shell colour polymorphism, to ascertain if this mechanism could represent an ancestral character in this subgenus that could be contributing to the maintenance of the colour polymorphism observed in each species. We collected or reanalysed from previous studies a sample of mating pairs of these three species from seven locations from NW Spain and NE Russia and estimated assortative mating using the I_{PSI} index. Our results suggest that all species and populations show a systematic tendency towards negative assortative mating when shell colour is grouped in the broad categories 'light' and 'dark'. Moreover, a more detailed analysis of each colour separately suggests that shell colour may not be the main target of assortative mating, but perhaps a physically-linked trait to the real target of selection. This hypothesis opens interesting new lines of research in *Littorina* snails.

Introduction

Colour is a key trait in evolution, usually contributing to individual survival (Olsson et al. 2013; Rolán-Alvarez et al. 2015a) and reproduction (Cordero 1989; Hugall and Stuart-Fox 2012), and therefore potentially influencing fitness. In addition, colour is a relatively easy trait to detect and measure, and so it has been thoroughly studied in several taxa (McKinnon and Pierotti 2010; Hugall and Stuart-Fox 2012; Svensson 2017; Cuthill et al. 2017). In particular, shell colour has been extensively studied in gastropod snails (reviewed in Williams 2017).

Of special interest in studies of colour is colour polymorphism, in which there exist various discrete and genetically determined colour phenotypes within a population (White and Kemp 2016). Colour polymorphisms are expected to be infrequent, given that directional natural selection and genetic drift tend to erode genetic variability over time, therefore reducing observed polymorphisms (Lewontin 1974; Nielsen 2005). However, under certain circumstances selection can maintain a polymorphism in a population, and even despite the tendency of genetic drift to fixation, a polymorphism can be transitorily maintained from a few to several generations depending on effective population size and gene flow (Wright 1977; Clarke 1979; Kimura 1983; Hartl and Clark 2006). This makes it rather difficult to identify the true causal factor responsible for the evolutionary persistence of colour polymorphism within a population. One of the most effective types of selection able to maintain a polymorphism in wild populations is negative frequency-dependent selection (Ayala and Campbell 1974; Fitzpatrick et al. 2007; Svensson and Connallon 2019). Negative frequency-dependent selection occurs when the fitness of a

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particular genotype correlates with its frequency, in a manner that a decrease in frequency increases its fitness relative to alternative genotypes (Ayala and Campbell 1974). Fitness will peak as frequency falls below a certain threshold, hence protecting the polymorphism from being lost in the population.

The genus *Littorina* has been shown to be a particularly useful model organism in the field of evolutionary ecology (Rolán-Alvarez et al. 2015a). Besides speciation being the main focus of the evolutionary research carried out in these species (e.g. Morales et al. 2019; Costa et al. 2020; Galindo et al. 2020), the maintenance of intrapopulation colour polymorphism has also been studied (Rolán-Alvarez and Ekendahl 1996; Rolán-Alvarez et al. 2012, 2015b; Johannesson and Butlin 2017; reviewed in Rolán-Alvarez et al. 2015a). Recently, it has been shown that shell colour polymorphism in populations of the intertidal gastropod *Littorina fabalis* (Turton 1825) can be maintained by negative frequency-dependent sexual selection via mate choice (Estévez et al. 2020). Another two species of the subgenus *Neritrema* (genus *Littorina*), *L. saxatilis* (Olivi 1792) and *L. obtusata* (Linnaeus 1758) have also shown colour polymorphism within certain populations (Reid 1996; Rolán-Alvarez et al. 2015a). These three species are direct developers (Reid 1996), lacking a planktotrophic larval stage, and therefore showing a potential for fast adaptation to local conditions (reviewed in Johannesson 2003; Galindo and Grahame 2014; Rolán-Alvarez et al. 2015a). Another important characteristic in these (dioecious) *Littorina* species is the possibility to observe mating pairs, which allows to readily estimate assortative mating in the wild (reviewed in Rolán-Alvarez et al. 2015a). In this context, assortative mating is understood as the phenotypic correlation between mates (Jiang et al. 2013; Janicke et al. 2019), which can be measured by isolation indexes in the case of qualitative traits, like shell colour (reviewed Gilbert and Starmer 1985; Rolán-Alvarez and Caballero 2000). Assortative mating for colour and its evolutionary consequences have been widely studied in several species over the last few years (Calderon et al. 2010; Pryke 2010; De Lanuza et al. 2013; Gade et al. 2016; Hedrick et al. 2016; Yang et al. 2016; Fargevieille et al. 2017). Besides, frequent biases in the estimation of assortative mating as well as its potential solutions have been recently identified (Rolán-Alvarez et al. 2015b).

Although there are alternative explanations (see Taborsky et al. 2009; Kopp et al. 2018), assortative mating in simple organisms such as marine gastropods is presumably caused by mate choice (Ng et al. 2019; Estévez et al. 2020). On the other hand, negative assortative mating can be directly related to negative-frequency dependent selection, as it has been shown both theoretically (Pusey and Wolf 1996; Hedrick et al. 2016, 2018) and empirically in several species (Takashi and Hori 2008; Field and Barrett 2012; Holman et al. 2013; Hedrick et al. 2016, 2018; Estévez et al. 2020). This, in turn, makes it plausible to assume that negative assortative mating is a relatively good proxy for both mate choice and negative-frequency dependent selection. Here we study assortative mating for shell colour in new populations of *L. fabalis* and two other related species with a similar colour polymorphism, *L. obtusata* and *L. saxatilis.* The aim of this study is to determine the existence of a negative assortative mating pattern in these two new species, as well as to corroborate its presence in the new populations of *L. fabalis*. Given that previous results strongly suggest a causal link between negative assortative mating and both mate choice and negative frequency-dependent selection, we suggest that if a negative assortative mating pattern is found in these two new species, an ancestral plesiomorphic behavioural trait for mate choice may exist within the subgenus *Neritrema*, which encompasses *L. fabalis*, *L. obtusata* and *L. saxatilis*. This could pave the way to design future studies to directly test the existence of mate choice and negative frequency-dependent selection in the *Littorina* clade.

Materials and methods

Study species and trait

We selected three colour polymorphic species from the genus *Littorina* (*L. obtusata*, *L. fabalis* and *L. saxatilis*), subgenus *Neritrema.* This subgenus is the only one with non-planktotrophic development within the genus, and it is formed by ten species with a Northern distribution (Reid 1996; Reid et al. 2012). Within this subgenus, we find the group of flat periwinkles composed by *Littorina fabalis and L. obtusata*, which are two sibling species (split 0.85 Mya, Sotelo et al. 2020) that live on intertidal brown macroalgae (e. g. *Fucus* sp.). *L. fabalis* feeds on diatoms and other microalgae that grow on the seaweed surface, while *L. obtusata* feeds directly on the macroalgae (Reid 1996). They also differ in habitat, *L. fabalis* commonly dwelling lower on the shore while *L. obtusata* lives in the upper part of the intertidal range. In certain localities their distribution partially overlaps, whereas in others only one species is found depending on the environmental conditions of the shore. In NW Spain they can be observed in true sympatry living on the same *Fucus vesiculosus* plants (Rolán-Alvarez and Ekendahl 1996). On the other hand, they are both oviparous, laying egg masses on the algae from which juveniles hatch as miniature snails (Reid 1996).

The rough periwinkle, *L. saxatilis* (split with *L. fabalis* 2.5 Mya, Panova et al. 2011), is the only ovoviviparous species in this subgenus (Reid 1996). The embryos are developed in a brood pouch inside the female until eventually

hatching as miniature snails. *L. saxatilis* lives on the rock surfaces across the whole intertidal range, feeding on diatoms, microalgae and detritus (Otero-Schmitt et al. 1997). The three species have very similar distribution across NE Atlantic coasts (Reid 1996) and in many shores they can be found overlapping, although their microhabitats might differ.

Shell colour in these species has been typically studied by visual (qualitative) codification (reviewed in Reid 1996; Rolán-Alvarez et al. 2015a) but at least in one case (*L. fabalis* from Abelleira) the qualitative classes have been validated by reflectance spectrometry (Rolán-Alvarez et al 2015b). Shell colour variation in the three species has been systematically attributed to one or a few genes (reviewed in Rolán-Alvarez et al. 2015a). Shell colours have not been commonly associated to selective pressures, but there are a few exceptions (reviewed in Reid 1996; Rolán-Alvarez et al. 2015a). For example, unknown predators cause colour-based selective mortality in *Littorina sp.* from Iceland (Johannesson and Ekendahl 2002), and yellow-painted *L. obtusata* specimens were cooler and survived better than black-painted ones (Phifer-Rixey et al. 2008). On the other hand, survival differences between colours in *L. saxatilis* has been associated with physiological differences to resist salinity gradients (Sokolova and Berger 2000).

Experimental methods

To study assortative mating for shell colour in *L. fabalis*, *L. obtusata* and *L. saxatilis*, the experimental design consisted in the capture of mating pairs and the surrounding unmated individuals in natural populations following previous studies (Rolán-Alvarez and Ekendahl 1996; Rolán-Alvarez et al. 2012, 2015b; Estévez et al. 2020). These three species show a similar mating strategy, where males usually perform an active search of females, following them by their mucus trails, and typically mate by size similarity (Saltin et al. 2013; Ng et al. 2019). In the field, a mating pair can be detected when the male is located over the female in a certain position on the right-hand side of the shell (with the axis of coiling of the two snails forming a 'V'-like pattern) and the penis is inserted in the mantle cavity of the female (Fig. 1A, B). Associated to every mating pair we collected the four closest unmated individuals of the same species (typically within 5 cm of the mating pair), and these six snails were defined as an individual microarea for further analyses (see Fig. 2 in Estévez et al. 2020). These samplings were done for several days during the summer for each year since snail density and matings increase during this season (Rolán-Alvarez and Ekendahl 1996).

All the snails were taken alive to the laboratory and stored at − 20° C. The mating pairs were dissected to discard male-male matings from the analyses, as they are known to be quite frequent in *L. saxatilis* (Rolán-Alvarez et al. 1999). Following these dissections, we discarded around 10% for *L. saxatilis* and 12% for flat periwinkles. Then, the background shell colour of each individual was scored using a colour printed model following Rolán-Alvarez et al. (2015b) and classified into the categories described in Fig. 1. The banding pattern present in some shells was ignored in further analyses, as its frequency differed between species and it

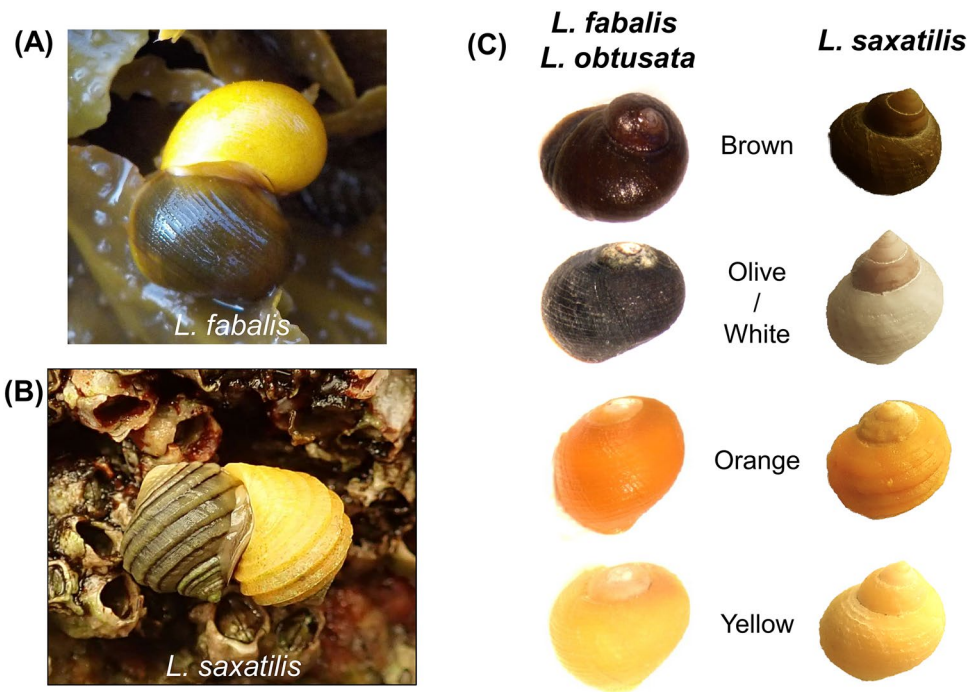

Fig. 1 **A** *Littorina fabalis* mating pair (yellow male and brown female). **B** *L. saxatilis* mating pair (yellow male and brown female). **C** Shell colour pattern in *L. fabalis, L. obtusata* and *L. saxatilis*

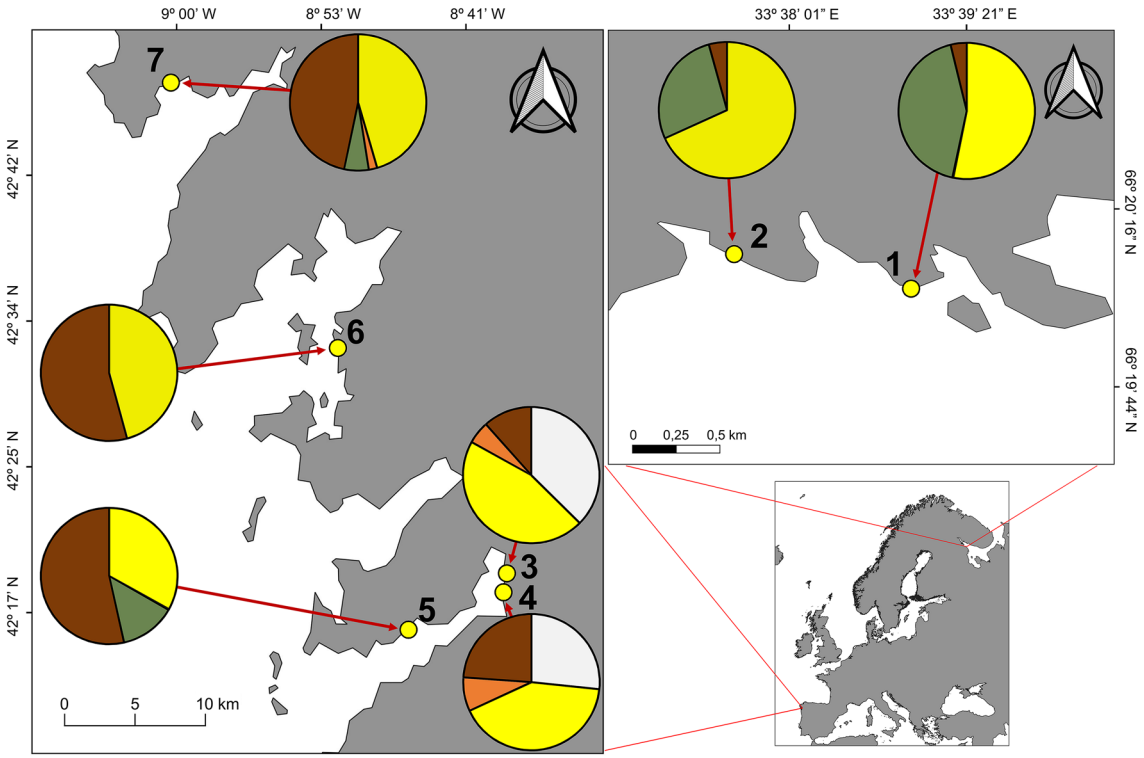

Fig. 2 Geographical location of the different localities sampled for *L. fabalis* (1, 2, 5, 6, and 7), *L. obtusata* (6) and *L. saxatilis* (3 and 4). The pie charts represent the observed frequency of shell colours in each locality. Notice that in Castelete both *L. fabalis* and *L. obtusata* samples showed similar colour frequencies (Supplementary Table S1), and so only the *L. obtusata* pie chart is presented. Russia: 1 (WS-1) and 2 (WS-2). Spain: 3 (Portela), 4 (Soutoxuste), 5 (Cangas), 6 (Castelete) and 7 (Abelleira)

also varied in fadedness. The three studied species shared yellow, orange and brown shells while olive is only present in flat periwinkles and white only in *L. saxatilis* (Fig. 1C, D), for the studied populations. As in previous studies of *L. fabalis* (Rolán-Alvarez et al. 2015b; Estévez et al. 2020) these colours were classified into two categories: light (white, yellow, orange) and dark (olive, brown). In addition, in this study we have also performed the analyses based on the individual colours, comparing each colour against the rest, on a binary manner (see Table S1).

The analyses were carried out with the colour of the mated and unmated individuals using a previously published dataset on *L. fabalis* (Estévez et al. 2020) and newly collected data on the three species (data available at: https://doi.org/10.6084/m9.figshare.13295873.v1). Estévez et al. (2020) data is composed by samples from 7 consecutive years (2011–2017) from Abelleira (Ria of Muros-Noia; NW Spain; 42°47' 45.9" N–9°01'19.1" W) and two close sites (< 1 km apart) from the White Sea (NE Russia; WS-1, 66°20' 9.80" N–33°38'41.29" E; WS-2, 66°20'10.73" N–33°37'53.00" E) collected in 2015. Then a new population of *L. fabalis* was sampled in Cangas NW Spain (Ria of Vigo; NW Spain, 42°15' 43.7" N–8°45' 28.4" W) in 2018. Another locality, Castelete, was sampled for *L. fabalis* and *L. obtusata* (Ria of Arousa; NW Spain, 42°32' 36.9" N–8°49′57.0″W) in 2016. In this case, both species were differentiated based on penis morphology following Reid (1996) and assuming the lack of interspecific mating pairs. For *L. saxatilis* two close by localities were sampled in the Ria of Vigo (NW Spain), Portela (42°17' 17.74" N–8°37' 21.87" W) and Soutoxuste (42°19' 22.30" N–8°36' 57.44" W) in 2020. The number of collected mating pairs for each species and locality is shown in Table 1.

Statistical analyses

Assortative mating for qualitative traits can be estimated by several indices (reviewed in Rolán-Alvarez and Caballero 2000), from which the I_{PSI} index is the most robust under different sampling scenarios (Pérez-Figueroa et al. 2005). This index can be estimated in a hypothetical population with two colours (brown, B; yellow, Y) as,

$$I_{PSI} = \frac{\left(PSI_{BB} - PSI_{BY} - PSI_{YB} + PSI_{YY}\right)}{\left(PSI_{BB} + PSI_{BY} + PSI_{YB} + PSI_{YY}\right)}.$$

Being PSI_{BB} the ratio between the observed number of brown female–brown male mating pairs and the expectation

Table 1 Estimates of assortative mating (I_{PSI}) for shell colour in natural populations of *L. fabalis*, *L. obtusata* and *L. saxatilis*

Species	Locality	*N*	I_{PSI}					
			Dark/light	White	Yellow	Orange	Olive	Brown
L. fabalis	Abelleira[a]	811	− 0.29*** ± 0.068	–	− 0.31** ± 0.021	− 0.03 ± 0.632	0.07 ± 0.403	− 0.30** ± 0.032
	Cangas	223	− 0.24 ± 0.158	–	− 0.23 ± 0.165	− 0.16 ± 0.291	0.41 ± 0.304	− 0.31? ± 0.133
	Castelete	12	− 0.55 ± 0.403	–	–	–	–	–
	WS-1 (Russia)[a]	93	− 0.44? ± 0.202	–	− 0.35* ± 0.132	-	− 0.45* ± 0.145	-
	WS-2 (Russia)[a]	77	− 0.27 ± 0.403	–	− 0.23 ± 0.165	-	− 0.40* ± 0.117	− 0.35
			− 0.36* ± 0.123		− 0.28** ± 0.060	− 0.09 ± 0.092	− 0.09 ± 0.409	− 0.32** ± 0.026
L. obtusata	Castelete	44	− 0.59*** ± 0.117	–	–	–	–	–
L. saxatilis	Soutoxuste	74	− 0.36** ± 0.035	− 0.16 ± 0.291	− 0.26 ± 0.269	− 0.17 ± 0.247	–	− 0.56 ± 0.262
	Portela	114	− 0.31? ± 0.113	− 0.11 ± 0.156	− 0.34* ± 0.104	− 0.24 ± 0.339	–	− 0.33 ± 0.085
			− 0.33* ± 0.025	− 0.14 ± 0.035	− 0.30? ± 0.057	− 0.20 ± 0.049	–	− 0.44 ± 0.163

The studied localities for each species and the number (*N*) of microareas (mating pair + unmated individuals) are indicated. The first estimate of I_{PSI} takes into account the colours grouped into dark (olive, brown) or light (white, yellow, orange). The second estimate is carried out for each separate colour vs the rest of the colours (e.g. white vs yellow/orange/olive/brown). For each I_{PSI} the standard deviation is shown

?$P < 0.10$; *$P < 0.05$; **$P < 0.01$; ***$P < 0.001$

[a]From Estévez et al. (2020)

of this type of mating pair under random mating (using exclusively the mated individuals to estimate random mating; see further details in Carvajal-Rodríguez and Rolán-Alvarez 2006 and references therein). The remaining *PSI* values are calculated accordingly. The I_{PSI} index ranges from -1 (maximum negative assortative mating) to 1 (maximum positive assortative mating), being 0 complete random mating. This index has been used successfully in previous studies of assortative mating in *Littorina* (Rolán-Alvarez et al. 2015b; Estévez et al. 2020). The I_{PSI} index was calculated for the trait shell colour, categorized as a binary trait in two different analyses. The first one (pooling analysis), grouped colours into light (white, yellow, orange) and dark (olive, brown) as mentioned in the experimental design and following previous studies. The second analysis (individual analysis) considered every colour separately against the rest of the colours grouped. These two analyses were carried out for each species and population datasets.

The I_{PSI} index gives an estimate of assortative mating for qualitative traits similar to Pearson's correlation coefficient for quantitative traits (Rolán-Alvarez et al. 2015b). However, these authors also showed that the existence of trait heterogeneity at a smaller scale than the scale of mate choice would bias assortative mating towards positive values (i. e. the scale-of-choice effect sensu Rolán-Alvarez et al. 2015b). Therefore, to reduce the potential bias towards positive values of assortative mating, we also performed an analysis subdividing each dataset (species/population) into three different homogeneous groups of microareas (mating pair + four unmated individuals) based on the colour frequencies of the microareas (pooling analysis: light vs dark; individual analysis: one colour vs the rest) as follows: low frequency (0–33.3% light or other colour), intermediate frequency (33.4–66.6%) and high frequency (66.7–100%). Then the I_{PSI} was estimated within each of these three homogeneous groups of microareas and then averaged across groups (Rolán-Alvarez et al. 2015b; Ng et al. 2016; Estévez et al. 2018, 2020). The significance of the I_{PSI} indices was accomplished by *t* tests (with null value of 0) using the three different homogeneous groups (low, intermediate, high frequency), and for both analyses (pooling and individual). On the other hand, due to the low sample size obtained in Castelete, the significance has been estimated by bootstrapping in JMATING (Carvajal-Rodríguez and Rolán-Alvarez 2006) following Rolán-Alvarez et al. (2015b).

Results and discussion

Even though directional natural selection and genetic drift would preclude the persistence of genetic diversity (Lewontin 1974; Nielsen 2005), colour polymorphisms exist in many invertebrate and vertebrate species. The same colour polymorphism can even sometimes be found in closely related species, suggesting the persistence of a shared evolutionary mechanism maintaining polymorphism across species radiation (Jamie and Meier 2020). We have studied a group of marine gastropod species within the subgenus *Neritrema* (genus *Littorina*) that share a similar shell colour polymorphism (Fig. 2; Table S1). The results of the pooling analyses of assortative mating (light vs dark) using mating pairs from natural populations of the three studied species (*Littorina fabalis*, *L. obtusata* and *L. saxatilis*) show a consistent negative assortative mating across all populations and

species (Table 1). *L. obtusata* showed the greatest negative assortative mating ($I_{PSI} = -0.59$) and *L. fabalis* from Cangas, the lowest (− 0.24). A similar result is obtained from the I_{PSI} estimates for the homogeneous groups of colour frequency (Supplementary Table S2), which have been used to correct for the scale-of-choice effect (Rolán-Alvarez et al. 2015b). These estimates support previous results obtained for *L. fabalis* in Abelleira (Rolán-Alvarez and Ekendahl 1996; Rolán-Alvarez et al. 2012, 2015b; Estévez et al. 2020) and are also similar to those found in other species (Takashi and Hori 2008; Field and Barrett 2012; Holman et al. 2013; Hedrick et al. 2016, 2018). Even though our results cannot unravel the evolutionary mechanisms responsible for the observed patterns, they strongly suggest that negative assortative mating for shell colour in *Neritrema* species is caused by mate choice and therefore it could be at least partially responsible for the maintenance of the colour polymorphism within population via negative frequency-dependent sexual selection. There are at least two different pieces of evidence that support this claim, as we have already suggested: first, theoretical analysis have shown that any mate-choice based negative assortative mating can render a negative frequency-dependent sexual selection that could be able to maintain the polymorphism (Pusey and Wolf 1996; Hedrick et al. 2016, 2018). Second, recently Estévez et al. (2020) have shown that this relationship between negative assortative mating, mate choice and negative frequency-dependent selection holds for *L. fabalis* (see similar evidences for other model organisms: Takashi and Hori 2008; Field and Barrett 2012; Holman et al. 2013; Hedrick et al. 2016, 2018). These two lines of evidence, in addition to the results presented in this study, suggest that an ancestral plesiomorphic behavioural mechanism could exist for mate choice within the *Neritrema* subgenus. We find the ancestral behavioural mechanism hypothesis plausible, since, from an evolutionary point of view, it is reasonable to expect similar behavioural mechanisms operating under similar ecological conditions in closely related species. This is akin to what in evolutionary studies of cognition has been dubbed the "evolutionary parsimony" principle (de Waal 1999). Thus, if *L. fabalis*, *L. obtusata* and *L. saxatilis* show a similar behavioural pattern, live in the same ecosystem, and they all share a (relatively) recent common ancestor, it is reasonable to expect a common behavioural mechanism behind this behavioural pattern. Nevertheless, this hypothesis, although plausible, needs to be experimentally tested in the future before concluding its validity.

In any case, there are a few potential caveats in the present study that need to be addressed to dispel doubts about our results. First, the I_{PSI} and similar indexes allow us to estimate assortative mating, and are frequently used as a proxy of mate choice, at least in laboratory conditions (reviewed in Gilbert and Starmer 1985; Rolán-Alvarez and Caballero 2000; Pérez-Figueroa et al. 2008). However, their capability to estimate mate choice has been discussed and several sources of bias have been identified (Rolán-Alvarez and Caballero 2000; Pérez-Figueroa et al. 2008). In any case, when they have been exclusively used to estimate assortative mating in the laboratory, these indices performed appropriately (Conde-Padín et al. 2008; Rolán-Alvarez et al. 2015b). On the other hand, it has been shown that correlation indexes could be potentially inflated towards positive values when estimated directly in the wild, especially in species with low mobility, as well as when the study species shows great variability (e. g. colour) at a scale smaller than the scale at which the samples were collected (Rolán-Alvarez et al. 2015b). This bias could be corrected by estimating the I_{PSI} in homogeneous groups (e. g. by similar colour), as it has been shown by simulations (Rolán-Alvarez et al. 2015b) and has been carried out here. In fact, this correction has been used previously in cases of negative and positive assortative mating (Rolán-Alvarez et al. 2015b; Ng et al. 2016; Estévez et al. 2018, 2020). Thus, the correction used here does not correct, a priori, towards either negative or positive values of assortative mating. For example, the same correction was used to estimate the real scale at which mate choice occurs in two species of marine gastropods (Estévez et al. 2018), one characterised by positive assortative mating (*Echinolittorina malaccana*) and the other by negative assortative mating (*L. fabalis*), highlighting that this method is suitable to estimate both types of assortative mating. Therefore, the correction that we applied in this study assures a more reliable estimate of assortative mating in natural populations.

Second, another potential bias of our results could be caused by the actual observation of the mating pairs during the collection of the samples. For example, mating pairs in flat periwinkles (*L. fabalis, L. obtusata*) are commonly found among a dense canopy of *Fucus vesiculosus*, typically dark brown, allowing for a possible sampling bias towards the most visible mating pairs (light-light, dark–light). Although, to avoid this potential bias, we performed the search of mating pairs very close to the algae (~ 30 cm) and for each sampled microarea we checked all the individual algae, so it was unlikely to miss dark-dark pairs more often than the rest. Moreover, this bias in the sampling procedure has been rejected in previous studies as light-light mating pairs, which are highly noticeable on the *Fucus* canopy, appear at significantly lower frequencies than expected under random mating (Rolán-Alvarez et al. 2015b; Estévez et al. 2020). This potential sampling bias is even more unlikely in *L. saxatilis,* given that in this case all mating pairs are collected on the rock surface, as this species grazes on microalgae that grows on the bare rock. Therefore, in the light of the current results it seems plausible a hypothesis where a mating strategy of negative assortative mating for shell colour could represent an ancestral character within the subgenus *Neritrema.*

Recently, Estévez et al. (2020) have investigated the behavioural mechanism responsible for the negative assortative mating observed in *L. fabalis* from Abelleira (NW Spain) and the White Sea (NE Russia) populations. They found indirect evidence that mate choice is involved in assortative mating, with males showing preference for females with different shell colours. This, in turn, points to negative frequency-dependent selection as the evolutionary mechanism maintaining colour polymorphism in this species. However, their results did not provide any evidence regarding the potential contribution of colour to fitness, so that it would be favoured by mate choice. Their results neither clarify if colour, and not any other genetically linked trait, is the real target of selection. Here, we present a complementary result that could help to suggest the relationship between shell colours and mating in these species. In the analysis of each colour separately (Table 1 and Supplementary Table S2), we found an interesting pattern given that, although in general all colours showed the exact same trend as in those cases in which they are grouped in dark and light colours, there was an exception. The olive shell colour in *L. fabalis* changes from a pattern of random (positive but not significant estimates) mating in Spain to a pattern of negative assortative mating in the Russian populations (Table 1 and Supplementary Table S2; Fig. 3). This result is less probable if the shell colour itself is the target of the putative behavioural mechanism causing the assortative mating, but it could be expected if the colour is linked to another adaptive trait, physically or through pleiotropy. A relevant question is, therefore, what circumstances make a particular mating behaviour to be present in one location but not in others. One hypothesis is that the colour genes are associated with several other genes in a chromosomal inversion. In such scenario, different alleles of the colour gene could be associated with different chromosome variants (inversed or not). In support of this view is the fact that colour vision has not been detected and olfactory cues seems more relevant than visual cues in determining movement in aquatic gastropods (Seyer 1992; Wyeth 2019). Actually, Cephalopods, the animals with the most complex visual system within the phylum Mollusca, have only one visual pigment and therefore cannot see in colour (Nilsson 2013). If cephalopods do not have colour vision despite having a complex visual system, it seems implausible that gastropods could have it. Because of this, previous hypotheses about mate choice for shell colour in gastropods were based on the possibility that they can smell different colours when males follow mucus trails of females in the context of mate searching (see Estévez et al. 2020). Actually, a relationship between assortative mating for colour forms and sex chromosomes have been claimed in one finch species (Pryke 2010). Chromosomal inversions have been recently detected and associated with presumably adaptative traits in several organisms (Wellenreuther and Bernatchez 2018), including littorinids (Westram et al. 2018; Faria et al. 2019), and in at least one case has been related to negative assortative mating too (Hedrick et al. 2018). This could mean that the trait responsible for the pattern of negative assortative mating would be linked to the colour gene within the inversion, but could not be the colour itself. Although speculative, this hypothesis can be tested in the future and it opens a potentially fruitful new line of research.

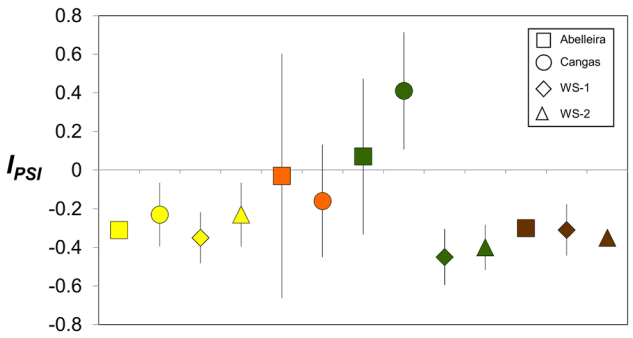

Fig. 3 Estimates of corrected (for the scale-of-choice effect) I_{PSI} values (with standard deviations) for the different populations of *L. fabalis* (from Table 1). Colours in the figure represent different shell colours. Notice that estimates for olive shell colour differ between Spain and Russia (Russia samples are significantly different from 0; see text). Notice that standard deviations for frequent colours in Abelleira are so small that fall inside symbols

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Declarations

Conflict of interest The authors declare that they have no conflict of interest.

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Consent to participate All included authors have given their consent to participate.

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Supplementary materials – Chapter 2

Table S1. Observed percentages of shell colours in the studied populations for *L. fabalis*, *L. obtusata* and *L. saxatilis*. A few individuals with rare colours (Others) were observed and excluded from the assortative mating analyses, like a tessellated pattern in *L. fabalis* in Castelete and a banded pattern in *L. saxatilis* in Soutoxuste. N is the total number of specimens studied (including mating pairs from Table 1 and surrounding unmated individuals).

			Light			Dark		
Species	Locality	N	White	Yellow	Orange	Olive	Brown	Others
L. fabalis	Abelleira[#]	4764	–	45.5	2	5.8	46.7	–
	Cangas	1331	–	33.1	0.2	13.2	53.5	–
	Castelete	71	–	45.1	–	–	53.5	1.5
	WS-1 (Russia)[#]	558	–	53.2	0.2	42.8	3.8	–
	WS-2 (Russia)[#]	462	–	68.2	–	27.5	4.3	–
L. obtusata	Castelete	263	–	45.2	–	–	53.2	1.4
L. saxatilis	Soutoxuste	444	37.5	45.6	5.4	–	11.5	0.2
	Portela	683	26.6	41.4	8.1	–	23.9	–

Taken from Estévez *et al*. (2020)

Table S2. Estimates of I_{PSI} for different subsets of microareas, as a general strategy to correct for scale-of-choice biases (see Rolán- Alvarez et al. 2015a). The averaged of these values (Dark, Intermediate and Light) are shown in Table 1 with corresponding t test. The significance and standard deviation are calculated by bootstrapping in JMATING (see Carvajal-Rodríguez and Rolán-Alvarez, 2008) except for Abelleira that they are calculated by a t test on seven years. The studied localities for each species and the number (N) of microareas (mating pair + unmated individuals) are indicated.

			Light microareas			White microareas		
Species	Locality	N	Dark	Intermediate	Light	Rest	Intermediate	White
L. fabalis	Abelleira[#]	811	-0.31** ± 0.159	-0.29*** ± 0.069	-0.30** ± 0.172	–	–	–
	Cangas	223	-0.07 ± 0.109	-0.38* ± 0.148	-0.28 ± 0.158	–	–	–
	Castelete	12	–	–	–	–	–	–
	WS-1 (Russia)[#]	93	-0.52* ± 0.140	-0.21* ± 0.184	-0.28 ± 0.158	–	–	–
	WS-2 (Russia)[#]	77	-0.07 ± 0.170	-0.73* ± 0.228	0	–	–	–
L. obtusata	Castelete	44	–	–	–	–	–	–
L. saxatilis	Soutoxuste	74	–	-0.33 ± 0.334	-0.38 ± 0.346	-0.04 ± 0.215	-0.04 ± 0.228	-0.50 ± 0.526
	Portela	114	–	-0.23 ± 0.156	-0.39 ± 0.286	–	0.0 ± 0.165	-0.22 ± 0.219

Taken from Estévez *et al.* (2020); * p<0.05; ** p<0.01; *** p<0.001

Table S2 (Continuation-1).

			Yellow microareas			Orange microareas		
Species	Locality	N	Dark	Intermediate	Yellow	Rest	Intermediate	Orange
L. fabalis	Abelleira[#]	811	-0.32*** ± 0.138	-0.33*** ± 0.057	-0.29** ± 0.149	-0.03 ± 0.632	–	–
	Cangas	223	-0.28 ± 0.209	-0.37* ± 0.150	-0.05 ± 0.110	–	–	–
	Castelete	12	–	–	–	–	–	–
	WS-1 (Russia)[#]	93	–	-0.35* ± 0.139	-0.52* ± 0.146	–	–	–
	WS-2 (Russia)[#]	77	–	-0.45* ± 0.223	-0.07 ± 0.171	–	–	–
L. obtusata	Castelete	44	–	–	–	–	–	–
L. saxatilis	Soutoxuste	74	-0.41 ± 0.431	0.05 ± 0.149	-0.42 ± 0.455	-0.35 ± 0.519	0.0 ± 0.435	–
	Portela	114	-0.40 ± 0.424	-0.40 ± 0.107	-0.22 ± 0.221	0.0 ± 0.197	-0.48 ± 0.337	–

Taken from Estévez *et al.* (2020) * p<0.05; ** p<0.01; *** p<0.001

Table S2 (Continuation-2).

			Olive microareas			Brown microareas		
Species	Locality	N	Dark	Intermediate	Olive	Rest	Intermediate	Brown
L. fabalis	Abelleira[#]	811	-0.22 ± 0.221	0.35 ± 0.354	–	-0.29** ± 0.134	-0.34*** ± 0.070	-0.28* ± 0.217
	Cangas	223	–	0.63$^{?}$ ± 0.283	0.20 ± 0.138	-0.23* ± 0.113	-0.47** ± 0.112	-0.25$^{?}$ ± 0.151
	Castelete	12	–	–	–	–	–	–
	WS-1 (Russia)[#]	93	-0.45 ± 0.360	-0.31* ± 0.129	-0.60$^{?}$ ± 0.244	–	–	–
	WS-2 (Russia)[#]	77	-0.42 ± 0.276	-0.27* ± 0.180	-0.50 ± 0.446	-0.35 ± 0.513	–	–
L. obtusata	Castelete	44	–	–	–	–	–	–
L. saxatilis	Soutoxuste	74	–	–	–	-0.38 ± 0.317	-0.75 ± 0.469	–
	Portela	114	–	–	–	-0.39 ± 0.233	-0.27 ± 0.167	–

Taken from Estévez *et al.* (2020) * p<0.05; ** p<0.01; *** p<0.001; ? p<0.10

Chapter 3

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Experimental evidence of mate choice as the driving mechanism behind negative assortative mating for shell colour in a marine snail

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ABSTRACT

How colour polymorphisms are maintained in natural populations constitutes a key subject of study for evolutionary ecologists. One of the mechanisms that has been put forward to explain this phenomenon is negative frequency-dependent selection, which can be caused by different mechanisms. Among these mechanisms is mate choice, or the selection as a potential mate of the rare or dissimilar colour morph by the choosy sex of a given species. In the context of colour polymorphism, mate choice predicts the occurrence of a negative assortative mating pattern, whereby different colour morphs mate more often than expected by chance alone. However, negative assortative mating can also result from other mechanisms, making it, at best, a fallible indicator of a mate choice-driven negative frequency-dependent selection. For this reason, experimental evidence explicitly connecting such a mating pattern to mate choice is necessary to claim that a particular colour polymorphism is being maintained by negative frequency-dependent selection. Here we present experimental evidence backing this relationship for the colour polymorphic marine gastropod *Littorina saxatilis* from the Ría de Vigo (NW Iberian Peninsula), although how specifically these organisms choose their mates remains unclear. This calls for further experimental efforts to clarify this issue and its relationship to the maintenance of colour polymorphism in *L. saxatilis*.

Keywords: colour polymorphism; *Littorina saxatilis*; mate choice; negative assortative mating; negative frequency-dependent selection; shell colour

INTRODUCTION

The evolution and maintenance of colour polymorphism is a classic topic of study in evolutionary ecology (Ford 1945, 1979, Bond 2017, Svensson 2017). Most major animal taxa contain at least some species that show intrapopulation colour diversity (Gray and McKinnon 2007, White and Kemp 2016), which prompts the question of how this condition is preserved in spite of the homogenizing forces of natural selection and genetic drift (Lewontin 1974, Nielsen 2005). The presence of colour polymorphism has been linked to several important evolutionary processes, such as speciation and diversification (Gray and McKinnon 2007, Hugall and Stuart-Fox 2012, Kusche *et al.* 2015, Brock *et al.* 2022), or population persistence (Forsman *et al.* 2008). Coloration in animals may have three different (and nonexclusive) fitness-related functions: communication (with either conspecifics or predators, or both), crypsis, and thermoregulation (Endler 1978). These functions, in conjunction with microhabitat or spatial heterogeneity, can lead to the evolution of colour polymorphisms through divergent selection (Gefaell *et al.*, 2023a). For instance, there is evidence that the shell colour polymorphism shown by the populations of the marine snail *Littorina obtusata* from the Gulf of Maine (which live under different thermal regimes) is maintained by selection on thermal stress resistance (Phifer-Rixey *et al.* 2008).

Another possibility regarding the evolution and maintenance of colour polymorphism—or for that matter, of any other kind of genetic polymorphism—is negative frequency-dependent selection, in which the rarer colours (or their corresponding alleles) are protected from extinction because of their greater relative fitness (Ayala and Campbell 1974, Futuyma and Kirkpatrick 2017,

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Brisson 2018, Sætre and Ravinet 2019). Negative frequency-dependent selection, in turn, can take place through different ecological and/or behavioural mechanisms, such as resource competition, apostatic selection, or mate choice of rare or dissimilar individuals (Kokko *et al.* 2007, Chouteau *et al.* 2017, Brisson 2018, Estévez *et al.* 2020). The role of apostatic selection as the driving force of negative frequency-dependent selection for colour variation has been widely studied over the last 60 years, with the classic model genus *Cepaea* standing out in this regard (e.g. Clarke 1962, Tucker 1991, Cook 2017). In such a system, apparently individuals with the least frequent colour patterns go unnoticed by at least some of their predators (particularly the song thrush, *Turdus philomenos*), which seem to prey by forming search images of their most common prey (Tucker 1991; but see Cook 2017). The mechanism of mate choice has also been investigated, and some putative examples have been described in nature (Hughes *et al.* 2013, Wellenreuther *et al.* 2014, Estévez *et al.* 2020, Potter *et al.* 2023). According to the mate choice hypothesis, the individuals of the choosy sex of a given species would show a consistent behavioural preference either for those conspecifics of the opposite sex that bear the least frequent colours and/or alleles, or for those that are simply different (i.e. dissimilar) from themselves, depending on whether the choice is for rarity or dissimilarity. In the mid- to long term, this preference would lead to an (inter)sexual selective advantage of the rare and/or dissimilar colours and, thereby, to their protection from extinction (Wellenreuther *et al.* 2014, Estévez *et al.* 2020).

An indicator of mate choice-driven negative frequency-dependent selection in a population can be negative assortative mating (also known as disassortative mating), in which the frequency of matings between individuals with different phenotypes or genotypes is higher than that expected by chance alone (Follett *et al.* 2007, Jiang *et al.* 2013). Assortative mating and its relationship to colour polymorphism have also been studied in several taxa (Calderón *et al.* 2010, Pryke 2010, Pérez i de Lanuza *et al.* 2013, Fargevieille *et al.* 2017). Additionally, support for the link between negative frequency-dependent selection and negative assortative mating has been obtained at both the empirical and the theoretical levels (Pusey and Wolf 1996, Takahashi and Hori 2008, Field and Barrett 2012, Holman *et al.* 2013, Hedrick *et al.* 2016, 2018, Estévez *et al.* 2020), which means that the presence of a negative assortative pattern in a population could be counted as evidence, if only weak, of negative frequency-dependent selection. However, it is also well known that (negative) assortative mating can result from other population processes or mechanisms (Hendry and Day 2005, Taborsky *et al.* 2009, Snowberg and Bolnick 2012, Wellenreuther *et al.* 2014, Kopp *et al.* 2018). For this reason, to argue for a mate choice-driven negative frequency-dependent selection, it is not enough to corroborate the existence of a negative assortative mating pattern for colour, but also to provide direct and independent evidence that this mating pattern is brought about by an active choice of the rare or dissimilar colour morph(s) by the choosy sex. Taken together, both pieces of evidence (negative assortative mating and mate choice) would constitute a more robust suggestion of negative frequency-dependent selection driving the maintenance of a colour polymorphism.

Littorina saxatilis is a marine gastropod that lives in the supralittoral zone of intertidal ecosystems across the northern American and European coasts of the Atlantic Ocean (Reid 1996, Rolán-Alvarez *et al.* 2015a). This species has been widely used as a model organism for the study of evolutionary processes, as its ovoviviparous mode of reproduction curbs its dispersal capability, leading to the isolation of its populations and to the evolution of (sometimes recurrent) local adaptations (Rolán-Alvarez 2007, Galindo and Grahame 2014, Rolán-Alvarez *et al.* 2015a, Johannesson 2016). *Littorina saxatilis*, like other species of the genus *Littorina*, also shows a remarkable and genetically based shell colour polymorphism, with shell colours ranging from whitish cream to black, to which several patterns can be added (lines, tessellations, and bands) (Reid 1996, Rolán-Alvarez *et al.* 2015a, Johannesson and Butlin 2017, Gefaell *et al.* 2023a). A particularly remarkable area in terms of the colour polymorphism shown by *L. saxatilis* is the Ría de Vigo, a coastal inlet in the NW Iberian Peninsula that shows both intrapopulation and clinal colour diversity in the populations of this species (Sacchi 1979, Reid 1996, Gefaell *et al.* 2023a, b). More specifically, the populations of *L. saxatilis* living in the wave-sheltered localities of this Ría show a recurrent intrapopulation shell colour polymorphism, with snails coexisting that can be divided into whitish or greyish-hazel, yellow, and brown colour morphs (Gefaell *et al.* 2021, 2023; see Fig. 1A, B and Supporting Information, Fig. S1A). *Littorina saxatilis* and other species of this genus constitute an excellent system for studying sexual selection-related processes, as the matings of this species are easily detectable in the field (during low tide) and show a relatively high prevalence in most populations (Rolán-Alvarez *et al.* 2012, 2015a, b). In *Littorina*, males appear to be more active and selective in the search for potential mates, and females apparently do not exert too much choosiness in their mating behaviour, showing multiple paternity (Panova *et al.* 2010, Saltin *et al.* 2013, Ng *et al.* 2019).

In a previous study, evidence was obtained using a validated protocol (e.g. Rolán-Alvarez *et al.* 2015b, Ng *et al.* 2016, Estévez *et al.* 2018) that *L. saxatilis* from the wave-sheltered localities of the Ría de Vigo, as well as their sister species *Littorina fabalis* and *L. obtusata* from other ecologically similar populations, show a consistent negative assortative mating pattern in the wild, with lighter coloured individuals mating significantly more often with darker ones than expected by chance (Gefaell *et al.* 2021). This result was interpreted as suggesting that (i) an ancestral behavioural mechanism of mate choice for colour dissimilarity could be driving the negative assortative mating pattern found in these *Neritrema* species (the subgenus comprising *L. saxatilis*, *L. fabalis*, and *L. obtusata*, as well as other related species); and (ii) that this would bring about the maintenance of their shell colour polymorphisms through negative frequency-dependent selection. However, in accordance with the recommendation outlined above, these conclusions would not be warranted unless we could find independent evidence of mate choice in at least one of these species. This evidence has been found in wild populations of *L. fabalis* (Estévez *et al.* 2020), but no further investigations have been carried out so far in *L. saxatilis* or in a laboratory setting.

In the present study, we employ an experimental laboratory paradigm consisting of mating balls to establish whether the negative assortative mating pattern detected in the wild populations of *L. saxatilis* from wave-sheltered localities can be attributed to a mate

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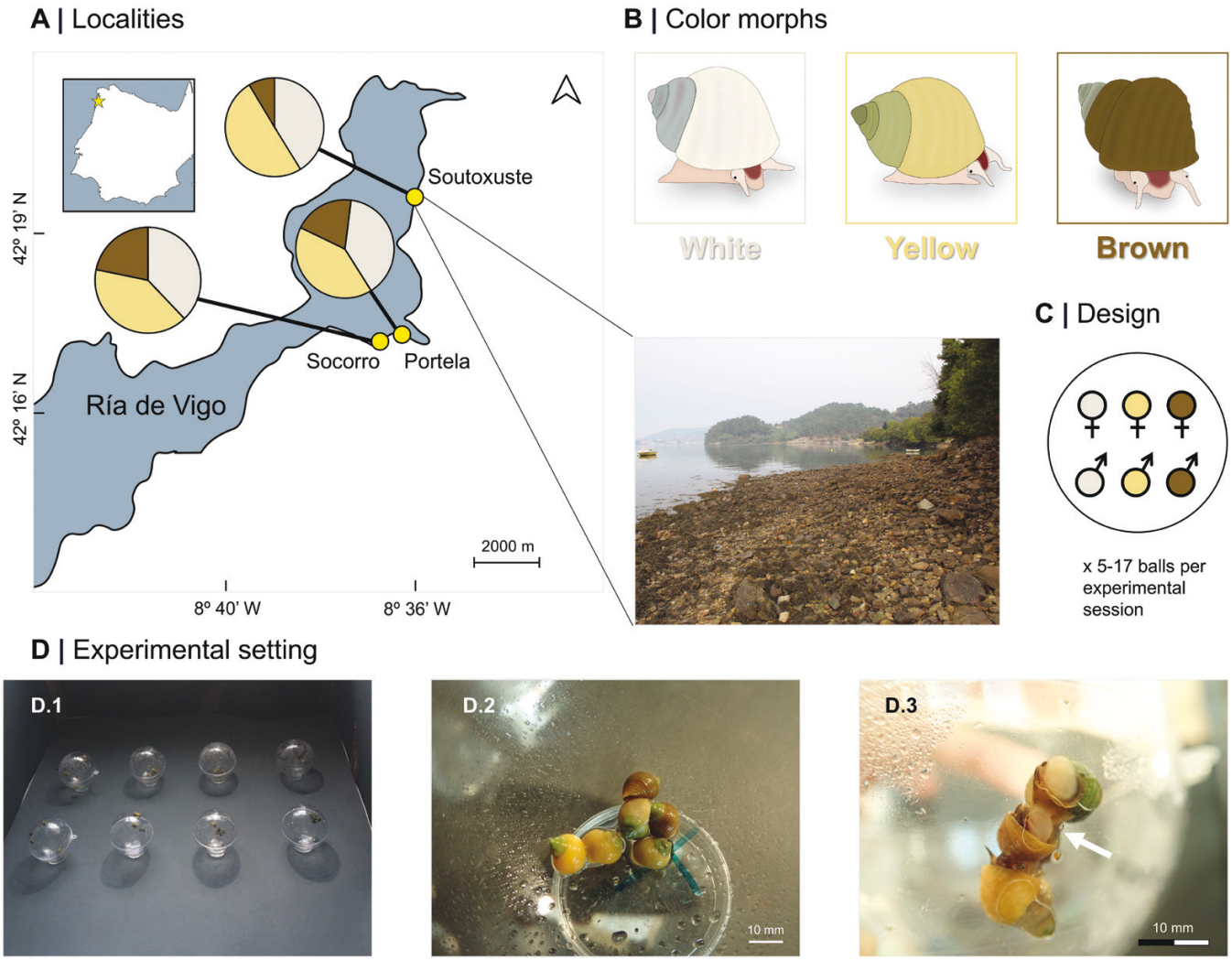

Figure 1. A, location of the localities studied within the Ría de Vigo with their corresponding frequencies of colour morphs (taken from Supporting Information, Table S1), and a photograph of Soutoxuste as representative of the three localities (map taken and modified from https://www.d-maps.com). B, drawings of the three colour morphs involved in this study (white, yellow, and brown). C, our adaptation of the experimental paradigm employed, which consists of a series of Christmas (mating) balls in which individuals from each sex (male, female) and colour (white, yellow, and brown) are introduced to record their mating behaviours. D, experimental setting devised using the mating balls. D.1 shows the balls in an experimental session, inside a cardboard box that had been covered with a black paper box and illuminated with artificial light. D.2 shows the six snails (two sexes from each of the three colours) inside a mating ball at the beginning of an experimental session. The yellow ones are on the left, followed by the whites and the browns. D.3 shows an instance of mating as seen from below. The penis of the male (white arrow) is entering the aperture of a female (diagnosed by the absence of a penis), which according to the operational definition outlined in the text counts as a mating.

choice behaviour for colour dissimilarity. If so, that would constitute a further piece of evidence in favour of the negative frequency-dependent selection scenario for the maintenance of shell colour polymorphism in these localities. In accordance with the natural history of this species, as well as our previous results on this matter (Gefaell *et al.* 2021), our hypothesis for this experiment was that males of *L. saxatilis* would show a consistent behavioural preference for females (i.e. mate choice) with a different shell colour that would result in a negative assortative mating pattern.

MATERIALS AND METHODS

Localities studied and sampling procedure

The snails used in the laboratory experiment were collected during the summer (end of June and the whole of July) of two consecutive years (2021 and 2022) in three different localities of San Simón Bay, in the inner zone of the Ría de Vigo and close to the mouth of the Verdugo River: Soutoxuste, Portela, and Socorro (see Fig. 1A). The first two localities (Soutoxuste and Portela) were also sampled for a previous study in 2020 (Gefaell *et al.* 2021), and for the present one both were sampled in 2021 and 2022, while the third locality (Socorro) was sampled only in 2022. Soutoxuste, Portela, and Socorro constitute wave-sheltered spots around 100 m long and 15 m wide formed by small alkaline granitoid rocks no larger than 20–30 cm in size (see the photo in Fig. 1A). Although these localities do not experience any swell, they are fully covered at high tide (especially during spring tides), so the sampling sessions always took place at low tide. Soutoxuste is separated from Portela by 7380 m of coast and from Socorro by 7960 m, which means that Portela

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and Socorro are around 580 m apart (see Table 1 for GPS coordinates). From the point of view of *L. saxatilis*, however, these distances are large enough to consider the populations from these localities as at least partially genetically isolated (Rolán-Alvarez *et al.* 2004, Tirado *et al.* 2016). All three localities were selected based on their high density of *L. saxatilis*, while the season for conducting the experiment (i.e. summer) was chosen because of the higher mating propensity of this species at this time of year (in part because the weather is better; E. Rolán-Alvarez, personal observation). The populations of *L. saxatilis* from Soutoxuste, Portela, and Socorro, as with the rest of populations from other wave-sheltered localities in the Ría de Vigo, are composed of three main colour morphs bearing different background colorations and that are present at relatively stable frequencies (see pie charts in Fig. 1A and Supporting Information, Table S1): whitish or greyish-hazel (hereafter, white), yellow, and brown (in the last case, sometimes with thin slight lines along the grooves of the shell) (see Fig. 1B and Fig. S1A).

These three colour morphs were used for our experiment. In other publications studying the colour polymorphism of *L. saxatilis* from the Ría de Vigo, a different colour classification system than the one employed for this study has been used (Sacchi 1979, Gefaell *et al.* 2023b). Under this system, most of the snails labelled 'white' in the present study would fall under the category '*fulva*', which would also cover the lightest yellow and brown individuals. For the present study, however, we stuck to the colour classification system employed by Gefaell *et al.* (2021) to have a common basis from which to make meaningful comparisons between both studies (something that philosophers of science call 'commensurability'; Kuhn 1982). Although occasional assignation errors may have occurred when classifying the snails for this study, major or systematic errors were avoided by picking up only the clearest and distinctive white, yellow, and brown-shelled individuals (see Supporting Information, Fig. S1A), and discarding those showing less vivid or more intermediate colorations between the three selected classes. Snails with microalgae growing on them, or with worn and/or damaged shells were also excluded.

Our experiment was divided into discrete experimental sessions, so the samples for each of these sessions were always collected in the field the day before. The snails were released to the locality where they were collected either after the last experimental session of the season (in 2021), or immediately after finishing their corresponding experimental session (usually 2 days after being collected; in 2022). This last release strategy drastically reduced the stress and mortality rate of the snails used. To avoid resampling (and consequently, retesting) previously used snails, in each sampling session at a given locality we only picked individuals from a small spatial range (4–5 m^2). The selection of snails was also carried out taking into account shell size, since positive assortative mating for this trait has been previously described in *L. saxatilis* and other species of the family Littorinidae (Saltin *et al.* 2013, Fernández-Meirama *et al.* 2017, Lau *et al.* 2021). For this reason, only snails converging around the mean shell size of the population (e.g. shell length of ~10 mm) were collected. This resulted in relatively homogeneous samples per locality. For transportation from the field to laboratory (and vice versa), all snails collected were introduced into separate plastic boxes by colour (white, yellow, and brown) that were previously arranged so that they simulate the natural habitat of the snails, with small rocks and brown algae (mainly *Ascophyllum nudosum*, *Pelvetia canaliculata*, and *Fucus* sp.) taken from the localities (see Supporting Information, Fig. S1B). Since *L. saxatilis* is not a protected species and is present in large numbers throughout the Ría de Vigo, no special permissions were needed to conduct this experiment. However, we tried to adhere to the ASAB guidelines (ASAB Ethical Committee/ABS Animal Care Committee 2023) during all steps of the process.

Sex determination and laboratory storage

The live snails were taken to the laboratory facilities of the Toralla Marine Science Station (ECIMAT), ~30 min from the localities by car (35–40 km), avoiding aggressive movements or sounds, and maintaining a moderate temperature (~18°C) by using the car's air conditioning. Once at ECIMAT, the individuals were sexed and stored in separate water tanks (by colour and sex) recreating a natural environment (again, with small rocks and brown algae from their localities) and filled with constant running water from the ocean. Sex determination was carried out by visual inspection of the snails with the aid of a stereo microscope (see Supporting Information, Fig. S1C). Males were diagnosed by the presence of a large protuberance (the penis) in the right dorsolateral portion of their bodies, while females were diagnosed by the absence of such a trait. The presence of immature individuals was ruled out since the range of shell sizes of the snails used in this experiment corresponds to that of mature individuals for this species. Once the sex determination process ended, the snails were stored in their water tanks for between

Table 1. Summary of the sample characteristics of each locality studied (with their corresponding GPS coordinates).

	GPS coordinates	Sample characteristics			
		n_1/n_2	N_1/N_2	Male size	Female size
Soutoxuste	42°19′21.19″N, 8°36′55.73″W	33/13	198/78	9.2 ± 0.53	9.5 ± 0.56
Portela	42°17′17.87″N, 8°37′22.16″W	57/48	342/288	10.3 ± 0.88	10.6 ± 0.91
Socorro	42°17′16.16″N, 8°37′37.53″W	38/38	228/228	10.7 ± 0.95	11.1 ± 0.90

n_1 is the number of mating balls per locality used in this experiment for which there was at least one mating event (total $N = 128$), while n_2 is the subsample of those balls for whose snails the shell size was measured (most of them corresponding to the 2022 experimental sessions; total $N = 99$). N_1 is the number of individuals per population that participated in laboratory experiments, and N_2 is the subsample of those in which shell size was recorded (note that $N_1 = n_1 \times 6$ and $N_2 = n_2 \times 6$; as there were six individuals per mating ball or experimental unit; see text for more details). Mean ± standard deviation of male and female size (mm) for that subsample of snails in which shell length was measured (N_2) is also shown, also grouped by locality.

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4 and 31 h for acclimation, with a natural day/night light cycle until their corresponding experimental sessions began (see Fig. S1D).

Mating ball experiment design

For the mate choice experiment, we followed an experimental paradigm similar to that of Lau *et al.* (2021) to measure positive assortative mating for shell size in *Echinolittorina malaccana*. In that experiment, male and female individuals of different sizes were placed in transparent plastic Christmas balls for a predefined time period and the matings between them were recorded. This experimental paradigm allows male snails to chase females by following their mucus trails (as is usual in the family Littorinidae; Erlandsson and Kostylev 1995, Ng *et al.* 2011, 2019, Saltin *et al.* 2013), so that it is possible to link the resulting mating pattern to a behavioural mechanism of (male) mate choice.

In our adaptation of the experimental paradigm of Lau *et al.* (2021), snails of each colour and sex ($N = 6$ individuals, 3 per colour × 2 per sex) were introduced into a mating ball (diameter = 10 cm; see Fig. 1C, D, as well as Supporting Information, Fig. S1E), and their mating behaviours during the next 3 h were recorded following a scan sampling protocol (Bateson and Martin 2021) with behavioural recordings every 3 min, as the available data suggest that the most common matings in this species usually last around 4 min (Perini *et al.* 2022). This 'multiple-choice' design, in which three different males (white, yellow, and brown) are placed together in a single ball with three different females of equivalent colours, was favoured over 'single-choice' (one male and one female of each possible colour combinations) or 'male-choice' designs (one male of a given colour and the corresponding three different coloured females), as the former scenario is more biologically realistic (*L. saxatilis* tends to live and mate at high densities of both males and females). Furthermore, in their experiment, Lau *et al.* (2021) revealed that only the multiple-choice paradigm (but not the single- or male-choice paradigms) resembled the mating patterns observed in the wild in *Echinolittorina malaccana*, which shares both its taxonomic family (Littorinidae) and its ecological niche with *L. saxatilis*. For our experiment, the working hypothesis was that, if a male mate choice behaviour for dissimilar females does exist (as revealed in the field; Gefaell *et al.* 2021), then the mating pattern observed in the laboratory should also be disassortative (i.e. negative assortative).

Throughout the whole experiment, we employed a total of 192 mating balls. These were divided into different 18 experimental sessions of 5–17 mating balls each, depending on the number of snails collected and sexed each sampling day. Since males are the choosy sex, females were always introduced into the mating balls first. As for the colour morphs (white, yellow, brown) of both sexes, their order of introduction was randomized. The size of all snails introduced into each ball was homogenized to account for positive assortative mating for size, selecting those with a similar shell length under visual inspection. However, taking into account that, despite the overall positive assortative mating for size, previous results also have shown a preference of male littorinid snails for slightly larger females (Saltin *et al.* 2013, Ng *et al.* 2019, Lau *et al.* 2021, Clancey *et al.* 2022), we also tried to choose females that, despite their overall size similarity to the males, were slightly larger than these to control such an effect in our experiment. In addition to the size homogenization, for most of the 2022 sessions the shell length of all snails used was also recorded to the nearest 0.1 mm (from apex to the tip of the last whorl) using a vernier caliper to formally rule out the effect of size when explaining the resulting mating pattern. Before the start of each experimental session, all balls were sprayed with ocean water (sometimes containing pulverized brown algae) to recreate the natural environment of the snails. This spraying was repeated once every hour during the 3-h experimental session to keep the balls moist. To minimize the influence of extraneous environmental variables, all balls employed in each experimental session were placed on a cardboard box covered with a black paper board and illuminated with a constant artificial light (Fig. 1D.1).

We determined that a male had chosen a mate when we observed a copulation. We defined a mating when two requirements were met: (i) when inspecting the snails from above, one snail (usually the male) was placed with the aperture of the shell facing up such that it overlapped with the aperture of a second snail (usually the female), which was located face down, with both snails forming a V-like pattern (Gefaell *et al.* 2021); and (ii) when close inspection from below revealed that the penis of the male (faced up) was inside the aperture of the female (faced down) (see Fig. 1D.3). Behavioural recordings were carried out by a single experimented observer (Juan Gefaell) to avoid intersubject reliability problems. No blinding procedure was implemented, as it was strictly necessary to see the colours involved in the matings. Although we recorded all matings that took place during the experimental sessions, only the first one in each ball was initially considered for the statistical analyses, as the subsequent matings cannot be considered independent observations (see further details in the section below). No simultaneous matings occurred when it came to the first mating, so this was always defined as the one that occurred first at a given ball. Once the experimental session was over, the snails were removed from the mating balls and stored in separate tanks, prior to being carried back to the field (see Supporting Information, Fig. S1F). Before a new experimental session, all mating balls used were cleaned with fresh water to remove the mucus trails from the previous session.

Statistical analyses

For data analysis, the mating ball was treated as the experimental unit. In each ball, there could be a total of $3^2 = 9$ possible mating types, corresponding to the combinations of each colour (white, yellow, brown) and sex (male, female); these, in turn, can be divided into 'homotypic' (with both sexes showing the same colour) or 'heterotypic' (with different colour) matings. Three possible outcomes were possible for each experimental unit (i.e. mating ball): (1) no matings during the 3 h; (2) just one mating; and (3) more than one mating. All the experimental units in which outcome (1) occurred were discarded, keeping only those that met outcomes (2) and (3). Male–male matings, relatively common in this species (Rolán-Alvarez *et al.* 1999), were also excluded to avoid possible confounding factors. As previously stated, for balls with outcome (3) only the first mating was incorporated into our initial analyses, as it is the only one that meets the assumption of the independence of data (i.e. the ensuing matings

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could be somehow influenced by the previous ones, especially when the same male individual mated more than once per experimental session; Bateson and Martin 2021). The frequencies of the nine possible mating types can be expressed through a nine-cell mating table; a homogeneity G likelihood test (Sokal and Rohlf 1995) was used to compare the frequencies of the different mating types obtained in the first matings of outcomes (2) and (3) to those obtained in the subsequent matings of outcome (3). When the frequencies in the former test were clearly homogeneous for a given locality ($P > .25$), these were nonetheless grouped together to increase the sample size.

Of the total of 192 mating balls, 128 (67%) met criteria (2) or (3) outlined above (see Table 1). These were then divided into two separate groups based on whether the size of the snails in each of these balls was measured or not (mostly corresponding to the 2022 and 2021 series, respectively; see Table 1). This allowed for two independent analyses of matings: a logistic regression analysis in which size was included as an explanatory factor for the type of mating (homotypic vs. heterotypic; see paragraph below), and a second analysis in which the variable size was ignored and the mating pattern was studied using the I_{PSI} estimator (Pérez-Figueroa *et al.* 2005). All analyses were carried out separately for each locality (Soutoxuste, Portela, Socorro).

Despite the fact that the size of the snails was homogenized both when collecting the snails in the field and during the setup of the mating balls for the experimental sessions, it could still be conceivable that this parameter could exert some influence over the observed mating pattern. Therefore, a logistic regression analysis was carried out to explicitly rule out this possibility. For this analysis, only the subsample for which shell size was measured was used. The first mating observed in each ball was employed as the dependent variable, and coded either as a homotypic mating ('1') or as heterotypic ('0'). The size of the snails involved in the matings, as well as their interaction, were employed as predictor variables in the logistic regression model.

The I_{PSI} analysis was performed using the JMATING v.1.0 software (Carvajal-Rodriguez and Rolan-Alvarez 2006) to examine the overall direction of the pattern of matings recorded. The I_{PSI} estimator is an index of assortative mating that ranges between −1 (complete negative assortative mating) and 1 (complete positive assortative mating), with 0 being complete random mating. The I_{PSI} index has proven to be a robust and reliable estimator of assortative mating for qualitative traits under different sampling assumptions (Pérez-Figueroa *et al.* 2005, Rolán-Alvarez *et al.* 2015b, Estévez *et al.* 2020). In the present experiment, the I_{PSI} index was calculated from a table summarizing all actual matings between colour morphs in each locality (normally the first ones except when the homogeneity tests allowed the inclusion of the subsequent matings, which happened for Soutoxuste and Portela, but not Socorro; see Supporting Information, Table S2) under two different scenarios: each colour and sex plotted separately against each other, and every single colour plotted against the rest of the colours grouped together (e.g. white vs. yellow and brown grouped together). Statistical significance for I_{PSI} values was calculated through bootstrapping, resampling 10 000 times the frequencies of observed mating combinations for the different colour morphs. The existence of a negative assortative mating pattern was studied through a maximum likelihood G test, comparing the frequency of observed matings of each kind with their corresponding expected values in a scenario of random mating for colour. The absence of 'size' as a potential variable contributing to the observed pattern in this test was countervailed by the previously described logistic regression analysis. All statistical analyses were carried out in Excel and SPSS v.24.0 (IBM Corp., USA).

Mate choice estimation through model selection

The previous analyses were complemented with an estimation of mate choice made by connecting the empirical data with explanatory mathematical models. This was done using the InfoMating software v.0.4 for model selection and parameter estimation (Carvajal-Rodríguez 2020), using the same samples of matings as for the I_{PSI} analysis. The methodology behind this estimation is that InfoMating works with several alternative models for the parameters of mate choice and/or mating propensity (in our case it considers up to 23 alternative models). These mechanisms (mate choice and mating propensity) have traditionally been considered the main explanations for selective mating under controlled laboratory conditions (O'Donald 1980, Carvajal-Rodríguez 2018, Lau *et al.* 2021). InfoMating calculates the maximum likelihood values of the parameters of each model. These values are then used to rank each model according to its fit to the data following the Akaike information criterion corrected for low sample size (AICc; Carvajal-Rodríguez 2018, 2020). The purpose of this analysis is to detect mate choice coefficients (C) in the nine-cell mating table of different mating types—i.e. values that are distinct from the random mating expectation coefficient (corresponding with a value of 1). If $C < 1$ values are shown by the model in the diagonal of homotypic mating combinations (those with identical male and female colours), then that would represent a negative assortative mating pattern (for formal details, see Carvajal-Rodríguez 2018, 2020). Different models propose different C coefficients ($C_1, \ldots, C_n$) to account for the various mating types. A model that presents a unique C for each of the nine possible mating types in the nine-cell mating table (C_{1-9}) would represent the same coefficients as the PTI (pair total isolation) statistic, as they stand for the maximum likelihood model with the same number of parameters as cells in the mating table (Carvajal-Rodríguez 2020). The PTI also constitutes the deviation of observed mating frequencies in each of the nine possible mating types, with respect to the random mating expectation based on initial colour frequencies (Rolán-Alvarez and Caballero 2000).

RESULTS

The logistic regressions for each locality (Soutoxuste, Portela, and Socorro) are summarized in Table 2. Male and female shell sizes (as well as their interactions) were not significant in any of the logistic regression models, as shown by their low R^2 values and their lack of significant contribution to the explanation of the dependent variable. This points to an absence of effect of the variable 'size' on the overall mating pattern and supports our procedural effort to homogenize the shell size of the snails used in the experiment. For this reason, the subsequent analyses ignored size as an explanatory factor.

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Table 2. Logistic regression of snail size parameters (male size, female size, and their interaction) as potential predictors of mating type ('Homotypic mating or not', with same colour–homotypic–mating = 1 and different colour–heterotypic–mating = 0; see the text for a detailed explanation).

Dependent	Locality	n_2	R^2	Variables studied	*B*	SD	*P*
Homotypic mating or not	Soutoxuste	13	0.205	Constant	440.3	726.2	.544
				Male size	−45.3	75.6	.56
				Female size	−49.3	75.6	.514
				Male size × Female size	5.1	8	.528
Homotypic mating or not	Portela	48	0.04	Constant	−1.7	61.3	.978
				Male size	0.5	6.1	.929
				Female size	0.3	5.9	.965
				Male size × Female size	−0.1	0.6	.901
Homotypic mating or not	Socorro	38	0.08	Constant	76.9	101.7	.449
				Male size	−8.2	9.4	.381
				Female size	−7.1	8.8	.419
				Male size × Female size	0.7	0.8	.363

R^2 is the value of the Cox–Snell *R* value in the regression model. *B* is the Wald regression coefficient of each variable with its corresponding standard deviation (SD) and probability of the Wald test (*P*). n_2 is the sample size used for this analysis, which corresponds to the number of mating balls (experimental units) for which there was at least one mating and in which snail shell size was measured (as in Table 1; see text for further details).

Table 3. Estimates of assortative mating.

	Soutoxuste					Portela					Socorro				
	I_{PSI}			$G_{Isolation}$		I_{PSI}			$G_{Isolation}$		I_{PSI}			$G_{Isolation}$	
	Estimate	SD	n_1	*G* value	d.f.	Estimate	SD	n_1	*G* value	d.f.	Estimate	SD	n_1	*G* value	d.f.
All colours	−0.09	0.171	33	0.76	4	−0.19*	0.091	57	14.6**	4	−0.74***	0.157	38	18.2**	4
White	−0.06	0.177	33	0.1	1	−0.32***	0.087	57	10.5**	1	−0.46**	0.141	38	7.7**	1
Yellow	−0.07	0.185	33	0.2	1	−0.01	0.099	57	0.1	1	−0.37*	0.168	38	3.5	1
Brown	−0.04	0.134	33	0.1	1	−0.08	0.092	57	0.8	1	−0.56**	0.122	38	9.4**	1

Two complementary approaches are presented. First, the I_{PSI} value is given both for the whole set of data ('All colours') and the data subdivided by pairs of colours (one given colour vs. each one of the remaining colours; 'White', 'Yellow', and 'Brown'). The probability of the estimate and the standard deviation (SD) are calculated by bootstrapping the original data (see Supporting Information, Table S3). Second, the maximum likelihood test ($G_{Isolation}$) compares the expected mating frequencies (assuming random mating from marginal mated frequencies) and observed matings by JMATING software (Carvajal-Rodriguez & Rolan-Alvarez, 2006). n_1 represents the total number of experimental units (mating balls) used in the experiment (as in Table 1). *P < .05; **P < .01; ***P < .001.

The results in relation to the pattern of assortative mating (estimated through I_{PSI}), as well as their deviation from random mating (evaluated through G tests), are shown for the three localities in Table 3. All I_{PSI} estimates were negative (inclined towards the negative assortative mating side), but they were statistically significant only for all colours and white vs. each of the rest of the colours in Portela and for all possible combinations in Socorro. G likelihood tests yielded very similar results to those obtained through bootstrapping estimates (except for the yellow colour in Socorro). Taken together, these show a general trend towards negative assortative mating in all three populations, even though only in two of them (Portela and Socorro) was this trend statistically significant.

For the two localities showing statistically significant negative assortative matings (Portela and Socorro), the distribution of matings was studied using PTI estimates (Rolán-Alvarez and Caballero 2000), which represent the maximum likelihood estimate of the success of a given mating type (see Table 4). PTI measures deviations from random mating in each of the nine possible mating types. In both localities, there is at least one diagonal cell (homotypic) that presents a value significantly lower than 1, which can be interpreted as evidence in favour of some kind of mate choice for negative assortative mating (i.e. homotypic matings having lower occurrence probability than that expected by chance). In Portela, this effect is seen in white (♀) vs. white (♂) matings, whereas in Socorro that would affect both yellow (♀) vs. yellow (♂) and brown (♀) vs. brown (♂). When the model comparison tool is applied, in both locations a preference for models that incorporate a coefficient of mate choice (C_1 and/or C_2) is selected. This suggests that choosing a model with only two *C* coefficients fits better to the data than a model considering all PTIs as well as the rest of the alternative models. In Portela, the preferred model incorporates two choice coefficients (C_1 and C_2), whereas in Socorro, despite showing a stronger choice trend, the preferred model shows only one choice coefficient affecting all diagonal matings (C_1), which could be interpreted as an a priori simple choice mechanism.

DISCUSSION

Overall, this experiment provides additional support for the hypothesis that the negative assortative mating pattern for shell colour seen in wild populations of *L. saxatilis* is driven by a male mate choice mechanism for dissimilarity (Gefaell *et al.* 2021).

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Table 4. Analysis of the whole set of mating pairs using PTI (pair total isolation) estimates (calculated using JMATING), and the best mate choice model estimated by InfoMating for the localities with significant assortative mating (Portela and Socorro).

		Portela				Socorro		
		White (♂)	**Yellow (♂)**	**Brown (♂)**		**White (♂)**	**Yellow (♂)**	**Brown (♂)**
PTI	White (♀)	0.39***	1.42	1.23	White (♀)	0.47	2.13	0.95
	Yellow (♀)	0.77	0.91	1.04	Yellow (♀)	0.71	0.24**	0.95
	Brown (♀)	1.49	0.71	1.04	Brown (♀)	2.37	1.18	0***
Model	White (♀)	C_1	1	1	White (♀)	C_1	1	1
	Yellow (♀)	C_2	C_1	1	Yellow (♀)	1	C_1	1
	Brown (♀)	1	C_2	C_1	Brown (♀)	1	1	C_1
Estimate	White (♀)	0.6	1	1	White (♀)	0.17	1	1
	Yellow (♀)	0.57	0.6	1	Yellow (♀)	1	0.17	1
	Brown (♀)	1	0.57	0.6	Brown (♀)	1	1	0.17
		C_1 ± SD = 0.60 ± 0.171				C_1 ± SD = 0.17 ± 0.131		
		C_2 ± SD = 0.57 ± 0.085						
		Weight = 36%				Weight = 30%		

The analyses were conducted on the corresponding summary tables shown in Supporting Information, Table S3. PTI includes the effects of sexual selection and sexual isolation in the population. Statistical significance is calculated by bootstrapping 10 000 times the frequencies of observed mating pairs. A model is chosen by InfoMating in each locality with their corresponding parameters (C_1 and C_2). The weight of the chosen model is also given for each locality. **$P < .01$; ***$P < .001$.

This, in turn, would support the negative frequency-dependent selection scenario for the maintenance of shell colour polymorphism in the wave-sheltered localities of the Ría de Vigo. We have found that, in the populations of two out of three of the studied localities (Portela and Socorro), a negative assortative mating pattern for shell colour like the one detected in the field is replicated under controlled laboratory conditions. The I_{PSI} and PTI estimates support such a scenario, while the logistic model rules out the potential contribution of shell size to the obtained results. Taking into account the natural history and ecology of *L. saxatilis* (Reid 1996, Saltin *et al.* 2013), the experiment described in this study allows us to conclude that a male mate choice is responsible for the mating pattern observed. Furthermore, our model selection analysis shows that the best explanatory model for mate choice in two localities (Portela and Socorro) incorporates a choice coefficient (C_1) lower than that expected by random mating alone, affecting the whole diagonal of homotypic matings, as well as another coefficient lower than 1 (C_2) which affects only two types of matings in Portela. These coefficients mean that those particular mating combinations show a tendency to mate below the random expectation (which equals 1). This pattern of choice coefficients is only compatible with a negative assortative mating, as all homotypic (between identical colours) mating pair combinations show an occurrence probability < 1 (below random mating), thus backing the main conclusion obtained by the other analyses carried out.

However, despite the fact that we believe that the above conclusion—namely, that the negative assortative mating pattern detected is brought about by mate choice—is warranted, a few caveats also exist. The mating pattern observed for the snails of Soutoxuste, despite being slightly negative, cannot be distinguished from random mating (Table 3). This means that such a result departs not only from what is seen in the Portela and Socorro samples used for this experiment but also from what was detected in the field in such a locality. In fact, Gefaell *et al.* (2021) found a statistically significant negative assortative mating pattern for shell colour in the wild populations of Soutoxuste and Portela. Several potential explanations might be put forward to account for this inconsistency. One of these has to do with the effects of laboratory handling and storage of the organisms involved in the study. Discrepancies between laboratory and field results are a well-known phenomenon for ethologists and behavioural ecologists, with laboratory experiments usually entailing extreme manipulations that sometimes alter the behaviour of the studied organisms with respect to the one they display in the wild (Ruxton and Colegrave 2006, Bateson and Martin 2021). Moreover, even though the laboratory conditions are normally much more homogeneous than those of the field, there could still be a great deal of variation in at least some parameters, which in turn might affect specific subsets of organisms employed in an experiment. In our case, this could mean that certain extraneous variables might have affected the Soutoxuste snails more than those of the two other localities, even if we are not aware of any of these. *Littorina saxatilis* is extremely sensitive to environmental variation, particularly regarding its mating behaviour under artificial conditions. For instance, in a laboratory experiment with this species, Rolán-Alvarez *et al.* (2004) had to study mating attempts because the observed number of actual matings was too low to be representative. Although the experimental protocol for obtaining matings in the laboratory has been greatly improved since the introduction of the mating balls paradigm (Lau *et al.* 2021), this species and behaviour still constitute a troublesome case.

The second set of possible explanations for the observed discrepancies has to do with temporary starting differences in the natural populations of each locality. It could have been that the nutritional or parasitic status of the Soutoxuste snails at the time and year when sampled was worse than that of the remaining two populations, such that when they were taken to the laboratory, this influenced their mating behaviour. It is widely known that

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parasites influence the mating behaviour of their hosts (Moore 2002). Moreover, there is evidence that infection by trematodes alters the sexual and nonsexual behaviour of at least another species of the genus *Littorina* (*L. littorea*; Saur 1990, Erlandsson and Kostylev 1995, Davies and Knowles 2001, Seaman and Briffa 2015). For instance, Erlandsson and Kostylev (1995) observed that male mucus trail following was greatly reduced in a male *L. littorea* following a highly infected female. It cannot be ruled out that this might have affected the behaviour of the Soutoxuste snails in this experiment as well. The fact that the total number of mating balls in which there was at least one mating was lower in Soutoxuste than in the mean of all three localities (M = 60.8%, SD = 3.5 in Soutoxuste vs. M = 67%, SD = 6.8 in the three populations; see Supporting Information, Table S4) supports this possibility. However, both this and the former explanations are mere guesses for which there is no conclusive evidence and that should be studied in the future.

Finally, a third potential explanation could be that the fitness advantage of being a rare colour in Soutoxuste might not be as great compared to the other two localities (Portela and Socorro), perhaps because shell colours in Soutoxuste may also have an impact on fitness in other ways in this specific population of *L. saxatilis*. Although we neither can rule out this possibility, currently there is no direct data in support of such a hypothesis. Quite the contrary, this contradicts the results obtained in the field study carried out in Soutoxuste and Portela, which found negative assortative mating in both localities as well (Gefaell *et al.* 2021). Furthermore, all known ecological variables in Soutoxuste, as well as all the relevant morphological traits of the snails of that locality, are virtually identical to those of Portela and Socorro, thereby making it difficult to come up with plausible scenarios regarding the potential factors that could be exerting specific fitness effects on the snails of Soutoxuste.

The fact that there is an incongruence in the results obtained for one out of three localities suggests that the best way to settle this issue would be to conduct a mate choice experiment in the field. That would remove any possible extraneous laboratory factors from affecting the behaviour of the organisms while providing more ecological validity to the results obtained (Bateson and Martin 2021). If the same overall pattern as the one observed both in Gefaell *et al.* (2021) and in this study persists, then we would expect a corroboration of male mate-driven negative assortative mating for shell colour.

Even in the face of the limitations of our laboratory experiment, we nonetheless provide evidence that the colour polymorphism seen in the wave-sheltered localities of the Ría de Vigo is probably maintained through negative frequency-dependent selection driven by mate choice for dissimilarity. This raises the question of how the colour of females is detected by male *L. saxatilis* individuals. So far, all evidence suggests that snails of the genus *Littorina* (including *L. saxatilis*) do not have colour vision (Gefaell *et al.* 2021), which means that, if there is any mate choice for shell colour, there should be an indirect way for male snails to detect and select those females with a dissimilar coloration. It has been posited that there might be differences in the metabolic composition of the mucus trails between the colour morphs of *Littorina* snails (Estévez *et al.* 2020), but this hypothesis has not been tested so far. In addition to this problem, the existence of mate choice for dissimilar shell colours also prompts the question of what fitness benefits derive from shell colour. As we have argued, shell coloration might contribute to fitness by providing important information to either conspecifics and/or predators (communication), better camouflage (crypsis), or a more efficient control of body temperature (thermoregulation). Apparently, there are no a priori obvious fitness differences between the colour morphs of *L. saxatilis* from the wave-sheltered localities of the Ría de Vigo when it comes to crypsis or any selective pressure directly related to shell colour per se (E. Rolán-Alvarez, personal observation). This opens the door to the possibility of shell colour just being a trait correlated with other fitness-enhancing traits (Gefaell *et al.* 2023a). In that respect, a previous study on *L. saxatilis* from Sweden (Koch *et al.* 2021) was able to associate one of the colours present in these populations to a specific chromosomal inversion, which was also linked to other adaptive shell traits. Therefore, a plausible hypothesis could be that the true target of (sexual) selection might be a trait genetically linked to shell colour that the snails somehow manage to detect, but not shell colour itself. For this reason, future studies should not only be aimed at settling whether a male mate choice-driven negative assortative mating for shell colour really exists in these populations of *L. saxatilis* but also at studying its potential genetic underpinnings.

SUPPLEMENTARY DATA

Supplementary data are available at *Biological Journal of the Linnean Society* Journal online.

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CONFLICT OF INTEREST

The authors declare that they have no conflict of interest.

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DATA AVAILABILITY

Raw data are available in the Figshare repository: https: //doi.org/10.6084/m9.figshare.23515020.v1.

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Supplementary materials – Chapter 3

Figure S1. Photos of all steps of the experimental protocol. A: Three exemplars in the field showing clear and distinctive white, yellow, and brown colorations. B: Plastic boxes used for storing and transporting each color morph, prepared with rocks and brown algae taken from the localities. C: Stereo microscope used for sexing the snails at the Toralla Marine Science Station (ECIMAT). D: Water tanks used for acclimation of each color morph and sex, with constant running ocean water and also containing rocks and algae from the localities. E: Mating ball with their corresponding six individuals (one per sex and color) at the beginning of an experimental session. Notice the moist caused by the spraying of the balls. F: Water tank with used snails before taking them back to the field.

Table S1. Frequencies of colors in each locality (June 2023). The percentages were obtained from two random samplings in each population (mean *N* = 83, SD = 11).

Sample 1

	White	Yellow	Brown	N
Soutoxuste	39	42	10	91
Portela	41	40	18	99
Socorro	34	33	16	83

Sample 2

	White	Yellow	Brown	N
Soutoxuste	32	44	5	81
Portela	29	33	17	79
Socorro	23	27	16	66

% Sample 1

	White	Yellow	Brown
Soutoxuste	42,9	46,2	11,0
Portela	41,4	40,4	18,2
Socorro	41,0	39,8	19,3

% Sample 2

	White	Yellow	Brown
Soutoxuste	39,5	54,3	6,2
Portela	36,7	41,8	21,5
Socorro	34,8	40,9	24,2

% Average

	White	Yellow	Brown
Soutoxuste	41,2	50,2	8,6
Portela	39,1	41,1	19,9
Socorro	37,9	40,3	21,8

Chapter 4

Gefaell J, Vigo R, González-Vázquez AH, Galindo J & Rolán-Alvarez E. 2024. Temporal stability and directional change in a color cline of a marine snail from NW Spain. *Current Zoology* 70: 163–173. [Open Access]

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Original Article

Temporal stability and directional change in a color cline of a marine snail from NW Spain

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Abstract

The evolution and maintenance of color clines is a classic topic of research in evolutionary ecology. However, studies analyzing the temporal dynamics of such clines are much less frequent, due to the difficulty of obtaining reliable data about past color distributions along environmental gradients. In this article, we describe a case of decades-long temporal stability and directional change in a color cline of the marine snail *Littorina saxatilis* along the coastal inlet of the Ría de Vigo (NW Spain). *L. saxatilis* from this area shows a clear color cline with 3 distinct areas from the innermost to the more wave-exposed localities of the Ría: the inner, protected localities show an abundance of fawn-like individuals; the intermediate localities show a high diversity of colors; and the outer, wave-exposed localities show populations with a high frequency of a black and lineated morph. We compare data from the 1970s and 2022 in the same localities, showing that the cline has kept relatively stable for at least over half a century, except for some directional change and local variability in the frequency of certain morphs. Multiple regression analyses and biodiversity measures are presented to provide clues into the selective pressures that might be involved in the maintenance of this color cline. Future research avenues to properly test the explanatory power of these selective agents as well as the possible origins of the cline are discussed.

Key words: clinal variation, color polymorphism, ecological gradient, *Littorina saxatilis*, natural selection, wave exposure.

Genetically dependent intraspecific color variation, or color polymorphism, is a common phenomenon in a wide variety of animal groups, including insects (Dearn 1990; Suárez-Tovar et al. 2022), arachnids (Oxford and Gillespie 1998), anurans (Hoffman and Blouin 2000), squamates (Abalos 2021), birds (Galeotti et al. 2003), bivalves (Cook 2017), or gastropods (Gefaell et al. 2023). A special case of color polymorphism is clines, in which there is intraspecific and geographical gradual color variation along an environmental gradient (Mayr 1963; Endler 1977; Smith and Smith 2012). Color clines can either consist of gradual changes in the intensity of a continuous color variation (e.g., going from brighter to less bright individuals), or gradual changes in the frequencies of discrete color morphs (see Davison et al. 2019 for a discussion of the continuous vs. discrete variation controversy in gastropod color polymorphism). From an evolutionary point of view, clines are normally caused by selection or neutral diffusion via interbreeding and dispersal (Sætre and Ravinet 2019), and they can be of interest to evolutionary biologists inasmuch as they can provide clues into the putative adaptive significance of color variation. Color clines have also been described in several terrestrial and marine taxa (e.g., Dearn 1981; Phifer-Rixey et al. 2008; Antoniazza et al. 2010; Cooper 2010; Takahashi et al. 2011; Köhler et al. 2017; Magnúsdóttir et al. 2018).

A particularly understudied topic in the evolutionary ecology of color clines is its temporal stability and change across prolonged periods (i.e., decades-long). So far, not many studies have systematically explored this issue, in part because it is usually hard to obtain reliable data on past dynamics of color variation (de Jong and Brakefield 1998; Cook et al. 2002; Mullen and Hoekstra 2008; Saccheri et al. 2008; Brakefield and de Jong 2011; Evans et al. 2018; Hantak et al. 2021). Particularly important in this realm are the studies conducted in the model organism *Cepaea nemoralis* (e.g., Wolda 1969; Cook and Pettitt 1998; Cook et al. 1999; Ożgo and Schlthuizen 2012; Cameron et al. 2013; Ramos-Gonzalez and Davison 2021). This kind of investigation is of uttermost importance not only because of the insights it can provide on how evolutionary forces shape biodiversity in the medium-to-long run but also because it can yield interesting information on how recent anthropogenic processes such as climate change can affect these evolutionary dynamics (e.g., de Jong and Brakefield 1998; Brakefield and de Jong 2011; Ożgo and Schlthuizen 2012; Roulin 2014; Consentino et al. 2017; Evans et al. 2018).

The rough periwinkle (*Littorina saxatilis*, Olivi 1792) is a marine gastropod species known for its high phenotypic diversity and its role as a model organism for evolutionary ecology (e.g., Johannesson 2015; Rolán-Alvarez et al.

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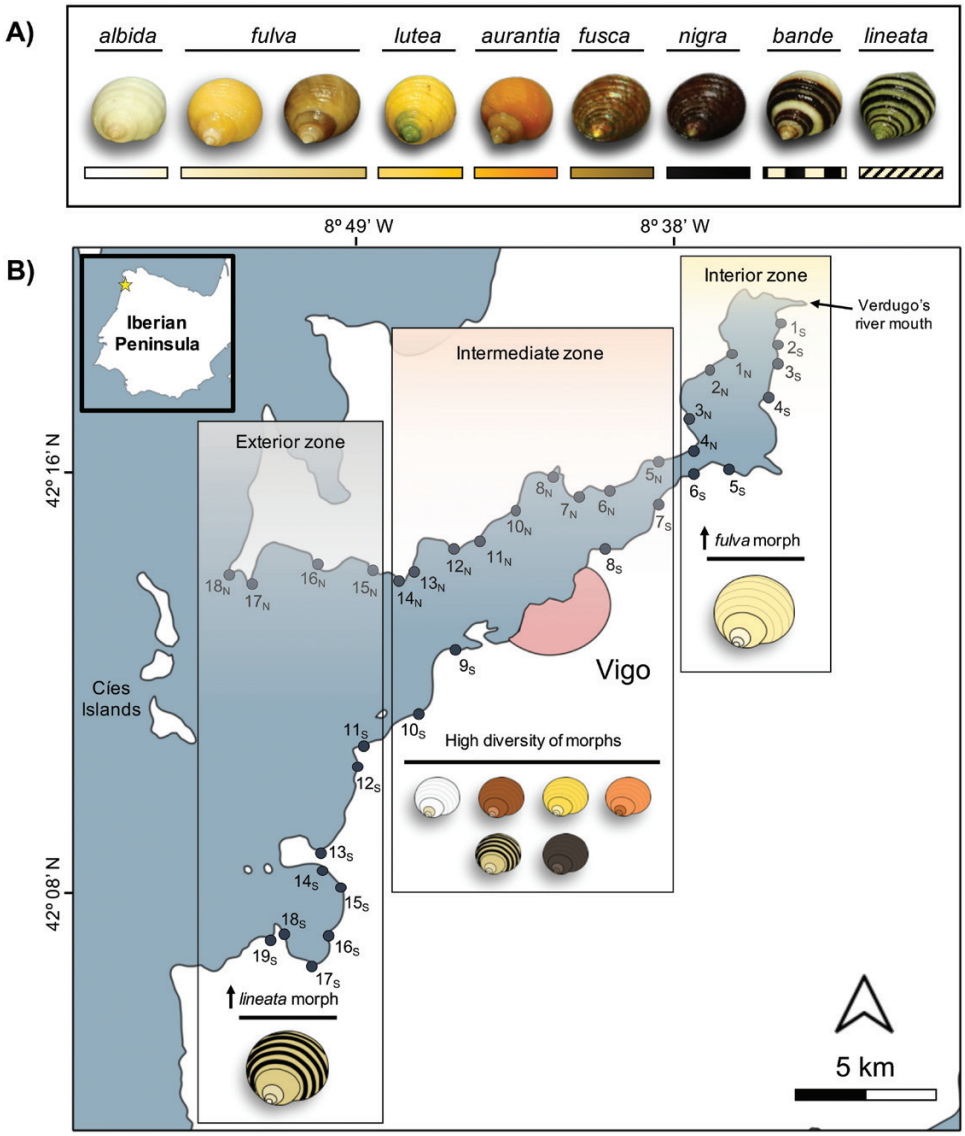

Figure 1 (A) Color categories used by Sacchi (1979). *Tessellata* and *sanguinea* colors are not shown either because they are not representative of the composition of populations (due to low frequencies; *tessellata*) or are absent from the Ría de Vigo (*sanguinea*). (B) Color cline hypothesis for the Ría de Vigo as derived from Sacchi (1979; see also Reid 1996). According to this hypothesis, 3 different zones exist along the Ría de Vigo, each with its own color distribution: an interior zone, with high frequencies of fawn-like ("*fulva*") individuals; an intermediate zone, with high color diversity; and an exterior zone, with high frequencies of a lineated ("*lineata*") morph. Dots and numbers represent the different localities explored by Sacchi in 1974–1976 and re-sampled by us in 2022. Note that in the north there is a leap from 8_N to 10_N, as no snails were found in the 9_N location ("Moaña, entro il porto" in Sacchi 1979; see the main text). The light pink area represents the approximate urban surface of the city of Vigo. The maps were taken and modified from d-maps (https://www.d-maps.com).

2015). *L. saxatilis* lives in the rocky intertidal ecosystems all over the American and European coasts of the North Atlantic Ocean, where it feeds on microalgae that grow on the surface of rocks (Reid 1996). In part due to its ovo-viviparous mode of reproduction (that significantly limits its dispersal capability), as well as to the heterogeneity of the intertidal ecosystem, *L. saxatilis* shows a high tendency to evolve local adaptations and to present an enormous intraspecific phenotypic variation (reviewed in Rolán-Alvarez et al. 2015). Indeed, several different ecotypes have been described in *L. saxatilis*, among which the Crab and Wave ecotypes stand out. These ecotypes—present at least in the Swedish, British, and Spanish coasts—show specific morphological and behavioral adaptations to the different parts of the intertidal ecosystem, and molecular data show that there exists partial reproductive isolation between them (Rolán-Alvarez 2007; Johannesson 2015; Johannesson et al. 2017).

In addition to the Crab and Wave ecotypes, of particular interest within the *L. saxatilis* system is the high diversity of color morphs that it can present (see Figure 1A). These color morphs can be found either within the same populations or between populations in geographically adjacent areas, sometimes giving rise to color clines (Reid 1996 and references therein; Gefaell et al. 2023). A region where a high diversity of color morphs of *L. saxatilis* exists is the Rías Baixas (Galicia, NW Spain). The Rías Baixas constitute a series of 4 coastal inlets that originated through the immersion of river valleys that are open to the sea (Méndez and Vilas 2005). In at least 3 out of 4 of these Rías, *L. saxatilis* presents a purported

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color cline, described decades ago but not verified since then (Fischer-Piette and Gaillard 1961; Fischer-Piette et al. 1961; Sacchi 1979; Sacchi and Malcevschi 1983; Reid 1996). More specifically, such a color cline would have 3 relatively well-differentiated zones: an interior zone, comprising the innermost localities closest to the mouth of the main river of each Ría, and dominated by fawn-like colors ("*fulva*"); an intermediate zone, corresponding to the midway localities of the Rías and in which high color diversity is found in terms of the number of color morphs; and an exterior zone, corresponding to the most wave exposed localities and mostly dominated by a lineated ("*lineata*") morph (see Figure 1B).

Of these 4 Rías, the Ría de Vigo (the southernmost one; with around 41 km of coast on the north and 57 km on the south) has been the most studied in the past. Sacchi (1979) analyzed the frequency of the different color morphs in several localities across this Ría, corroborating the existence of the above-mentioned color cline, previously described by Fischer-Piette and Gaillard (1961) and Fischer-Piette et al. (1961). Color-frequency data provided by Sacchi (1979), readily available within that article, constitute a unique opportunity to analyze the patterns of temporal stability and variation of a color cline in an intertidal organism. In this article, we attempt to resurvey exactly the same populations as Sacchi surveyed in the 1970s to (1) corroborate the existence of such cline; (2) determine to what extent it has changed or remained stable for the last few decades; and (3) provide a preliminary examination of the potential evolutionary mechanisms behind the medium-to-long-term dynamics of such a color cline, to come up with future ways to experimentally test their explanatory power.

Material and Methods

Sample collection and color assignment

In this study, 2 distinct samples were used: "1979" and "2022." The "1979" data were taken from Sacchi's original publication (Sacchi 1979: Tables 1–5). These localities were collectively called "1979" due to the date of Sacchi's article, but data on such populations were collected in August–September 1974 and 1976 (Sacchi 1979). Sacchi took samples from 42 different localities across the Ría de Vigo (average $N = 1020$), avoiding the urban area of the city of Vigo (see light pink area in Figure 1B). For our analyses, from these original 42 populations, 5 corresponding to Cíes Islands (an archipelago of 3 islands at the mouth of the Ría de Vigo) were excluded, as these populations do not show geographical continuity with the remaining populations. Moreover, many of these populations are associated with the Cíes Lagoon, in which a different *L. saxatilis* ecotype exists (Galindo et al. 2020), and this could confound the cline analysis. This exclusion of localities from Cíes Islands yielded a total of 37 valid localities along the Ría of Vigo for the "1979" sample, ranging from the innermost areas (close to the mouth of the Verdugo River, the Ría de Vigo's main river) to the outermost parts (Cabo Home, N; and Baiona, S). The average distance between localities was 2.61 km (SD = 1.99 km). Based on previous results on genetic differentiation in *L. saxatilis* from this area (Rolán-Alvarez et al. 2004; Tirado et al. 2016; Rivas et al. 2018), these distances are big enough to consider populations from different localities as partially genetically isolated.

Samples for the 2022 data were collected in 5 nonconsecutive days between February and March 2022. Based on Sacchi's naming of the different localities and their positions as shown in Figure 1 of that article (Sacchi 1979:9), the coordinates of these localities were inferred (see Table 1). Snails were collected in 36 out of 37 localities (see locality dots and numbers in Figures 1B and 2). The remaining population, corresponding with Sacchi's "Moaña, entro il porto" (Moaña, within the port; 9_N in our coding system; located between 8_N and 10_N in Figures 1B and 2), was also visited, but no *L. saxatilis* were found there, probably due to the expansion of the port since Sacchi's samplings. Although *L. saxatilis* does not constitute an endangered species and it is very common in intertidal ecosystems throughout the Ría of Vigo, we followed a "reduction" strategy for our sample size (see Russell and Burch 1959), thus significantly reducing the number of snails used (average $N = 57$) in comparison with Sacchi.

The sample was collected using a sampling quadrat (diameter = 60 cm) and following a stratified method. For this, a characterization of the different microhabitats in the localities was performed based on rock type and intertidal zone. Then, the sampling quadrat was placed randomly at different points of the representative littorinid microhabitat, usually the supralittoral zone (most of the times irrespective of rock type). Sampled individuals were picked up from the sampling quadrat, using as many replicates as were necessary until reaching an average number of ~50 snails. The samples were taken alive from the field to the laboratory and then stored at −20 °C. The determination of the color of the individuals was based on Dautzenberg and Fischer (1912), as shown in Sacchi (1979; see Figure 1A). According to this classification, 10 different color categories would exist; from these, 9 would be present in the Ria of Vigo: *albida* (white individuals), *fulva* (a wide class of colorations, from grayish-hazel to yellow-straw—or even light brown), *lutea* (yellow, from citrine to golden), *aurantia* (orange), *fusca* (brown to blackish), *nigra* (brilliant-black), *bande* (shells with 2 large dark main bands on a lighter background), *tessellata* (reticulated or checkered tint), and *lineata* (dark lines coinciding with the grooves of the sculpture). The remaining category, *sanguinea*, described as "blood red," does not exist in the Ría de Vigo. The available evidence for the species in the genus *Littorina* suggests that shell color (both background, bands, and line patterns) is genetically based, with different background colors and ornamentations coded by different alleles in Mendelian unlinked loci (Ekendahl and Johannesson 1997; Kozminsky 2011; Johannesson and Butlin 2017; Gefaell et al. 2023 and references therein). Genomic analyses have revealed that color traits in the Swedish system of *L. saxatilis* are influenced by several chromosome inversions (Westram et al. 2018; Koch et al. 2021), although the details and reach of such influences are yet not known.

Out of the previous qualitative descriptions, RGB categories for each color category were created to aid the assignment process (except for *bande*, *tessellata*, and *lineata*, which are not unicolor and can be diagnosed based on observed ornamentations) (see Sacchi 1979:12). To avoid potential biases derived from frequency expectations for the different 1979 localities, the 2022 samples were classified without knowing the precise frequencies and distributions of colors provided by Sacchi (1979).

Ecological data sampling

Ecological data for each locality were collected to provide clues on potential environmental factors associated with the

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Table 1 Localities analyzed by Sacchi (1979) and resampled again in 2022, with their corresponding codes and GPS coordinates. Data from the locality 9_N are not shown, as we did not find snails there (see main text). Distance (m) is the distance from each locality to the river mouth. *N* is the sample size of the color samplings (1979 or 2022). Size (mm) is the average size (± *SD*) in 15 representative specimens. *H'* is the Shannon index and D_{Roger} is the divergence between studies in color frequencies by Roger's distance. χ^2 is the chi-squared test of color frequencies as measured by D_{Roger} between color samplings. "ns" indicates non-statistical significance, $^{**}P < 0.01$, $^{***}P < 0.001$

Locality	Code	GPS coordinates	Distance (m)	N_{1979}	N_{2022}	Size (mm)	H'_{2022}	H'_{1979}	D_{Roger}	χ^2
Punta Cabalo	1_N	42°19'36.64"N 8°38'10.68"W	6,960	462	58	9.1 ± 2.20	0.86	1.10	0.05	1.9^{ns}
Punta Pereiró	2_N	42°19'19.26"N 8°38'50.07"W	8,093	554	52	10.2 ± 1.97	0.80	0.62	0.08	5.9^{ns}
San Adrián	3_N	42°18'05.1"N 8°39'20.3"W	11,495	1,368	50	9.4 ± 3.70	1.32	0.90	0.26	395^{***}
Costa San Adrián	4_N	42°17'30.05"N 8°39'30.42"W	12,905	486	50	11.2 ± 3.80	1.01	1.36	0.17	53^{***}
Punta Collo de Bestia	5_N	42°17'26.07"N 8°40'14.29"W	14,040	1,028	53	11.3 ± 1.49	1.16	0.69	0.12	150^{***}
Borna	6_N	42°16'50.75"N 8°41'48.43"W	17,008	1,804	51	6.8 ± 0.94	1.28	1.60	0.25	141^{***}
Punta Arroás	7_N	42°16'40.42"N 8°42'25.56"W	18,066	852	57	8.4 ± 0.74	1.78	1.63	0.11	45^{***}
Meira	8_N	42°17'0.05"N 8°43'9.46"W	21,307	752	59	10.5 ± 1.94	1.67	1.40	0.30	132^{***}
Praia de Con	10_N	42°16'14.13"N 8°44'24.69"W	24,450	846	50	4.8 ± 0.78	2.03	2.11	0.20	101^{***}
Punta Niño Corvo	11_N	42°15'47.70"N 8°45'18.60"W	26,019	1,000	64	7.0 ± 1.06	1.19	2.23	0.28	40^{***}
Punta Rodeira	12_N	42°15'29.04"N 8°45'56.67"W	27,237	1,250	56	7.0 ± 1.28	0.79	1.62	0.20	57^{***}
Punta Balea	13_N	42°15'6.62"N 8°47'17.89"W	30,212	1,220	58	5.5 ± 0.96	0.89	1.78	0.25	62^{***}
Punta Borneira	14_N	42°14'48.60"N 8°47'35.73"W	31,018	1,174	48	5.0 ± 0.89	0.29	0.02	0.03	36^{***}
Punta Salgueira	15_N	42°15'3.12"N 8°48'15.04"W	32,368	1,064	58	5.5 ± 0.98	0.32	0.08	0.05	61^{***}
Punta Igreixiña	16_N	42°15'14.73"N 8°50'3.75"W	36,235	1,200	57	5.7 ± 0.79	0	0	0	-
Punta Subrido	17_N	42°14'51.56"N 8°51'47.65"W	40,007	1,822	50	5.3 ± 1.01	0	0	0	-
Cabo Home	18_N	42°15'2.18"N 8°52'18.21"W	41,632	1,704	58	5.4 ± 1.03	1.03	0.00	0.19	388^{***}
Muelle Arcade	1_S	42°20'17.73"N 8°36'48.03"W	858	1,252	59	9.6 ± 3.57	1.43	0.95	0.31	59^{***}
Punta Puntal	2_S	42°20'6.31"N 8°36'52.21"W	1,221	332	73	10.1 ± 1.15	0.71	0.79	0.04	2.9^{ns}
Punta Pesqueira	3_S	42°19'46.01"N 8°36'51.01"W	1,982	832	50	8.5 ± 2.95	1.10	0.94	0.11	4.7^{ns}
Cesantes	4_S	42°18'57.52"N 8°37'7.82"W	3,915	1,853	63	10.1 ± 1.15	0.93	1.25	0.17	10^{ns}
Punta Soutelo	5_S	42°17'17.01"N 8°38'17.26"W	11,631	495	57	10.5 ± 1.93	1.22	1.05	0.31	170^{***}
Cabanas	6_S	42°16'53.95"N 8°39'46.13"W	14,253	788	63	9.9 ± 1.71	2.07	1.79	0.26	104^{***}
Punta Cubillo	7_S	42°16'08.4"N 8°40'23.8"W	16,061	680	60	9.5 ± 1.23	1.52	1.46	0.53	212^{***}
Areiño	8_S	42°15'35.34"N 8°41'56.07"W	18,791	726	48	6.3 ± 1.07	1.62	1.42	0.65	592^{***}
Alcabre	9_S	42°13'25.85"N 8°46'19.56"W	28,311	2,118	61	6.7 ± 1.54	1.18	1.13	0.07	139^{***}
Coruxo	10_S	42°12'3.21"N 8°47'7.28"W	31,474	1,190	60	5.4 ± 0.98	1.12	0.94	0.06	47^{***}
Canido	11_S	42°11'24.75"N 8°48'35.26"W	34,158	852	63	6.0 ± 0.88	1.32	1.95	0.21	69^{***}
Cabo Estai	12_S	42°10'57.47"N 8°48'49.55"W	35,432	1,322	54	8.1 ± 1.50	1.53	1.28	0.48	761^{***}
SE Monteferro	13_S	42°9'13.61"N 8°50'4.35"W	39,827	696	59	5.7 ± 1.28	0.41	0.70	0.05	4.5^{ns}
SW Monteferro	14_S	42°8'52.07"N 8°49'56.00"W	43,863	718	60	5.4 ± 1.18	0.24	0.21	0.02	14^{**}
Panxón	15_S	42°8'32.82"N 8°49'33.62"W	44,911	1,030	61	6.8 ± 1.06	0.00	0.39	0.04	3.2^{ns}
Praia América	16_S	42°7'32.02"N 8°49'27.14"W	47,616	844	56	4.3 ± 0.56	1.14	1.54	0.34	118^{***}
Santa Marta	17_S	42°6'56.26"N 8°50'13.00"W	53,976	1,160	64	7.4 ± 2.10	1.84	1.01	0.18	161^{***}
Punta Tenaza	18_S	42°7'34.25"N 8°50'56.89"W	56,305	372	56	6.1 ± 0.88	0	0.72	0.12	8.7^{ns}
Cuncheira	19_S	42°7'23.57"N 8°51'6.54"W	57,405	786	53	7.3 ± 1.53	0	0	0	-
Average				1,020 ± 435	57 ± 6		1.0 ± 0.60	1.0 ± 0.64	0.2 ± 0.15	

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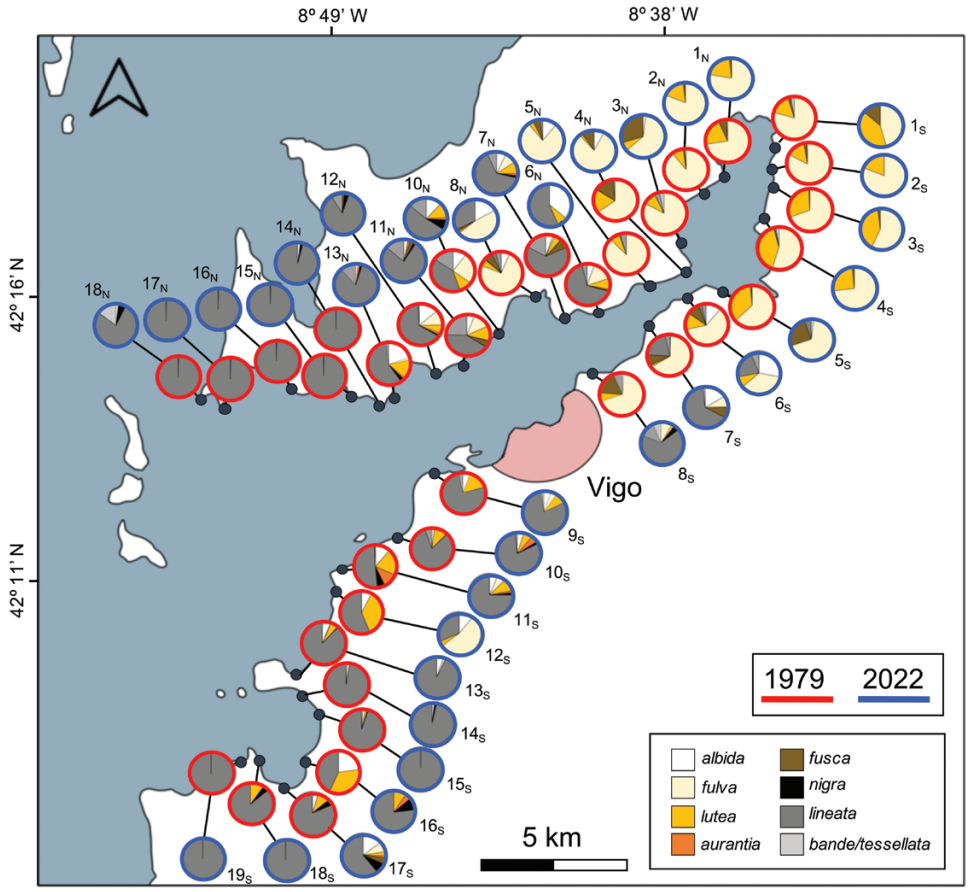

Figure 2 Pie charts comparing the color frequencies of the different localities along the Ría de Vigo in 1979 and 2022 samples. The map was taken and modified from d-maps (https://www.d-maps.com).

color cline. These data were taken on 4 consecutive days during October 2022 using a sampling quadrat (diameter = 60 cm) and following a semiquantitative procedure similar to the Ballantine scale (used to estimate wave exposure; Ballantine 1961; see also Raffaelli and Hawkins 1996; Little et al. 2009). For this, semiquantitative scales and presence/absence classes for key ecological parameters of the intertidal ecosystem were created based on relevant parameters for the Rías Baixas as derived from key published sources (Rolán 1990; Méndez and Vilas 2005; Muñoz Sobrino et al. 2012). Semiquantitative procedures such as the Ballantine scale constitute a useful approach when dealing with large geographical scales (such as the Ría de Vigo; ~41 km bordering the northern coast, and ~57 km the southern coast), and can provide insights into the main ecological trends at play in intertidal ecosystems (Raffaelli and Hawkins 1996; Burrows et al. 2008; Little et al. 2009).

The recorded variables and their corresponding scales for the ecological sampling are presented in Supplementary Table A1 (see online material available). These tried to capture either physical properties (e.g., type of rock, rock size) or ecological parameters like absence/presence or relative abundance of different intertidal species (*Melarhaphe neritoides*, *Monodonta lineata*, *Littorina fabalis/obtusata*, *L. littorea*, barnacle species—mostly *Semibalanus balanoides* and *Chthamalus* sp.—, *Enteromorpha* sp., and *Lichina pygmaea*). These species are unevenly distributed across the Ria, varying mostly according to the degree of wave exposure (Rolán-Alvarez, personal observation). The distance of each locality to the river mouth, as well as the distance between localities, was also calculated using Google Earth Pro (v. 7.3.6.9285) bordering the coast from the mouth of the Verdugo River to each locality. Given the linearity of the Ría de Vigo, this parameter can be used as an indicator of the degree of wave exposure.

Additionally, the salinity of each locality was estimated using a refractometer (averaged over 2 technical replicates), as previous studies show a clear gradient for this variable from the river mouth to the outermost parts of the Ría de Vigo (Torres López et al. 2001; Souto et al. 2003; Ibánhez et al. 2021). The average shell size (i.e., columella length) was also measured to the nearest 0.1 mm using vernier calipers in a set of 15 specimens selected at random from the 2022 data. Even though Sacchi didn't measure such parameter, size was chosen as a potential indicator of local phenotypic adaptation of *Littorina* species to distinct microhabitats (either by natural selection or phenotypic plasticity; reviewed in Rolán-Alvarez et al. 2015). Previous studies suggest that size might constitute an important trait that affects mortality and is involved in ecotypic evolution (e.g., Boulding et al. 2017). Raw data on color frequencies and ecological parameters are available online (https://doi.org/10.6084/m9.figshare.21771152.v1).

Statistical analyses

Variation in shell color along the Ría de Vigo as well as its potential association with environmental parameters was studied both by comparing shell color frequencies between 1979 and 2022 samples through classical chi-squared tests

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and by using a stepwise multiple linear regression of physical and ecological variables on frequencies of the 2 most representative colors of the cline (*fulva* and *lineata*), as well as on the average shell size of *L. saxatilis* for each population. Early in situ inspection revealed that the *fulva* and *lineata* morphs represent the dominant morphs in the innermost and outermost localities of the Ria (respectively), so they can be useful in characterizing the shape of the cline. The stepwise multiple regression used different criteria to introduce ($P \leq 0.05$) or exclude ($P \geq 0.10$) variables into the model.

The Shannon index (*H'*; Shannon 1948; Jain et al. 1975) was used to calculate color-frequency variation at each locality separately for the 1979 and 2022 samples. Shannon index describes the species (or color morph) richness of a community (or a population) as,

$$H' = \sum_{i=1}^{S} Pi * Log_2 Pi$$

where S stands for the number of different color morphs, and P_i is the relative frequency of each color morph. The greater the number of colors at relatively intermediate frequencies, the higher the Shannon index (*H'*). Conversely, monomorphic populations (all individuals showing the same color morph) have *H'* = 0. Between-year differences for *H'* were analyzed using a fixed-factor ANOVA.

The color-frequency divergence over time (1979 vs. 2022) was analyzed using Roger's genetic distance (D_{Roger}; Rogers 1972; Nei 1987). D_{Roger} can also be used to measure discrete variation in a given phenotypic trait, such as color. Roger's distance (D_{Roger}) measures color divergence between populations as,

$$D_{\text{Roger}} = \frac{1}{L}\sqrt{\frac{\sum_u (X_u - Y_u)^2}{2}}$$

where L stands for the total number of loci-traits (assuming one trait), X and Y are the analyzed populations, and u is the frequency of a given color. $D_{Roger} = 0$ is obtained when there is an identical color frequency between populations, and $D_{Roger} = 1$ when there exists a complete frequency divergence between them. The significance of D_{Roger} was evaluated through chi-squared tests (Sokal and Rohlf 1995). The contribution of each particular color to the former overall distance was estimated using the particular quadratic difference of the previous D_{Roger} algorithm per locality and averaged across localities (non-normalized). Differences between colors across the 36 localities were evaluated by one-way ANOVA and the Student–Newman–Keuls (SNK) a posteriori test. All the ecological variables were summarized by means of a principal component analysis (PCA) to test whether they could be interpreted as proxies of wave exposure by latter correlating them with the distance to the river mouth. Statistical analyses were carried out with SPSS v.24 (IBM Corp., USA), and plots were made using R version 4.2.0, with ggplot2 version 3.4.0 for data visualization (Wickham 2016; R Core Team 2022).

Results

Color frequency, for both 1979 and 2022 samples, shows a clear clinal pattern along the Ría de Vigo from the innermost to the outermost localities (Figure 2). The innermost localities (sheltered and with low salinity) show a high frequency of *fulva* individuals, with frequencies > 50% in most of them. The outer localities (wave-exposed and relatively high salinity; around 35–36 g of salt per liter/kg of water) show a high frequency of the *lineata* morph, even reaching monomorphism (100%) in many of them. Finally, intermediate localities show high color diversity both in terms of the number of color morphs present at each locality (including *albida*, *lutea*, *aurantia*, or *nigra*) and in terms of the relative frequency of each of these morphs. A significant Pearson' *r* correlation exists between the color frequencies of *fulva* and *lineata* morphs and each locality's distance to the river mouth (used as a simple proxy for the degree of wave exposure), irrespective of the data series (1979 vs. 2022): positive for *lineata* 1979 ($r = 0.83$; $df = 35$; $P < 0.001$) and *lineata* 2022 ($r = 0.83$; $df = 35$; $P < 0.001$), and negative for *fulva* 1979 ($r = -0.82$; $df = 35$; $P < 0.001$) and *fulva* 2022 ($r = -0.73$; $df = 35$; $P < 0.001$) (Figure 3). Taken together, these results both support the existence of a color cline in *L. saxatilis*, with 3 different color zones across the Ría de Vigo, as well as its overall maintenance over time.

The mean color biodiversity (*H'*) is also similar between both samples (1979 vs. 2022; Table 1; $F_{ANOVA} = 0.1$; $df = 1/70$; $P = 0.875$). In addition, there is a high correlation across localities between data samples ($r = 0.74$; $df = 35$; $P < 0.001$). However, putting aside the global pattern of similarity (i.e., stability), some differences also exist between samples, as revealed by D_{Roger} and chi-squared tests (Table 1). Averaged divergence per locality between periods is about 20% of the maximum possible ($D_{Roger} = 0.20 \pm 0.15$), and the frequencies differ significantly in most chi-squared comparisons (25 out of 36; Table 1). The contribution of each color to the D_{Roger} distance suggests that this divergence between samples affects preferentially the main clinal colors (i.e., *fulva* and *lineata*; Figure 4), as there were significant differences between colors by ANOVA ($F = 5.08$; $df = 8/315$; $P < 0.0001$). The SNK test identifies 2 significant groups (alpha = 0.05): *fulva* and *lineata* vs. the rest of the colors (Figure 4). In turn, this could be interpreted as an indication that, despite the stability of the overall pattern, the cline has changed during the studied period, resulting in an increased frequency of *lineata* along several intermediate localities (mostly southern) while decreasing that of *fulva* (see Figure 2).

Several potential predictive variables of the clinal color distribution (estimated through *fulva* and *lineata* frequencies) and the average shell size (a characteristic adaptive trait in this species; e.g., Rolán-Alvarez et al. 2015; Boulding et al. 2017) were explored using the stepwise linear multiple regression (Supplementary Table A2, see online material available) to gain insight into potential causal factors. This analysis shows that the 2 colors (both in northern and southern localities) exhibit a strong association with several variables that capture different physical and/or ecological characteristics of the environment (from 1 to 4 variables). We can interpret such variables as proxies of the degree of wave exposure and position along the Ría, as a PCA including all of them yields a PC1 that explains 41% of the variance and that shows a significant correlation with the river mouth ($r = 0.79$, $P = 0.002$; see Burrows et al. 2008 for an analogous use of PCA). In this sense, multiple regression analysis shows that *lineata* frequency is almost fully explained ($R^2 = 0.97$) by 3 variables in the north (including the presence of *Monodonta lineata* and the abundance of *Lichina pygmaea*) but only one (presence of *Melarhaphe neritoides*; $R^2 = 0.69$) in the south. On the other hand, *fulva*

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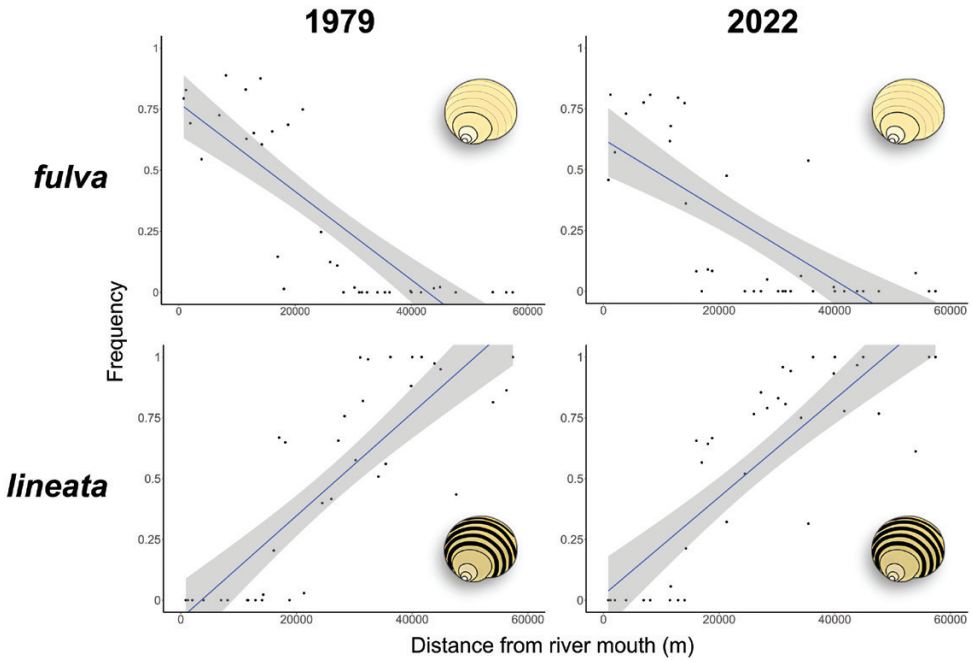

Figure 3 Frequencies of *fulva* (up) and *lineata* (down) for the 1979 (left) and 2022 (right) samples in relation to the distance from the mouth of the Verdugo River. R^2 coefficients and *P*-values for the linear regression corresponding to each figure: for *fulva* 1979 ($R^2 = 0.67$; $P < 0.001$), *fulva* 2022 ($R^2 = 0.53$; $P < 0.001$), *lineata* 1979 ($R^2 = 0.68$; $P < 0.001$), and *lineata* 2022 ($R^2 = 0.69$; $P < 0.001$).

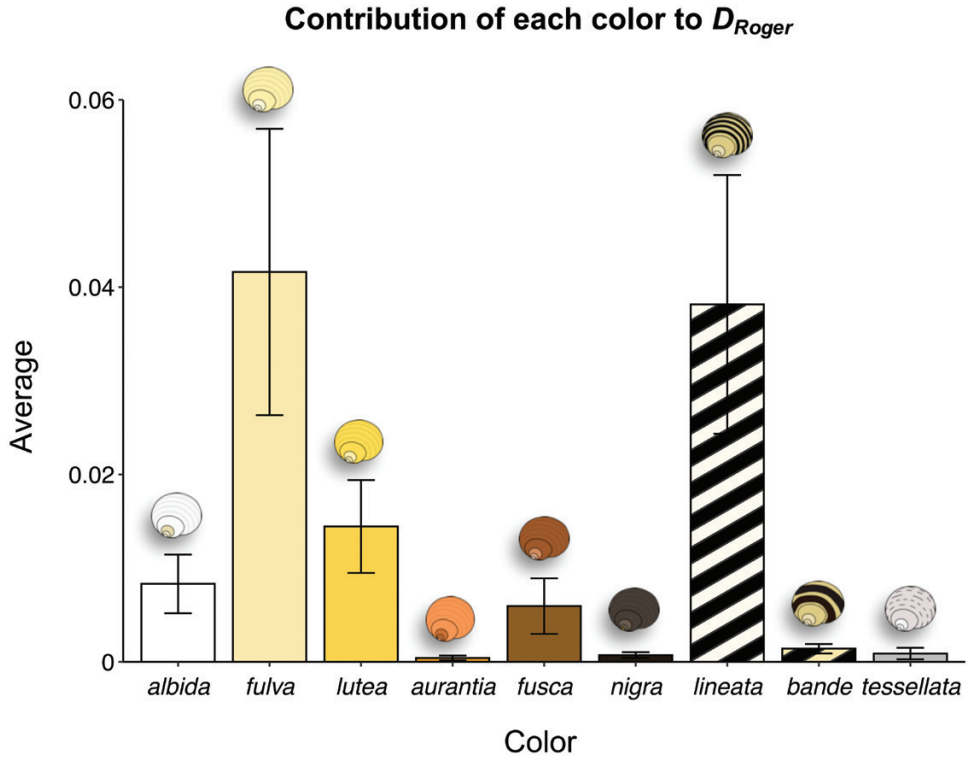

Figure 4 The contribution of each color to D_{Roger}, or the average squared difference in color frequency between the 1979 and 2022 samples for each single color morph. The bars represent standard errors.

frequency is fully explained ($R^2 = 0.98$) by 4 variables in the north (absence of *Monodonta lineata*, mean salinity, average rock size, and *Enteromorpha* sp. abundance), while in the south a large portion of variance ($R^2 = 0.83$) is explained by (low) salinity. Interestingly, a trait typically associated with local adaptation—i.e., size—is fully explained ($R^2 = 0.97$) by 2 variables (presence of *Melarhaphe neritoides* and *Enteromorpha* sp. abundance) in the north, and explained ($R^2 = 0.83$) by 2 variables in the south (presence of *Melarhaphe neritoides* and *L. littorea*). All explanatory

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variables in multiple regression analyses show a clear environmental gradient along the Ría de Vigo, corresponding to the degree of wave exposure.

Discussion

Our study corroborates the existence of a decades-long color cline in *L. saxatilis* from the Ría de Vigo (Sacchi 1979; Reid 1996). Lab cultures suggest that the Swedish populations of *L. saxatilis* become sexually mature around 6–8 months (Johannesson and Butlin 2017), and although no reliable data exists for the Galician populations, previous simulation studies assume that *L. saxatilis* from this area has a generation time of around 6 months (Fernández-Meirama et al. 2022). This means that it is reasonable to assume that the 46–48 years between both samples analyzed in this study (1979 vs. 2022) are equivalent to around 92–94 generations of *L. saxatilis*. Taken into account that Fischer-Piette and collaborators (Fischer-Piette and Gaillard 1961; Fischer-Piette et al. 1961) already described this color cline in the early 1960s (although they do not provide actual data for comparison with the time series presented in this study; see also Reid 1996), the temporal stability shown in these populations may have lasted for more than 60 years, equivalent to around 120 generations of *L. saxatilis*. Previous studies analyzing the temporal dynamics of color clines suggest that this number of generations might constitute time enough for evolutionary processes to act on populations. For instance, Brakefield and de Jong (2011) show a significant decay in the steepness of a color cline in the 2-spot ladybird (*Adalia bipunctata*) over the course of 50 generations, presumably driven by a genetic response to thermal changes induced by climate change along a 115 km transect in the Netherlands. Cameron et al. (2013) found that in about half the number of these generations, banding morphs of *Cepaea nemoralis* did not change over a cline, while the frequency of the yellow morph increased all over it. Therefore, depending on whether the putative selective pressures acting on a cline vary or not over a given timespan, and whether drift and migration are strong enough to influence the evolutionary outcomes, we might expect changes or stability in the shape of clines across generations.

The other remarkable phenomenon revealed in this study is the variability observed in some populations, despite the overall trend remaining stable over the past decades. We can distinguish between 2 different types of variability: what might be called "local variability" and "directional change." By local variability we mean the divergence in color frequencies found in specific populations, such as 10_N or 12_S (see Table 1 and Figure 2). Conversely, directional change refers to predictable and recurrent patterns of variation in the frequency of certain morphs. When we compare data from 1979 and 2022, a systematic and recurrent pattern of variation in the *lineata* and *fulva* morphs is observed (*lineata* typically displaces *fulva* in several intermediate-to-inner localities; Figures 2 and 4).

Occasional sampling errors (errors in the geographical correspondence of the sampling sites studied in 1979 and 2022, since Sacchi's article does not provide any geographic coordinate system to identify exact sampling points), assignation errors, or genetic drift could potentially explain local variability. However, although it cannot be known whether the 2022 sampling points exactly correspond to those of 1979, or whether the assignation process was done using the exact same criteria, we believe that the temporal stability and directional change observed in this cline between different time series cannot be simply attributed to either sampling or assignation errors, or genetic drift. As for assignation errors, it is virtually impossible to confuse *lineata* and *fulva* ecotypes (the morphs mostly involved in both the temporal stability and directional change of the cline). As for sampling errors or genetic drift, such factors would preferentially affect rare colors (due to the fact that random errors tend to affect lower frequencies most), while the contribution of individual colors to Roger's distance (D_{Roger}) shows the opposite pattern (more pronounced in *lineata* and *fulva*, the most frequent morphs in the outer and inner localities, respectively; Figure 4). Another possible caveat with respect to the observed patterns has to do with potential seasonality effects. Our 2022 data were taken in different seasons (autumn–winter) than these of 1979 (summer). There are only a few studies so far that have analyzed the seasonal changes in the frequency of snails from the genus *Littorina*, yielding negative results. Ekendahl (1994, 1995) didn't find any seasonal changes in the frequency of color morphs of the Swedish populations of *L. fabalis*. This researcher found, however, seasonal changes in microhabitat choice. Nevertheless, Carballo et al. (2005) found no seasonal changes in the microdistribution of the Crab and Wave ecotypes of *L. saxatilis* in a 3-year study close to the Ria de Vigo. We cannot formally rule out that seasonal microhabitat changes might have affected the clinal color frequencies compared in this study, but given the above-mentioned study, we believe this is highly unlikely. Therefore, the evidence presented in this study suggests that both the temporal stability observed in the cline and the directional changes detected in certain populations are real, demanding mechanistic explanations.

Regarding the directional change, the *lineata* morph is mostly associated with ecological parameters related to wave exposure and relatively high salinity (in a way similar—but not restricted—to the Crab ecotype; Rolán-Alvarez et al. 2007). The expansion of this color morph to the intermediate zone of the Ría (with a corresponding decrease of *fulva*) could indicate a change in any of such parameters from the outer to the mid localities over the past 46–48 years associated with an expansion of Vigo's main city of Ría de Vigo. However, no systematic analyses of this possibility have yet been carried out, so future studies should systematically investigate this and related scenarios.

As for the temporal stability of the color cline of *L. saxatilis* in the Ría de Vigo, 2 main hypotheses could be invoked as potential explanatory factors: either the combination of migration and genetic drift or some kind of natural selection along one or more ecological gradients (e.g., Mayr 1963; Endler 1977; Ridley 2004; Sætre and Ravinet 2019). We believe that such stability in a clinal pattern is unlikely to be explained through the combination of migration and genetic drift, as such mechanisms predict a much greater degree of stochasticity. Moreover, the cline described in this study shows not only temporal stability but also spatial replication. Considering the low dispersal capacity of *L. saxatilis* (Rolán-Alvarez et al. 2015), it is highly unlikely that there is strong genetic connectivity between its northern and southern populations. This suggests that both clinal trends (north and south) are maintained independently through non-stochastic forces, and may qualify as separate spatial replicas. Furthermore, such a clinal pattern has been described in other Rías from the Rías Baixas (Reid 1996; Fischer-Piette and Gaillard 1961; Fischer-Piette

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et al. 1961; Rolán-Alvarez, personal observation). Although the genetic structure and connectivity between the northern and southern populations of *L. saxatilis* from the Ría de Vigo should be studied through different genetic markers to formally rule out the migration plus drift hypothesis (e.g., as in Westram et al. 2018), we believe that this is an a priori less convincing scenario.

A much more plausible explanatory mechanism might involve some form of natural selection. In addition to the temporal stability and spatial replication of the cline (which by its own right might already suggest the action of selective forces; Endler 1986), the main reason for this is that there are strong environmental gradients (e.g., temperature, salinity, CO_2, and wave exposure) along the Ría de Vigo that go parallel to the color cline (Ruiz-Villareal et al. 2002; Gago et al. 2003; Muñoz Sobrino et al. 2012; Ibánhez et al. 2021). These environmental gradients might give rise to differential selective pressures that end up maintaining the cline in a relatively stable form.

Furthermore, multiple regression analyses show that several key ecological variables can explain the observed color distribution, thereby strengthening the case for natural selection. The *lineata* and *fulva* color clines are clearly explained by a few ecological or physical variables that show gradients from inner to outer localities. For example, *Monodonta lineata*, *Melarhaphe neritoides*, and *Lichina pygmaea* tend to be associated with moderate to highly wave-exposed rocky shore areas (Ballantine 1961; Rolán 1990; Burrows et al. 2008), and they show concomitantly positive regression coefficients when explaining the *lineata* clines, whereas, for instance, *Monodonta lineata* shows a negative regression coefficient when explaining the *fulva* ones. The degree of wave exposure might be linked to differences in temperature or salinity, which in turn might constitute powerful selective agents. In the case of temperature, for instance, Phifer-Rixey et al. (2008) provide experimental evidence that thermal stress tolerance might be driving a color cline in *L. obtusata* from the Gulf of Maine. As for salinity, which also shows a concentration gradient over the Ría (e.g., Muñoz Sobrino et al. 2012; this study), its positive regression coefficients with *lineata* frequency and negative with *fulva* frequency suggest that it constitutes another selective pressure operating along the cline. In this sense, previous experimental evidence exists that the different color morphs of *L. saxatilis* from the White Sea have different saline-stress tolerance (Sokolova and Berger 2000), although the molecular mechanisms for such association are not known yet (see also Minton and Gunderson 2001 for a similar case in a different gastropod species). If that would be the case, then shell color would not be the primary target of selection, but only a correlated trait genetically associated with saline-stress resistance (or other biotic or abiotic factors) genes. This genetic association would not be surprising, as genomic data suggest that in *L. saxatilis* chromosomic inversions may contain several loci involved in adaptive traits, thereby contributing to low recombination between them (Koch et al. 2021).

As for the explanation of cline formation, 2 different mechanisms have been put forward: primary differentiation and secondary contact (Endler 1977). Spatial heterogeneity of a formerly homogenous environment leads to cline formation by primary differentiation. A similar primary differentiation scenario can be achieved through the colonization of a novel environment with distinctive characteristics by a species hitherto confined to a homogeneous area. In our case, the latter primary differentiation scenario is a priori supported by the historical geomorphology of the Ría de Vigo: before the early Holocene sea-level rise (12,000–7,000 years ago), sea level at this location was about 30–35 m below the current level, with the Ría de Vigo forming a river valley and the coast being located to the west of the Cíes Islands (Costas et al. 2009). As the sea level rose, marine water inundated the river valley, thereby gradually giving rise to the different zones (exterior, intermediate, and interior; in that order) of the Ría. It is possible, therefore, that from that point on populations of *L. saxatilis* began to colonize the intermediate and inner zones of the Ría, thereby being exposed to different selective pressures. An alternative hypothesis on the formation of the color cline observed in this study is viewing it as the product of a secondary contact between 2 distinct ecotypes (*lineata* and *fulva*) that differ in shell color and a suit of related traits, and that are locally adapted to the different ends of a temperature, salinity, and/or wave exposure gradient. Under this scenario, both ecotypes would show at least partial reproductive isolation between them, and the intermediate zone of the Ría would constitute a hybrid zone with a more or less steep cline in traits and allele frequencies, depending on the strength of divergent selection (as in the Crab and Wave ecotypes; see Westram et al. 2018). Although this latter scenario fits worst with historical geomorphology than the former, future studies on DNA markers and experimental settings should be put in place to reveal which of them is the correct one and which selective agents might constitute the causal mechanisms of this decades-long clinal pattern in a marine snail.

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Conflict of Interest

The authors declare that they have no conflict of interest.

Supplementary Material

Supplementary material can be found at https://academic.oup.com/cz.

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Supplementary materials – Chapter 4

Table A1. Ecological and physical variables recorded in our adaptation of the Ballantine scale to the Rías Baixas.

Variables	Categories	Meaning
Granite rock	Yes No	Defines whether the rocks present in the intertidal ecosystem are made of granite or not.
Schist rock	Yes No	Defines whether the rocks present in the intertidal ecosystem are made of schist or not.
Melarhaphe neritoides	Presence Absence	Defines whether *Melarhaphe neritoides* is present in the location sampled (after 15 min sampling).
Monodonta lineata	Presence Absence	Defines whether *Monodonta lineata* is present in the location sampled (after 15 min sampling).
L. fabalis/*L. obtusata*	Presence Absence	Defines whether *Littorina fabalis* and/or *L. obtusata* is present in the location sampled (after 15 min sampling).
L. littorea	Presence Absence	Defines whether *Littorina littorea* is present in the location sampled (after 15 min sampling).
Rock size	Very small surface rocks (approx. diameter < 20 cm) Small surface rocks (approx. diameter = 21-50 cm) Medium surface rocks (approx diameter = 51 cm to 1 m) Big surface rocks (approx diameter > 1 m) Bedrock surface (all covered by very big rocks)	Defines the size of the intertidal rocks of the sampled locality.

Table A1 (Continuation).

Variables	Categories	Meaning
Barnacles	Absent (no exemplars in a quadrat of diameter = 60 cm) Rare/occasional (covering less than ~10% of the surface rock in a quadrat of diameter = 60 cm) Frequent (covering between 11% and 30% of the surface rock in a quadrat of diameter = 60 cm) Common (covering between 31% and 50% of the surface rock in a quadrat of Ødiameter = 60 cm) Abundant (covering between 51% and 80% of the surface rock in a quadrat of diameter = 60 cm) Super-abundant (covering more than 81% of the surface rock in a quadrat of diameter = 60 cm)	Defines the relative covering of rocks by barnacles (mostly *Chthalamus* sp. and *Semibalanus balanoides*) in a quadrat of Ø = 60 cm placed in a representative area where *L. saxatilis* is present in the sampled locality.
Enteromorpha sp.	Absent (no exemplars covering the rocks in the 10 m^2 surrounding the sampling point) Rare/occasional (covering less than ~5% of the surface rock in the 10 m^2 surrounding the sampling point) Frequent (covering approximately between 20% and 40% of the surface rock in the 10 m^2 surrounding the sampling point) Common (covering approximately between 41% and 60% of the surface rock in the 10 m^2 surrounding the sampling point) Abundant (covering approximately between 61% and 80% of the surface rock in the 10 m^2 surrounding the sampling point) Super-abundant (covering between 81% and 100% of the surface rock in the 10 m^2 surrounding the sampling point)	Defines the relative covering of rocks by *Enteromorpha* sp. (green algae) in a quadrat of Ø = 60 cm placed in a representative area where *L. saxatilis* is present in the sampled locality.
Lichina pygmaea	Absent (no patches in areas where *L. saxatilis* is present) Rare/occasional (less than 1 patch/10 m^2 in with *L. saxatilis* on them) Frequent (between 2 and 5 patches/10 m^2 in with *L. saxatilis* on them) Common (between 6 and 10 patches/10 m^2 in with *L. saxatilis* on them) Abundant (between 11 and 15 patches/10 m^2 in with *L. saxatilis* on them) Super-abundant (most surface rock in 10 m^2 covered by patches of *Lichina* with *L. saxatilis* on them)	Defines the presence of patches of *Lichina pygmaea* (black lichen) with *Littorina saxatilis* in the sampled locality.
Salinity	-	Two measurements of the salinity of the water as grams of salt per liter/kg of water (‰) of the locality sampled with a refractometer.

Table S2. Stepwise multiple regression results for the frequency of the two main morphs (*lineata* and *fulva*), as well as for size.

Trait	Latitude	ANOVA	R^2	Included Variables	Beta	Probability
lineata	North	170***	0.97	Constant	0.24	0.029
				Monodonta	0.57	0.001
				Lichina	0.49	0.001
				Granite	-0.20	0.001
	South	41***	0.69	Constant	0.13	0.128
				Melarhaphe	0.84	0.001
fulva	North	187***	0,98	Constant	1.82	0.001
				Monodonta	-0.77	0.001
				Salinity	-0.56	0.001
				Rock size	0.34	0.001
				Enteromorpha	-0.14	0.003
	South	28***	0,60	Constant	1.00	0.001
				Salinity	-0.79	0.001
Size	North	43***	0,97	Constant	10	0.001
				Melarhaphe	0.47	0.001
				Enteromorpha	0.5	0.006
	South	44***	0,83	Constant	9.90	0.001
				Melarhaphe	0.41	0.001
				L. littorea	0.41	0.022

*** $P < 0.001$

Table S2. Homogeneity G likelihood tests for the mating data grouped by locality, year, and first vs. subsequent matings.

Comparison	Mating(s)	Subgroup	G value	d.f.	Probability
2021 vs. 2022	First	Soutoxuste	8,1	8	0,427
		Portela	8,8	8	0,358
		Socorro[§]	-	-	-
	Subsequent	Soutoxuste	7,3	8	0,253
		Portela	16,8	8	0.032*
		Socorro[§]	-	-	-
	All Matings	Soutoxuste	8,3	8	0,406
		Portela	14,8	8	0,063
		Socorro[§]	-	-	-
First Mating vs. Subsequent Matings	All Years	Soutoxuste	6,8	8	0,563
		Portela	2,6	8	0,959
		Socorro	24	8	0,002

*Half of the cells showed an expected value lower or or equal to 1, which tends to result in false positives.

[§]Socorro cannot be compared between years (2021 vs. 2022) as it was studied only in 2022.

Table S3. Summary tables of the number of different mating types obtained, separated by locality.

		Soutoxuste White (M)	Yellow (M)	Brown (M)
		33	33	33
White (F)	33	4	7	14
Yellow (F)	33	3	3	6
Brown (F)	33	4	8	12

		Portela White (M)	Yellow (M)	Brown (M)
		57	57	57
White (F)	57	6	22	19
Yellow (F)	57	12	14	16
Brown (F)	57	23	11	16

		White (M)	Rest (M)
		33	66
White (F)	33	4	21
Rest (F)	66	7	29

		White (M)	Rest (M)
		57	114
White (F)	57	6	41
Rest (F)	114	35	57

		Yellow (M)	Rest (M)
		33	66
Yellow (F)	33	3	9
Rest (F)	66	15	34

		Yellow (M)	Rest (M)
		57	114
Yellow (F)	57	14	28
Rest (F)	114	33	64

		Brown (M)	Rest (M)
		33	66
Brown (F)	33	12	12
Rest (F)	66	20	17

		Brown (M)	Rest (M)
		57	114
Brown (F)	57	16	34
Rest (F)	114	35	54

Table S3 (Continuation).

		Socorro		
		White (M)	Yellow (M)	Brown (M)
		38	38	38
White (F)	38	2	9	4
Yellow (F)	38	3	1	4
Brown (F)	38	10	5	0

		White (M)	Rest (M)
		38	76
White (F)	38	2	13
Rest (F)	76	13	10

		Yellow (M)	Rest (M)
		38	76
Yellow (F)	38	1	7
Rest (F)	76	14	16

		Brown (M)	Rest (M)
		38	76
Brown (F)	38	0	15
Rest (F)	76	8	15

Table S4. Total experimental units (i.e., mating balls) used, and mating success per locality.

	Total Experimental Units			Experimental Units w/ ≥1 Observed Matings			Experimental Units w/ >1 Observed Mating		
	2021	2022	Total	2021	2022	Total	2021	2022	Total
Soutoxuste	24	30	54	14	19	33	6	13	19
Portela	13	74	87	9	48	57	3	32	35
Socorro	NA	51	51	NA	38	38	NA	28	28

Table S4 (Continuation).

	Percentage of Success (≥1)			Percentage of Success (>1)		
	2021	2022	Total	2021	2022	Total
Soutoxuste	58.3	63.3	61.1	25.0	43.3	35.2
Portela	69.2	64.9	65.5	23.1	43.2	40.2
Socorro	NA	74.5	74.5	NA	54.9	54.9
		Average (all)	67			
		SD	6.8			
		Average (Soutoxuste)	60.8			
		SD	3.5			

Chapter 5

Gefaell J, Vigo R & Rolán-Alvarez E. Under review. Does differential predation of *Littorina saxatilis* color morphs contribute to the maintenance of a color cline? Insights from a field tethering experiment.

Does differential predation of *Littorina saxatilis* color morphs contribute to the maintenance of a color cline? Insights from a field tethering experiment

Juan Gefaell, Ramón Vigo & Emilio Rolán-Alvarez

Abstract: Color clines constitute an interesting topic of study for evolutionary ecologists as they allow for testing different hypotheses about the adaptive value of color variation. One possible explanation for the selective advantage of coloration is crypsis, or the chromatic concealment of prey from visual predators. In *Littorina saxatilis* from the Rías Baixas (northwest Iberian Peninsula), a temporally stable and spatially recurrent color cline has been described. This cline extends from wave-sheltered localities to the most wave-exposed areas, the latter bearing monomorphic populations of a lineated morph. As the ecological communities, and therefore the background colors, change gradually along the wave-exposure continuum, a plausible explanation for this cline is selection for crypsis, by which less cryptic morphs are weeded out from the population, presumably by crabs. This would be especially intense at the monomorphic localities along the cline. In the present study, we describe a tethering experiment at a representative wave-exposed locality from the Rías Baixas (Cabo Silleiro) aimed at testing this hypothesis, in addition to a series of complementary analyses on Cabo Silleiro and a nearby polymorphic locality (Aguncheiro), from which most of the snails used in the experiment were collected. The analysis of Aguncheiro revealed slight differences in the distribution of shell scars (reflecting past crab attacks) across color morphs, without clear support for the cryptic selection hypothesis. In the tethering experiment, predation rates by crabs were recorded for three different transplanted color morphs from Aguncheiro, along with the resident morph from Cabo Silleiro. The experiment yielded suggestive but inconclusive results, as the supposedly less cryptic morph was more likely to be predated upon in only one of the two sessions, assuming that small breaks in the shells constitute predation attempts. However, the study's power was limited by low predation rates, underscoring the challenges inherent to field experiments.
Keywords: color polymorphism, marine gastropods, crypsis, tethering experiment, reflectance spectrometry.

Introduction

Clines, or the gradual progression in a given trait over a geographic transect, have long been a predilect topic of study for evolutionary ecologists (Huxley, 1938; Endler, 1977; Barton & Hewitt, 1985; Futuyma & Kirkpatrick, 2017). Clines provide insights into how the contrasting forces of genetic drift and spatially varying selection on the one hand, and gene flow on the other, shape populations' genetic and phenotypic diversity. While genetic drift and, most importantly, spatially varying selection, favor the genetic differentiation of populations, gene flow prevents these populations from getting genetically isolated, thus hindering the speciation process and conserving the species' unity (Futuyma & Kirkpatrick, 2017). An interesting type of clines is color clines, or clines that affect the chromatic properties of the integument of an organism (Antoniazza *et al.*, 2010; McLean & Stuart-Fox, 2014; Köhler, Samietz & Schielzeth, 2017; Gefaell *et al.*, 2024). Color clines can either result from a gradual change in the value of a quantitative chromatic variable, or a progressive variation in the relative frequencies of a series of discrete morphs (i.e., a morph ratio cline *sensu* Wunderle, 1981). Both kinds of color clines have been described in most animal groups, from birds and mammals (Sumner, 1926; Mullen & Hoekstra, 2008; Antoniazza *et al.*, 2010; Cosentino, Vanek, & Gibbs, 2023) to insects and mollusks (Saccheri *et al.*, 2008; Brakefield & de Jong, 2011; Gefaell, Galindo & Rolán-Alvarez, 2023).

The color of an animal can serve at least three different purposes: thermoregulation, communication with conspecifics, or predation avoidance (Endler, 1978). This suggests that color clines might result from gradual changes in the thermal or chromatic profiles of localities along a geographic transect, such that these changes generate differential selective pressures that ultimately drive the evolution and maintenance of the color cline. Color clines driven by thermal variation have been described in many species, corroborating what is usually referred to as the thermal melanism hypothesis (Clusella-Trullas, van Wyk & Spotila, 2007). Likewise, a study on lizards suggests that selection for increased detectability by mating partners can sometimes induce the evolution of a color cline (McLean, Stuart-Fox & Moussalli, 2015). Finally, differences in the chromatic properties of substrates along a geographic transect have also been shown to favor the establishment of cryptic color clines in prey species attacked by visual predators (Sumner, 1926; Mullen & Hoekstra, 2008).

Focusing on this last case, color clines putatively maintained by selection for crypsis can be studied through different methodologies (see Endler, 1986). Among these, frequentist-observational studies have traditionally been the most used, especially in mollusks (for a review in marine gastropods, see Gefaell *et al.*, 2023). In this approach, the frequencies of color morphs along a gradient are analyzed and compared to the different substrates on which they are found.

If the dominant morph at a given locality shows a chromatic concordance with its habitat, it is concluded that selection for crypsis is most likely driving the maintenance of the cline.

This frequentist-observational approach has provided suggestive evidence for the role of crypsis in the maintenance of many marine gastropod color clines. However, its inferential power, that is, its capacity to generate robust causal inferences on the mechanisms at work, is limited (Endler, 1986). For this reason, when feasible, either lab or field experiments are demanded to test whether a particular visual predator acting under a chromatically diverse transect can indeed bring about the evolution of a color cline in a prey (gastropod) species. Both in mollusks and other taxa, there are some examples of successful laboratory and, especially, field experiments aimed at testing the selective importance of cryptic coloration (Ekendahl, 1998; Webster *et al.*, 2008; Vignieri, Larson & Hoekstra, 2010; de Alcantara Viana *et al.*, 2023). Nonetheless, field experiments, despite their high ecological validity, are particularly difficult to conduct due to the unpredictability and lack of control over extraneous variables inherent in working in the wild. Additionally, these experiments rely on numerous tacit assumptions about the natural history and ecology of the species involved. While evolutionary ecologists devote significant efforts to corroborating some of these assumptions (Lloyd, 1987), many remain untested. This means that if any of these untested assumptions do not hold, the field experiment can fail.

A marine gastropod widely used in evolutionary ecology studies is *Littorina saxatilis* (Olivi, 1872) (Rolán-Alvarez, Boulding & Austin, 2015; Johannesson, 2016). *L. saxatilis* inhabits the supralittoral zone of intertidal ecosystems along the northern coasts of the Atlantic Ocean, occupying the ecological niche of a microalgae-grazing feeder (Reid, 1996). Partly due to its ovoviviparous breeding system, *L. saxatilis* is particularly prone to evolving local adaptations (Rolán-Alvarez, 2007; Butlin *et al.*, 2014; Rolán-Alvarez *et al.*, 2015). This mode of reproduction curbs its dispersal capabilities, thereby significantly limiting the erosion of genetic adaptations to specific environments that gene flow can produce (i.e., gene swamping; Futuyma & Kirkpatrick, 2017). Plus, the diversity of intertidal ecosystems in terms of species and substrate composition generates diverging selective pressures to which local populations of *L. saxatilis* adapt (Rolán-Alvarez, 2007).

Among the many traits exhibiting variation in *L. saxatilis*, shell color stands out (Johannesson & Butlin, 2017; Gefaell *et al.*, 2023a, 2024). This species presents both intrapopulation and geographic color variation, the latter usually taking the form of morph ratio clines (Sacchi, 1979; Reid, 1996; Gefaell *et al.*, 2024). An area where these clines are particularly noticeable is the Rías Baixas, a series of four coastal inlets that originated through the immersion

of four river valleys following the Holocene transgression (Arce-Chamorro, Vidal-Romaní & Sanjurjo-Sánchez, 2022). The Rías Baixas show a nearly continuous wave-exposed gradient from their inner parts (less wave-exposed) to their outer regions (more wave-exposed) (Gefaell *et al.*, 2024). This gradient is accompanied by a concomitant gradual progression in the species composition of their intertidal ecosystems (with different algae and sessile and mobile animals), as well as a steady change in the frequencies of the two main *L. saxatilis* color morphs living there: a fawn-yellowish morph that abounds in the sheltered zones, and a lineated morph (*lineata*) whose incidence grows as we approach the most exposed parts, until eventually reaching monomorphism. At intermediate localities, high color diversity (white, yellow, orange, black, etc.) and less predictable patterns usually occur (Sacchi, 1979; Reid, 1996; Gefaell *et al.*, 2024).

It has been recently shown that the morph ratio cline of *L. saxatilis* from the Rías Baixas has remained relatively stable over the last forty years in at least one of these four Rías, the Ría of Vigo, the southernmost and most studied (Gefaell *et al.*, 2024). The temporal stability of this cline, in addition to its spatial replication in the four Rías (Reid, 1996), suggests that it is being maintained by spatially varying selection (Endler, 1986). The specific selective pressures at work, though, are not yet known. Of the three functions color might have, mate recognition and conspicuousness are highly unlikely due to the lack of color vision in *L. saxatilis* (see Gefaell *et al*. 2024b for an experimental exploration of sexual selection on *L. saxatilis* color morphs). Thermal stress, another possibility, should be studied independently due to its physiological complexity and will be the focus of an upcoming work.

The remaining candidate, therefore, is selection for crypsis, which has been described in many marine gastropods (Gefaell *et al*. 2023 and references therein). An informal observation of many *L. saxatilis* color morphs in their natural environments already suggests crypsis (JG, personal observation). Furthermore, most of *L. saxatilis*' predators (Pettitt, 1975), particularly shorebirds, crabs, and intertidal percomorph fishes (such as blennies and gobiids), rely on vision to varying extents when hunting for prey. If the chromatic properties of localities along the Rías Baixas are heterogeneous due to the different ecological communities they contain, this can favor the evolution of a cryptic color cline in a prey species such as *L. saxatilis*. This selective pressure may be especially intense in areas exhibiting monomorphic populations, such as those of the *lineata* morph in wave-exposed localities, where it can presumably weed out all individuals not displaying the most cryptic coloration. The *lineata* morph, in contrast, might be highly cryptic against the barnacle-covered and bare alkaline granitoid rocks characteristic of these localities, thus more efficiently avoiding visual predators.

To test this hypothesis, we conducted a tethering experiment in a representative *lineata* monomorphic wave-exposed locality from the Ría of Vigo (Cabo Silleiro) using three different transplanted (i.e., allochthonous) *L. saxatilis* morphs of varying degrees of presumed crypsis (yellow, black, and *lineata*) from a nearby polymorphic population (Aguncheiro). To these, *lineata* individuals from the experiment's locality (resident –i.e., autochthonous– and supposedly highly cryptic) were added to account for transplantation effects. Snails from all these morphs were tethered at points of seemingly high marbled rock crab (*Pachygrapsus marmoratus*) density to record successful or failed predation attempts by this species, as revealed by the resulting marks on the snails' shells. If cryptic selection is at work, we would expect increased predation of the less cryptic morphs (yellow and black, especially the former) compared to the *lineata* snails (particularly the resident ones). This would help explain why wave-exposed populations of *L. saxatilis* tend to be monomorphic for *lineata*, thus providing a partial account of the color cline observed in the Rías Baixas. As a complementary study on the role of selection for crypsis, the frequencies of scars (reflecting past unsuccessful crab attacks) on a sample of the morphs from the source locality (Aguncheiro) were counted, and a series of key shell traits were measured. Particularly, the degree of crypsis of the different morphs used in the experiment was assessed using reflectance spectrometry.

Reflectance spectrometry corroborated that *lineata* snails (particularly residents) are more cryptic than the black and yellow morphs in the experiment's locality (Cabo Silleiro). However, the pattern of shell scars on the different morphs at their source locality (Aguncheiro) did not yield clear support for the selection for crypsis hypothesis. In the first session of the tethering experiment, there was significant evidence of differential predation between color morphs, with the supposedly less cryptic morph (yellow) experiencing more predation. However, in the second session of the experiment, there was no significant evidence, but power was reduced due to low rates of predation overall. These results provide some limited support for selection for crypsis as a contributing mechanism to the maintenance of the *L. saxatilis* color cline in the Rías Baixas. Nevertheless, they also highlight the challenges of using field approaches to study predation and adaptation.

Materials and methods

Localities, predator species, and morphs studied

The tethering experiment was carried out in the Cabo Silleiro locality (Oia, Galicia, Spain; 42.10467, -8.89885; see Fig. 1A). This constitutes a 200 meter-long and 80 meter-wide dark brown alkaline granitoid bedrock with small rocks on it that experiences intense swell throughout

the year (Fig. 1D). The lower parts of the bedrock surface are fully covered with barnacles (*Semibalanus balanoides* and *Chthamalus* spp.), with this barnacle coverage gradually receding as we approach its upper part. In terms of color, and excluding the low shore *Wave* ecotype, the population of *L. saxatilis* inhabiting Cabo Silleiro is monomorphic for *lineata*, with snails showing a white to creamish background with superimposing black lines or stripes that run perpendicular to the lines of shell growth (Fig. 1B.1). This morph, which in wave-exposed localities can be equated to the *Crab* ecotype in the *L. saxatilis* ecotype literature, is presumably cryptic in dark alkaline granitoid rocks (with or without high barnacle coverages), but this hypothesis has not been previously tested. The *L. saxatilis* population from Cabo Silleiro is relatively dense, and previous investigations (e.g., Boulding *et al.*, 2017) suggest that it is subjected to intense crab predation.

One main agent responsible for such predation is the marble rock crab (*P. marmoratus*; Fig. 1C), present at high densities in Cabo Silleiro, where they are especially abundant in the upper part of the ecosystem. These crabs constitute one of the main selective pressures driving the phenotypic and genetic differentiation of the *Crab* and *Wave* ecotypes of *L. saxatilis* in the Galician coasts and have thereby been the subject of numerous investigations over the last few decades (e.g., Rolán-Alvarez, 2007). However, their potential role in maintaining the *lineata* color monomorphism seen in the wave-exposed localities of the Rias Baixas through the eradication of less cryptic color morphs of *L. saxatilis* has not yet been assessed.

A crucial question is whether *P. marmoratus* can see colors, something for which there is no direct evidence. Nevertheless, neurophysiological and behavioral studies on other brachyuran species suggest that this group of animals has a chromatic vision. For example, both the blue crab (*Callinectes sapidus*) and the green shore crab (*Carcinus maenas*) have two photoreceptors, one with a peak sensitivity at 508 nm (green), and another with a peak sensitivity at 440 nm (blue; only in the ventral portion of the eye) (Martin & Mote, 1982; Baldwin & Johnsen, 2012). This implies that these crabs can detect chromatic signals through the opponency of the green and blue photoreceptors. This has been confirmed in behavioral experiments, which show that at least *C. sapidus*, as well as other crab species, can use visual information to carry out decisions (Detto, 2007; Baldwin & Johnsen, 2009, 2012; Kawamura *et al.*, 2020). Furthermore, molecular evidence in the sand fiddler crab (*Uca pugilator*) suggests that this species could even be trichromat, as its members have three opsin-encoding genes that appear to have different wavelength sensitivities (Rajkumar *et al.*, 2010). Taken together, this information makes it likely that *P. marmoratus* can see colors. But even if they don't, or if this ability is of little use when hunting because they tend to be more active at night (as has been shown in Portuguese

populations of this species; Cannicci, Paula, & Vannini, 1999), *P. marmoratus* could still respond to achromatic cues based on the signal intensity of its most abundant (or only) photoreceptor. Extrapolated to the present experiment, a plausible assumption is, therefore, that *P. marmoratus* can distinguish between snail color morphs based on either chromatic (via opponency) or achromatic (intensity) signals, or both.

For the experiment, different color morphs of *L. saxatilis* present at a nearby locality, Aguncheiro (Oia, Galicia; 42.050963, -8.885492; Fig. 1A), were used. This locality, right at the mouth of the Mougás River (around 6 km from Cabo Silleiro), is composed of small to medium (~1 meter in diameter) yellowish alkaline granitoid rocks with black-reddish patches showing a full barnacle coverage in its lower parts that decreases to around 30-50% in the upper areas (see Supplementary materials Fig. S1A). Despite being surrounded by highly wave-exposed areas, the population of *L. saxatilis* from this locality is made up of four color morphs: white or creamish with superimposed lines (*lineata*), black or very dark brown (*nigra*), yellow (*lutea*), and white (*albida*) (see Fig. 1B.2 for the first three morphs; all the color morph nomenclatures in brackets – used henceforth– are from Sacchi, 1979). Of the four color morphs present at Aguncheiro, three of them were used in the experiment: *lineata*, *nigra*, and *lutea*. *Albida* was discarded because of its low frequency. To these, a subsample of *lineata* from Cabo Silleiro (resident, i.e., autochthonous, in the locality in which the experiment was carried out) was also used to account for the effects of transplantation and autochthony of the snails.

Tethering experiment design and procedure

The differential predation of snail morphs by crabs was tested with a tethering experiment. The tethering paradigm consists of attaching snails to nylon threads using a water-resistant instant epoxy glue, which are in turn tethered to screws that had been previously installed at spots with high crab predation (one snail per screw). After an (ideally preestablished) amount of time, the snails tethered to the screws are then recaptured and the shell marks indicating successful or unsuccessful crab predation attempts are counted. Despite their shortcomings (such as potential inflation of mortality rates in predator-outrunning prey; Barbeau & Scheibling, 1994; Zimmer-Faust *et al.*, 1994; Peterson & Black, 1994; Baker & Waltham, 2020), tethering experiments are one of the few experimental designs that allow for measuring predation rates of Littorinids and other marine organisms in the wild (Aronson & Heck, 1995; Yamada & Boulding, 1996; Boulding, Holst & Pilon, 1999; Boulding *et al.*, 2017).

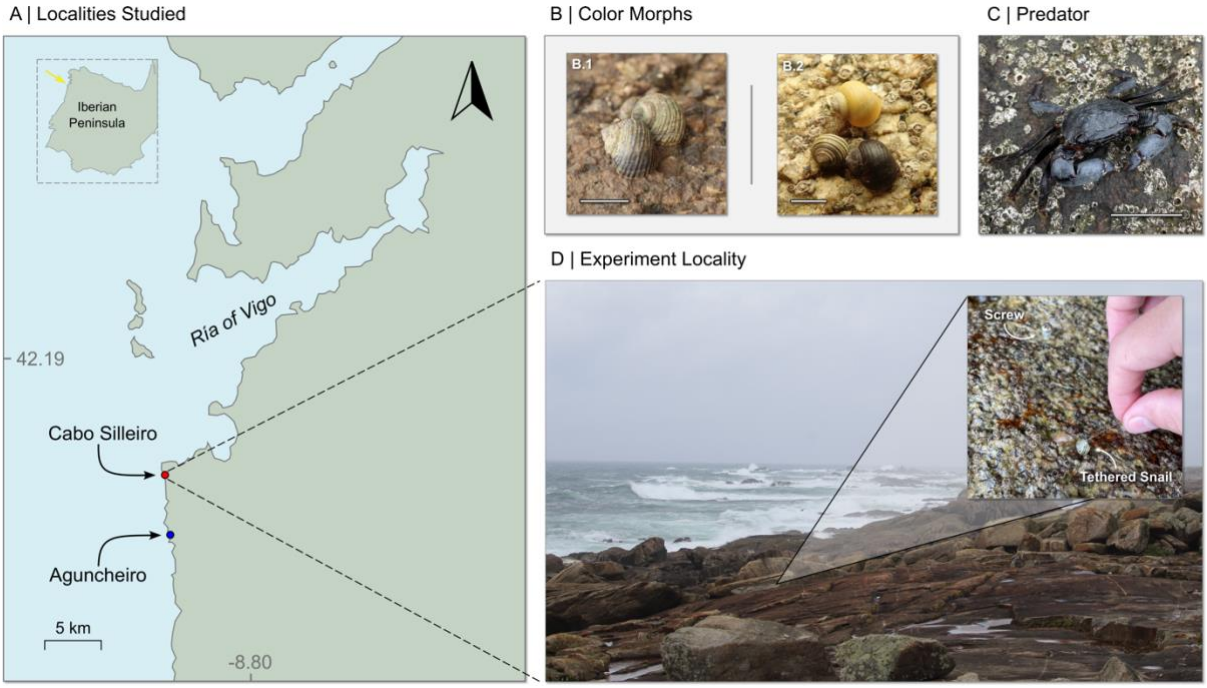

Figure 1. Methodology of the tethering experiment. A. Localities studied within the context of the Ría of Vigo. B. Color morphs of *L. saxatilis* used in the tethering experiment, as well as in the shell measurements carried out for the present study (thickness, geometric morphometrics, and reflectance spectrometry). B.1 shows the autochthonous *lineata* (from Cabo Silleiro), and B.2 the *lineata* (allochthonous), *nigra*, and *lutea* morphs from Aguncheiro. C. The main predator species studied, *P. marmoratus*. D. Cabo Silleiro locality –where the experiment took place– with a *lineata* allochthonous (from Aguncheiro) snail tethered to a screw in the inset. Scale bars: B.1 = 7.0 mm; B.2 = 7.0 mm; C = 4.5 cm.

A total of 284 snails, 71 per morph (*lineata* autochthonous and transplanted –i.e., allochthonous, *nigra*, and *lutea*) were collected in two batches for the two sessions that comprised the experiment. The samplings were carried out on two non-consecutive days in September and October 2022, on the days before each experiment session. The size of the snails was standardized as much as possible during the sampling sessions, collecting only intermediate snails in a range of less than 0.5 mm in shell length (7-7.5 mm from apex to the tip of the last whorl). Smaller snails (like many utilized by Boulding *et al.*, 2017) were discarded as the glue employed to tether them, despite being transparent, would otherwise cover most of the shell, potentially masking their original color. The integrity of the shell of the snails collected was also ensured. Once picked up, the snails were taken for processing to the facilities of the University of Vigo (25-30 km from the localities). Transportation was done in plastic boxes with rocks from their respective locations to ensure the well-being of the snails. All the specimens were handled and prepared alive for the experiment (see Supplementary materials Fig. S1B) and stored for 16 hours in a fridge until the experiment sessions began.

A total of eleven transects of either 12 or 28 screws (with at least 1 meter of separation between screws), each situated at different spots of the Cabo Silleiro locality, were used in the tethering experiment (exact locations of transects available at:

https://www.google.com/maps/d/edit?mid=1CFlEfOUPsXpst7W7tYs3xFD2RDuAEbA&usp=sharing). Of these, eight transects were specifically installed for this experiment (see Supplementary materials Fig. 1C), while the three remaining ones were reused from the tethering experiment described in Boulding *et al*. (2017) (Supplementary materials Fig. S1D). All transects were installed near crevices where *P. marmoratus* had been previously spotted, along the continuum from the mid-lower to mid-upper part of the ecosystem (corresponding to what has been found in *P. marmoratus* from southern Portugal, which is mostly present at the eulittoral and littoral fringes; Cannicci *et al.*, 1999).

The snails used in the experiment were tethered alive to the screws of each transect (inset on Fig. 1D and Supplementary materials Fig. S1E). Most of the resident snails present in the nearby surroundings to the screws were allocated outside the transects to increase the chance of the tethered snails being preyed upon. The order of the color morphs was semi-random, always maintaining the *lineata* autochthonous and allochthonous separated by either one *nigra* or *lutea* snail to avoid sensory biases for the crabs but allowing the order of these last two color morphs to vary randomly. The length of the nylon threads used was 30 cm, so the inflated mortality bias of tethering experiments –according to which mortality rates are inflated because the prey cannot outrun or outmaneuver their predators– was minimized. However, Littorinid snails are apparently not affected by this bias, as no effect was found of tether length (3 vs. 30 cm) on predation rates in two species of the genus *Littorina* (Rochette & Dill, 2000). Immediately after tethering each snail, its shell length was measured to the nearest 0.1 mm using a vernier caliper to add a statistical control to the procedural standardization of this variable carried out during the samplings (Supplementary materials Fig. S1F).

The experiment was divided into two sessions conducted between September and October 2022. The season of the experiment was chosen following Boulding *et al*. (2017), who obtained a relatively high percentage of predation of *L. saxatilis* by *P. marmoratus* in a tethering experiment carried out during the Autumn at the same locality. Based on a pilot experiment in which the three Aguncheiro morphs (*lineata* allochthonous, *nigra*, and *lutea*) were used, it was predetermined that the experiment would last four days; due to bad weather and lack of predation during the first days, however, the first session had to be extended one more week.

Once each of the experiment sessions was over, the outcome of the tethered snails was ranked *in situ* using a series of pre-established operational categories based on Boulding *et al*. (2017) (see Supplementary materials Table S1). The categories raking the various outcomes covered the full range of possibilities for the snails. Of these, six categories were the most relevant: ‘Alive Intact’ (alive and without any marks), ‘Preyed Alive’ (alive with predation mark),

'Preyed Dead' (dead with predation mark), 'Chipped' (the outer lip of the shell showing small or smooth –yet clearly noticeable– breaks, but of a magnitude that cannot be readily attributed to a predation attempt by a normal-sized crab –i.e., of 3-4 cm in caparace length), 'Empty Shell' (shell intact without snail, possibly due to predation by any animal that can remove the snail without breaking the shell, like a percomorph fish), and 'Epoxy Only' (only the tethering line with the epoxy glue found –i.e., snail detached) (Supplementary materials Fig. S2A-E, respectively except for 'Empty Shell').

Once the outcome of the snails was assessed, all the alive specimens (including 'Preyed Alive') were untethered from the screws, their shells cleaned up from epoxy glue remainings and carried alive in plastic boxes with rocks to their original localities (Aguncheiro in the case of *lineata* allochthonous, *nigra*, and *lutea*, and the exact spots of Cabo Silleiro where they were originally picked up in the case of *lineata* autochthonous). All the tethering lines were also removed from the screws.

Frequency of scars and shell measurements of the studied color morphs

Samplings of the four morphs inhabiting Aguncheiro (*lineata*, *nigra*, *lutea*, and *albida*), as well as of *lineata* from Cabo Silleiro were also carried out for the present study after the tethering experiment was carried out but previous to its formal analysis (March 2024). These samplings were aimed at independently exploring the predation intensity of each locality, as well as the susceptibility of the different morphs to predation. This was done by counting the number of individuals of each morph at both localities with at least one shell scar. Shell scars, defined as deformations of the line of growth below which new shell material grows, can provide clues on predator abundance in a locality (Butlin *et al.*, 2014; Stafford, Tyler & Leighton, 2015). Additionally, if systematic biases exist in the frequency of scars between a series of coexisting color morphs, then these scars can deliver information on differential predation rates, and therefore fitness, of the different morphs. The sampling procedure involved selecting a series of random patches in the intertidal zone where *L. saxatilis* settles, using a 60 cm diameter sampling circle to demarcate each area, and collecting all the snails within the circle (i.e., a stratified random sampling procedure). Once picked up, the snails were *in situ* color-classified, and possible shell scars on any of their whorls were inspected. After the sampling finished, the snails were immediately returned to their initial spots to not compromise the population's viability.

Additionally, several snails representative of each color morph used in the experiment (*lineata* autochthonous and allochthonous, *nigra*, and *lutea*; $n = 40$, 10 per morph) were collected from Cabo Silleiro and Aguncheiro and taken to the lab to measure their shell thickness, shape,

and reflectance spectrometry. These measurements aimed to rule out possible differences between color morphs in thickness and shape that could account for potential differences in predation propensity, as well as to formally assess their degree of crypsis. The specimens picked up were size-standardized as much as possible, choosing individuals between 7 and 7.5 mm in length to replicate the sampling process of the tethering experiment. These exemplars were euthanatized using cold temperature (i.e., cold anesthesia) by putting them into the freezer upon arrival at the University of Vigo facilities. Shell shape was analyzed using two-dimensional geometric morphometrics based on photographs of each individual (see Supplementary materials for an extended description of the geometric morphometrics protocol followed). Shell thickness was measured at the lip of the shell (analogous position in *L. saxatilis* to the one measured in the girdled dogwhelk in Martin, Clusella-Trullas & Robinson, 2022) to the nearest 0.01 mm using an electronic gauge. Previous evidence suggests that there are no differences in shell thickness in *L. saxatilis* between color morphs of a given population (Raffaelli, 1979). However, as in the present experiment samples from two populations were used (Cabo Silleiro: *lineata* autochthonous; Aguncheiro: *lineata* allochthonous, *nigra*, and *lutea*), measurements of this variable were collected to formally evaluate such a possibility.

Lastly, the reflectance spectra of these very same specimens, as well as of random points of two representative rocks from each locality (Cabo Silleiro and Aguncheiro) were recorded to assess the degree of crypsis shown by each of the morphs in their autochthonous and allochthonous habitats. These measurements were primarily used to verify a tacit assumption of our experiment, namely, that in the wave-exposed localities of the Rias Baixas (such as Cabo Silleiro, where the experiment was carried out), the degree of crypsis of the different color morphs of *L. saxatilis* is *lutea* < *nigra* < *lineata*, so that crabs would probably detect and prey *lutea* more than *nigra* and *lineata* (both allochthonous and autochthonous). Additionally, the crypsis of the Aguncheiro morphs against their original background was also assessed to aid in the interpretation of shell scars in this locality. The reflectance spectra of the shells and the rocks were obtained with an OceanOptics DH-mini, UV-VIS-NIR Lightsource reflectance spectrometer (Ocean Optics Inc., Dunedin, FL, USA), calibrated with a white diffuse reflectance standard and measured at a standardized distance of 3 mm under constant lighting conditions. The relative angle of the measured objects to the probe was 90º. The snails were measured in the central area of the main whorl, while the rocks in randomly selected points (three measures per rock) to avoid selection biases and because classic definitions of crypsis involve concealment against a random sample of the background substrate (Ender, 1978). Integration time was 2800 ms, and the spectra were averaged to 6 scans, with a boxcar width of

10. The resulting reflectance spectra were processed with the R package 'pavo' (Maia *et al.*, 2013). The spectra acquired were smoothed fitting a LOESS regression (span = 0.7), and negative values were corrected by converting them to 0.

Statistical analyses

All the statistical analyses were carried out in R version 4.3.3 (R Core Team, 2023). The frequencies of shell scars in the color morphs from the source locality (Aguncheiro) were studied using independence chi-square and Fisher's exact tests (Sokal & Rohlf, 1995). Chi-square was preferentially used; if the contingency table contained cells with relatively low expected values (~5), which can produce unreliable results, Yates' continuity correction was applied by default to the chi-square test to reduce the chance of a false positive (but see Sokal & Rohlf, 1995). For 2 × 2 or larger tables with very low expected values (well below 5), Fisher's exact test was used instead. Fisher's exact test is typically used for 2 × 2 contingency tables with small sample sizes but can be applied to larger tables as well, estimating the resulting *P*-value with Monte Carlo simulations to speed up the computation process. Monte Carlo simulations generate data under the null hypothesis of independence and compare the observed test statistic to the distribution of test statistics generated via the simulations (Poldrack, 2023).

The predictive power of several potential explanatory variables on the main outcomes of the experiment was explored by fitting generalized linear models with binomial error structure and logit link function (i.e., logistic regressions; Sokal & Rohlf, 1995). For these models, dichotomous variables were created for each main outcome (e.g., variable 'Chipped': 1 = Chipped, 0 = Not Chipped –the remaining outcomes). A 'Preyed' outcome variable was obtained from grouping the 'Preyed Alive' and 'Preyed Dead' outcomes, as both represent instances of crab predation irrespective of whether the snail survived the attack. An additive category of 'Preyed + Chipped' was created by granting both 'Preyed' and 'Chipped' outcomes '1' and the remaining outcomes '0'. This variable assumes the hypothetical scenario that 'Chipped' represents a predation attempt –probably by a small crab. For the different models, cases whose original outcome was 'Epoxy Only' or 'Lost' were considered NAs and excluded from the analyses, since we don't know whether these snails were preyed upon or not due to their loss (thus avoiding the 'missing as predation' bias in tethering experiments; Baker & Waltham, 2020). As for the predictors, these included the color and source locality of the snails (fused into a single variable so as not to overfit the models), their shell length, the position of the transect along the intertidal ecosystem ('Lower' vs. 'Upper' transects, based on the distance to the ocean and barnacle coverage), and finally, the session of the experiment, as the weather was different

between the two sessions and this could have influenced the predatory performance of *P. marmoratus*. The color morph variable was converted to a factor, with *lineata* from Cabo Silleiro (autochthonous) used as a reference level to assess the impact of the remaining color morphs on the different outcomes. All the models were fitted using maximum likelihood estimation, and convergence was studied using the default Fisher Scoring Iterations. The fit of the different models was evaluated with the Akaike Information Criterion (AIC). The significance of the potential predictors was studied through *Z*- and *P*-values (significance threshold, $P < 0.05$). Statistical power in the logistic regression models was assessed by calculating the odds ratio for each predictor or level, as this measure is commonly used to evaluate effect size for binary variables (Poldrack, 2023). All the models were fitted using the 'glm' function from the 'stats' package. The results from the logistic regressions were further explored through various chi-square and Fisher's exact tests following the same logic described above.

Ethical standards

Since *L. saxatilis* is not a protected species and shows relatively dense populations all along the Rias Baixas, no special permissions were required for the present experiment. Nevertheless, we acted in compliance with the ASAB guidelines when handling the snails (ASAB Ethical Committee/ABS Animal Care Committee, 2023). The experiment was deliberately planned to return all the used specimens to their original localities or spots after the sessions, and due to the relatively low predation rate we obtained, most of the snails survived the experiment. The samplings of Aguncheiro and Cabo Silleiro aimed at determining scar frequencies were explicitly designed so as not to compromise the viability of the populations. Finally, the sample of snails collected for the shell measurements –which entails their death– was the lowest possible to ensure a minimum of statistical power.

Results

Frequencies of scars at the source and experiment localities (Aguncheiro and Cabo Silleiro)

A total of 89 (Aguncheiro) and 123 (Cabo Silleiro) snails were collected in six random samplings. Cabo Silleiro was, as expected, monomorphic for *lineata*, while Aguncheiro showed the following frequencies: *lineata* (49.44%) > *nigra* (35.96%) > *lutea* (12.36%) > *albida* (2.25%) (see Table 1). In Aguncheiro, the frequency counts of each morph (*albida*, *lutea*, *nigra*, and *lineata*) were homogeneous between the six samplings ($P = 0.493$, Fisher's exact test with *P*-value based on 2000 Monte Carlo simulations; henceforth used always in Fisher's exact tests),

so can be grouped for further analyses. As for predation intensity, indirectly assessed through the frequency of snails with shell scars, both localities have very similar values (Aguncheiro = 24.7%, Cabo Silleiro = 27.6%; χ^2 on count data = 0.10, df = 1, P = 0.750).

Table 1. Frequencies of morphs and scars, as well as their corresponding percentages, in Aguncheiro (the locality where most of the snails for the tethering experiment were collected), and Cabo Silleiro (where the experiment was conducted).

Locality	Color Morph	Morph Counts	Scar Counts	Relative Incidence of Scars
Aguncheiro	*Lineata*	44 (49.44%)	9 (20.45%)	40.9%
	Nigra	32 (35.96%)	6 (18.75%)	27.27%
	Lutea	11 (12.36%)	5 (45.45%)	22.73%
	Albida	2 (2.25%)	2 (100%)	9.09%
	Sample size (*n*)	89	22 (24.72%)	–
Cabo Silleiro	*Lineata*	123 (100%)	34 (27.64%)	–

In Aguncheiro, the percentages of shell scars follow an almost exact reverse pattern than the frequencies of color morphs: *albida* (100%) > *lutea* (45.45%) > *lineata* (20.45%) > *nigra* (18.75%). Figure 2 shows the ratio of the relative incidence of scars of a morph (i.e., of the total snails with scars, how many of them belong to a given morph) to its relative frequency in the population, excluding *albida*, whose sample size (n = 2) is too low to draw robust inferences. Values around 1 indicate that the number of scars in a given morph is the one that would be expected by its frequency. We see this in both *lineata* and *nigra*, with values slightly lower than 1. Conversely, values significantly over 1 imply a higher rate of scars than that expected by its frequency, as seen in *lutea*. The difference in the frequency of scar counts between color morphs was statistically significant (Fisher's exact test, P = 0.031). However, if we exclude *albida*, which has a very low frequency of snails, the test is not significant (Fisher's exact test, P = 0.216).

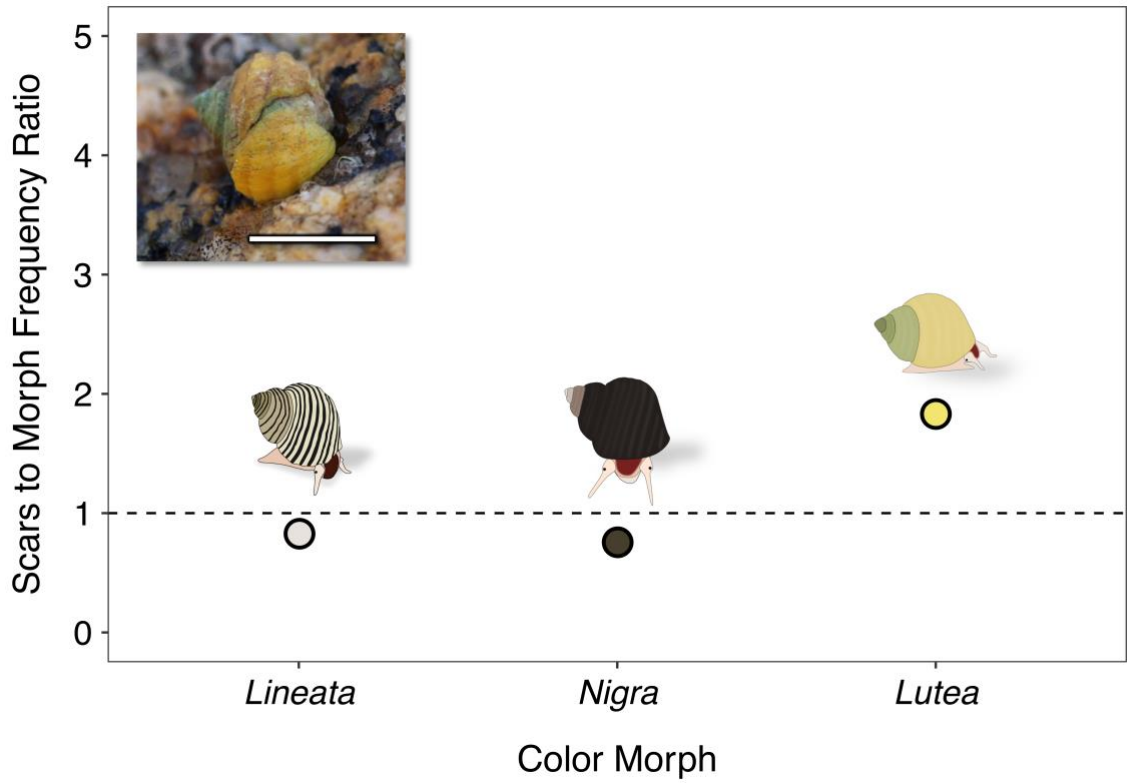

Figure 2. Scars to morph frequency ratio for each color morph of Aguncheiro (excluding *albida*, which was not used in the tethering experiment and had a low sample size in the samplings). Values well above 1 indicate a higher incidence of scars than the one expected by the frequency of the morph in the population alone. The difference in scars between these morphs, however, is not statistically significant. The inset picture shows a *lutea* snail with a shell scar. Scale bar in the picture: 7.0 mm.

Shell measurements of the color morphs

No differences between the color morphs used in the experiment (*lineata* autochthonous and allochthonous, *nigra*, and *lutea*) were found in shell shape, assessed using RW1 (ANOVA, $F(3) = 0.76$, $P = 0.524$) (see Supplementary materials Figs. S3 and S4). Conversely, a Kruskal-Wallis test for shell thickness between color morphs yielded near statistically significant values ($H(3) = 7.31$, $P = 0.062$), with *lutea* showing slightly thicker shells (see Supplementary materials Fig. S5). Effect size, measured through a non-parametric Eta-squared for Kruskal-Wallis (η^2_H), was calculated to assess whether the previous result could be due to the low sample size used ($n = 40$), which can lead to a false negative (Bateson & Martin, 2021). This test yielded a moderate effect ($\eta^2_H = 0.12$), suggesting that there might be real differences in shell thickness between the color morphs.

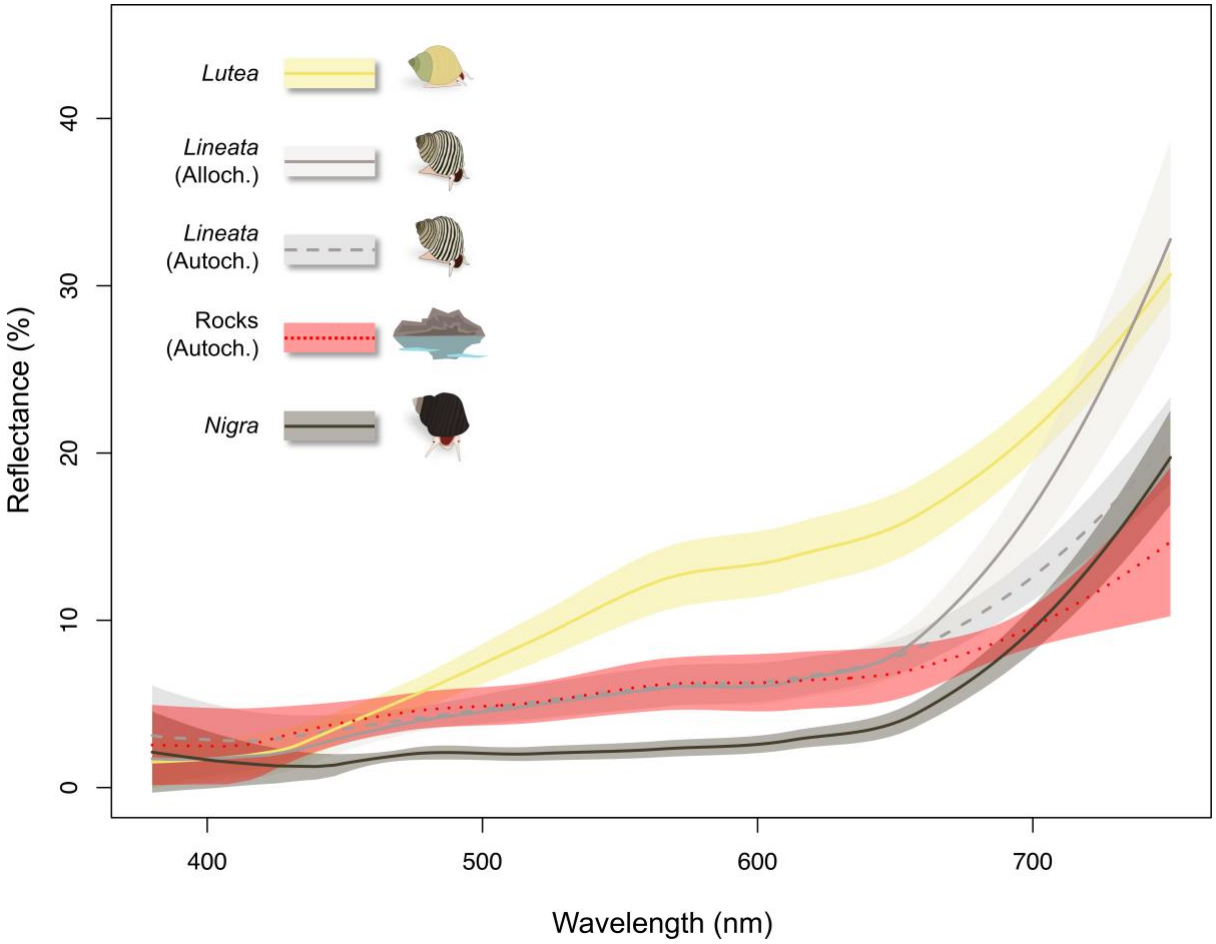

Figure 3. Spectra of objects studied in the experiment. Comparison of the spectra of a sample of snails from the same color morphs as those studied in the experiment (*lineata* autochthonous and allochthonous, *nigra*, and *lutea*) and a random sample of rocks from Cabo Silleiro, 'Rocks (autoch.)'. Each line represents the mean of all measured objects within a class, and the shaded area their corresponding 95% confidence intervals.

The comparison between the spectra of the rocks and snails from Aguncheiro suggests that in this locality, *lutea* and *lineata* might be more cryptic than *nigra*, as inferred from the overall shape of the spectra lines and their relative reflectance intensities (see Supplementary materials Fig. S6). At the locality of the experiment (Cabo Silleiro), however, *lutea* is much more conspicuous than *nigra* and *lineata* (both allochthonous and autochthonous, especially the latter) (see Fig. 3). The *lineata* morphs (both allochthonous and autochthonous) are the ones that overlap the most with the rocks (from around 380 nm up to 650 nm, the point at which the spectrums start to diverge, with *lineata* autochthonous diverging less dramatically). The *lutea* morph stands above the spectrum of the rocks at most wavelengths (higher intensity; thus being brighter), while that of *nigra* stands below it (i.e., less intensity; darker). The distance from these spectra to that of the rocks, however, is apparently less dramatic in *nigra* than in *lutea*.

Tethering experiment

Table 2 shows a summary of the characteristics and outcomes of the snails used in the two sessions and the whole experiment (Session 1 and 2 pooled). In the whole experiment, all but 27 snails were recovered and ranked, with the remaining being lost due to unknown reasons or bearing a highly ambiguous outcome (original NAs = 4). The most frequent outcome in the experiment was 'Alive', which groups all snails that completed the experiment alive and with their shells undamaged ('Alive Intact' + 'Recovered Alive Intact' pooled). The frequencies of the remaining outcomes were comparatively low. As for the 'Preyed' outcome, only 9% of snails in Session 1 ([10/111] × 100) and 6.61% in Session 2 ([8/121] × 100; in both cases excluding 'Lost (All)' cases, which group 'Epoxy Only', 'Lost', and NAs) were attacked by crabs (results pooled = 7%).

The outcomes were different in each session of the experiment (Fisher's exact test, P = 0.029; see also Figure 4). Pairwise comparisons revealed that irrespective of the morph, there were more 'Chipped' (χ^2 = 6.49, df = 1, P = 0.011) snails in Session 1 than in Session 2, but no differences in 'Preyed' (χ^2 = 0.73, df = 1, P = 0.395) or 'Empty Shell' (Fisher's exact test, P = 1). As for the added results of 'Preyed' and 'Chipped' (i.e., 'Preyed + Chipped'), a higher count was also observed in Session 1 than in Session 2 (χ^2 = 7.28, df = 1, P = 0.007). In addition to outcomes, the shell length of the snails involved in the experiment varied between sessions (Session 1 = 7.41 ± 0.62 mm, Session 2 = 7.19 ± 0.64; Wilcoxon Rank Sum Test, W = 11844, P = 0.009). Within each session, statistically significant differences in size existed as well between the color morphs (see Supplementary materials Fig. S7). Shell length was always higher in the *lutea* and *nigra* snails than in both *lineata* autochthonous and allochthonous.

Table 2. Main results of the tethering experiment. It shows the sample used (*n*), shell length (mean ± SD in mm), and main outcomes of the color morphs used in each experiment session. '*Lineata* (Autoch.)' is the resident snail from Cabo Silleiro, and '*Lineata* (Alloch.)' is the transplanted *lineata* from Aguncheiro. The outcome 'Alive' was obtained by summing the 'Alive Intact' and 'Recovered Alive Intact' outcomes. 'Preyed' resulted from adding 'Preyed Alive' and 'Preyed Dead' (both indicative of crab predation attempts). 'Lost (All)' groups those outcomes in which the snail was lost or not recovered at the end of the experiment. Adding the results of 'Preyed' and 'Chipped' a new category 'Preyed + Chipped' was created for the logistic regression analyses (see the main text for further details).

				Outcome				
Session	Color Morph	*n*	Shell Length	Alive	Preyed	Chipped	Empty Shell	Lost (All)
Session 1	*Lineata* (Autoch.)	33	7.15 ± 0.57	23	1	2	1	6
	Lineata (Alloch.)	33	7.28 ± 0.62	24	2	3	1	3
	Nigra	33	7.64 ± 0.50	20	1	3	1	8
	Lutea	33	7.58 ± 0.68	16	6	6	1	4
	Total Sample (Session 1)	132	7.41 ± 0.62	83	10	14	4	21
Session 2	*Lineata* (Autoch.)	38	6.81 ± 0.59	32	2	1	1	2
	Lineata (Alloch.)	38	6.96 ± 0.56	31	5	1	1	0
	Nigra	38	7.60 ± 0.61	34	1	0	2	1
	Lutea	38	7.41 ± 0.45	30	0	3	2	3
	Total Sample (Session 2)	152	7.19 ± 0.64	127	8	5	6	6
Sessions 1 & 2 (Whole Experiment)	*Lineata* (Autoch.)	71	6.97 ± 0.60	55	3	3	2	8
	Lineata (Alloch.)	71	7.11 ± 0.60	55	7	4	2	3
	Nigra	71	7.62 ± 0.56	54	2	3	3	9
	Lutea	71	7.49 ± 0.58	46	6	9	3	7
	Total Sample (Sessions 1 & 2)	284	7.30 ± 0.64	210	18	19	10	27

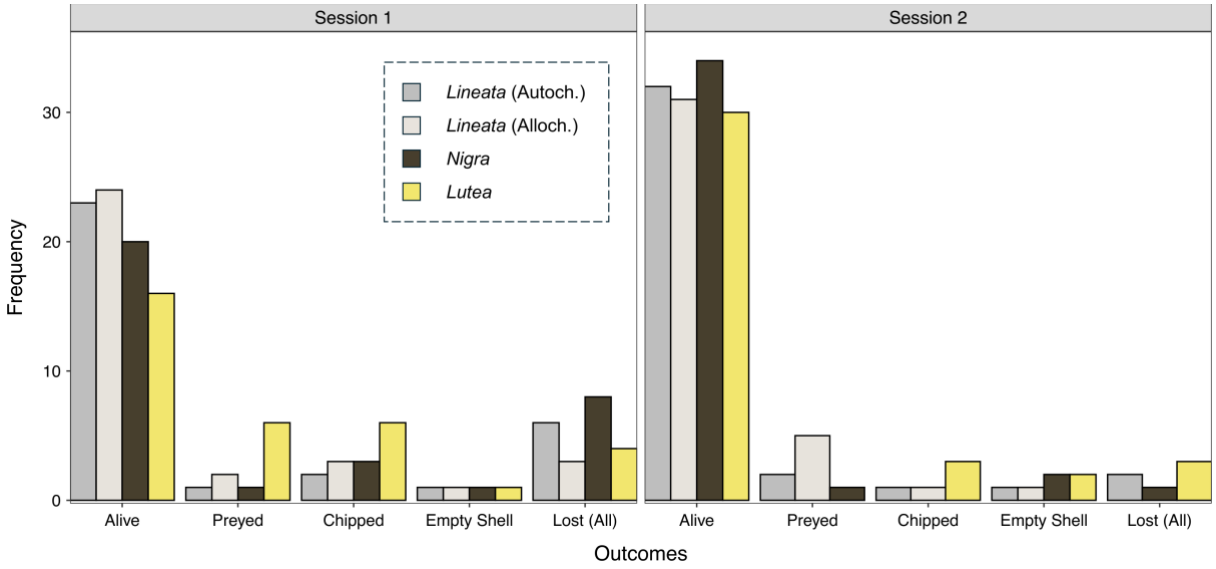

Figure 4. Histograms of main outcomes per color morph in each experiment session.

Since there were differences in the shell length of the snails between sessions, logistic regressions were used to explore the predictive power of this (i.e., 'Shell Length') and other potentially explanatory variables (particularly the color of the morphs, but also 'Session') on the probability of the different outcomes (see Table 3 and Supplementary materials Tables S2 to S4). The resulting models confirmed 'Session' and 'Shell Length' as explanatory variables of the outcome 'Chipped' ($P < 0.05$; Supplementary materials Table S3), but not of 'Preyed' or 'Empty Shell' (Supplementary materials Table S2 and S4). In the 'Chipped' model, 'Transect Position: Lower' (whether the transect was located in the lower zone of the intertidal ecosystem) yielded a near-significant result ($P = 0.086$). 'Shell Length' returned a positive relationship to the log odds of the probability of 'Chipped', with an odds ratio of 2.29 (95% confidence interval = 1.04–5.23), while 'Session 2' (reference category: 'Session 1') and 'Transect Position: Lower' (reference category: 'Upper') resulted in a negative relationship (the latter, however, with an odds ratio 95% confidence interval including 1). These results can be interpreted as larger snails, snails of Session 1, and perhaps snails of lower transects having a higher probability of being 'Chipped'. The contribution of 'Shell Length' may be related to the fact that the snails of Session 1 were generally larger.

None of the models except that of 'Preyed + Chipped' conceded the color morph of the snail a significant predictive role (Table 3). In that model, the color morph *lutea* stood close to statistical significance, with a positive slope ($\beta = 0.97$, $P = 0.077$; reference level: *Lineata* autochthonous). The odds ratio suggests that bearing this color increases the chance of being preyed on or having a small break in the shell by 2.65 times, but the corresponding 95%

confidence interval includes 1 (0.94–8.30), so this value should be interpreted cautiously. 'Session 2' also yielded a significant result (P = 0.014) in the 'Preyed + Chipped' model (using 'Session 1' as the reference category), with a negative estimate value and an odds ratio below 1 (95% confidence interval = 0.18–0.82), meaning that having participated in Session 1 increases the probability of this outcome. When 'Preyed + Chipped' regression models were run for each session separately (Supplementary materials Table S5), in Session 1 the color morph *lutea* also yielded a significant result (P = 0.04; odds ratio = 4.56, 95% confidence interval = 1.17–22.93), but not in Session 2 (P = 0.82; odds ratio = 1.23, 95% confidence interval = 0.19–8.12). While separate analyses of 'Preyed' and 'Chipped' counts by color morph in Session 1 didn't yield statistically significant results (Fisher's exact tests; 'Preyed', P = 0.136; 'Chipped', P = 0.503), the combined 'Preyed + Chipped' outcome did (Fisher's exact test; P = 0.035), with a higher count of *lutea* in the contingency table. When the experiment sessions were pooled, no differences were found between the different color morphs in the frequencies of outcomes (Fisher's Exact or chi-square tests, $P > 0.10$), except again in the additive category of 'Preyed + Chipped', which showed a near statistically significant result (χ^2 = 7.65, df = 3, P = 0.054).

Table 3. Results of the logistic regression fitted to the composite 'Preyed + Chipped' outcome of the experiment. The model used 'Shell Length', 'Session' (with 'Session 1' as reference level), 'Transect Position' (with 'Upper' as reference level), and 'Color Morph' (with '*Lineata* from Cabo Silleiro –Autochthonous–' as reference level) as predictors. The table provides the coefficient estimates (β), standard errors (SE), and *Z*- and *P*-values for each predictor or level, in addition to their corresponding odds ratios (e^{β}) and 95% confidence intervals (CI). The lower table displays the model fit statistics, which include the Akaike Information Criterion (AIC), the null and residual deviance, and the number of Fisher Scoring iterations. Statistically significant results are presented in bold font. The 'Preyed + Chipped' outcome was created by granting the snails with 'Preyed' or 'Chipped' outcomes a value of '1', and '0' to those with the remaining outcomes. In all models, 'Lost (All)' cases were considered NAs to avoid the 'missing as predation' bias.

Outcome: Preyed + Chipped

Predictor	Level	Estimate (β)	SE	*Z*	*P*	Odds Ratio	95% CI (Odds Ratio)
Intercept	-	-3.381	2.232	-1.52	0.1298	-	-
Shell Length	-	0.237	0.306	0.77	0.4392	1.267	0.692–2.313
Session (reference: Session 1)	**Session 2**	**-0.948**	**0.387**	**-2.45**	**0.0143**	**0.387**	**0.177– 0.817**
Transect Position (reference: Lower)	Upper	-0.233	0.388	-0.60	0.5473	0.792	0.362–1.675
Color Morph (reference: *Lineata* Cabo Silleiro – Autoch.)	*Lineata* Aguncheiro (Alloch.)	0.576	0.552	1.04	0.2965	1.778	0.618–5.565
	Nigra	-0.290	0.666	-0.44	0.6629	0.748	0.194–2.781
	Lutea	0.973	0.549	1.77	0.0765	2.646	0.935–8.297

Model Fit Statistics	
AIC	209.4
Null deviance	211.8
Residual deviance	195.4
Number of Fisher Scoring iterations	5

Discussion

Ever since Francis B. Sumner's classic research on the coat coloration of *Peromyscus* mice (Sumner, 1926), color clines have traditionally been investigated not only because coloration is a highly conspicuous and easily scorable trait but also because it poses interesting questions about the maintenance of color variation (McLean & Stuart-Fox, 2014). Several instances of color clines putatively maintained by selection have been described in mollusks and other animal groups (Antoniazza *et al.*, 2010; Takahashi *et al.*, 2011; Brakefield & de Jong, 2011). Similarly, the study of intrapopulation color polymorphism has enhanced our understanding of adaptation, divergence, and ultimately, speciation (Comeault *et al.*, 2015; Svensson, 2017; O'Connor *et al.*, 2021). Various methodologies have been devised to study color adaptation. Among these, frequentist-observational methods are typically the most used, with genomic approaches being added in recent decades (Takahashi, 2015; Cuthill *et al.* 2017; San-Jose & Roulin, 2017).

The frequentist-observational procedure has yielded valuable insights into the adaptive role of coloration, substantially increasing our knowledge of this phenomenon (Gefaell *et al.* 2023). However, more direct evidence is often needed to understand the precise ecological and evolutionary forces underlying variation in coloration, whether clinal or intrapopulation. In such cases, field experiments offer a powerful alternative. In field experiments, one or more potentially relevant causal factors are deliberately manipulated to assess and quantify their contribution to the phenomenon of interest. There are successful examples of field experiments in the evolutionary ecology of coloration (e.g., Ekendahl, 1998; Mullen & Hoekstra, 2008), but many of them involve several difficulties that can limit the validity of their results. First, many extraneous variables can rarely be controlled, potentially altering the results in unexpected ways. Second, these experiments rely on numerous untested assumptions about the natural history of the organisms involved. If any of these assumptions are not met, the results might be compromised. Even if the organisms taking part in the experiment have been thoroughly studied in the past, many relevant aspects of their behavior in the wild are unknown, which can ultimately jeopardize field experiments.

In the present study, a tethering experiment aimed at testing whether there was differential crab predation of *L. saxatilis* color morphs by *P. marmoratus* was conducted as a means of shedding light on the maintenance mechanisms of a color cline in the populations of *L. saxatilis* from the Rías Baixas. More specifically, this experiment sought to determine why the wave-exposed localities along the cline tend to be monomorphic for *lineata*. As a complementary study, a series of samplings in the source and experiment localities (Aguncheiro and Cabo

Silleiro), as well as a series of measurements of key shell parameters of the color morphs involved, were carried out. Several insights can be gained from this study.

It appears that at the source locality (Aguncheiro), not all morphs are equally attacked by the crabs, with *lutea*, despite being less common, having a higher predation propensity than *nigra* and *lineata*, as inferred from the frequency of scars. This effect, however, is not statistically significant. Of these morphs, together with *lineata* from Cabo Silleiro (autochthonous), *lutea* appears to have a slightly thicker shell. Although this result should be taken cautiously due to its near-significance, it is unexpected, as previous studies suggest that no differences in shell thickness exist between color morphs from the same locality (Raffaelli, 1979). Shell thickness has traditionally been related to predation attempts, either constituting a genetic adaptation or a plastic response against them (Seeley, 1986; Trussell, 1996; Bourdeau, 2009). In fact, healed predation attempts have been linked to the thickening of the shell, which might in turn reduce vulnerability to future crab attacks (Teck *et al.*, 2023). As for crypsis, the comparison of the spectra of the snails and rocks from Aguncheiro suggests that, in that locality, *lutea* and *lineata* might be more cryptic than *nigra*. Given the chromatic diversity of Aguncheiro (with rocks with yellowish and black-reddish spots), it could be that each morph is adapted to the different color patches (*lutea* to yellow patches, *lineata* to black-reddish ones), with *nigra* perhaps settling more on the bottom of the rocks, where light strikes less intensely and therefore can be more cryptic. If so, then the polymorphic population of Aguncheiro could be maintained by selection for crypsis in a heterogeneous environment. However, this hypothesis deserves further investigation to evaluate its veracity.

As for the tethering experiment, first, it is worth noting that this study provides some support for the relatively high conspicuousness of *lutea* (and to some degree *nigra*) in a representative wave-exposed locality from the Rías Baixas, where monomorphic populations of *lineata* abound. Although the degree of crypsis of the morphs should ideally be evaluated using visual models of the predator species (currently unavailable for *P. marmoratus*), the information provided by the spectra can be provisionally taken to back the hypothesis that, at Cabo Silleiro, the *lineata* morphs (especially the autochthonous one) are more cryptic than *lutea* and *nigra*. For instance, at around 508 nm (the peak sensitivity of the main photoreceptor class of crabs), the spectra of *lineata* coincide with that of the rocks, something not seen in the remaining morphs (especially in *lutea*). With all due caution, this might constitute frequentist-observational evidence for cryptic selection as the maintenance mechanism of the *lineata* monomorphic populations of *L. saxatilis* from the Rías Baixas. Reflectance spectrometry assessments of the degree of crypsis have rarely been used in *L. saxatilis* (see Gefaell *et al.*, 2023 and references

therein, as well as the pioneering study by Cruickshanks, 2006). However, they provide a useful way of testing a commonly assumed hypothesis in cryptic selection studies, namely, that the different morphs vary in their level of camouflage.

Apart from this, very low predation rates were observed in the experiment, substantially limiting the statistical power of the study. Several possible explanations exist for this, all related to the previously discussed difficulties inherent to field experiments: either the lack of control over extraneous variables or the reliance on untested assumptions about the natural history of the organisms involved. One possibility might relate to the size of the snails used in the experiment (between 7-7.5 mm in shell length), which could have been too large to be preyed upon by the crabs. However, as stated previously, using much smaller snails would likely have masked their original color, thus invalidating the experiment. Another possibility is related to our limited knowledge of the ecology and natural history of the predator species, *P. marmoratus*. It cannot be ruled out that the crabs were affected by some unknown factor that deterred them from preying on the snails. Alternatively, their densities might not have been as high as assumed during the weeks of the experiment. Measuring predation by crabs on *Littorina* snails, either in the lab or the field, is a demanding task that has faced many challenges in the past (Tucker, 1988; Ekendahl, 1998; Cruickshanks, 2006). For instance, in a well-designed laboratory experiment aimed at testing visual selection on *L. mariae* (now *L. fabalis*) color morphs by the European green crab (*Carcinus maenas*), Tucker (1988, chapter 4) found no evidence of differential selection. However, the predation rates obtained were very low, thus compromising the validity of the results.

Despite the low predation rates, however, the tethering experiment described in the present article found a selective disadvantage of the supposedly less cryptic morph, *lutea*, against predation by *P. marmoratus* but only in the first session of the experiment and provided that the 'Chipped' outcome is interpreted as an instance of predation. This requires consideration of these factors ('Chipped' and 'Session') separately. The 'Chipped' outcome has been previously interpreted as at least partly due to the snails being bashed against the rocks by the waves (Boulding *et al.*, 2017). In our case, this would predict more 'Chipped' instances in Session 1 (where there was bad weather during the first days) and in lower transects (as the snails tethered there would suffer more wave action). Evidence backs the first prediction (see the 'Chipped' logistic regression, Supplementary materials Table S3), but the second one finds, at best, partial support, as differences in the 'Chipped' outcome between transect locations are close to significance in both the regression models ($P = 0.086$) and a chi-square test ($\chi^2 = 3.06$, df = 1, $P = 0.080$). Alternatively, 'Chipped' can be considered an instance of predation attempt, most

likely by a small crab. Favoring this interpretation is the fact that the outcome 'Preyed', which under no circumstances can be taken as the result of bashing against the rocks, goes in the same direction as 'Chipped' in the *lutea* morph from Session 1 (see Figure 4). This doesn't happen with other morphs. Furthermore, if *lutea* is really a little bit thicker than *lineata* (both autochthonous and allochthonous) and *nigra*, then it would be expected that the shell breaks caused by the crabs in *lutea* would be smaller than those of the remaining morphs (thus contributing to a relatively higher frequency of 'Chipped' in this morph). As for why the presumed selective disadvantage of *lutea* in the tethering experiment only occurred in Session 1, it should be noted that, due to bad weather, Session 1 lasted a week more than Session 2, which would have given the crabs more time to prey upon the snails, as is seen in the comparison between the sum of the 'Preyed' and 'Chipped' outcome counts of both sessions (higher in Session 1). However, if it were only a matter of time, we would expect to see the same trend in both sessions, which we don't, as no *lutea* individuals were preyed upon in Session 2. So while it would be reasonable to assume that 'Chipped' constitutes a predation attempt, and that predation in Session 1 was higher because of the number of days it lasted, several uncertainties remain. It cannot be ruled out that 'Chipped' includes both predation attempts by small crabs and breaks caused by wave action.

To summarize, no clear evidence of selection for crypsis has been found in the polymorphic source population studied (Aguncheiro), as the asymmetrical distribution of shell scars between morphs is not statistically significant. Reflectance spectrometry has shown that in the monomorphic *lineata* population of Cabo Silleiro, this morph is more cryptic than the allochthonous morphs, especially *lutea*. However, the tethering experiment conducted there only provides partial support for selection for crypsis, as evidence of predation against *lutea* was only found in one session of the experiment. In any case, due to low predation rates, the statistical power of the present study was rather low. In the face of this, it is unwarranted to assert that *P. marmoratus* plays a role in maintaining the monomorphic populations of *lineata* seen in the wave-exposed localities from the Rías Baixas by pruning the less cryptic morphs based solely on the results of this study. However, this does not support the opposite conclusion either –that this crab species does not contribute to maintaining the cline– as some data from the tethering experiment suggests otherwise. Thus, the evidence provided here should be viewed as suggestive yet inconclusive, calling for a lab or mesocosm experiment aimed at definitely settling the question.

Data availability

All data and software codes are openly available at https://github.com/juangefaell/TetheringPredation.git.

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Supplementary materials – Chapter 5

Extended Materials and Methods (1): Methodological figures and tables

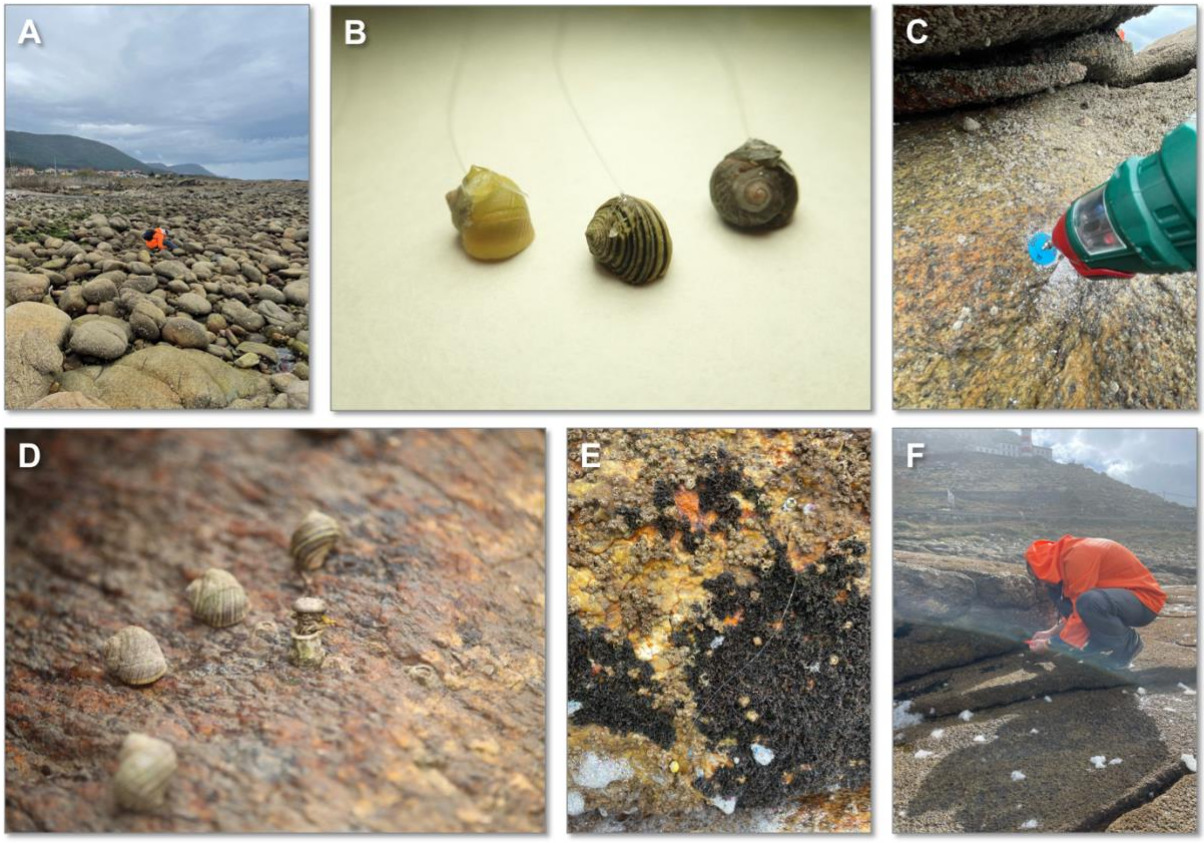

Figure S1. Main steps of the tethering experiment. A. Aguncheiro locality, where most of the snails (*lineata* allochthonous, *nigra*, and *lutea*) for the experiment, as well as for the study of scars and shell measurements, were collected. B. Processing of the snails (attachment of tethering lines using transparent instantaneous epoxy glue; the picture shows the three morphs from Aguncheiro). C. Installation of new screws. D. A screw from Boulding *et al*. (2017) that was reused in the present tethering experiment (the surrounding snails are free and did not participate in the experiment). E. *Lutea* snail just tethered to its corresponding screw, belonging to a transect in the lower part of the ecosystem. Patches of *Lichina pygmaea* like the one that appears in the picture were quite rare in the transects. F. Shell length measurement of the tethered snails right before the beginning of the experiment.

Table S1. Operational definitions of the main categories used in the tethering experiment (based on Boulding et al. 2017).

Outcome	Description
Alive Intact	Alive snail with its shell undamaged and still attached to the tethering line.
Chipped	Slightly chipped outer lip and snail alive. The outer lip of the shell aperture is visibly chipped or smoothly abrased, but the magnitude of the fracture is small, so cannot be readily attributed to a crab predation attempt. The difference between 'Preyed Alive' and 'Chipped' is the magnitude of shell breakage ('Preyed Alive' > 'Chipped'): if the shell shows a sharp, abrupt fracture, it should be ranked as 'Preyed Alive'; if it shows a smooth or small break, then it should be ranked as 'Chipped'.
Preyed Dead	Large or small fragments of shell attached to epoxy, which is still attached to the tethering line. No snail body within the shell.
Preyed Alive	Large fracture in the outer lip of the shell aperture, but the snail within still alive and attached to the tethering line. Much more dramatic shell break than in 'Chipped'.
Epoxy Only	Only the epoxy glue is attached to the tethering line and the snail is not found in the nearby surroundings.
Recovered Alive Intact	Tethering line and epoxy still attached to the screw ('Epoxy Only'), and the snail found 'Alive Intact' (i.e., undamaged) in the nearby surroundings.
Recovered Chipped	Tethering line and epoxy still attached to the screw ('Epoxy Only'), and the snail found 'Chipped' in the nearby surroundings.
Recovered Preyed Dead	Tethering line and epoxy still attached to the screw ('Epoxy Only'), and the snail found 'Preyed Dead' in the nearby surroundings.
Recovered Preyed Alive	Tethering line and epoxy still attached to the screw ('Epoxy Only'), and the snail found 'Preyed Alive' in the nearby surroundings.
Hole	Dead snail with a hole in body whorl of shell likely from predation by *Nucella lapillus*.
Lost	No line on the screw. Tethering line no longer attached to the screw, and the snail and line not found in the nearby surroundings.
Knot Only	Tethering line with no glue on the knot. Missing snail with the knot or tethering line visible.
Dead Intact	Dead snail attached to the tethering line but with shell undamaged.
Empty Shell	Empty undamaged shell. Likely preyed by an organism that somehow manages to remove the animal from the shell without breaking it.
NA	Missed data or highly ambiguous outcome.

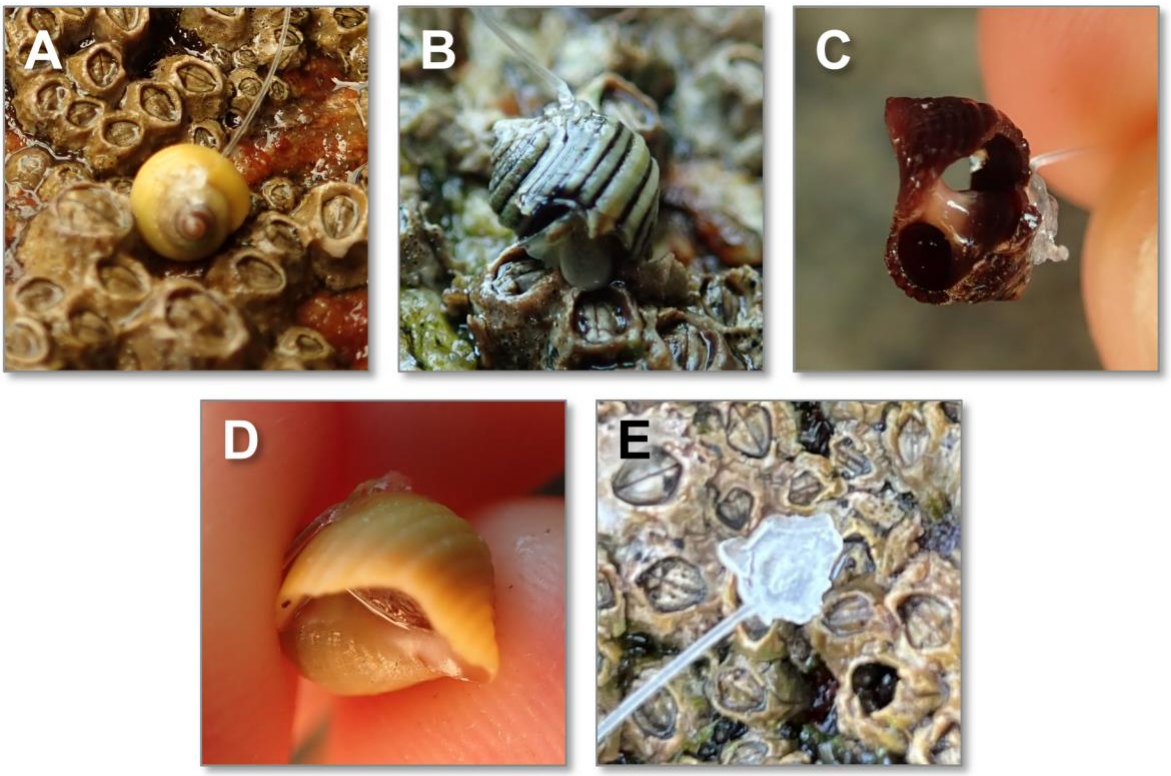

Figure S2. Illustrative examples of the main outcomes of the tethering experiment except 'Empty Shell'. A. 'Alive Intact'. B. 'Preyed Alive'. C. 'Preyed Dead'. D. 'Chipped'. E. 'Epoxy Only'. Note that the 'Chipped' category also includes less smoothed breaks than the one depicted in the figure (i.e. similar breaks to 'Preyed Alive' but of smaller magnitude).

Extended Materials and Methods (2): Geometric morphometric methods

Geometric morphometric analyses (e.g. Zelditch, Swiderski, & Sheets, 2012) were based on eleven two-dimensional landmarks placed at different key points of the shell of each studied specimen, previously photographed with a Panasonic Lumix DMC-FZ300 digital camera. The landmarks were positioned with the TPS software (TpsUtil v. 144 and TpsDig v.1.40; Rohlf, 2015), following Gefaell, Varela & Rolán-Alvarez (2020). This allowed us to assess both the size of each snail –measured with the Centroid Size– and a series of shape parameters –i.e. Relative Warps. The Centroid Size constitutes the square root of the sum of squared distances of each landmark to their centroid (Bookstein, 1992); the Relative Warps, in turn, represent the principal components of a series of individual partial warp scores that measure deviations from the reference configuration obtained through the minimization of squared distances between homologous landmarks across specimens (using a Procrustes generalized orthogonal method; Rohlf & Slice, 1990). A total of 18 Relative Warps were obtained out of the eleven landmarks, with RW1–RW8 explaining a cumulative variance of > 95%. Within these RW1, which explains 46.35% variance, has been traditionally interpreted in morphometric studies of *L. saxatilis* as the shell's relative aperture size and overall roundness (e.g., Gefaell *et al.*, 2020).

Extended Results (1): Geometric morphometrics of a representative sample of the color morphs studied in the experiment

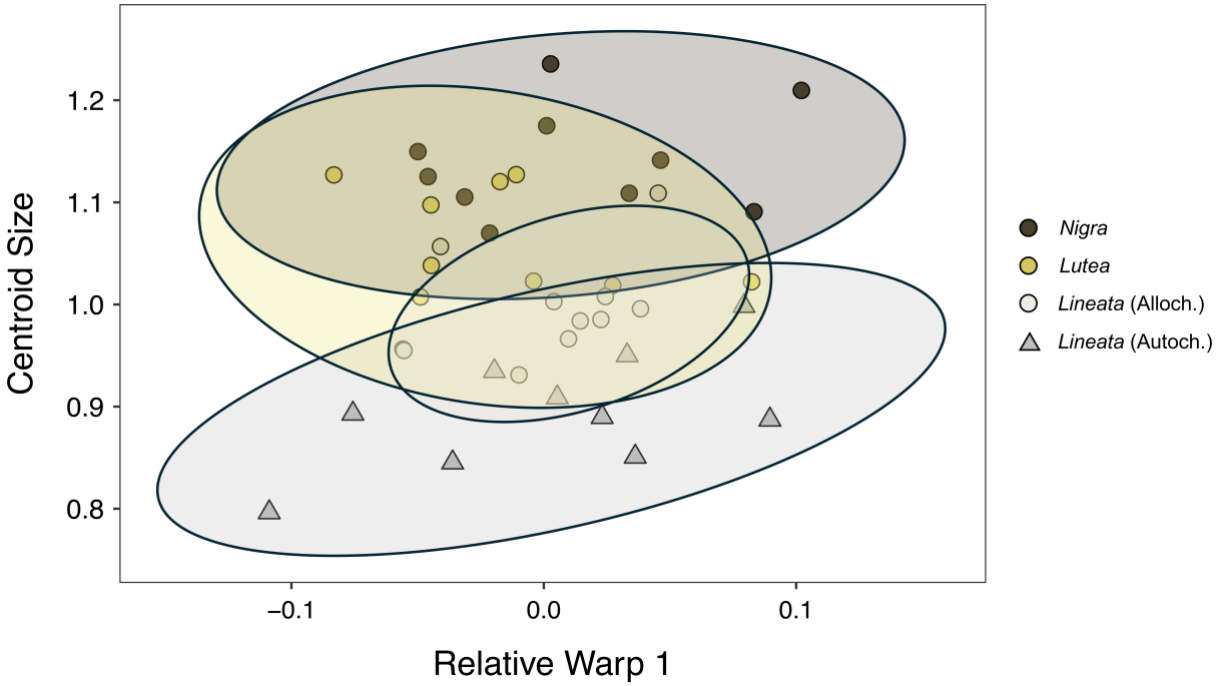

Figure S3. Plot of the Centroid Size and Relative Warp 1 of the measured snails (n = 10 per morph), with their corresponding 95% confidence interval concentration ellipses. There were differences in the Centroid Size, with *nigra* being larger than the rest (ANOVA, $F(3) = 34.35$, $P < 0.001$; confirmed with Tukey test for post hoc pairwise comparisons). This suggests that our size standardization efforts failed when taking the samples for shell measurements.

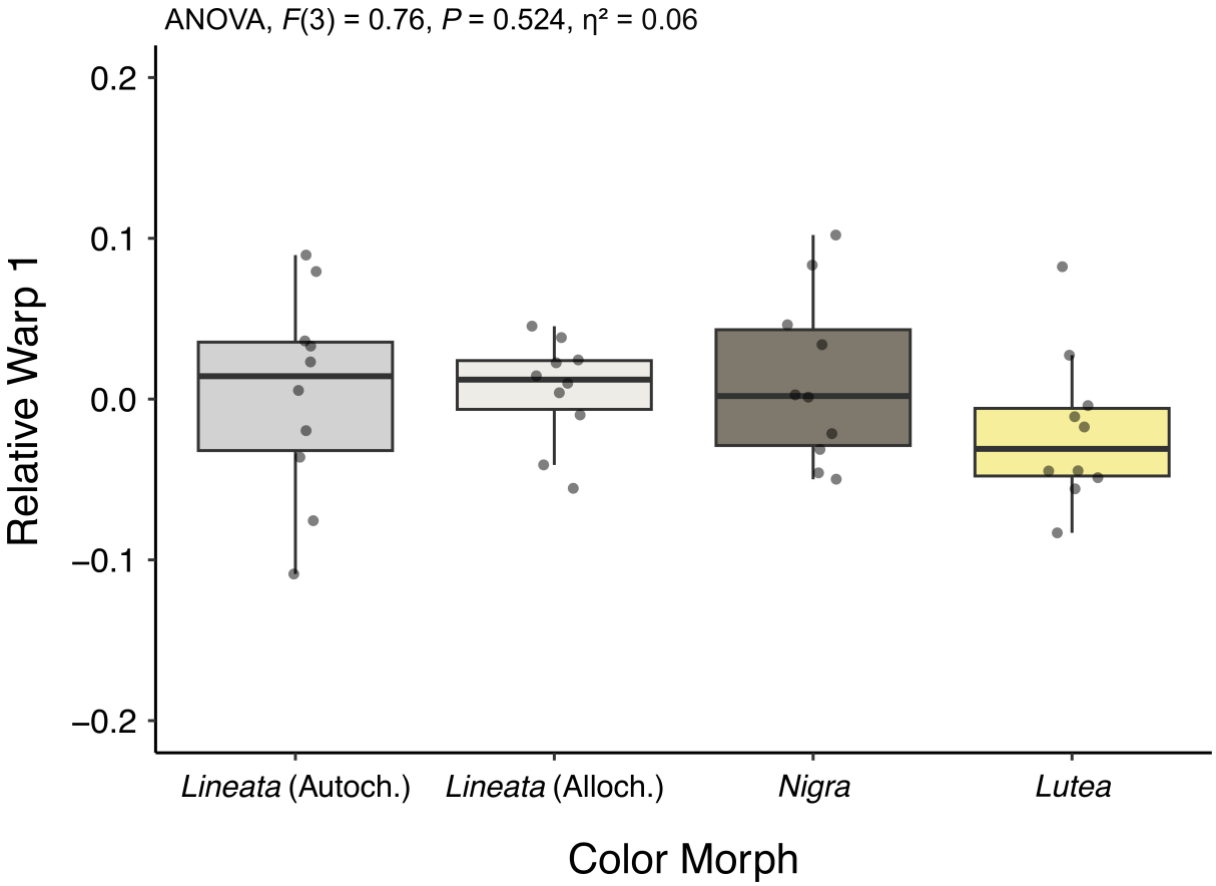

Figure S4. Boxplots of Relative Warp 1 for each color morph class, plus the ANOVA results (non-significant).

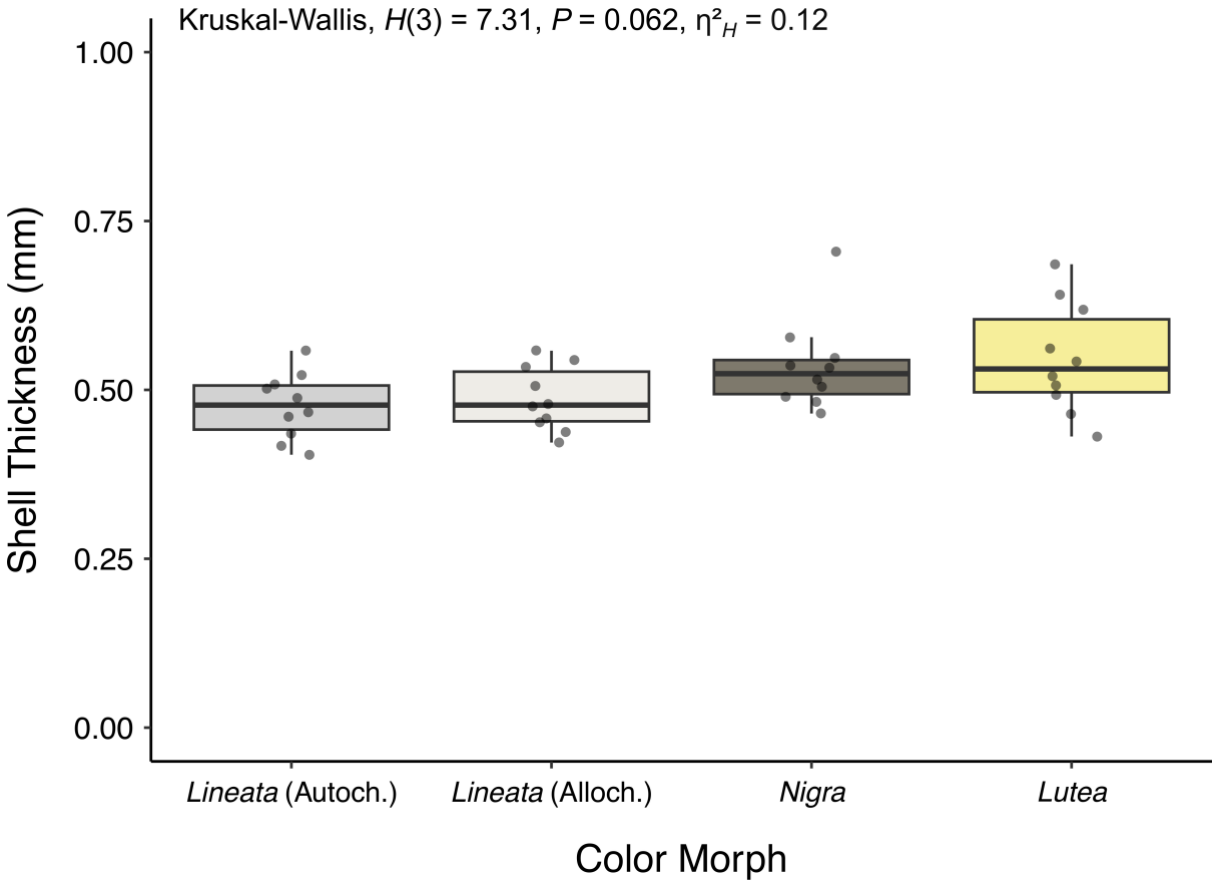

Figure S5. Boxplots of shell thickness (measured in mm) for each color morph class, plus the Kruskal-Wallis' test results (near statistically significant). The moderate effect size obtained (η^2_H) suggests there might be real differences in thickness between the morphs, despite near significance.

Extended Results (2): Reflectance spectra of Aguncheiro snails and rocks

To create the spectra of the rocks, the three random points measured on each rock were taken as independent replicas. This is not problematic, as *in situ* inspection of the rocks clearly shows that intra-rock chromatic variability is much greater than between-rock variability. All rocks have yellow spots interspersed with reddish-black patches. In Aguncheiro, however, one rock measurement was removed from the average as it could represent an artifact (caused by a higher probe-to-object distance than the rest of the measurements). The high confidence interval shown by the rocks of Aguncheiro likely results from pooling measures with clearly distinguishable chromatic properties.

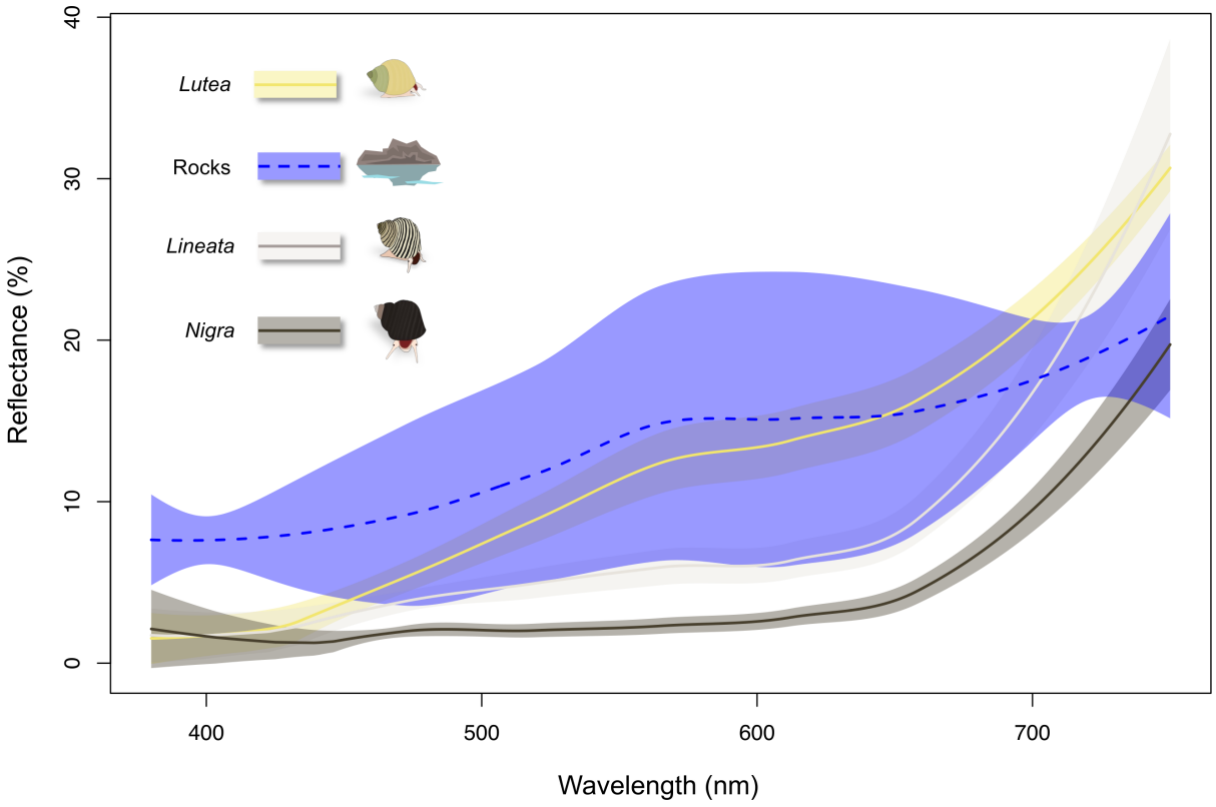

Figure S6. Comparison of the spectra of snails and rocks from Aguncheiro (mean spectrum for each class of snails and the rocks, with their corresponding 95% confidence intervals).

Extended Results (3): Further results from the tethering experiment

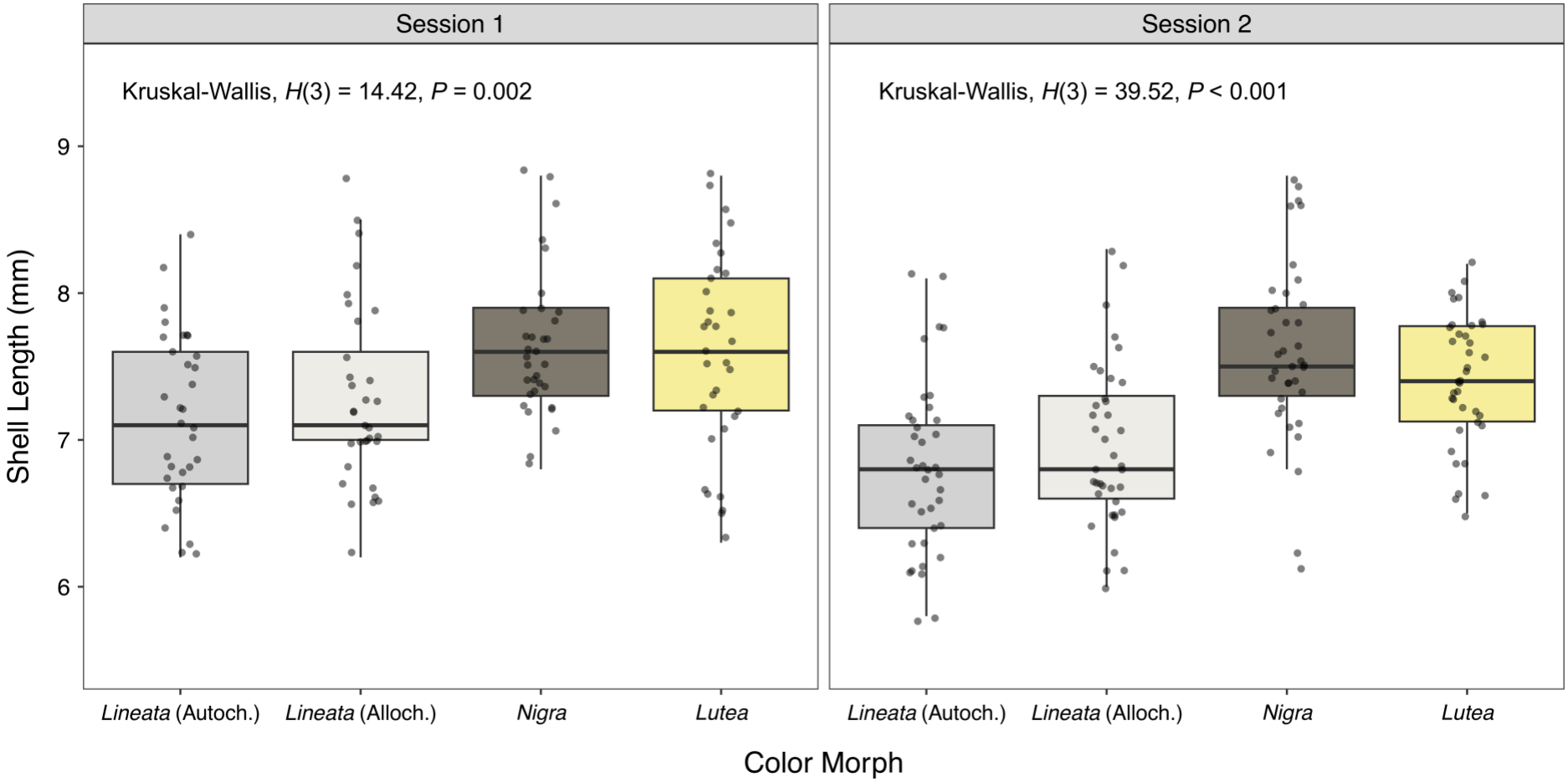

Figure S7. Boxplots of the shell length of the snails of each color morph in each experiment session.

Table S2. Results of the logistic regression model, using 'Shell Length', 'Session' (with 'Session 1' as reference level), 'Transect Position' (with 'Lower' as reference level), and 'Color Morph' (with '*Lineata* from Cabo Silleiro –Autochthonous–' as reference level) as predictors of the variable 'Preyed'. The table reports the coefficient estimates (β), standard errors (SE), and *Z*- and *P*-values for each predictor. Additionally, the table provides the odds ratios (e^{β}) with their corresponding 95% confidence intervals (CI). The lower table displays the model fit statistics, which include the Akaike Information Criterion (AIC), the null and residual deviance, and the number of Fisher Scoring iterations.

Outcome: Preyed

Predictor	Level	Estimate (β)	SE	*Z*	*P*	Odds Ratio	95% CI (Odds Ratio)
Intercept	-	0.140	3.092	0.05	0.964	-	-
Shell Length	-	-0.433	0.435	-1.00	0.32	0.649	0.267–1.489
Session (reference: Session 1)	Session 2	-0.694	0.517	-1.34	0.179	0.500	0.176–1.371
Transect Position (reference: Lower)	Upper	0.460	0.506	0.91	0.364	1.584	0.581–4.347
Color Morph (reference: *Lineata* Cabo Silleiro – Autoch.)	*Lineata* Aguncheiro (Alloch.)	0.897	0.720	1.25	0.213	2.453	0.640–11.921
	Nigra	-0.156	0.968	-0.16	0.872	0.856	0.104–5.738
	Lutea	0.937	0.769	1.22	0.223	2.552	0.595–13.363

Model Fit Statistics	
AIC	137.9
Null deviance	130.4
Residual deviance	123.9
Number of Fisher Scoring iterations	6

Table S3. Results of the logistic regression model for the 'Chipped' outcome. Statistically significant results are highlighted using bold font.

Outcome: Chipped							
Predictor	Level	Estimate (β)	SE	*Z*	*P*	Odds Ratio	95% CI (Odds Ratio)
Intercept	-	**-8.125**	**3.054**	**-2.66**	**0.0078**	-	-
Shell Length	-	**0.827**	**0.409**	**2.02**	**0.0434**	**2.287**	**1.035–5.232**
Session (reference: Session 1)	**Session 2**	**-1.106**	**0.560**	**-1.98**	**0.0483**	**0.331**	**0.101–0.945**
Transect Position (reference: Lower)	Upper	-1.041	0.606	-1.72	0.0855	0.353	0.094–1.067
Color Morph (reference: *Lineata* Cabo Silleiro – Autoch.)	*Lineata* Aguncheiro (Alloch.)	0.057	0.811	0.07	0.9441	1.059	0.217–5.785
	Nigra	-0.394	0.888	-0.44	0.6577	0.675	0.110–4.105
	Lutea	0.888	0.734	1.21	0.2268	2.429	0.621–12.093

Model Fit Statistics	
AIC	129.8
Null deviance	135.5
Residual deviance	115.8
Number of Fisher Scoring iterations	6

Table S4. Results of the logistic regression model for the 'Empty Shell' outcome.

Outcome: Empty Shell

Predictor	Level	Estimate (β)	SE	*Z*	*P*	Odds Ratio	95% CI (Odds Ratio)
Intercept	-	-1.800	4.038	-0.45	0.656	-	-
Shell Length	-	-0.219	0.567	-0.39	0.699	0.803	0.253–2.378
Session (reference: Session 1)	Session 2	0.193	0.675	0.29	0.775	1.212	0.327–4.981
Transect Position (reference: Lower)	Upper	-0.632	0.712	-0.89	0.374	0.532	0.111–2.000
Color Morph (reference: *Lineata* Cabo Silleiro – Autoch.)	*Lineata* Aguncheiro (Alloch.)	-0.040	1.021	-0.04	0.969	0.961	0.112–8.293
	Nigra	0.606	1.009	0.60	0.548	1.833	0.255–16.208
	Lutea	0.538	0.975	0.55	0.582	1.712	0.253–14.315

Model Fit Statistics	
AIC	97.0
Null deviance	84.5
Residual deviance	83.0
Number of Fisher Scoring iterations	6

Table S5. Results of the logistic regressions for the 'Preyed + Chipped' outcome per experiment session. Statistically significant results are highlighted in bold font.

	Outcome: Preyed + Chipped (per Session)							
	Predictor	Level	Estimate (β)	SE	*Z*	*P*	Odds Ratio	95% CI (Odds Ratio)
Session 1	**Intercept**	-	**-6.199**	**2.8307**	**-2.19**	**0.0285**	-	-
	Shell Length	-	0.5864	0.3811	1.539	0.1239	1.798	0.859–3.863
	Transect Position (reference: Lower)	Upper	-0.3396	0.5362	-0.633	0.5265	0.712	0.235–1.972
	Color Morph (reference: *Lineata* Cabo Silleiro – Autoch.)	*Lineata* Aguncheiro (Alloch.)	0.3731	0.7936	0.47	0.6383	1.452	0.313–7.817
		Nigra	0.1696	0.8428	0.201	0.8406	1.185	0.224–6.832
		Lutea	**1.5163**	**0.7385**	**2.053**	**0.04**	**4.556**	**1.168–22.929**
Session 2	Intercept	-	-0.4234	3.775	-0.11	0.911	-	-
	Shell Length	-	-0.2807	0.553	-0.51	0.611	0.755	0.245–2.178
	Transect Position (reference: Lower)	Upper	-0.1706	0.592	-0.29	0.773	0.843	0.254–2.714
	Color Morph (reference: *Lineata* Cabo Silleiro – Autoch.)	*Lineata* Aguncheiro (Alloch.)	0.7782	0.758	1.03	0.304	2.178	0.519–11.192
		Nigra	-0.9595	1.256	-0.76	0.445	0.383	0.017–3.744
		Lutea	0.2095	0.922	0.23	0.82	1.233	0.192–8.120

Table S5 (Continuation). Model fit statistics.

	Model Fit Statistics			
	AIC	Null deviance	Residual deviance	Number of Fisher Scoring iterations
Session 1	116.5	115.9	104.5	4
Session 2	95.1	87.7	83.1	6

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Chapter 6

Gefaell J. Unpublished manuscript. A potential case of parallel evolution of a color cline in the populations of *Littorina saxatilis* from the Rías Baixas: Preliminary analyses.

A potential case of parallel evolution of a color cline in the populations of *Littorina saxatilis* from the Rías Baixas: Preliminary analyses

Juan Gefaell

Introduction

To what extent evolution is repeatable stands as one of the most fascinating questions in modern evolutionary biology (Blount, Lenski, & Losos, 2018). This problem is not only scientifically relevant but also has deep metaphysical implications: is the cosmos governed by chance, making evolutionary trajectories unique and unrepeatable, or does determinism take precedence, causing the various paths of evolution to often converge on the same outcomes? Bringing the question to the human sphere, we might ask whether our species is the product of a specific and irreproducible evolutionary route, or if an organism analogous to us would have evolved if we hadn't. The first possibility emphasizes contingency, while the second highlights determinism.

The view that chance has played a major role in the evolution of life was famously defended by Stephen J. Gould (1941–2002) in his book *Wonderful Life* (Gould, 1989). In it, Gould invites us to consider what would happen if we could rewind the tape of life and play it a second time; that is if we could go back in time and watch how evolution unfolds again. According to Gould, if we were to do that, we would probably see completely different organisms than the ones roaming the Earth today, given the role chance plays in determining the evolutionary outcomes of many species and lineages. His main evidence for this was the extinction patterns seen in the Cambrian fauna: while all these organisms seemed perfectly adapted to their environments, only a fraction survived. If these creatures were equally fit, as they appear from the fossil record, only contingency can account for their differential evolutionary success. If slightly different conditions had prevailed during the Cambrian period, evolution would have likely taken a completely different path. Importantly, Gould's interpretation of contingency was not blind randomness but rather the concatenation of unpredictable historical events, most of them totally unrelated to biological fitness, although he was not particularly clear in his use of the term (e.g., Beatty, 2006; Losos, 2017; Blount *et al.*, 2018).

Gould's views have been disputed by many prominent evolutionary biologists. Paleontologist Simon Conway Morris, in particular, has offered a different interpretation of the Cambrian fauna, emphasizing the role of determinism (Conway Morris, 1998). Ever since Gould's

Box 1. Contingency versus determinism in Western thought

Throughout the history of Western thought, numerous thinkers have leaned toward either determinism or contingency, with determinism usually taking precedence. The Presocratic philosopher Heraclitus (c. 6th century BCE–5th century BCE) was one of the first determinists. According to the doxographer Aetius, Heraclitus held that 'everything happens according to fate, and (…) this is the same as necessity' (Barnes, 1982: 100). Leucippus (5th century BCE) and Democritus (c. 460–c. 370 BCE), contemporaries of Socrates and early proponents of atomism, also espoused a deterministic view of the universe. While, for instance, Democritus mentioned chance, he was probably referring to the absence of purpose (Barnes, 1982). Within Hellenistic philosophy, the stoics were staunch determinists too, holding that events in the universe follow a cyclical pattern where everything that has happened in the past will happen exactly the same way after a new rebirth of the universe (White, 2003). Conversely, Epicurus (341–270 BCE), another Hellenistic philosopher, allowed for indeterminism in his view of the grand scheme of things. Although influenced by Leucippus and Democritus, Epicurus believed that atoms could randomly swerve, deviating from their pre-established paths and thus disrupting the deterministic course of the universe (Sharples, 1996).

In the modern era, determinism was also a widespread philosophy, with thinkers such as Thomas Hobbes (1588–1679) or Baruch Spinoza (1632–1677) standing out (LeBuffe, 2021). Meanwhile, indeterminism, despite not being a very popular view during the 17th and 18th centuries, began to gain a foothold during the 19th and 20th centuries, partly due to the emergence of statistics and probability, and to the formulation of quantum theory (Hacking, 1990; Faye, 2024). Modern indeterminist thinkers include Charles S. Peirce (1839–1914), who coined the term *tychism* (from Greek *tychē*, meaning chance or luck) to refer to the view that chance is an active force in the universe (see Hacking, 1990: chap. 23), and Karl Popper (1902–1994), with his idea of an open universe (Popper, 1985). Others, like Jacques Monod (1910–1976), adopted intermediate views between both camps (Monod, 1972).

book, many researchers have adopted an empirical approach to this debate, trying to provide evidence for the contrasting predictions that derive from each perspective (Blount *et al.*, 2018). This evidence comes from various sources. One prominent source is laboratory evolution experiments, where equivalent populations are introduced into enclosures with known ecological properties, and their changes over time are monitored. An example of this is Richard Lenski's *E. coli* long-term evolution experiment (Lenski & Travisano, 1994; Blount *et al.*, 2018). Another source is natural evolution experiments, where researchers take advantage of specific evolutionary trajectories that are particularly suited to addressing questions about contingency and determinism in evolution. These include adaptive radiations, where a species rapidly diversifies into several different species to fill the numerous empty ecological niches that often result from an extinction or colonization event (Schluter, 2000; Grant & Grant, 2008). Notable examples of this are the anole lizards from the Caribbean islands (Losos, 2009), the cichlid fishes

from the Great African Lakes (Muschick, Indermaur, & Salzburger, 2012), or the threespine sticklebacks from the coastal lakes of British Columbia (Rundle *et al.*, 2000). Contemporary investigations often favor mixed stances, like, for instance, the idea that determinism might be the rule within low-level taxa, while contingency prevails in high-level taxa (Blount *et al.*, 2018).

In these investigations, evolutionary convergence, or parallel evolution, is considered an indicator of determinism, while molecular or morphological disparity suggests contingency. A major force leading to convergence is natural selection (Blount *et al.*, 2018): if selective pressures are similar and intense, populations can evolve analogous solutions. Indeed, the very presence of similar adaptations in related or unrelated species living in comparable environments is often seen as indirect evidence for natural selection (Endler, 1986). Another potential cause of evolutionary convergence is developmental bias (Laland *et al.*, 2015), though its relative significance remains disputed (see Gefaell & Rolán-Alvarez, 2024). Conversely, genetic drift is a major source of contingency (Blount *et al.*, 2018), as seen in phenomena like the founder effect or population bottlenecks (Futuyma & Kirkpatrick, 2017). Genetic drift can fix random mutations, thereby contributing to stochastic species divergence. Thus, a useful way to frame the chance versus determinism debate is as a struggle between genetic drift and natural selection: genetic drift would favor random divergence, while selection on similar environments would cause convergence. Though this conception might diminish some of the problem's metaphysical allure, it makes it much more empirically tractable than alternative formulations.

A phenomenon in which the interplay between selection and drift –plus migration– might be particularly evident is the repeated evolution of clines. As we've discussed throughout this dissertation, *L. saxatilis* from the Rías Baixas putatively exhibits clinal variation from its inner to its outer zones. This phenomenon was first described by early malacologists in the latter half of the 20th century but has only recently been systematically studied, with most efforts focusing on the Ría de Vigo. For example, in Chapter 4 of this dissertation, corresponding to Gefaell *et al.* (2024), it is shown that in the Ría de Vigo, the cline has remained relatively stable over the past forty years. Furthermore, Gefaell *et al.* (2024) provide suggestive evidence that clinal variation in the *fulva* and *lineata* morphs of *L. saxatilis* likely depends on a series of ecological parameters reflecting the degree of wave exposure in different localities. Likewise, in Chapter 5 a possible mechanism underlying the *lineata* monomorphic populations of the Ría de Vigo (selection for crypsis) is tested, with suggestive but inconclusive results.

Despite these advances, very little is known about whether this clinal pattern holds in the remaining rias, as these have been much less studied. Based on their geomorphology, the Rías Baixas provide a potential natural evolution experiment for parallel or convergent evolution. They

all have a recent origin, were gradually formed by inundation (favoring steady colonization), and have relatively similar shapes. This might result in analogous distributions of ecosystem types (wave-exposed vs. wave-sheltered). Additionally, *L. saxatilis* is an exceptional organism to study evolutionary convergence, as cases of parallel evolution have already been documented in this species (e.g., the *Wave* and *Crab* ecotypes; Butlin *et al.*, 2014). Its ovoviviparous mode of reproduction, remarkable phenotypic diversity (likely dependent on standing genetic variation), and high number of chromosomal inversions (see Koch *et al.*, 2021, 2022) probably contribute to the evolution of repeated phenotypes in response to recurrent environmental conditions.

With that in mind, the present chapter describes a fraction of an ongoing project aimed at describing in detail the clinal variation of *L. saxatilis* from the Rías Baixas. For this project, over 3000 snails were collected from 55 localities across the four rias, many of which had been sampled in previous years, allowing for inter-annual comparisons (not presented here). Detailed information about the geophysical and ecological characteristics of these localities was also gathered. Measurements of several snail traits were recorded to test hypotheses about patterns of color variation and their relationship to environmental factors. This information is ultimately expected to shed light on whether the color clinal pattern of *L. saxatilis* from the Rías Baixas has evolved in parallel, and thus, whether this case corroborates or runs counter to the deterministic view of evolution. If analogous *lineata* and *fulva* morphs exist along the wave-exposed and wave-sheltered localities of the four rias, this would suggest that this cline has evolved in parallel, given what we know about the dispersal patterns of *L. saxatilis* and the historical geomorphology of the Rías Baixas.

The material presented here constitutes the very first steps in this project, which involve four main objectives: first, providing precise information about the rocky intertidal ecosystems of the Rías Baixas to determine to what extent these can be considered equivalent in their geophysical and ecological characteristics. As the philosopher of biology Elizabeth Lloyd has pointed out (Lloyd, 1987), much of the work in evolutionary biology consists of verifying tacit assumptions in evolutionary models. In our case, a key assumption is that the wave-sheltered and wave-exposed localities of the different rias are analogous in terms of species composition and overall ecological characteristics. Only if this is the case can the variation in color in *L. saxatilis* from the four rias be meaningfully compared. The second objective involves determining whether the pattern of color clinal variation described in the Ría de Vigo by Gefaell *et al*. (2024) also holds in the remaining three rias. The third objective, in turn, consists of establishing to what extent variation in other shell traits follows a similar clinal pattern to that of color. This can provide clues as to whether color variation is ultimately due to (parallel) ecotypic variation. Related to the

previous objective, this chapter aims to establish the extent of covariation between shell color and other fitness-important shell traits, in order to generate hypotheses about the selective regime to which shell color is subjected, as well as about its genetic basis. To provide insights on these topics, only population-level analyses are presented here, as a basis for more advanced individual-level analyses.

Based on the early descriptions of color variation in the Galician population of *L. saxatilis* and our informal observations, we hypothesize that the different localities of the Rías Baixas are analogous in terms of their ecological characteristics, which mainly depend on their degree of wave exposure; that the frequencies of the two most common color morphs, *fulva* and *lineata*, are predicted by wave-sheltered and wave-exposed ecosystems, respectively, and that at least shell shape and sculpture follow equivalent clines to that of color. Regarding the extent of covariation between shell traits and shell color, associations between them are expected, but no specific conjectures were made about their intensity.

Materials and methods

Localities studied

To describe the color cline of *L. saxatilis* from the four Rías Baixas (from south to north: Ría de Vigo, Ría de Pontevedra, Ría de Arousa, and Ría de Muros-Noia), a total of 55 rocky intertidal localities (13 in Vigo, 13 in Pontevedra, 17 in Arousa, and 12 in Muros-Noia) were surveyed during spring 2023 (Figure 1B). These localities are situated at varying distances from the mouths of the main rivers that flow into each Ría (Verdugo in Vigo, Lérez in Pontevedra, Ulla in Arousa, and Tambre in Muros-Noia), with an average distance between them of 9380 ± 6473 m. All these localities had been previously sampled by our group in 2021 or 2022, and those of Vigo and Arousa were also studied by early researchers in the 1970s and 1980s (Sacchi, 1979; Torelli, 1983). For the Ría de Vigo, we selected a subset of 13 localities (6 north, 7 south) from the original 37 localities sampled by Sacchi (1979) and resampled in 2022 by Gefaell *et al.* (2024). This subset of localities was chosen based on their overall representativeness of the clinal pattern, their density of snails, and their proximity to an urban area (to facilitate the samplings). The selection of 6 localities on the north and 7 on the south was due to the larger size of the southern coast of the Ría de Vigo. In Arousa, we resurveyed all the original 22 localities sampled by Alberto Torelli in 1978 (Torelli, 1983). Of these, we only found snails in 17 localities, which nonetheless cover the entire ria. For the remaining two rias (Pontevedra and Muros-Noia), we used Vigo as a template, sampling a total of 25 localities (13 in Pontevedra and 12 in Muros-Noia). As the northern coast of Pontevedra is much larger than the southern coast, we surveyed one location more in the north

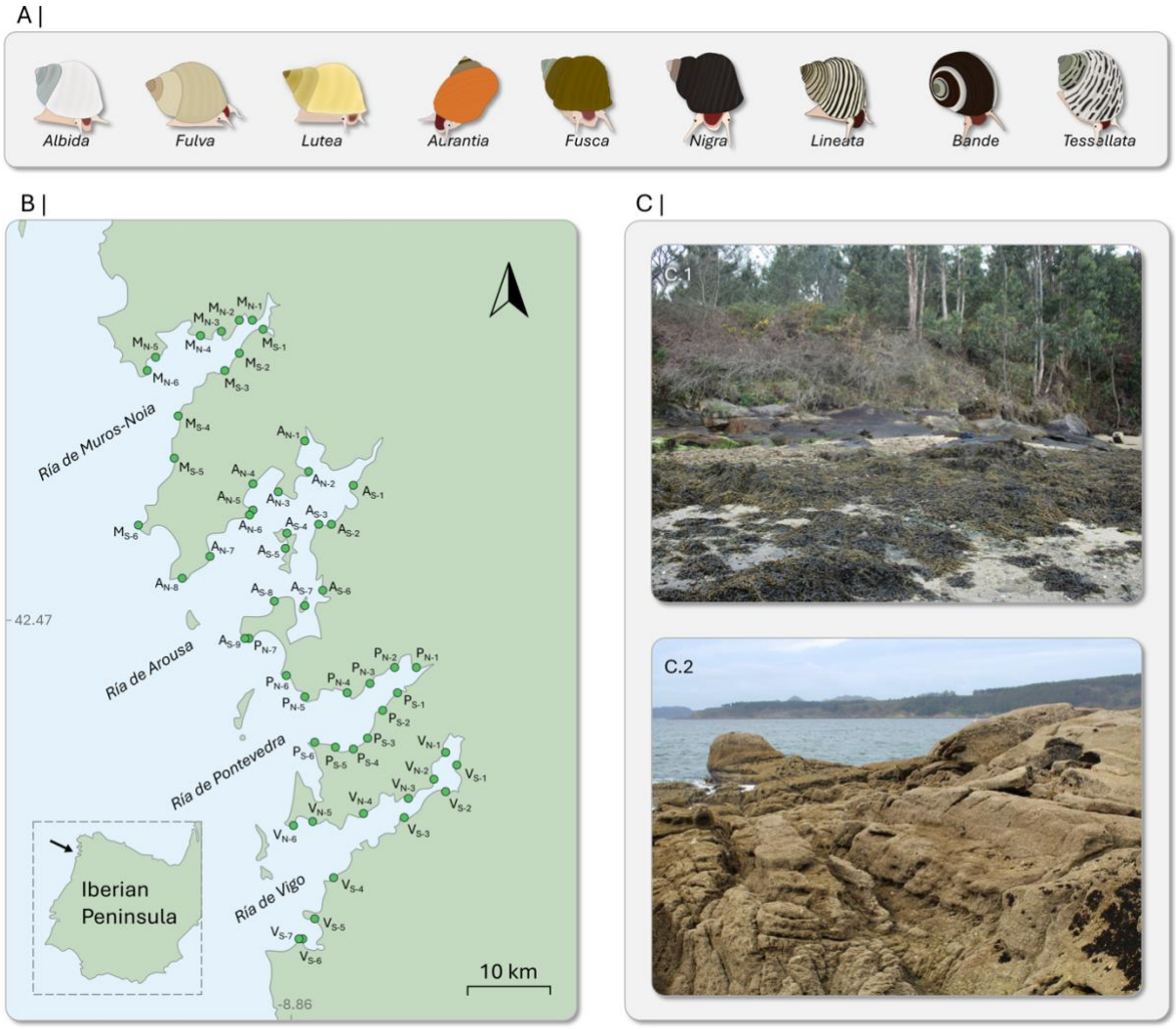

Figure 1. Localities and color morphs studied to describe the clinal pattern of *L. saxatilis* from the Rías Baixas. A: Main color morphs shown by *L. saxatilis* from this area. B: Localities studied in each ria, with their corresponding codes (see Supplementary materials, Table S1 for further information). C: Examples of two localities sampled. C.1: Punta Cabalo (V_{N-1}), a paradigmatic wave-sheltered locality. C.2: Punta Igreixiña (V_{N-5}), a representative wave-exposed locality.

than in the south. A complete list of the localities sampled, their corresponding GPS coordinates, and the number of snails collected in each can be found in Supplementary materials, Table S1.

Geophysical and ecological surveys

The localities sampled were geophysically and ecologically characterized to test the equivalence of their ecosystems. Geophysical data were collected in situ or taken from online repositories and included the distance from the river mouth, the type and size of the rocks forming the ecosystem, wave action, salinity, and temperature of the rocks where *L. saxatilis* settles. The distance from the mouth of each ria's main river to the localities was obtained using Google Earth Pro (v. 7.3.6.9285), tracing a route along the coastline following a pre-established protocol. Rock types were assessed in situ and confirmed using regional geological maps. Rock

size was semi-quantitatively evaluated using an ordinal scale from very small rocks (Ø < 20 cm) to bedrock surfaces. Wave action was determined by averaging the maximum and minimum significant wave heights (Hs, in meters) estimated for each locality based on online data from MeteoGalicia (the Galician Weather Agency). Salinity (grams per liter or kg of water, denoted ‰) was measured by averaging two in situ recordings taken during the sampling sessions using a refractometer. To obtain spatially and temporally accurate data on the thermal profile of the localities, a total of 12 loggers (Electric Blue Envloggers) were installed in 12 out of 13 localities from the Ría de Vigo on the same rocks where the *L. saxatilis* snails were sampled. The loggers were installed during the summer of 2023 and were left running for an entire year, recording temperature measurements every 30 min (data not presented here).

Data on key rocky intertidal algae, lichen, and animal species were also collected during the samplings to characterize the degree of wave exposure of the localities at an ecological level (see Figure 1C for typical wave-sheltered and wave-exposed ecosystems). The survey protocol adapted the Ballantine scale (Ballantine, 1961) to the ecosystems of the Rías Baixas, with feedback from Gefaell *et al.* (2024), Burrows, Harvey, & Robb (2008), and various local intertidal biota guides (e.g., Rolán & Otero-Schmitt, 1996). The Ballantine scale is a semi-quantitative instrument measuring the species composition of rocky intertidal ecosystems with varying degrees of wave exposure. Our adaptation included both binary (presence/absence) and ordinal (relative abundance) items. Data from Gefaell *et al.* (2024) and local guides assisted in identifying species that are particularly indicative of the localities' degree of wave exposure. Burrows *et al.* (2008) helped include additional variables, incorporating species that correlated > 0.4 with a principal component summarizing wave exposure. This resulted in a list of 13 species, including the brown algae *Ascophyllum nudosum*, the lichen *Lichina pygmaea*, and the animals *Mytilus galloprovincialis*, and *Phorcus lineatus*, among others (see complete list and definitions in Supplementary materials, Table S2). This list was heavily biased towards gastropods to eventually test ecological character displacement in *L. saxatilis* (Grahame & Mill, 1989; Losos, 2000; analyses not presented here).

L. saxatilis sample collection

A sample of *L. saxatilis* from each of the 55 surveyed localities was collected following a stratified sampling method inspired by Gefaell *et al.* (2024). First, the different intertidal zones of the localities were identified based on species composition. Once at the supralittoral zone where *L. saxatilis* settles, and having located individuals of this species, random samples were taken using a 20 cm-diameter sampling quadrat. Replicates were taken as needed to reach a sample

size of between 40-70 snails, depending on the observed or expected degree of color polymorphism (around 40 for monomorphic and 50-70 for polymorphic populations). The sample range was chosen to prevent compromising the viability of the populations in the mid to long term. Juvenile specimens (approximate shell length < 4 mm) were avoided, as shell color is not fully developed at this ontogenetic stage (Johannesson, Johannesson, & Rolán-Alvarez, 1993). This yielded a total sample size of n = 3085 snails (56 ± 7.3 individuals per population). The snails were taken alive to the laboratory facilities of the University of Vigo in Falcon tubes and euthanized using cold anesthesia. Although *L. saxatilis* is not an endangered species and has relatively dense populations in most intertidal ecosystems of the Rías Baixas, we obtained specific sampling permissions from the regional authority (Consellería do Mar, Xunta de Galicia).

Finally, the density of *L. saxatilis* in each locality was assessed in situ by counting the number of snails within a 60-diameter sampling quadrat in three random samples. The sampling quadrat used for this estimate was larger than that used for snail collection to avoid aggregation biases. In some low-abundance localities during the central hours of the day, *L. saxatilis* tends to cluster in small crevices, which, if measured without considering the whole potential area of distribution, might lead to overestimating its abundance. Density measurements were collected to investigate their relationship with color diversity, as a way of indirectly assessing the potential role of drift in color morph maintenance –under a neutral evolution scenario, higher color diversity would be expected in populations with greater abundance.

Sample processing, color assignment, and measurement of shell traits

Before processing, the stored snail samples were thawed and immediately rinsed in a 50% water-50% bleach solution for 5 minutes to remove microalgae and other potential pollutants from the shells' surface. Snails from particularly dirty populations were rinsed for 7-8 minutes. The snails were then color-assigned following the protocol and color categories used in Gefaell *et al.* (2024). These color categories, originally from Dautzenberg & Fischer (1912) as provided by Sacchi (1979), include the following classes (see Figure 1A): *albida* (white), *fulva* (a wide range from greyish-hazel to fawn, yellow straw, and light brown), *lutea* (yellow, varying in intensity) *aurantia* (orange), *fusca* (brown), *nigra* (black), *bande* (defined by two or three large bands darker than the background color, usually black or orange), *tesellata* (checkered or reticulated), and *lineata* (with lines in the grooves of the sculpture and a white to creamish background color). A tenth color category, *sanguinea* (blood red) was excluded due to its absence in the Rías Baixas. A series of HEX codes were ascribed to these categories to aid the assignment

process. Color assignment was performed by a single experienced individual (JG), using practical guidelines to ensure consistency and avoid inter-subject reliability issues.

Additionally, a series of key shell trait parameters were recorded in each snail: shell sculpture, shell length, shell thickness, shell shape, shell wear, and the presence of scars. These were classified into two groups: characters with a putatively strong genetic basis (shell sculpture, shell length, shell thickness, and shell shape) and characters reflecting the ecological conditions in which the snails live (shell wear and scars). The sculpture of the shells was ordinally characterized by ranking their smoothness/ribbedness. For this, three categories from Reid (1996) were used: *rudis*, *rudissima*, and *jugosa*, with *rudis* indicating smooth shells, *rudissima* intermediate shells, and *jugosa* strongly ribbed shells (see Supplementary materials, Figure S1A for a visual guide to these categories). Shell length was measured as the shell columella length to the nearest 0.1 mm with a vernier caliper. Shell thickness was measured to the nearest 0.01 mm using an electronic gauge placed at the lip of the shell, following Chapter 5 of this dissertation. Shell shape was quantified using 2D geometric morphometrics, also following the protocol detailed in Chapter 5. In the total sample, the first five relative warps cumulatively explained 84.34% of the variance in landmark positioning. Of these, relative warp 1, accounting for 39.06% of the variance, has been traditionally interpreted in *L. saxatilis* studies as the aperture size and roundness of the shell (Gefaell, Varela, & Rolán-Alvarez, 2020; see Chapter 5), so was preferentially used as an indicator of overall shell shape. Lastly, shell scars and wear were coded as presence/absence variables (Supplementary materials, Figure S1B-C). Scars, defined as abrupt breaks in the shell's growth line followed by new growth lines, indicate past unsuccessful crab predation attempts (Butlin *et al.*, 2014; Stafford, Tyler, & Leighton, 2015), making them crucial for assessing the intensity of crab predation of morphs and populations. Shell wear, in turn, might reflect harsh environmental conditions such as increased salinity levels, more intense solar exposure, or higher wave action.

Statistical analyses

All analyses were performed in R version 4.3.3 (R Core Team, 2023). For the present chapter, only population-level analyses were conducted. The geophysical and ecological parameters of each ria were compared to evaluate their equivalence. Principal Component Analysis (PCA) was used to summarize the various ecological (i.e., biotic) variables recorded, and the resulting first principal component was examined for ecological significance. Since PCA assumes that the variables analyzed are both continuous and normally distributed –conditions which our data do not fulfill– a Multiple Correspondence Analysis (MCA) was also performed to corroborate the PCA results. Although according to some authors, violation of PCA assumptions

is legitimate if this analysis aims at exploring data rather than testing specific hypotheses (Dytham, 2011), a corroboration through MCA provides a more robust warrant. For MCA, only binary (i.e., presence vs. absence) variables were used. Those of ordinal nature were transformed into binary variables (coded as 'presence' if ≥1). For barnacles, 'presence' was coded as having > 2 values on a 0 to 5 scale rather than ≥1. This adjustment was made because low densities of barnacles are found in virtually every locality, so only higher coverages would allow for a meaningful distinction between localities. PCA was also used to reduce the dimensionality of the recorded shell traits to study their covariation with shell color. Dimensionality reduction analyses were performed using the packages *FactoMineR* and *factoextra* (Lê, Josse, & Husson, 2008; Kassambara & Mundt, 2020).

The variation of the first ecological principal component (PC) and other geophysical parameters along the coastline was studied using the distance from the river mouth as a proxy for the localities' position. This provides information on the extent to which the geophysical and ecological factors follow a clinal trend from the inner to the outer zones of the rias. Given that the rias differ in size (with Arousa being larger than the other three), the distance values were normalized by dividing them by the total shoreline length per coast (north or south). For this, the island localities ($A_{S\text{-}4}$, $A_{S\text{-}5}$) were excluded. Clinal variation in the first ecological PC and wave action was also assessed using Moran's I autocorrelation index per ria, which measures spatial clustering of parameter values (Fox, Negrete-Yankelevich, & Sosa, 2015). This index ranges from -1 to 1: if spatially close localities tend to share similar values (indicating spatial clustering), Moran's I will be biased towards positive values; if, conversely, close localities have dissimilar values (spatial anti-correlation), it will lean towards negative values. Values around 0 indicate spatially random dispersion. Moran's I was calculated using the package *spdep* (Bivand & Piras, 2015). For count data (i.e., rock types), Fisher's exact test was used to evaluate differences between rias (Sokal & Rohlf, 1995).

Based on the ordination of ecological PC1 values, the localities were categorized into three ecosystem types: the first 33% (first tertile) were labeled as 'wave-sheltered', the second 33% as 'wave-intermediate', and the third 33% as 'wave-exposed'. Tertiles rather than units of standard deviation from the mean were used to partition the data, as quantiles are less sensitive to asymmetries in the distribution of values. Species composition differences between ecosystem types were evaluated using the Sorensen Index (Sorensen, 1948) and Permutational Multivariate Analysis of Variance (PERMANOVA; Anderson, 2017) using the *vegan* package (Oksanen, 2010). The Sorensen index measures the ratio between the number of species present in all kinds of ecosystems compared and the sum of those found in only one of them (conferring

double weight to shared species). This results in an index ranging between 0 and 1, with 1 indicating total concordance between the species composition of the ecosystems compared and 0 denoting total discordance. PERMANOVA, on the other hand, is a non-parametric multivariate technique that compares distance matrices quantifying community differences between ecosystem types. These distance matrices are calculated using various similarity or dissimilarity indices (such as Bray-Curtis or Jaccard) between localities assigned to the different ecosystem types. PERMANOVA quantifies the amount of variance in the distance matrix attributable to the factor of interest (e.g., ecosystem type) and tests the null hypothesis that the centroids (multivariate means) of the groups defined by the factor are the same in the multivariate space (Anderson, 2017). If the null hypothesis is rejected, there are systematic differences in the community composition of the ecosystem types studied. PERMANOVA, as well as Kruskal Wallis or ANOVA tests, were also used to study the distribution of color morphs and other variables across ecosystem types.

Once the ecological cline was characterized, clinal trends in shell color and other snail characters were also studied. Correlations between morphs were studied using Spearman's rank correlations. Most analyses conducted, however, used only the *fulva* and *lineata* morphs, as these are the most common shell colors along the cline (Gefaell *et al.*, 2024). Shell trait parameters were summarized per population sampled using counts for categorical variables (such as color morphs or shell sculpture types) and mean ± SD for quantitative variables (such as shell thickness or length). Counts, in turn, were on many occasions converted to frequency data, thus allowing more direct comparisons between localities and rias.

Various statistical tests were used to model the relationship between geophysical, ecological, and snail population variables, depending on the assumptions of the data. Linear models (LMs) were used for normally distributed and homoscedastic variables and residuals, and their fit was assessed with the adjusted R^2 coefficient of determination. For data violating these assumptions, generalized linear models (GLMs) with appropriate link functions and error distributions were employed (Fox *et al.*, 2015). For frequency data, quasi-binomial GLMs, which are specifically designed for proportion data or data with overdispersion, were used. Quasi-binomial GLMs allow the fitting of sigmoid-like curves to the data, which might better reflect the clinal variation in color morphs from the Rías Baixas. In quasi-binomial GLMs, the goodness-of-fit of the model was assessed using McFadden's pseudo-R^2 (Darlington & Hayes, 2017). To account for the variable 'ria' as a random effect in some models, linear mixed models (LMMs) were used. Linear models and their alternatives were fitted using the packages *lm4* (Bates *et al.*, 2015) and *fitdistrplus* (Delignette-Muller & Dutang, 2015) to check distribution fit. When different

models were compared, model selection was based on Akaike's Information Criterion (AIC) and analogous criteria. Assumptions were checked through various diagnostic plots and statistical techniques, including the Shapiro-Wilk test to assess normality, the Non-Constant Variance (NCV) test to study the homoscedasticity of the residuals, or the Variance Inflation Factor (VIF) to evaluate multicollinearity.

To study covariation, only shell traits with a presumed strong genetic basis were used, since including traits that reflect the ecological conditions of the snails might provide misleading insights into their underlying genetic basis. First, a PCA on these shell traits was performed, and the first principal component was studied for a biologically meaningful interpretation. This first principal component was then used as a predictor for *fulva* or *lineata* frequency in linear regressions. The residuals from these models were extracted and used as dependent variables in a second series of linear regressions, with the first ecological PC as the predictor. This sequential approach aimed to assess the extent of covariation between shell traits and shell color and to determine whether there is variation in shell color that is independent of variation in shell traits (i.e., 'free' color variation) but still associated with ecological parameters. Finally, the shell color diversity of populations was studied using the Shannon diversity index (*H*'), which measures the number of morphs and their relative frequencies at a given locality: higher numbers of color morphs and more even frequencies result in higher *H*' values (see Gefaell *et al.*, 2024 and references therein).

Results

Ecological characterization of localities

Despite slight differences mostly driven by the geomorphological peculiarities of each ria, the four Rías Baixas can be considered analogous in their geophysical parameters, with wave action increasing gradually from the inner to the outer parts (See Supplementary materials, *Extended Results (1)*, Figures S2, S4–S7). This allows us to use the distance of a locality from the river mouth as a proxy for its degree of wave exposure. Regarding the biotic parameters, the PCA on ecological variables yielded a PC1 explaining 49.2% of the variance, with PC2 explaining 13.3% (see scree plot in Supplementary materials, Figure S8). A plot of sites by PC1 and PC2 values shows a clear clustering of localities across PC1, with wave-sheltered localities on the left (negative values), and wave-exposed localities on the right (positive values; Supplementary materials, Figure S9). Similarly, a biplot including the vectors of each analyzed variable reveals two clearly distinguishable groups along PC1 values: one corresponding to species characteristic of wave-sheltered areas and another one to species typical of wave-exposed zones (Supplement-

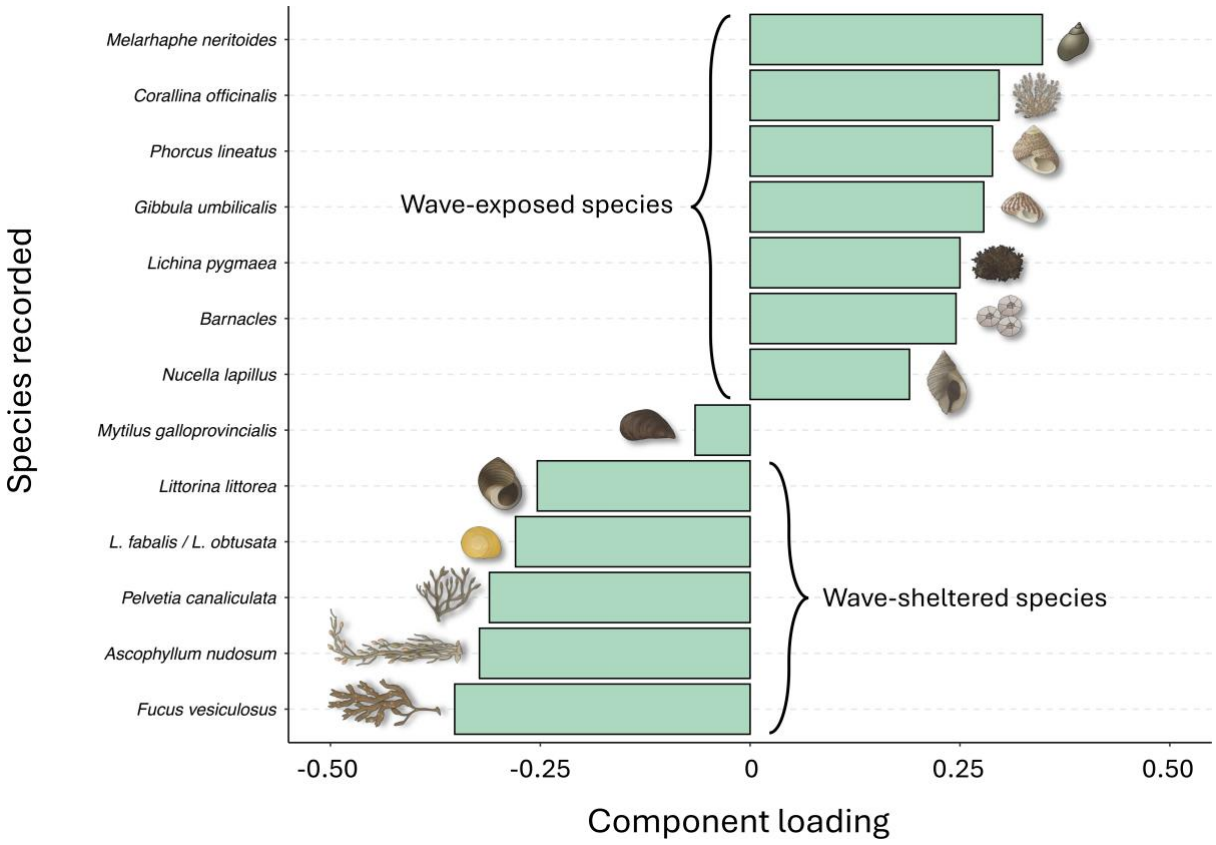

Figure 2. Plot of component loadings of each variable for PC1. Two clearly distinguishable clusters are observed: one comprising organisms characteristic of wave-sheltered environments, with the brown algae species *Fucus vesiculosus* and *Ascophyllum nudosum* standing out, and another consisting of species typical of wave-exposed zones, with the marine snail *Melarhaphe neritoides* on top.

ary materials, Figure S10). This is supported by the component loadings for PC1 (Figure 2). All these findings suggest that PC1 tracks the degree of wave exposure of a locality based on the presence and relative abundance of various animal and algae species, a result corroborated by the MCA (see Supplementary materials Figures S11–S13). More specifically, low PC1 values correspond to wave-sheltered localities, and high PC1 values to wave-exposed localities. Thus, PC1 is henceforth referred to as 'Ecological PC1'.

Ecological differences between rias

Pontevedra exhibits much less dispersion in Ecological PC1 values compared to the other rias, with two outlier wave-sheltered localities (low Ecological PC1 values) accompanied by several wave-exposed localities (high Ecological PC1 values) (Supplementary materials, Figure S14). A Kruskal-Wallis test indicates that the differences between rias in Ecological PC1 values are nearly statistically significant ($H(3) = 6.684$, $P = 0.08268$). Non-parametric effect size, however, suggests a small to moderate effect of rias on Ecological PC1 values ($\eta^2_H = 0.072$), implying that despite their differences, the rias can be considered ecologically equivalent. The

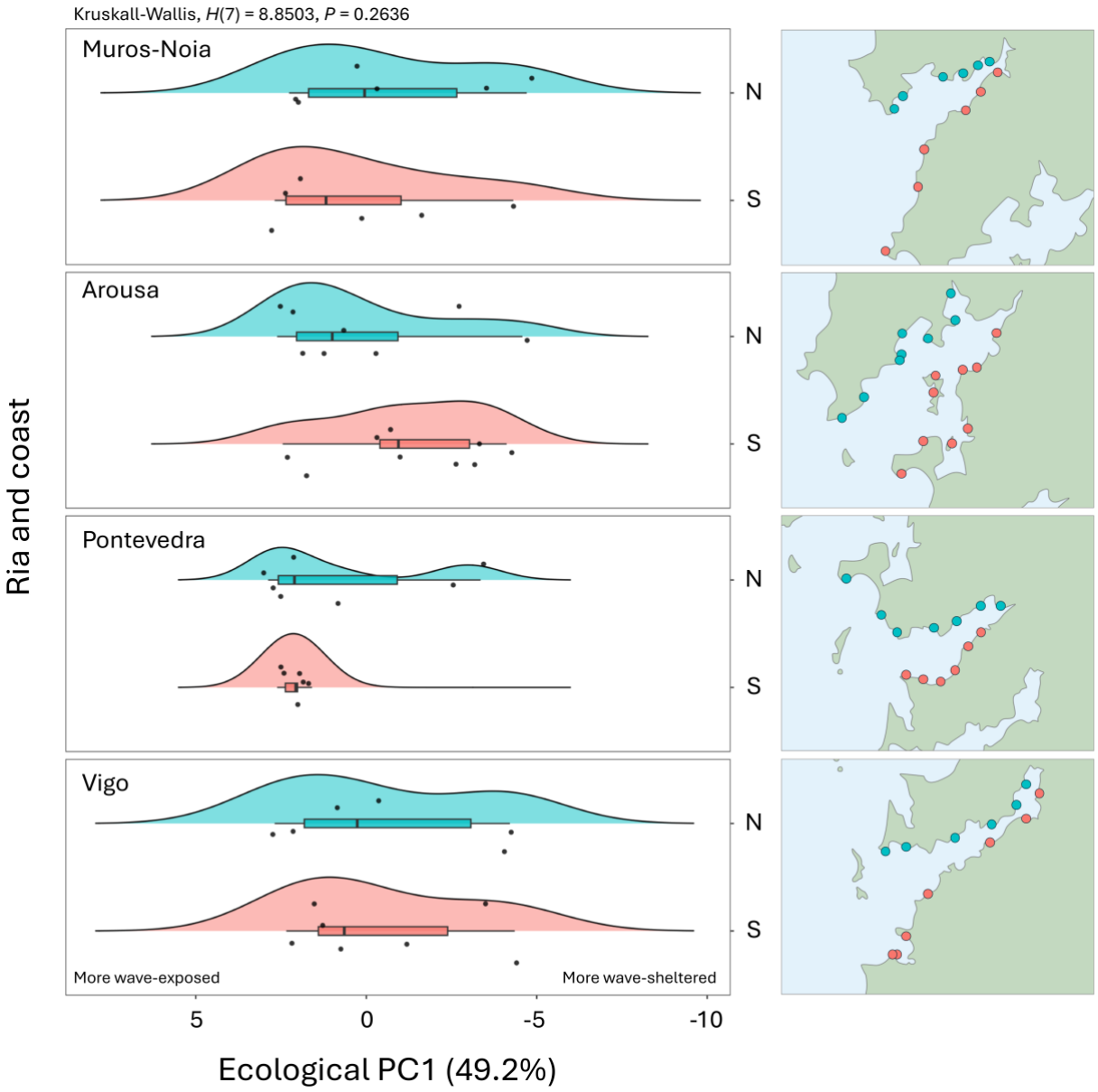

Figure 3. Boxplots and associated density plots of Ecological PC1 per coast (north –N– and south –S–) within each ria, with the localities studied in each of them (dots). Note that the x-axis values have been reversed to agree with the geographic pattern in the rias, where more wave-exposed localities are located to the west (left).

anomaly in the Ría de Pontevedra is caused by its southern coast, which lacks wave-sheltered localities (Figure 3). This is likely due to the city of Pontevedra, situated on the inner southern coast (see Supplementary materials, Figure S3). Additionally, it is worth noting that the southern coast of Arousa shows the lowest Ecological PC1 values, probably due to the influence of the Arousa island and the O Grove Peninsula (Supplementary materials, Figure S3). These peculiarities, however, are not large enough to yield statistically significant differences between coasts (Kruskal-Wallis, $H(7) = 8.8503$, $P = 0.2636$). Nevertheless, caution is advised due to the low sample size per coast.

Species composition differences between ecosystem types

A PERMANOVA using Bray-Curtis dissimilarity to create the distance matrix yielded a significant effect of ecosystem type on community composition (F = 56.498, P = 0.001), explaining around 68.5% of the variance (R^2 = 0.68484). When the same test was performed using the rias (Vigo, Pontevedra, Arousa, and Muros-Noia) as the independent variable instead of ecosystem type, no significant differences were found (F = 1.4935, P = 0.203, R^2 = 0.08076). Although this might suggest an effect of ecosystem type on community composition, caution is warranted, as a graphical representation of species composition using a Non-metric Multidimensional Scaling (NMDS) plot revealed differences in dispersion around the centroids between ecosystem types (Figure 4). Wave-exposed ecosystems exhibit much less dispersion than their wave-sheltered and wave-intermediate counterparts. An ANOVA on dispersion values between groups yielded statistically significant results (F = 21.482, P = 1.583e-07). As PERMANOVA is very sensitive to variation in dispersion around the centroids, it cannot be ruled out that this had also affected our previous statistically significant results. Beyond that, the NMDS plot indicates that in terms of species composition, wave-intermediate localities are closer to wave-exposed ecosystems than to wave-sheltered ones (as revealed by NMDS1).

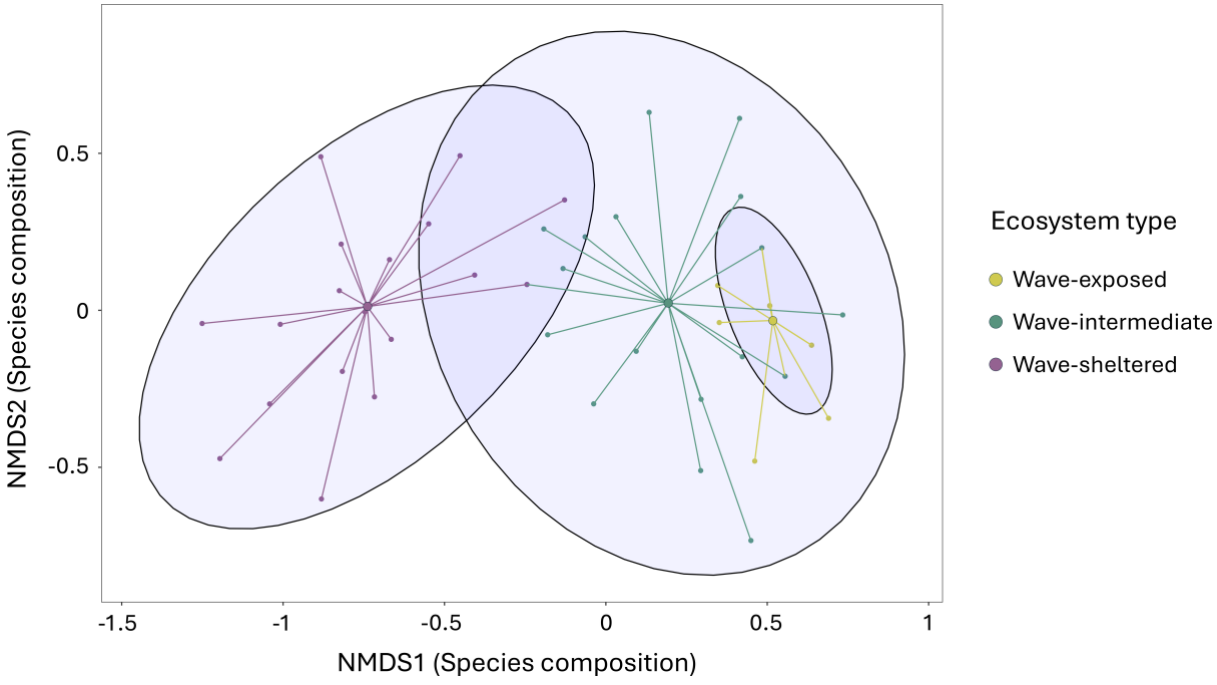

Figure 4. Non-metric Multidimensional Scaling (NMDS) plot of species composition across different ecosystem types (wave-sheltered, wave-intermediate, and wave-exposed). The largest outlined dot represents the centroid of each group. This plot illustrates how species composition varies for each locality compared to their corresponding group centroid.

With the rias pooled, the Sorensen index between wave-sheltered (bottom Ecological PC1 tertile) and wave-exposed (top Ecological PC1 tertile) localities yielded a value of 0.87, indicating that, in terms of species composition, both kinds of ecosystems are quite similar. Interestingly, the only species that differed between wave-sheltered and wave-exposed environments were the algae *Ascophyllum nudosum* and *Fucus vesiculosus*, and the gastropod *Littorina fabalis* or *Littorina obtusata* (as it is not possible to distinguish between them in situ), all three typical of wave-sheltered areas. When the Sorensen index was calculated per ria, subtle differences resulted, with Pontevedra and Arousa showing lower values (i.e., higher differentiation) than Vigo and Muros-Noia (See Supplementary materials, Figure S15).

Ecological PC1 along the space

A linear model using the distance from the river mouth as a predictor of Ecological PC1 was fitted to evaluate how this parameter varies along the space. The resulting model was statistically significant (F = 70.66, P = 3.387e-11), and explained around 57% of the variance (adjusted R^2 = 0.5726). A linear-mixed effect model using the distance from the river, wave action, and in situ salinity as predictors of Ecological PC1, and ria as a random factor, yielded a better model fit than the equivalent models without salinity on the one hand, and without salinity and wave action on the other (REML criterion = 183.4 < 191.2 < 195.5, respectively). For our purposes, however, the distance from the river mouth alone appears to be a good predictor of change in Ecological PC1 along the rias.

Despite the overall association between Ecological PC1 and the distance from the river mouth, substantial differences in the strength of the relationship exist between rias (Figure 5). The distance from the river mouth in Vigo explains around 86% of the variation in Ecological PC1 (adjusted R^2 = 0.8556, F = 72.1, P = 3.69e-06), approximately 44% in Pontevedra (adjusted R^2 = 0.4423, F = 10.52, P = 0.00783) and Arousa (adjusted R^2 = 0.4462, F = 12.28, P = 0.003881), and around 74% in Muros-Noia (adjusted R^2 = 0.736, F = 31.67, P = 0.0002192). The relationship between both variables is quite linear in Vigo and Arousa, but more sigmoidal in Pontevedra and Muros-Noia. While these differences might suggest using different models for each ria, the linear model has the advantage of being more generalizable than curves specifically tailored to each dataset (see Poldrack, 2023). The differences in the strength of the fit between rias and coasts are also evident when calculating Moran's I for Ecological PC1 (see Supplementary materials, Figure S16).

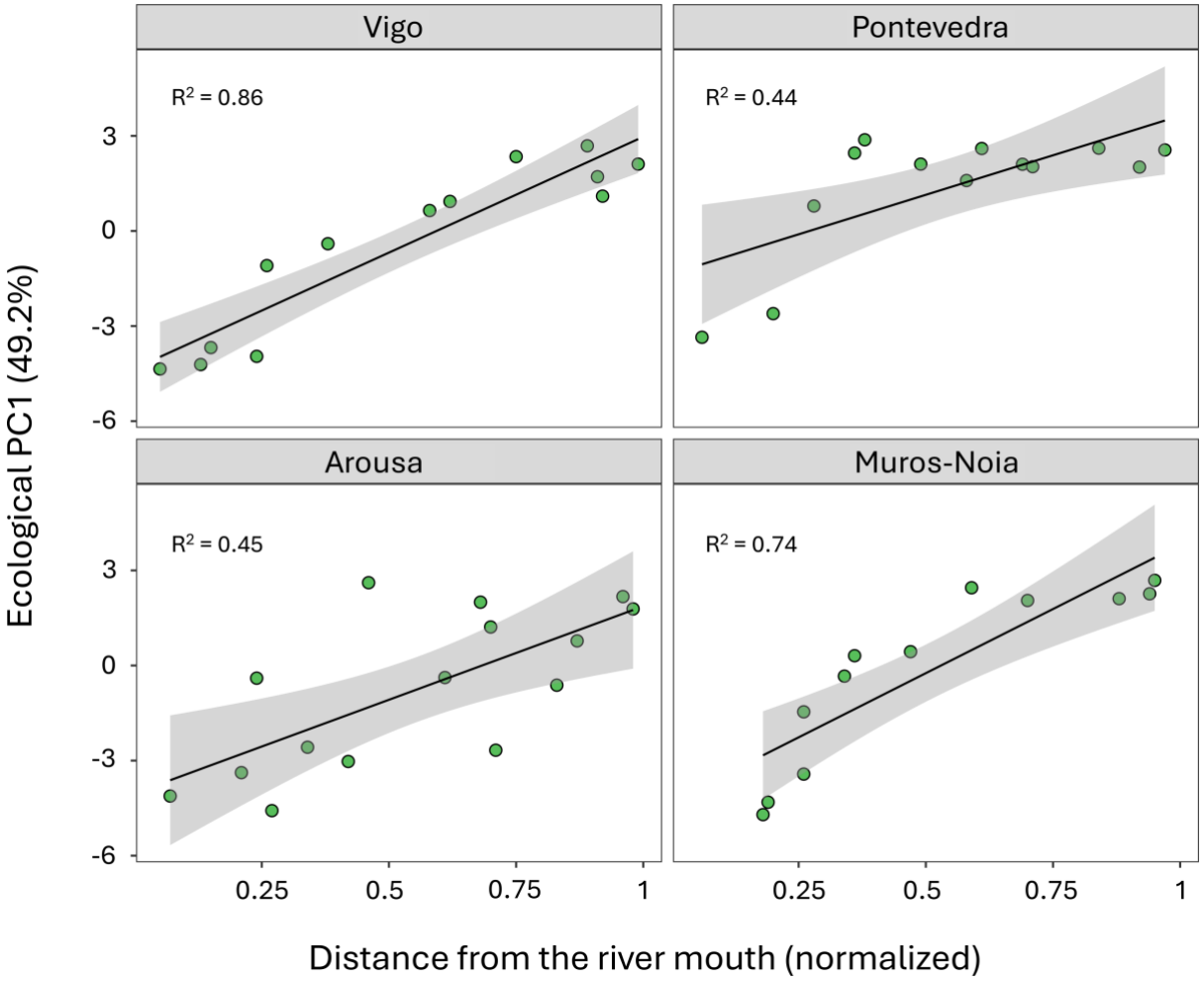

Figure 5. Scatter plot of Ecological PC1 values of each locality within each ria against the distance from the river mouth. Their least squares regression lines (with 95% confidence intervals) and corresponding R^2 values (adjusted) are also shown.

Shell coloration along the wave exposure gradient

There are differences in the total number of color morphs in the 55 localities (see Table 1). *Lineata* is the most abundant morph in all rias, followed by *fulva*. The next two most frequent morphs are *lutea* and *fusca*, which share similar percentages in Vigo and Pontevedra but differ in Arousa and Muros-Noia (where their frequencies are almost the inverse). *Lineata* shows the highest percentage in Pontevedra (63.07%) and the lowest in Arousa (34.05%). *Fulva*, conversely, follows the opposite trend.

Table 1. Total *n* and percentages (in brackets) of each morph in the four Rías Baixas. The most frequent morphs (*fulva* and *lineata*) are highlighted in bold.

	Vigo	Pontevedra	Arousa	Muros-Noia	All rias
Albida	23 (3.22)	14 (1.99)	74 (7.22)	11 (1.71)	122 (3.95)
Fulva	**148 (20.73)**	**89 (12.64)**	**283 (27.61)**	**166 (25.86)**	**686 (22.24)**
Lutea	56 (7.84)	22 (3.13)	96 (9.37)	76 (11.84)	250 (8.10)
Aurantia	5 (0.70)	61 (8.66)	53 (5.17)	7 (1.09)	126 (4.08)
Fusca	53 (7.42)	28 (3.98)	123 (12.0)	50 (7.79)	254 (8.23)
Nigra	1 (0.14)	0 (0)	17 (1.66)	0 (0)	18 (0.58)
Lineata	**386 (54.06)**	**444 (63.07)**	**349 (34.05)**	**324 (50.47)**	**1503 (48.72)**
Bande	27 (3.78)	11 (1.56)	18 (1.76)	0 (0)	56 (1.82)
Tessellata	15 (2.10)	34 (4.83)	12 (1.17)	8 (1.25)	69 (2.24)
Total	714	704	1025	642	3085

These morphs show a clear association with wave exposure values (see Figure 6). On the one hand, *fulva*, and to a lesser extent *lutea* and *fusca*, are more frequent in wave-sheltered localities. On the other hand, *lineata* is especially abundant in wave-exposed localities. This relationship holds for the four rias but is particularly evident in the Ría de Vigo and the Ría de Muros-Noia. The different distribution of color morphs across ecosystem types is confirmed by a PERMANOVA test on a morph count Bray-Curtis distance matrix. This analysis shows that ecosystem type accounts for around 63% of the variance in color morph composition (R^2 = 0.62962; F = 44.198, P < 0.001). In contrast, no statistically significant differences are found when comparing the color morph composition of the different rias (F = 1.7871, P = 0.135). However, as with our previous usage of PERMANOVA, there are differences in dispersion between ecosystem types (F = 3.7267, P = 0.03073; see also Supplementary materials, Figure S17), thus demanding caution.

Figure 6 suggests that *fulva*, *lutea*, and *fusca* tend to go hand in hand. This and other potential associations of color morphs were formally tested with Spearman's rank correlations on color morph counts, as for most morphs this variable was not normally distributed (Shapiro-Wilk normality test, W < 0.85, P < 0.001). A correlation matrix corroborates the positive association between *fulva*, *lutea*, and *fusca*, morphs which in turn exhibit a negative correlation with *lineata* (Figure 7). Additionally, significant correlations were found between *nigra* and *fusca* (positive), *albida* and *bande* (positive), and *tessellata* and *lutea* (negative).

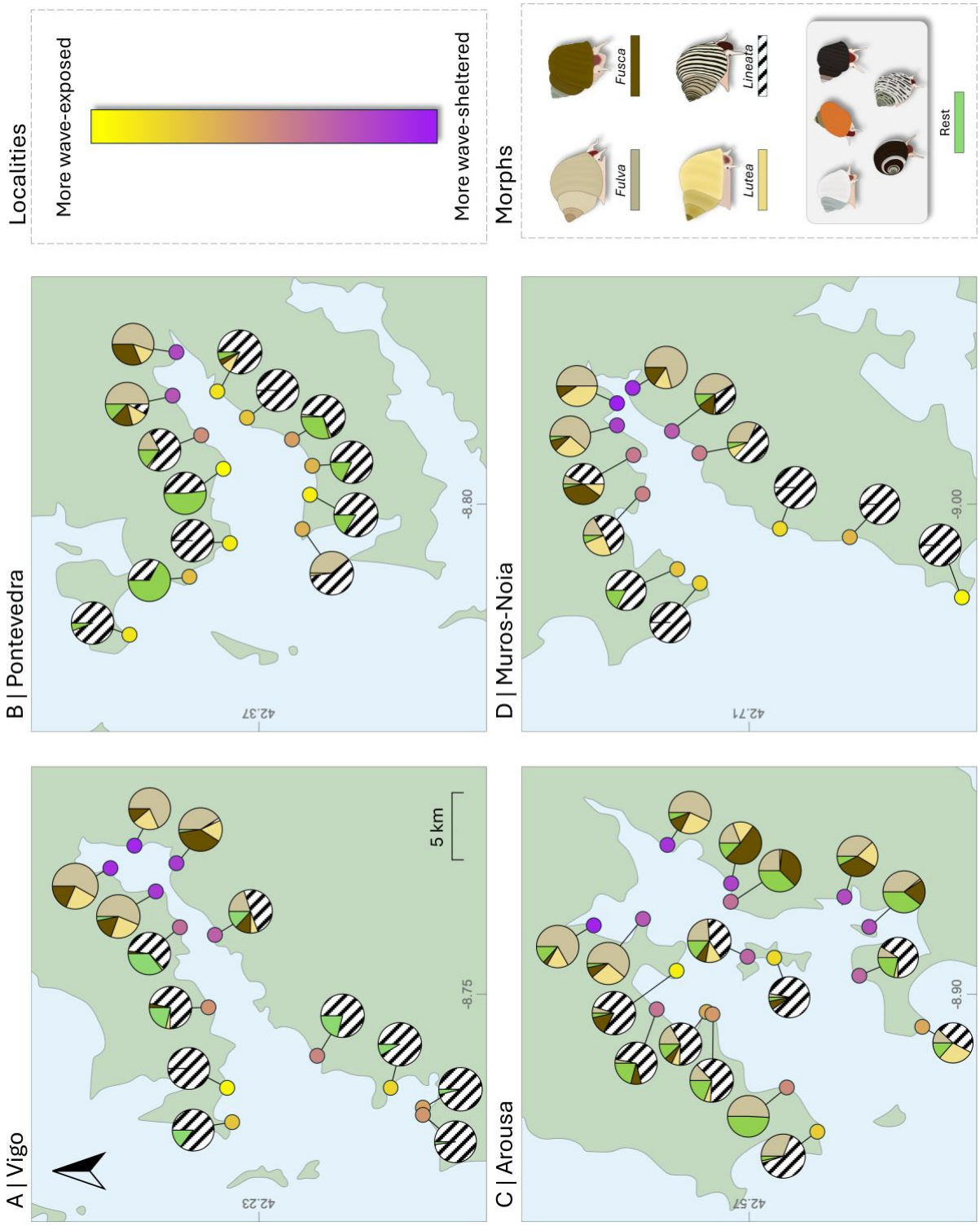

Figure 6. Map of localities sampled in each ria, from Vigo (A) to Muros-Noia (D). The color of the dots represents the degree of wave exposure of each locality, measured as their Ecological PC1 value on a color gradient scale. The minority morphs (*albida*, *aurantia*, *nigra*, *bande*, and *tessellata*) have been grouped into a single category. All rias are at the same scale and orientation, so the north arrow and scale of Vigo apply to the remaining three rias.

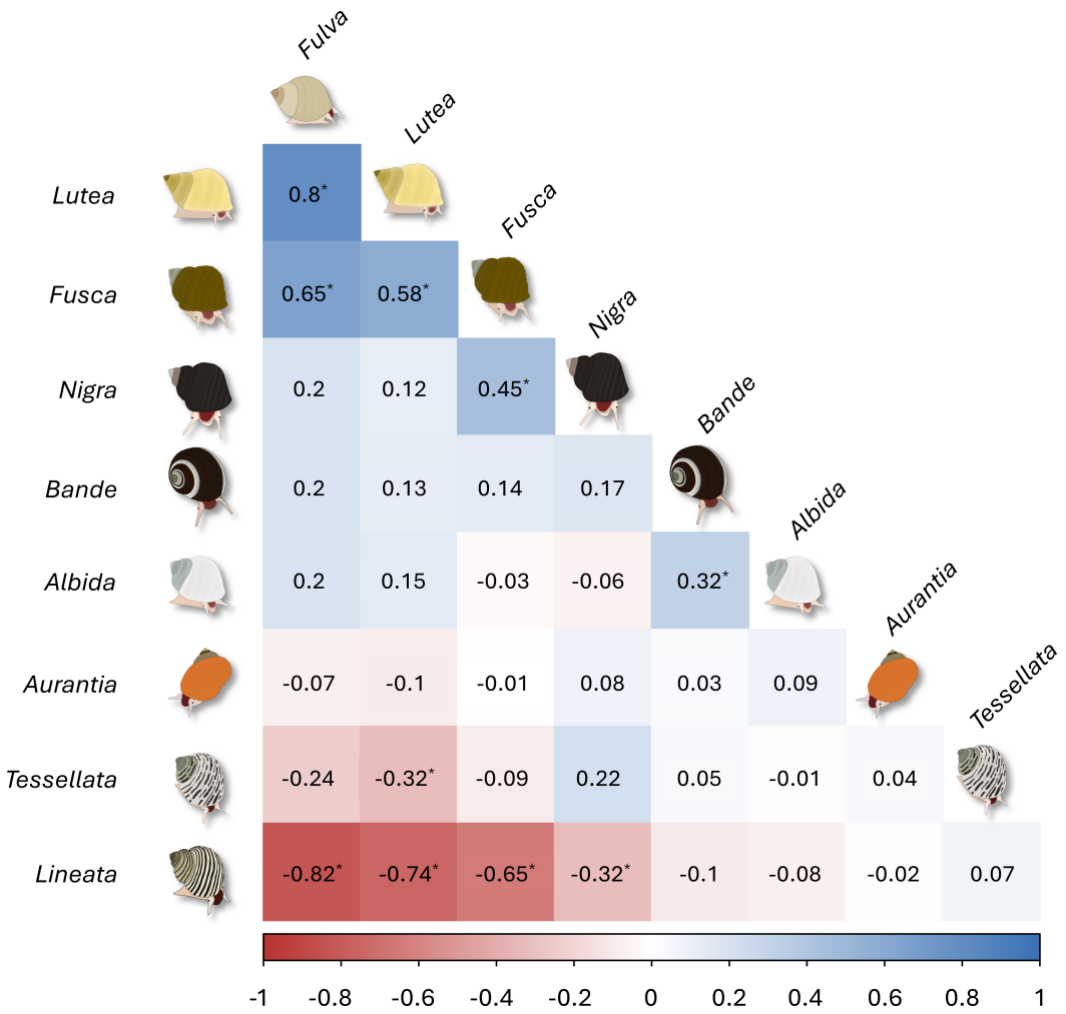

Figure 7. Correlation matrix based on Spearman's rank correlation coefficients between color morph counts in the 55 localities sampled. The color gradient represents the magnitude and direction of the correlation. * $P < 0.05$.

The spatial aggregation of the majority morphs of the cline, *fulva* and *lineata*, was also studied with Moran's I (see Supplementary materials, Figure S18). In the case of *fulva*, this test revealed substantial geographical aggregation on all coasts of the Rías Baixas except for the southern coasts of Pontevedra and, surprisingly, Arousa. These two coasts yielded slightly negative values (spatial anti-correlation), or near 0 values (random distribution). As for *lineata*, a similar result was found, with substantial spatial aggregation in all rias and coasts except for Pontevedra, where, again, either random distribution (northern coast) or spatial anti-correlation (southern coast) of *lineata* snails was found.

For all the rias pooled, a quasi-binomial GLM using Ecological PC1 as a predictor of *fulva* frequency yielded an Ecological PC1's β estimate value of -0.50757 (P = 9.3e-11), with the McFadden's pseudo-R^2 for the model being 0.6477326 (Supplementary materials, Figure S19A). As for *lineata*, the model resulted in a β estimate of 0.8115 (P = 6.12e-11), and a McFadden's pse-

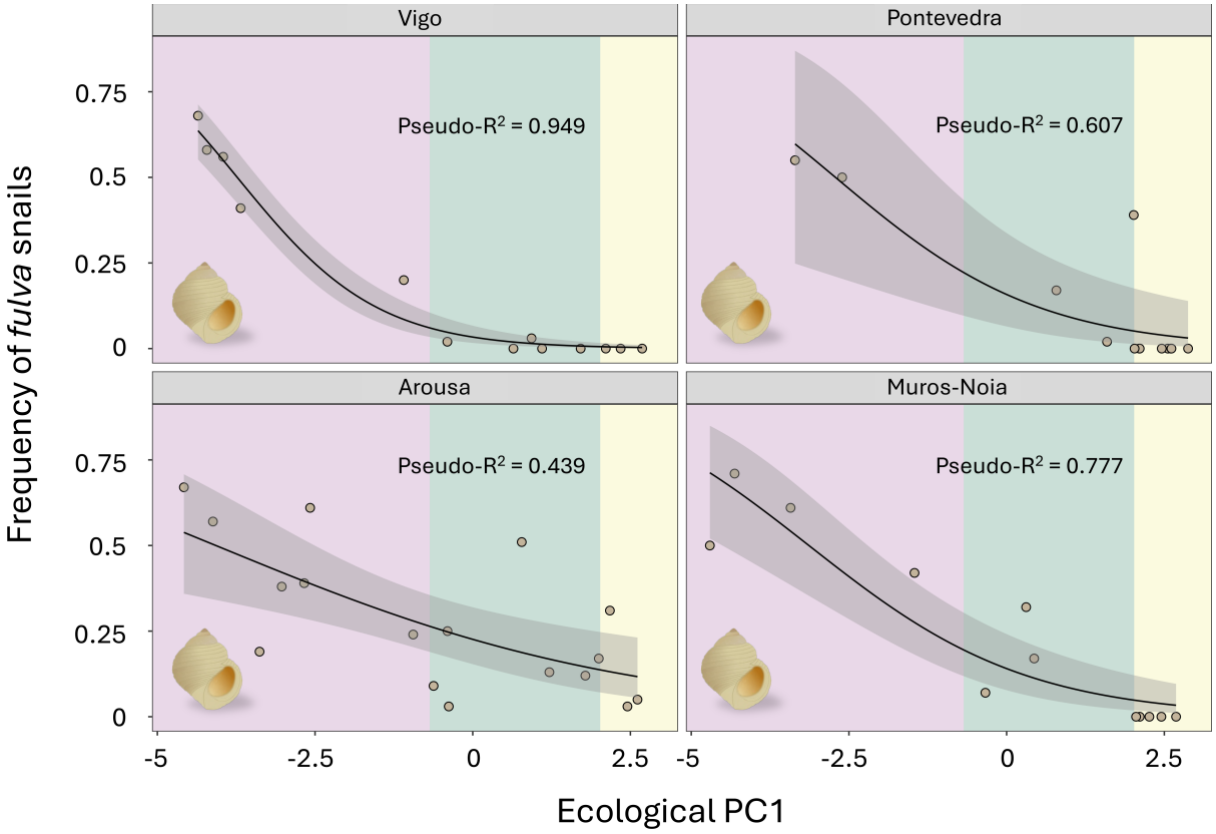

Figure 8. Frequencies of *fulva* snails in each locality as a function of Ecological PC1, fitted with quasi-binomial GLMs. The shaded areas represent the 95% confidence intervals, and pseudo-R^2s are McFadden's. The background colors represent the edge values of each Ecological PC1 tertile, used to create the different ecosystem types; pink-purple corresponds to wave-sheltered localities, green to wave-intermediate localities, and yellow to wave-exposed localities.

udo-R^2 of 0.6968109 (Supplementary materials, Figure S19B). Considering that McFadden pseudo-R^2 values are generally lower than ordinary R^2 values (Darlington & Hayes, 2017), these results corroborate Ecological PC1 as a robust predictor of the frequency of *fulva* and *lineata* snails, which show opposite trends: *fulva* frequency is higher at localities with low Ecological PC1 values (wave-sheltered), and *lineata* frequency is higher at localities with high Ecological PC1 values (wave-exposed).

Quasi-binomial GLMs for each ria separately yielded dissimilar adjustments. In the case of *fulva* frequency, Vigo shows an almost perfect adjustment (McFadden's pseudo-R^2 = 0.9486657), followed by Muros-Noia (McFadden's pseudo-R^2 = 0.7765695), Pontevedra (McFadden's pseudo-R^2 = 0.6069248), and Arousa (McFadden's pseudo-R^2 = 0.4389875) (Figure 8). An analogous result was obtained for *lineata* frequency in each ria, with again Vigo on top (McFadden's pseudo-R^2 = 0.9329454), closely followed by Muros-Noia (McFadden's pseudo-R^2 = 0.9148419), and Arousa and Pontevedra in a virtual tie (McFadden's pseudo-R^2s = 0.5764994 and 0.5664078, respectively) (Figure 9).

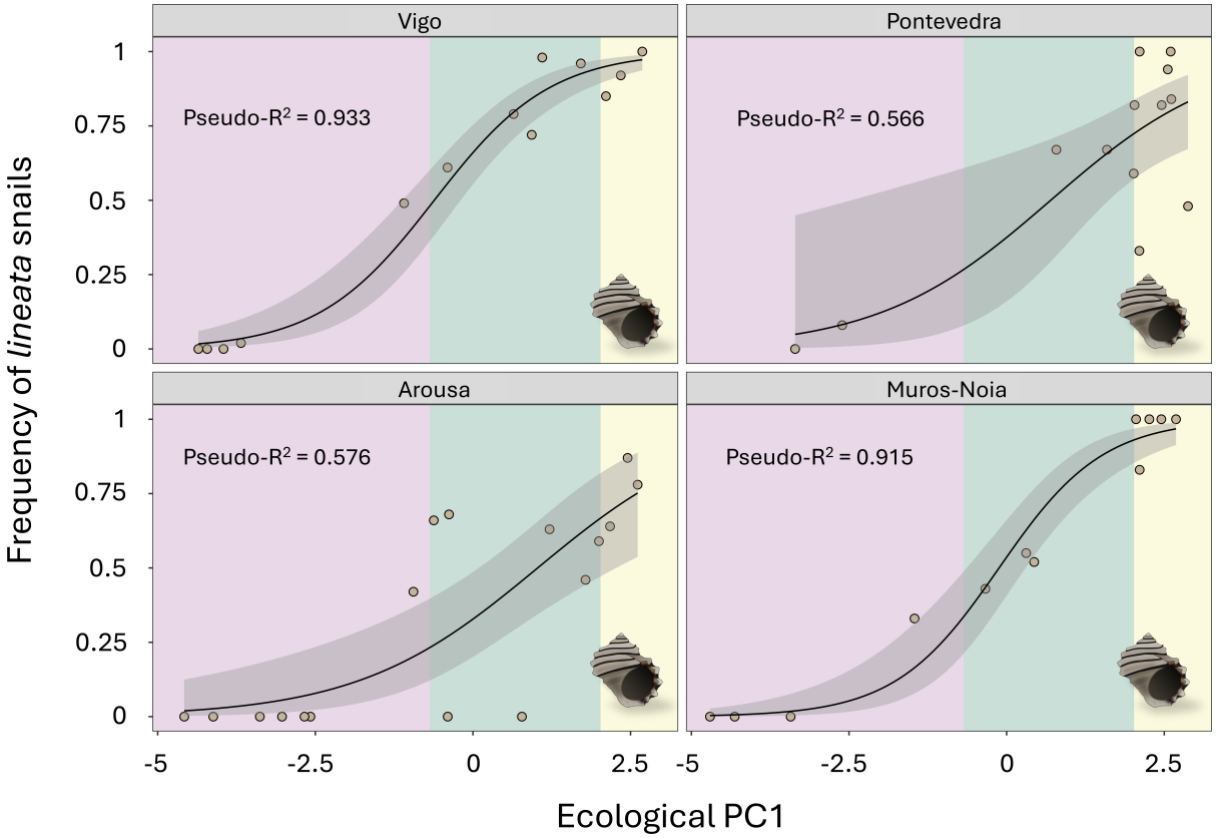

Figure 9. *Lineata* frequencies in each ria, fitted using quasi-binomial GLMs with Ecological PC1 as a predictor. Shaded areas, pseudo-R^2, and background colors have the same meaning as in Figure 8.

Other shell traits along the wave exposure gradient

The remaining shell traits measured also vary along the wave exposure continuum, except for shell thickness. Populations show a higher frequency of individuals with ribbed (*jugosa*) shell sculptures in wave-exposed areas (Figure 10A), and a higher frequency of smooth-shelled (*rudis*) snails in wave-sheltered localities (Figure 10B). On average, populations tend to have a spikier shape and a smaller aperture –low relative warp 1 values– at wave-sheltered zones and a rounder shape and a larger aperture –high relative warp 1 values– at wave-exposed areas (adjusted R^2 = 0.537, P = 1.249e-10) (Figure 11A). Additionally, snails are larger in wave-sheltered zones than in wave-exposed spots, with Ecological PC1 explaining around 56% of the variance in shell length (adjusted R^2 = 0.562, P = 2.808e-11) (Figure 11B). Finally, although shells appear to be slightly thicker on average at wave-sheltered localities, the percentage of variance explained by Ecological PC1 is negligible, around 7% (adjusted R^2 = 0.068, P = 0.03039) (Figure 11C). As with color, when these regressions are performed per ria, there is substantive variation in the quality of the adjustment between these traits and Ecological PC1 (see Supplementary materials, Figures S20-S23).

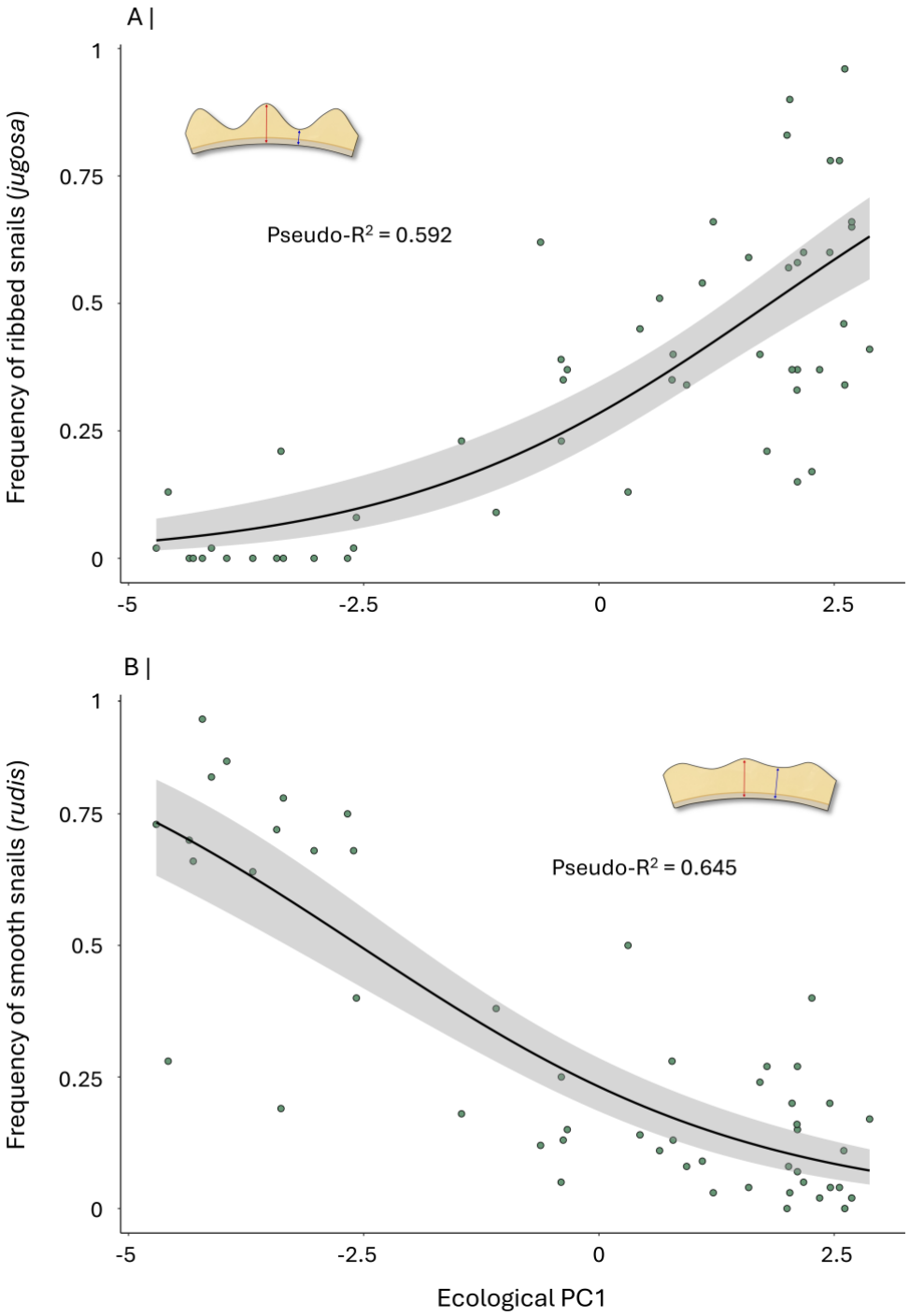

Figure 10. Quasi-binomial regressions using Ecological PC1 as a predictor of the frequency of either ribbed (*jugosa*; A) or smooth (*rudis*; B) shelled individuals. Pseudo-R^2 values are McFadden's. Note that, in addition to the categories displayed, an intermediate class, *rudissima*, exists (not used in the analyses).

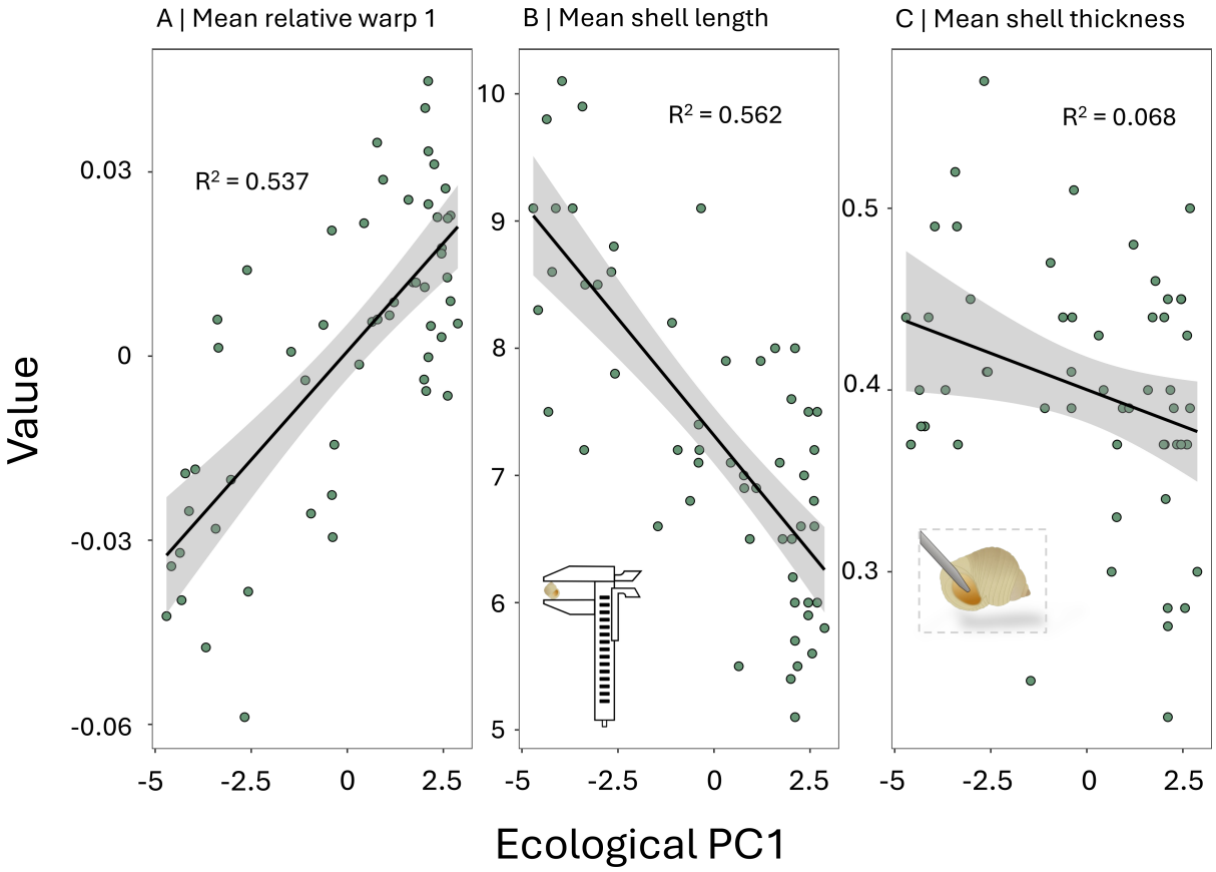

Figure 11. Linear models fitting Ecological PC1 on the mean relative warp 1 of the population (A), the mean shell length (B), and the mean shell thickness (C). R^2 represents the corresponding adjusted linear coefficient of determination. Note that the scales in the y-axes are specific to each parameter (relative warp 1 values in A, and mm in B and C).

As for traits supposedly providing information on the ecological conditions of snail habitats (i.e., shell scars and wear), they also show variation along the wave exposure gradient, with slight differences between rias (Supplementary materials, Figures S24-S26). In general, localities with low Ecological PC1 values (wave-sheltered) tend to exhibit more scarred and less worn-out snails than those with high Ecological PC1 values (wave-exposed). However, the predicted power of this variable, although significant, is much more limited than for shell color and other shell traits (shell scars McFadden's pseudo-R^2 = 0.406; shell wear McFadden's pseudo-R^2 = 0.258; see Supplementary materials, Figure S24).

Shell trait covariation

The previous results suggest covariation between shell color and other shell traits. A table of correlations (using Spearman's rank correlation test due to the non-normal nature of many variables) between the frequencies of *fulva* and *lineata* morphs per population and the frequencies or means of the various shell traits recorded also shows statistically significant associations between the morphs and all traits except shell thickness (Table 2).

Table 2. Spearman's rank coefficients for correlations between the main color morphs of the cline (*fulva* and *lineata*) and the various shell traits studied. *** $P < 0.001$

	Lineata frequency	*Fulva* frequency
Jugosa frequency	0.729***	-0.697***
Shell length (mean)	-0.606***	0.661***
Shell thickness (mean)	-0.214	0.253
Relative warp 1 (mean)	0.579***	-0.667***
Frequency of scarred snails	-0.527***	0.559***
Frequency of worn-out snails	0.494***	-0.525***

After checking the relevant assumptions (see Supplementary materials, *Extended Results (5)*, Figure S27), a PCA on the traits with a putatively strong genetic basis was performed (shell sculpture, shape, thickness, and length). Only one of the shell sculpture types was used (*jugosa*) to avoid multicollinearity issues. The PCA yielded a total of four PCs, with the first explaining 62.7% of the variance in shell parameters and the second 22.9% (Supplementary materials, Figure S28). A biplot and the component loadings for PC1 show that the *jugosa* sculpture and shell shape (as determined by relative warp 1) vary in a different direction than shell thickness and shell length, with the *jugosa* sculpture and relative warp 1 negatively correlating with PC1, and shell length and shell thickness correlating positively (Supplementary materials, Figures S29 and S30). This means that extreme negative values of PC1 correspond to populations with high frequencies of individuals with ribbed (*jugosa*), rounder and wide-apertured (i.e., high relative warp 1 values), smaller, and thinner shells. Conversely, positive PC1 values are represented by populations with low frequencies of *jugosa* snails, spikier and narrow-apertured (i.e., low relative warp 1 values), larger, and thicker shells. In this sense, PC1 effectively summarizes variation in shell traits and will henceforth be called 'Shell traits PC1'.

The distribution of Shell traits PC1 does not vary between rias (ANOVA, $F(3) = 1.935$, $P = 0.136$; see Figure 12), with this factor explaining only 10% of the variance in Shell traits PC1 ($\eta^2 = 0.10$). Furthermore, variation in this principal component along Ecological PC1 values (representing the wave exposure gradient) is remarkably similar across rias (Figure 13). In all of them, high Shell traits PC1 values concentrate around low Ecological PC1 values, although the magnitude of the adjustment, as revealed by their corresponding adjusted coefficients of determination (R^2), varies, despite being significant in the four rias ($P < 0.01$). Based on the previous interpretation of Shell traits PC1, populations in wave-sheltered localities tend to consist of snails with larger, thicker, smoother, spikier, and narrow-apertured shells. Conversely, populations in wave-exposed localities have snails with smaller, thinner, ribbed, and rounder and, wide-apertured shells (see also Supplementary materials, Figure S31).

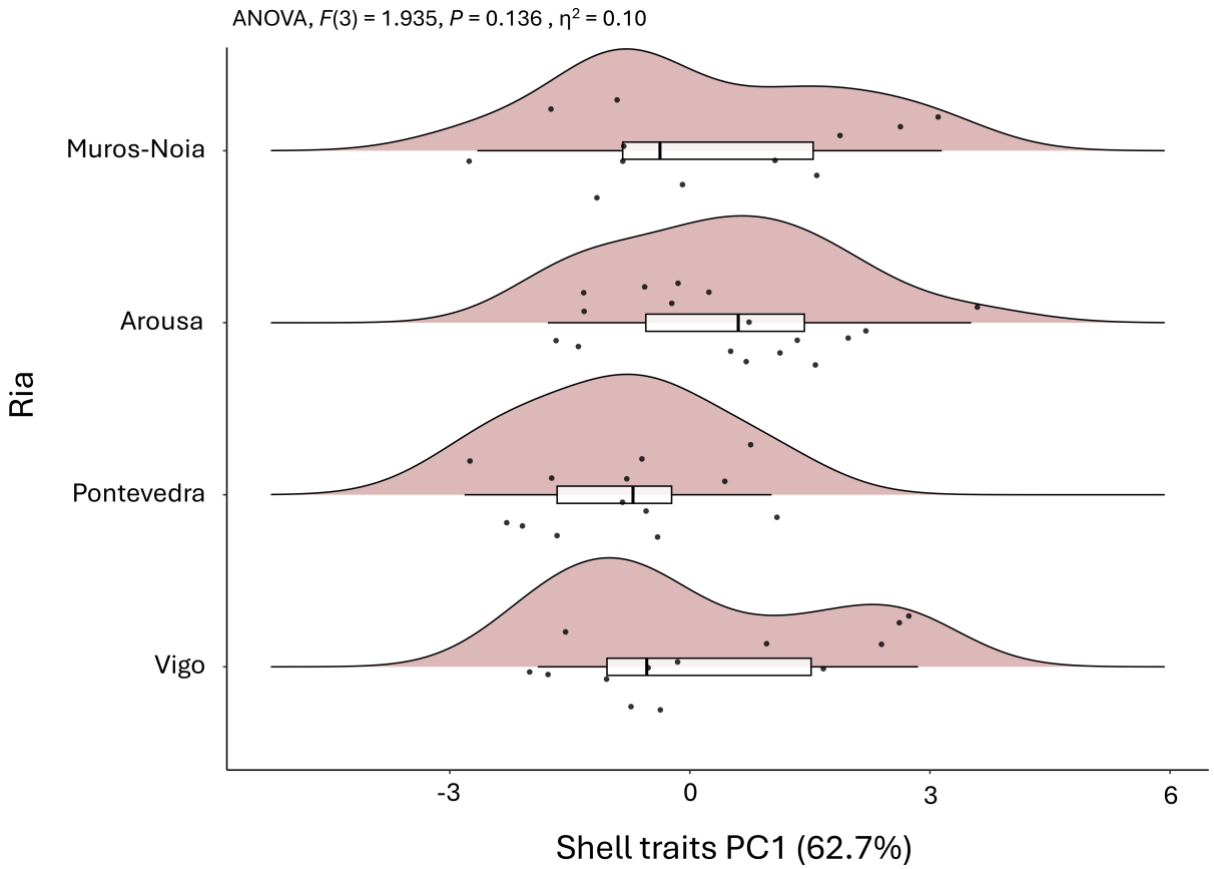

Figure 12. Boxplots with their associated density plots of Shell traits PC1 in each ria.

Shell traits PC1 explains a substantial percentage of the variation in *fulva* and *lineata* frequencies (Table 3, Models 1 and 2). This confirms a covariation between the main color morphs of the cline (*fulva* and *lineata*) and a suite of traits with a putatively strong genetic basis, at least at the population level. Shell traits PC1 accounts for approximately 51% of the variance in *fulva* frequency (adjusted R^2 = 0.5075, *P* = 6.425e-10), and around 52% of the variance in *lineata* frequency (adjusted R^2 = 0.5164, P = 3.924e-10). According to the corresponding β estimates from the regression models, high frequencies of *fulva* snails tend to be associated with larger, thicker, smoother, spikier, and narrow-apertured shells (β estimate = 0.10797, *P* = 6.42e-10), while high frequencies of *lineata* snails are associated with smaller, thinner, ribbed, rounder, and wide-apertured snails (β estimate = -0.17477, *P* = 3.92e-10).

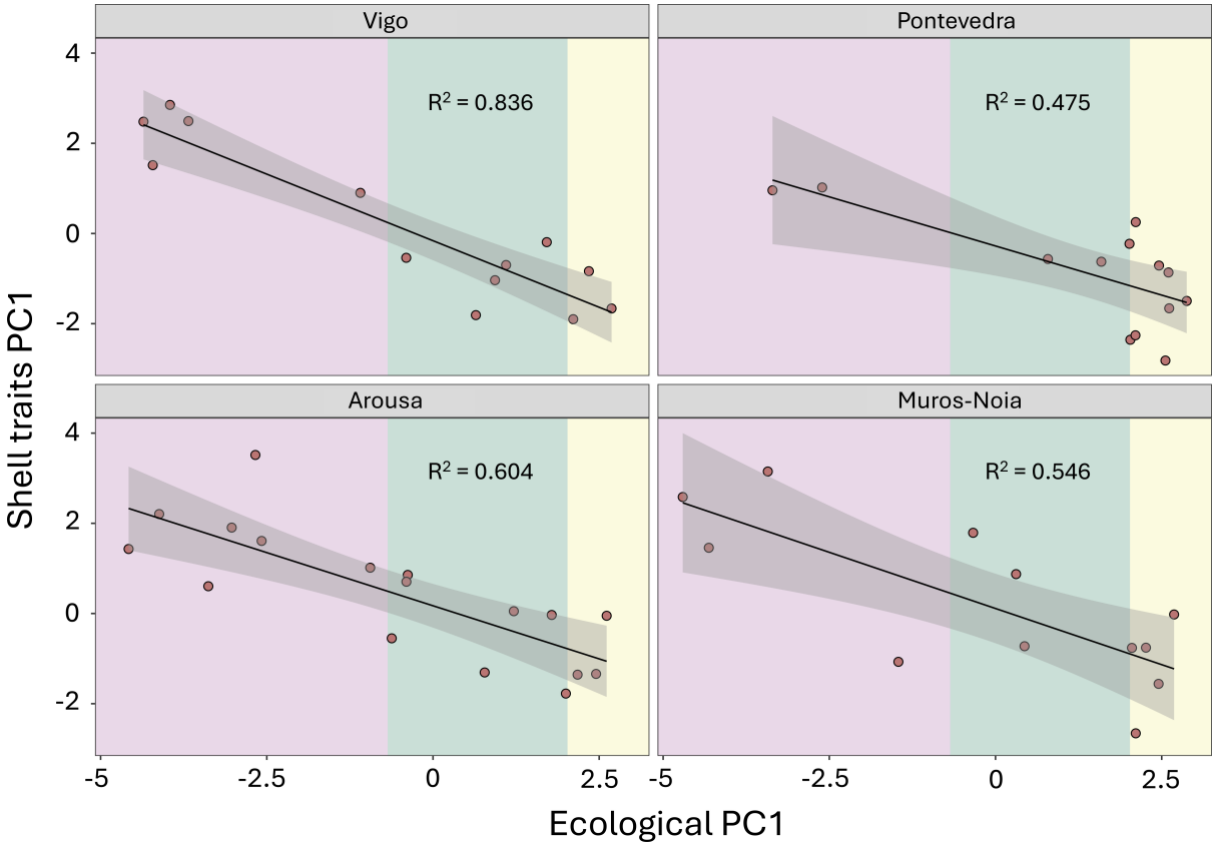

Figure 13. Plot of Shell traits PC1 values for each locality against their corresponding Ecological PC1 values, separated by ria. R^2 represents the adjusted coefficient of determination of the linear models using Ecological PC1 as a predictor of Shell traits PC1. Background colors indicate the Ecological PC1 values that define each ecosystem type: pink-purple for wave-sheltered ecosystems, green for wave-intermediate ecosystems, and yellow for wave-exposed ecosystems.

Table 3. Results of the regressions performed to study covariation in shell traits. Models 1 and 2 used Shell traits PC1 to predict *fulva* and *lineata* frequencies, respectively. Models 3 and 4 used the residuals from Models 1 and 2 as dependent variables, with Ecological PC1 as the predictor. These last two models inform us of the existence of independent shell color variation (i.e., variation not associated with other shell traits). ** $P < 0.01$, *** $P < 0.001$.

	Dependent	Predictor	β	SE	*T*	*P*	Adjusted R^2
Model 1	*Fulva* frequency	Intercept	0.21182	0.0225	9.412	6.70e-13	0.5075***
		Shell traits PC1	0.10797	0.01435	7.526	6.42e-10	
Model 2	*Lineata* frequency	Intercept	0.50855	0.03579	14.21	2.00e-16	0.5164***
		Shell traits PC1	-0.17477	0.02282	-7.66	3.92e-10	
Model 3	Model 1 Residuals (*fulva*)	Intercept	-8.85e-13	2.09e-02	0	1	0.1224**
		Ecological PC1	-2.44e-02	8.34e-03	-2.921	0.00511	
Model 4	Model 2 Residuals (*lineata*)	Intercept	1.51e-12	3.28e-02	0	1	0.1427**
		Ecological PC1	4.14e-02	1.31e-02	3.16	0.0026	

Subsequent regressions on the residuals of the previous models (i.e., the variance in *fulva* and *lineata* frequencies not explained by Shell traits PC1) using Ecological PC1 as a predictor yield statistically significant results (Table 2; Models 3 and 4). More specifically, Ecological PC1 explains approximately 12% of the variance in the residuals of *fulva* frequency (adjusted R^2 = 0.1224, P = 0.005113) and about 14% of the variance in the residuals of *lineata* frequency (adjusted R^2 = 0.1427, P = 0.002603). This indicates that there is a small, yet significant, portion of color variation in *L. saxatilis* that is not explained by other shell traits but is still dependent on ecological factors, representing 'free' or 'independent' shell color variation.

This is further confirmed by a series of linear models of Shell traits PC1 on *fulva* and *lineata* frequencies performed by ria (Supplementary materials, Table S5). These models result in substantial differences in the magnitude of the adjustment between rias and morphs. Both *fulva* and *lineata* frequencies are accurately predicted by Shell traits PC1 in Vigo, explaining around 82% of the variance (*fulva*: adjusted R^2 = 0.8189, P = 1.302e-05; *lineata*: adjusted R^2 = 0.8152, P = 1.462e-05). However, the explanatory power of this variable sharply decreases in other rias. For instance, the model is not significant or barely significant in Arousa for *fulva*, explaining only around 17% of the variance (adjusted R^2 = 0.1685, P = 0.05723), and in Pontevedra for *lineata* (adjusted R^2 = 0.1569, P = 0.0996). This confirms that although there is covariation in shell traits, these characters are largely independent and tend to co-occur only under certain circumstances.

To the previous results, it should be added that a preliminary test suggests that all the morphs appear to be adaptive rather than neutral (see Supplementary materials, *Extended Results (6)*).

Discussion

A major source of evidence for studying whether evolution is repeatable comes from natural experiments (Blount *et al.*, 2018). In these experiments, parallel or convergent evolution is often taken as evidence for the repeatability of evolution. A key assumption of natural parallel evolution experiments is that the ecosystems to which convergent populations adapt are equivalent. Thus, to properly test repeated evolution, a crucial first step is to determine the magnitude of the similarities between allegedly analogous areas or regions. Only after this can the populations of a given species living there be meaningfully compared. Once this has been done, characterizing the distribution of key traits in the allegedly convergent populations becomes the pivotal next step. This can provide clues as to whether selection has driven their supposedly parallel evolution. Finally, studying covariation between these traits and other

putatively genetically based characters can provide important insights into their genetic basis and selective regime.

The analyses presented in this chapter aimed to walk through these steps in the case of the *L. saxatilis* color cline from the Rías Baixas. Several conclusions can be drawn from the obtained data. First, the Rías Baixas are relatively similar at a geophysical and ecological level. All rias exhibit a wave action gradient from their inner zones to their outer areas. Although there are differences between the rias in rock composition and slight variations in their overall wave-exposure profiles –differences likely explained by the specific landforms of each ria– these do not obscure their general resemblance. The sampled localities vary along an ecological gradient of wave exposure, effectively summarized by Ecological PC1. The distribution of Ecological PC1 values is similar between the rias, except for the southern coast of Pontevedra. The wave-sheltered, wave-intermediate, and wave-exposed ecosystem types defined based on Ecological PC1 values differ in species composition but not between rias. Overall, these differences are not large. The only species that differ between wave-sheltered and wave-exposed ecosystems are *Ascophyllum nudosum*, *Fucus vesiculosus*, and *Littorina fabalis* or *Littorina obtusata*. Generally speaking, there is more variation in species composition in wave-sheltered and wave-intermediate ecosystems than in wave-exposed ecosystems. Ecological PC1 varies spatially, showing low values (wave-sheltered profile) at localities close to the river mouths and high values (wave-exposed profile) at distant localities. The strength of the association between distance from the river mouth and Ecological PC1 is greater in Vigo and Muros-Noia than in Pontevedra and Arousa. Putting all these threads of evidence together, then, this study shows that the Rías Baixas are more or less analogous in their intertidal ecosystems.

A second important finding is the confirmation of a clear connection between certain color morphs and different ecosystem types. In the four rias, the most abundant morphs, *fulva* and *lineata*, are associated with wave-sheltered and wave-exposed habitats, respectively. Additionally, *fulva* tends to co-occur with the *lutea* and *fusca* morphs in most localities, although the frequencies of the latter vary. Despite the overall trend, the strength of the association between *fulva* and wave-sheltered ecosystems, and *lineata* and wave-exposed ecosystems, differs among the rias. Vigo and Muros-Noia exhibit a tight association, while Pontevedra and Muros-Noia show a much looser, yet significant, link.

Lastly, the third main finding involves the extent of covariation between shell traits. Other shell traits, either with a presumably strong genetic basis or reflecting the ecological conditions in which the snails live, also vary along the wave-exposure continuum. Shell traits PC1 effectively captures variation in shell traits with a putatively strong genetic basis. All rias have similar

distributions of Shell traits PC1, and this composite parameter varies along the wave exposure gradient in much the same way as shell color. Shell traits PC1 explains around 51% of the variance in *fulva* and *lineata* frequencies, exhibiting opposite directions of change for each morph. The *fulva* morph covaries with larger, smoother, thicker, and spikier shells, while *lineata* shows the exact reverse tendency. Interestingly, however, both residual analysis and linear regressions per ria show that the magnitude of this covariation is different in each ria and that a substantial fraction of variation in color morph frequencies is independent of variation in other shell traits.

In sum, this study corroborates our working hypotheses about the color cline of *L. saxatilis* in the Rías Baixas: (1) that the four rias are equivalent in terms of their intertidal ecosystems; (2) that there is an analogous color cline in the shells of *L. saxatilis* from these estuaries; and (3) that, to a significant extent, variation in shell color co-occurs with variation in a suite of shell traits with a presumably strong genetic basis. This confirms the Rías Baixas and the color cline of their *L. saxatilis* populations a promising natural experiment to study repeated evolutionary trajectories, opening the door to a series of numerous exciting insights and questions about their ecology and evolution. Some of these are discussed in the General Discussion.

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Supplementary materials – Chapter 6

Extended Materials and methods: Methodological tables and figures

Table S1. Localities sampled with their corresponding number and code, GPS coordinates (decimal degrees), and number of snails collected (see continuation on next page).

Ria	Coast	Locality	Number	Code	GPS coordinates	Snails collected
Vigo	North	Punta Cabalo	1	V_{N-1}	42.327439, -8.636611	60
		San Adrián	2	V_{N-2}	42.301108, -8.655425	62
		Borna	3	V_{N-3}	42.281131, -8.697458	59
		Punta Niño Corvo	4	V_{N-4}	42.263236, -8.755239	61
		Punta Igreixiña	5	V_{N-5}	42.254017, -8.834264	46
		Cabo Home	6	V_{N-6}	42.2504, -8.871778	48
	South	Cesantes	7	V_{S-1}	42.316028, -8.618678	63
		Punta Soutelo	8	V_{S-2}	42.288058, -8.638128	44
		Areiño	9	V_{S-3}	42.259986, -8.698522	69
		Coruxo	10	V_{S-4}	42.201161, -8.785453	47
		Panxón	11	V_{S-5}	42.148131, -8.830506	59
		Punta Tenaza	12	V_{S-6}	42.126814, -8.849647	50
		Cuncheiro	13	V_{S-7}	42.123446, -8.85216	46
Pontevedra	North	Lourido	14	P_{N-1}	42.421172, -8.680689	51
		Chancela	15	P_{N-2}	42.422431, -8.717217	62
		Sinas	16	P_{N-3}	42.403028, -8.750881	60
		Areas	17	P_{N-4}	42.395417, -8.783575	58
		Paxariñas	18	P_{N-5}	42.390833, -8.844867	63
		Pociñas	19	P_{N-6}	42.413428, -8.871397	49
		Abelleira	20	P_{N-7}	42.452072, -8.924644	51
	South	Mogor	21	P_{S-1}	42.392336, -8.714067	50
		Loira	22	P_{S-2}	42.370006, -8.739614	45
		Santo do Mar	23	P_{S-3}	42.350775, -8.750575	54
		Agrelo	24	P_{S-4}	42.332497, -8.772503	62
		Beluso	25	P_{S-5}	42.336078, -8.798983	50
		Punta Cagadoiro	26	P_{S-6}	42.341161, -8.82595	49

Table S1 (Continuation).

Ria	Coast	Locality		Code	GPS coordinates	Snails collected
Arousa	North	Neixón	27	A_{N-1}	42.662892, -8.843661	60
		Portomouro	28	A_{N-2}	42.632428, -8.837792	62
		Cabo da Cruz	29	A_{N-3}	42.610458, -8.887486	58
		Escarabote	30	A_{N-4}	42.61885, -8.921622	62
		Torre Bermúdez	31	A_{N-5}	42.596747, -8.923539	66
		Punta de Cabio	32	A_{N-6}	42.586633, -8.924511	62
		Ribeira	33	A_{N-7}	42.542956, -8.984458	65
		Aguiño	34	A_{N-8}	42.517356, -9.022631	58
	South	Carril	35	A_{S-1}	42.621533, -8.773958	51
		Punta Laxe de Borreiros	36	A_{S-2}	42.578756, -8.805211	62
		Fuciño	37	A_{S-3}	42.574422, -8.830231	61
		Illa De Arousa (East)	38	A_{S-4}	42.566419, -8.874369	67
		Illa de Arousa (West)	39	A_{S-5}	42.551289, -8.874856	60
		Cambados	40	A_{S-6}	42.507711, -8.820236	53
		A Toxa	41	A_{S-7}	42.487889, -8.845525	61
		O Grove	42	A_{S-8}	42.491256, -8.891094	65
		San Vicente do Mar	43	A_{S-9}	42.4514, -8.924725	52
Muros-Noia	North	Broña	44	M_{N-1}	42.801908, -8.930581	62
		Lagoa	45	M_{N-2}	42.792831, -8.945153	57
		Esteiro	46	M_{N-3}	42.787333, -8.966569	60
		Raimundez	47	M_{N-4}	42.781186, -8.996733	64
		Bouga	48	M_{N-5}	42.75985, -9.061906	52
		Queixal	49	M_{N-6}	42.746006, -9.076656	47
	South	Testal	50	M_{S-1}	42.790456, -8.917828	58
		Hornanda	51	M_{S-2}	42.766433, -8.941461	60
		Langaño	52	M_{S-3}	42.745342, -8.960806	60
		Baroña	53	M_{S-4}	42.695061, -9.031647	40
		Sieira	54	M_{S-5}	42.656036, -9.041625	41
		Corrubedo	55	M_{S-6}	42.577489, -9.090133	41

Table S2. Ecological parameters sampled and their operational definitions.

Variables	Categories	Operational definition
Ascophyllum nudosum	Absent (no exemplars covering the rocks) Rare/occasional (covering of < 20% of the rocks) Frequent (covering approximately between 21% and 40% of the rocks) Common (covering approximately between 41% and 60% of the rocks) Abundant (covering approximately between 61% and 80% of the rocks) Super-abundant (covering > 81% of the rocks)	Relative covering of rocks by *Ascophyllum nudosum* in a 2 meters wide by 10 meters long (20 m^2) sample of the lower intertidal zone (usually below *Fucus vesiculosus*) at the sampled locality, closest to the *L. saxatilis* sampling points.
Barnacles	Absent (no exemplars covering the rocks) Rare/occasional (covering of < 20% of the rocks) Frequent (covering approximately between 21% and 40% of the rocks) Common (covering approximately between 41% and 60% of the rocks) Abundant (covering approximately between 61% and 80% of the rocks) Super-abundant (covering > 81% of the rocks)	Relative coverage of the surface of the rocks by barnacles (mostly *Chthamalus* sp. and *Semibalanus balanoides*) in a 2 meters wide by 10 meters long (20 m^2) sample of the low and mid-upper intertidal zones at the sampled locality, closest to the of *L. saxatilis* sampling points. Barnacles tend to show a uniform distribution, often increasing up to the point of covering the whole rock.
Corallina officinalis	Found Not found	*Corallina officinalis* is found in the lower intertidal zone after 15 minutes of sampling.
L. fabalis / *L. obtusata*	Found Not found	*Littorina fabalis* or *Littorina obtusata* are found in the lower intertidal zone of the sampled locality after 15 minutes of sampling. (These species are grouped together as they are very difficult to distinguish in situ.)
Fucus vesiculosus	Absent (no exemplars covering the rocks) Rare/occasional (covering of < 20% of the rocks) Frequent (covering approximately between 21% and 40% of the rocks) Common (covering approximately between 41% and 60% of the rocks) Abundant (covering approximately between 61% and 80% of the rocks) Super-abundant (covering > 81% of the rocks)	Relative coverage of rocks by *Fucus vesiculosus* in a 2 meters wide by 10 meters long (20 m^2) sample of the lower intertidal zone (usually above *Ascophyllum nudosum* and/or below *Pelvetia canaliculata*) of the sampled locality, closest to the *L. saxatilis* sampling points.
Gibbula umbilicalis	Found Not found	*Gibbula umbilicalis* is found in the mid intertidal zone of the sampled locality after 15 minutes of sampling.

Table S2 (Continuation).

Variables	Categories	Operational definition
Lichina pygmaea	Absent (no patches) Rare/occasional (1-5 patches) Frequent (6-10 patches) Common (11-16 patches) Abundant (17-21 patches) Super-abundant (most rocks in the upper zone of the intertidal covered by patches)	Relative coverage of rocks by patches of the black lichen *Lichina pygmaea* in a 2 meters wide by 10 meters long (20 m^2) sample of the upper intertidal zone of the sampled locality, closest to *the L. saxatilis* sampling points. Patches must be > 20x20cm (approx.) to count.
Mytilus galloprovincialis	Absent (no patches) Rare/occasional (1-5 patches; usually of small mussels) Frequent (6-10 patches) Common (11-16 patches) Abundant (17-21 patches) Super-abundant (most rocks in the mid-low zone of the intertidal covered by patches)	Abundance of patches of *Mytilus galloprovincialis* in a 2 meters wide by 10 meters long (20 m^2) sample of the mid-low intertidal zone of the sampled locality, closest to the *L. saxatilis* sampling points. Patches should contain groups of at least 10 (most of them mature) individuals.
Littorina littorea	Found Not found	*Littorina littorea* is found in the mid intertidal zone of the sampled locality after 15 minutes of sampling.
Melarhaphe neritoides	Found Not found	*Melarhaphe neritoides* is found in the upper intertidal zone of the sampled locality after 15 minutes of sampling.
Nucella lapillus	Found Not found	*Nucella lapillus* is found in the mid-low intertidal zone of the sampled locality after 15 minutes of sampling.
Pelvetia canaliculata	Absent (no exemplars covering the rocks) Rare/occasional (covering of < 20% of the rocks) Frequent (covering approximately between 21% and 40% of the rocks) Common (covering approximately between 41% and 60% of the rocks) Abundant (covering approximately between 61% and 80% of the rocks) Super-abundant (covering > 81% of the rocks)	Relative covering of rocks by *Pelvetia canaliculata* in a 2 meters wide by 10 meters long (20 m^2) sample of the lower intertidal zone of the sampled locality (above *Fucus vesiculosus*), closest to the *L. saxatilis* sampling points.
Phorcus lineatus	Found Not found	*Phorcus lineatus* is found in the mid-upper intertidal zone of the sampled locality after 15 minutes of sampling.

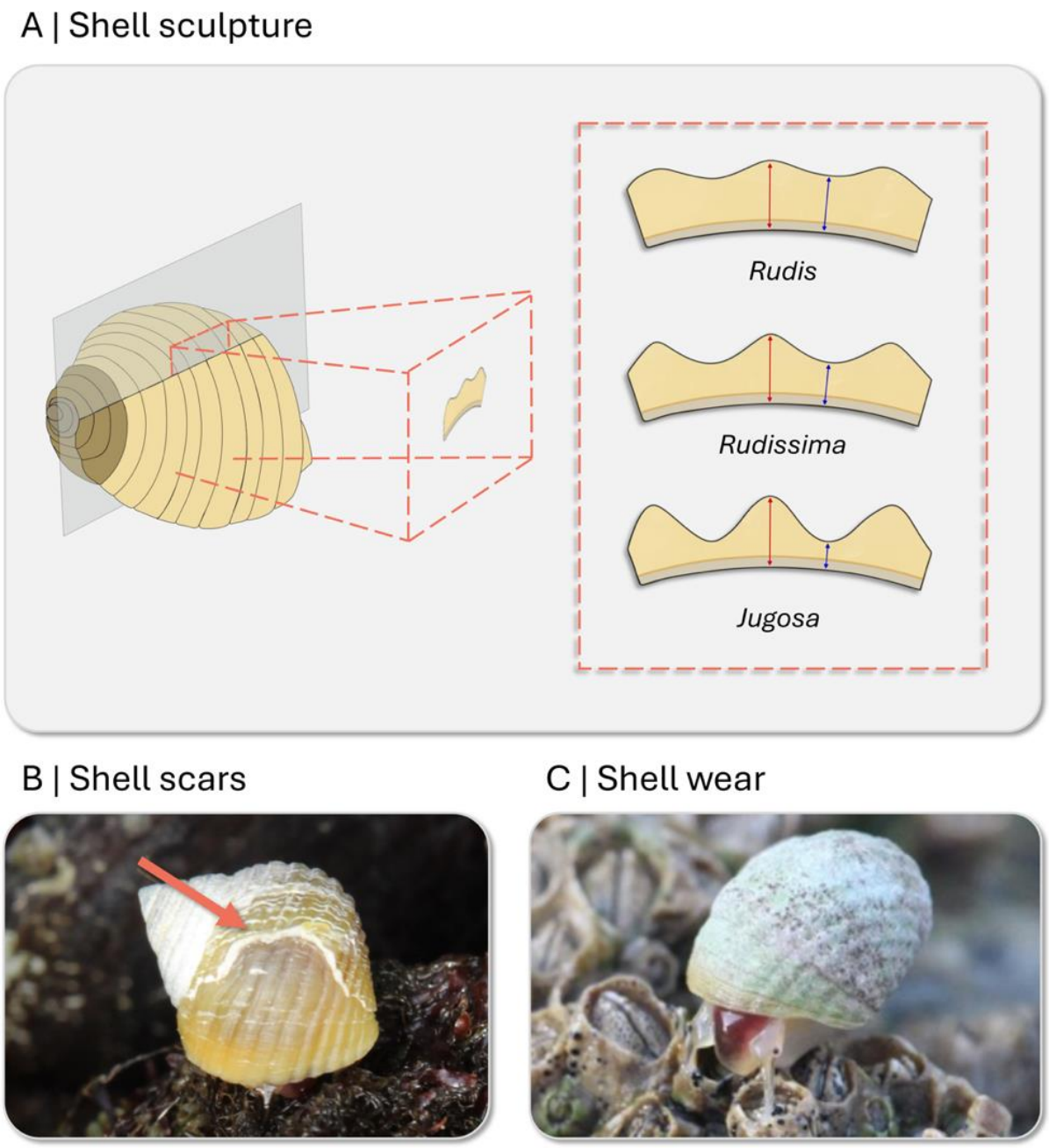

Figure S1. A visual guide to some of the measured shell traits. A: The different shell sculpture categories ranked. Note that the ranking of the snails was done by visually inspecting each individual's outer lip and main whorl, not by cutting the snail's shell (as might be interpreted from the figure). B: An example of a scarred shell. C: A worn-out snail. Note that the original shell color (likely *fulva*) is only noticeable along the line of shell growth.

Extended Results (1): Geophysical characterization of localities

Salinity does not show significant variation between rias (Kruskal-Wallis, $H(3) = 3.81$, $P = 0.2822$), with most localities in all four rias ranging between 30 and 36‰. An analysis considering the coasts confirms this result (Kruskal-Wallis, $H(7) = 9.892$, $P = 0.1948$) (Figure S2). Despite this non-significance, Muros-Noia and Vigo exhibit exceptions, with some localities showing quite low salinity values (around 20‰, and an outlier in Vigo below 10‰). This variation might be explained by geomorphology: both rias have relatively closed inner zones, with the San Simón Bay in Vigo standing out. This geomorphology might prevent ocean water from reaching these zones, thereby lowering their salinity values (see the main landforms of each ria in Figure S3).

Wave action, conversely, varies between rias (Kruskal-Wallis, $H(3) = 12.65$, $P = 0.005452$; $\eta^2(H) = 0.189$), with two clearly differentiated groups: Vigo and Arousa on the one hand, exhibiting lower average wave height values, and Pontevedra and Muros-Noia on the other, with higher average values. Again, this variation might be related to the shape of each ria: Pontevedra and Muros-Noia are wider on average than Vigo, and although Arousa is also very wide, it has a more meandering coastline and contains the O Grove Peninsula and the Arousa island (Figure S3), which likely block wave action in many localities. If the different coasts of each ria (north and south) are considered (see also Figure S2), the differences between rias marginally remain (Kruskal-Wallis, $H(7) = 13.90$, $P = 0.05296$). The southern coast of Muros-Noia contains localities with strong wave action, likely situated towards the exterior of the ria. However, caution is required due to the low sample size.

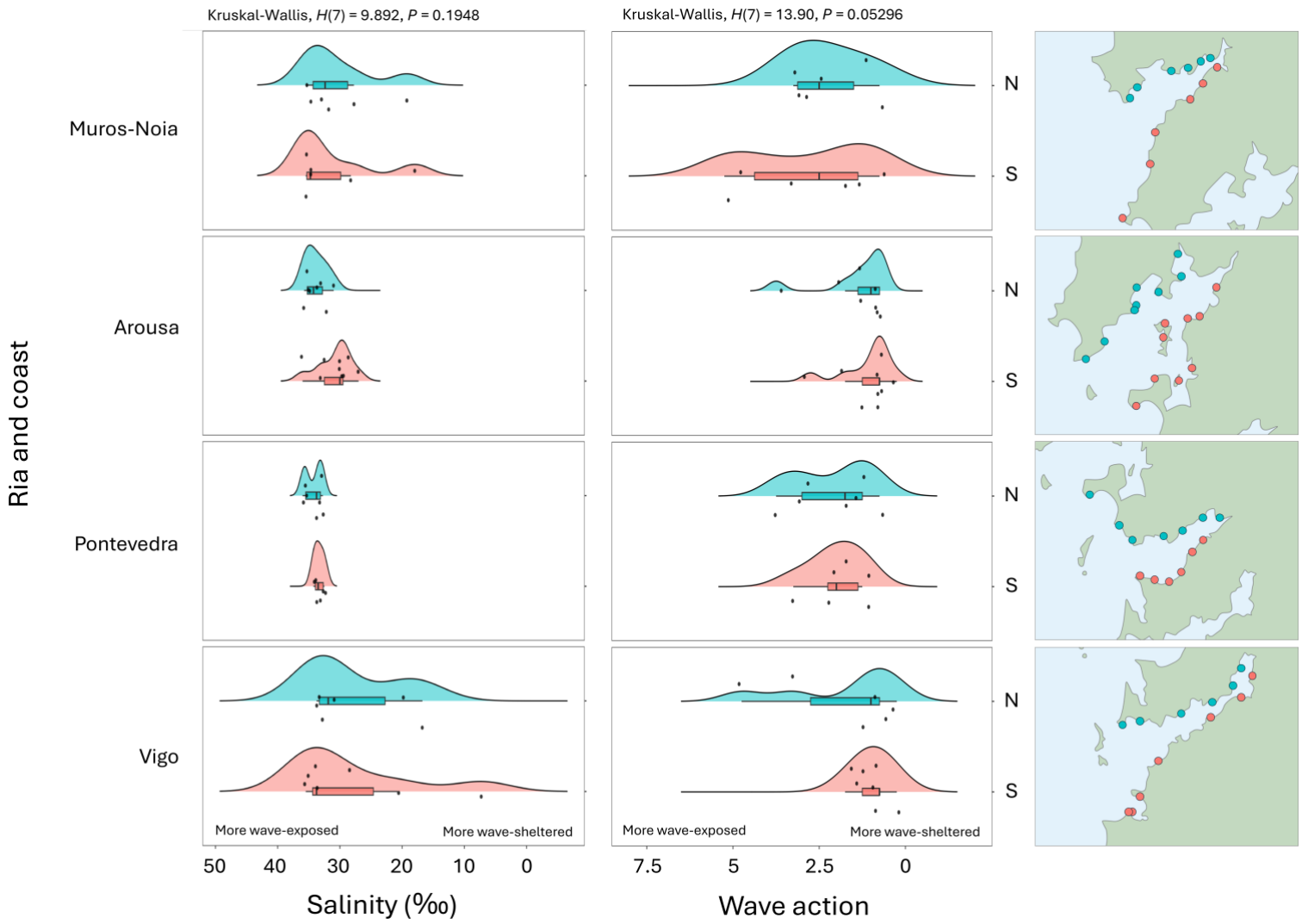

Figure S2. Boxplots with correspondent density plots for salinity and wave action comparison between the different coasts (north –N– and south –S–) of the rias, with the studied localities shown on the right. Note that the x-axes of both variables have been reversed to align with the direction of geographical variation in the rias, where the more wave-exposed localities are located towards the west (left).

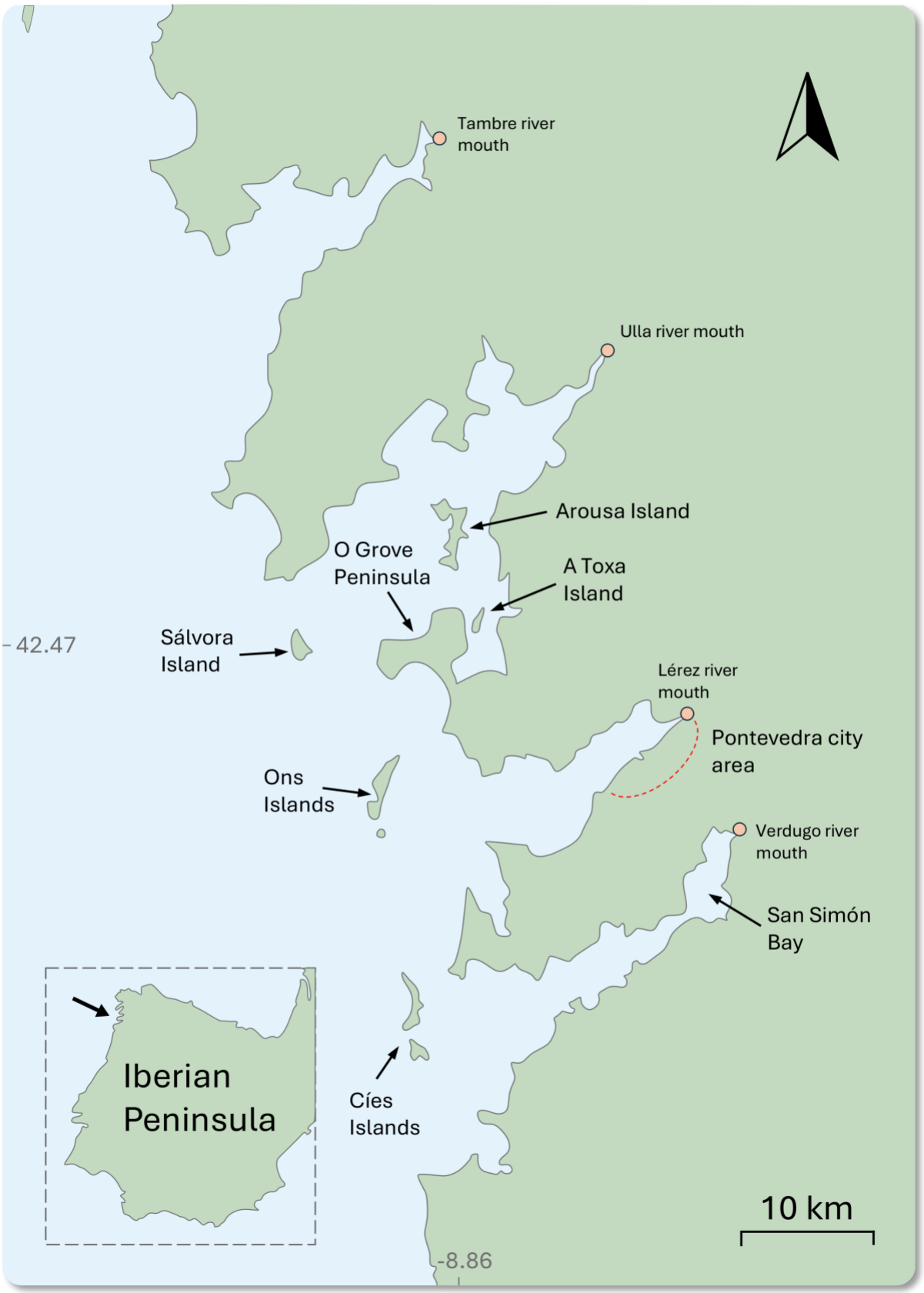

Figure S3. Main landforms of the Rías Baixas, which help explain some peculiarities of each ria. Note that only those landforms that are relevant to the results are shown.

Four main rock types make up the rocky intertidal ecosystems of the Rías Baixas: alkaline granitoid, calc-alkaline granitoid, gneiss, and schist. Within our sample, there are differences between the rias in terms of rock composition (Fisher's exact test for count data, $P = 0.000439$; Table S3): Vigo and Muros-Noia have more localities with alkaline granitoid rocks, while Arousa has more calc-alkaline localities than the other rias. However, the numbers of schist and gneiss rocks are quite similar across all rias. Rock size was excluded from the analyses, as most localities were composed of very large rocks (Ø > 1 m) or bedrock surfaces. Localities with small and very small rocks (Ø < 50 cm) were only typical in the inner zone of Vigo (i.e., the San Simón Bay; see Figure S3).

Table S3. Counts of localities made up of each rock type in the four rias.

	Vigo	Pontevedra	Arousa	Muros-Noia
Alkaline granitoids	9	4	1	7
Calc-alkaline granitoids	1	5	14	2
Gneiss	1	1	0	1
Schist	2	3	2	2

How salinity and wave action change along the rias was assessed by plotting them against the coastline distance of each locality from the river mouths (in meters). Salinity doesn't show a clear and gradual clinal trend from the inner to the outer localities (Figure S4). The curve that best describes this trend is logarithmic, but Arousa and Pontevedra barely show any change in salinity along their coasts. Rather than indicating a true trend, this suggests that our salinity data may not be accurate or might reflect a specific moment in time rather than the average annual conditions.

As for wave action, this variable was shown to be not normally distributed (Shapiro-Wilk normality test, $W = 0.85587$, $P = 9.956e\text{-}06$) and its residuals were not homoscedastic (Non-constant Variance Score Test, Chi-square = 9.904001, d. f. = 1, $P = 0.0016492$). For these reasons, it was log-transformed to achieve greater normality. A generalized linear model (GLM) was performed to evaluate the fit of the variable. Several goodness-of-fit statistics and criteria indicated that a gamma distribution fits this variable better (e.g., AIC = 57.80 vs. AIC > 61 for the normal and log-normal distributions; see Table S4).

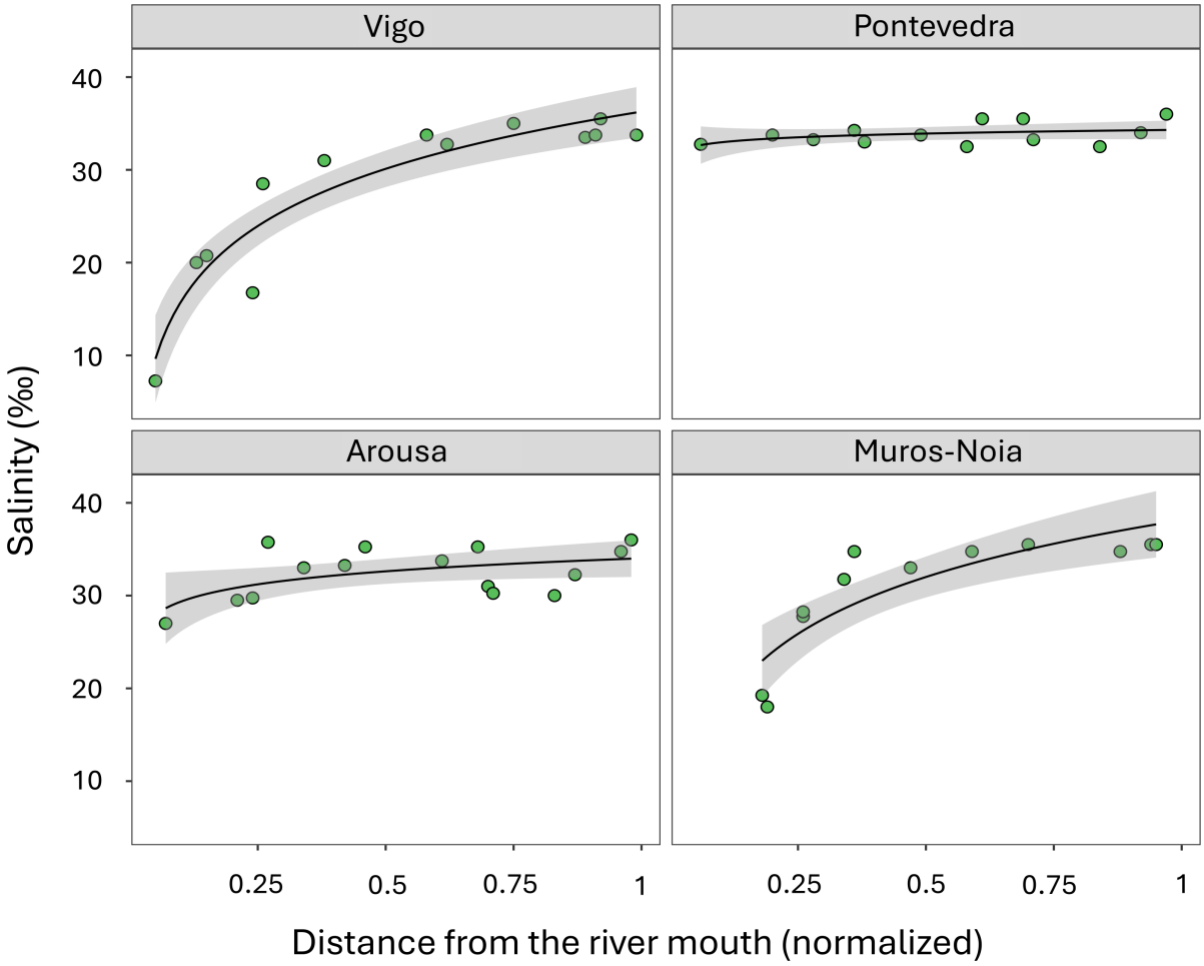

Figure S4. Variation in salinity concentrations along the coastlines of the different rias. The logarithmic lines fitted to the data (plus their 95% confidence intervals) should be taken cautiously, due to potential biases in the data (see the main text).

Table S4. Different goodness-of-fit statistics and criteria to determine the best-fit distribution for wave action log-transformed variable.

	Distributions		
Goodness-of-fit statistic or criteria	Normal	Log-normal	Gamma
Kolmogorov-Smirnov	0.1870219	0.1626261	0.1612878
Cramer-von Mises	0.318543	0.229837	0.2278471
Anderson-Darling	1.8425052	1.6077695	1.460068
Akaike's Information Criterion (AIC)	61.92128	61.92128	57.80318
Bayesian Information Criterion (BIC)	65.93594	65.36797	61.81784

A GLM using the distance from the river mouth (normalized) as a predictor of wave action (log-transformed) yielded a relatively good fit (Dist. River β estimate = 1.094, SE = 0.139, *P* = 2.19e-10; AIC = 23.41). This suggests that, in broad terms, the farther from the river mouth, the higher the wave action. This trend, however, is more pronounced in Muros-Noia than in the three remaining rias (Figure S5). The wave action Moran's I values for each ria (Figure S6) corroborates this pattern, which might have to do with Muros-Noia not having any island or archipelago at the mouth of the ria (see Figure S3 and pictures in Figure S6), thereby increasing the contrast between its inner and outer zones.

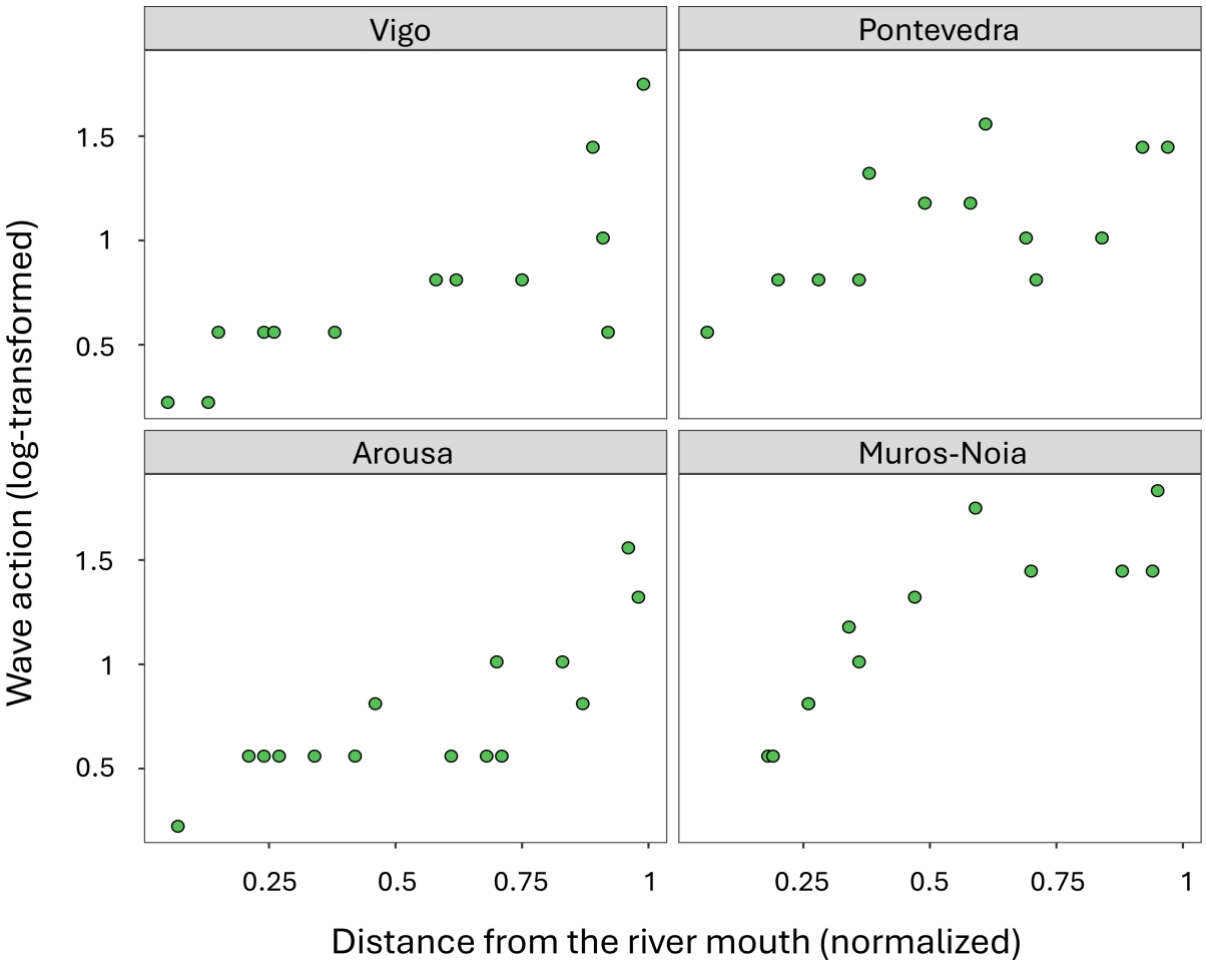

Figure S5. Plot of wave action (log-transformed) of each locality against their position along the coastline (i.e., its distance from the river mouth).

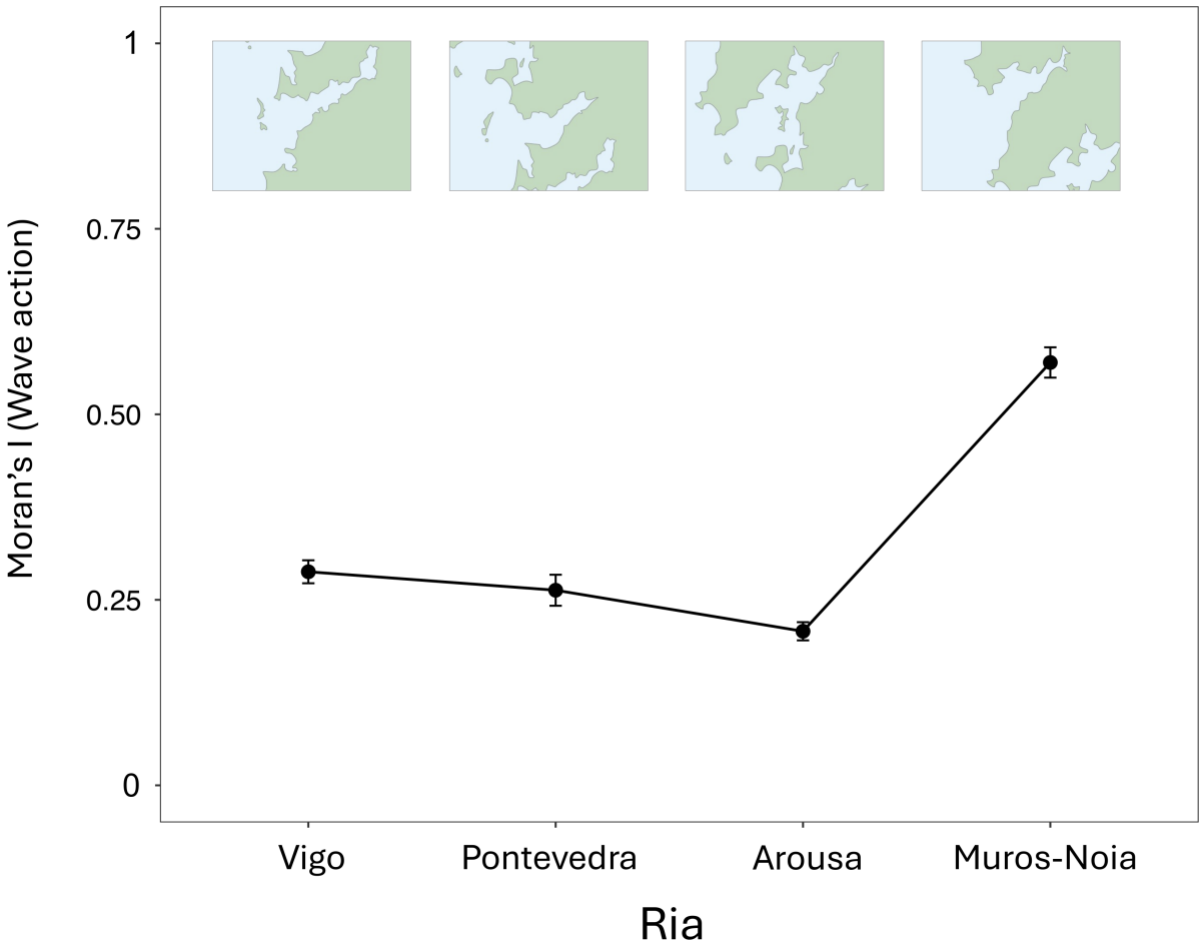

Figure S6. Wave action Moran's I values for each ria. Error bars represent the variances.

As expected, rock types don't show any clinal pattern (Figure S7). While alkaline and calc-alkaline rocks are more or less evenly distributed along the coast, gneisses appear to be restricted to the inner half of the rias. Schists, on the other hand, tend to be more common in the exterior half. This distribution might reflect the historical geomorphology of the Rias Baixas, but caution is needed when interpreting their distributions due to their low sample size.

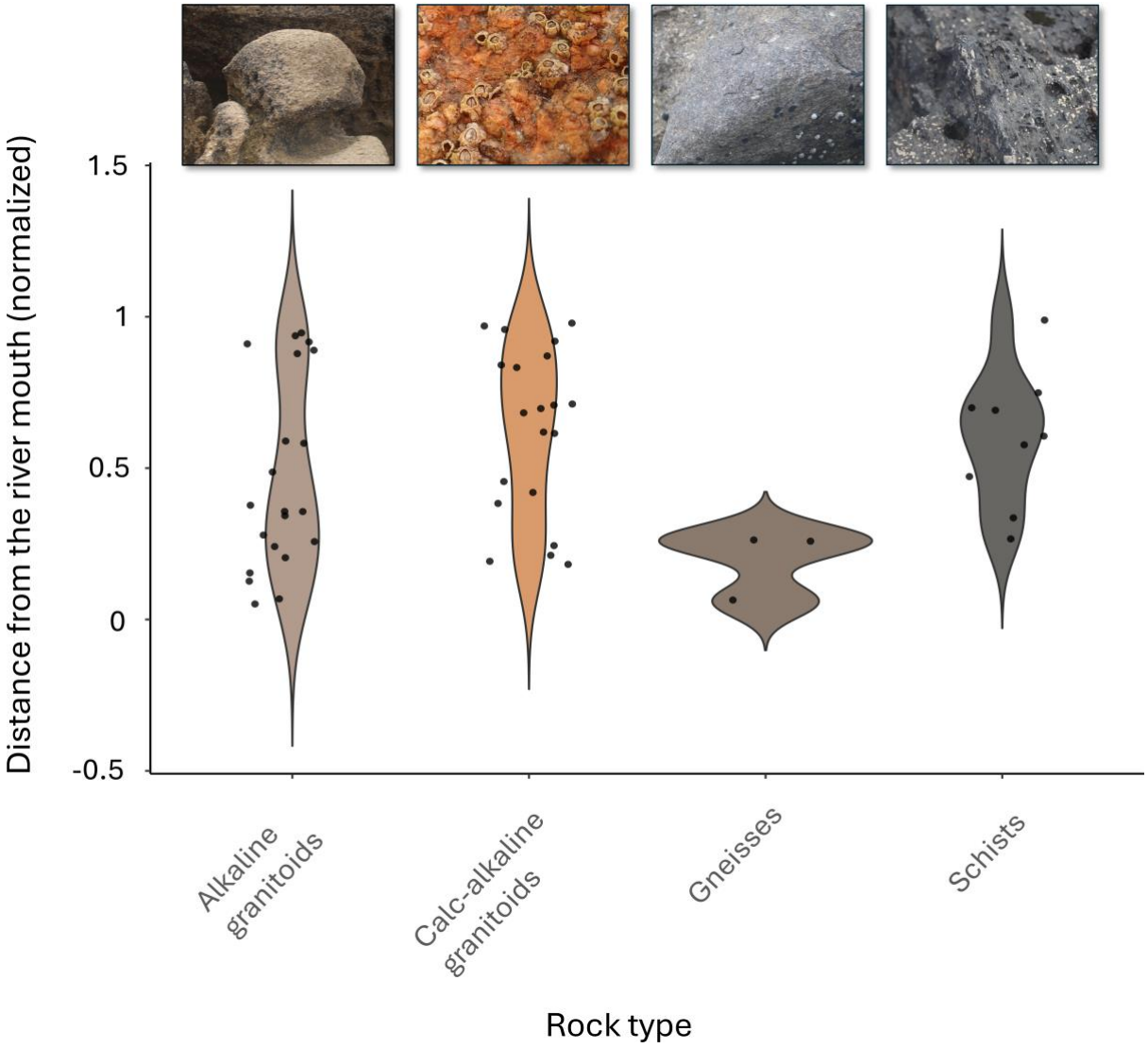

Figure S7. Violin plots showing the distribution of different types of rocks along the coastline (rias pooled). These plots serve visualization purposes only: they represent theoretical symmetrical density functions based on count data. Therefore, the plots extend beyond the edge values of distance from the river mouth (0 and 1), which is technically not possible. Furthermore, the distributions are sometimes based on very few data points, thus also demanding caution.

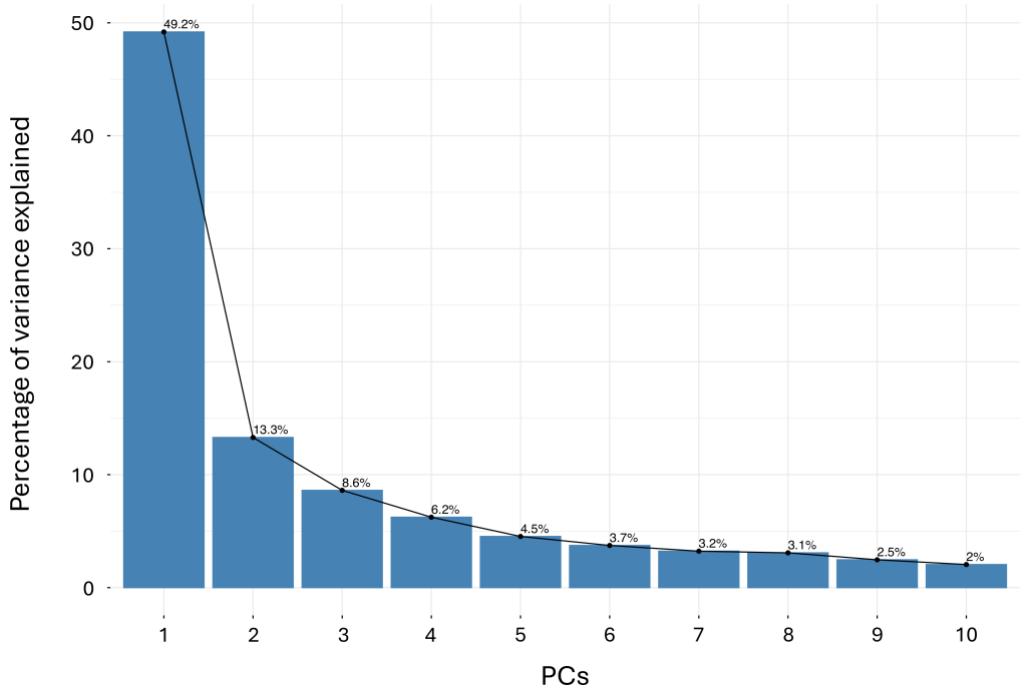

Figure S8. Scree plot of the PCA performed on the ecological variables, showing the percentage of variance explained by each major PC.

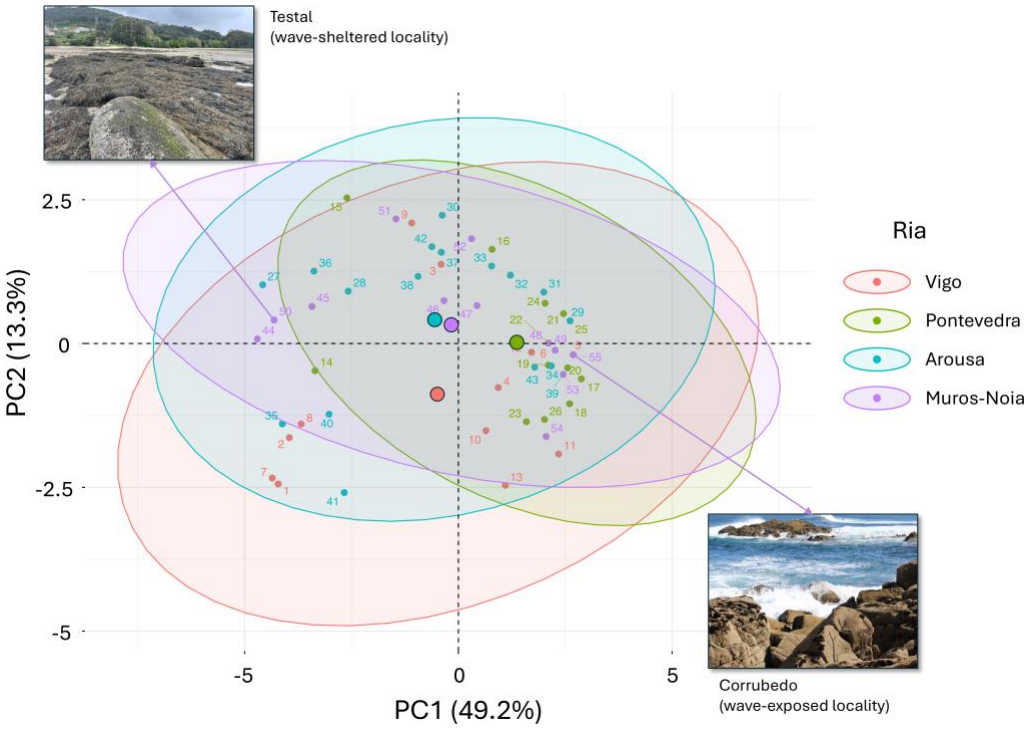

Figure S9. Plot of PC1 (x-axis) and PC2 (y-axis) values for each locality, with photographs of two extreme cases along PC1. Each color corresponds to a ria, with their corresponding 95% confidence ellipses, and each number to the one assigned to each locality studied (see Table S1). Large outlined dots represent the centroids of each ellipse.

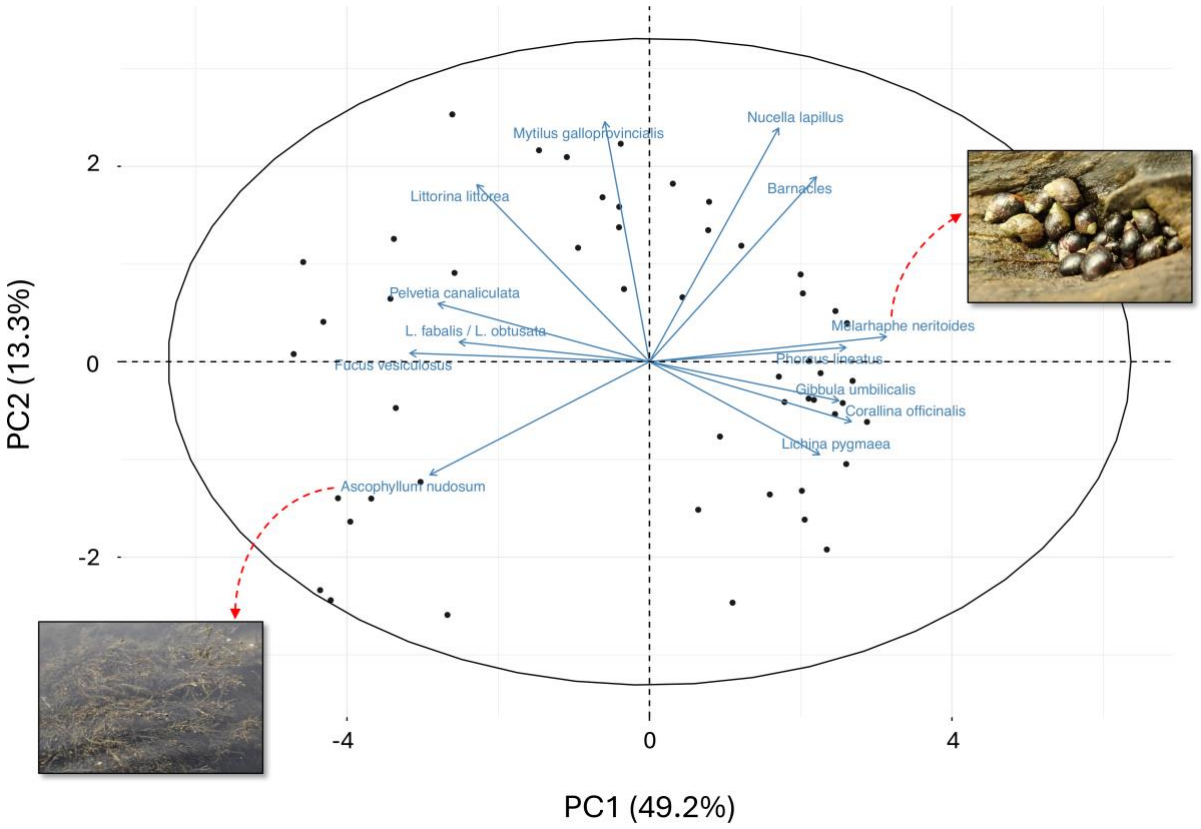

Figure S10. Biplot showing the directions of the vectors for each variable, with two clear groups: one pointing towards high values of PC1 and another pointing towards low values. The dots represent each locality studied. The red dashed lines point to photographs of *Ascophyllum nudosum* (bottom left corner) and *Melarhaphe neritoides* (top right), representative species of wave-sheltered and wave-exposed ecosystems, respectively.

The MCA yields a series of dimensions (equivalent to PCs in PCA) that explain very similar percentages of variance to those obtained in the PCA: Dimension 1 explains 47.2% of variance, while dimension 2 explains 13.5% (Figure S11). A plot of categories displaying each case for each of the variables studied is highly congruent with the equivalent graph in PCA (Figure S10) but follows the exact opposite trend (Figure S12). Species characteristic of exposed environments gather around the left edge of the plot (negative values), and species typical of wave-sheltered areas around the right edge (positive values). *Cos*^{2} (cosine square), which quantifies the contribution of each category to the MCA (the closer to 1, the higher the contribution), exhibits similar weights to the PC1 loadings (Figure 2 from the main text). This result is also confirmed in Figure S13, which shows that the absence of *Melarhaphe neritoides* (a gastropod species typical of wave-exposed ecosystems) and the presence of *Ascophyllum nudosum* (an algae species characteristic of the most wave-sheltered regions) contribute the most to Dimension 1. So taken together, the MCA corroborates the results obtained in PCA, which validates PC1 as an ecological index of wave exposure.

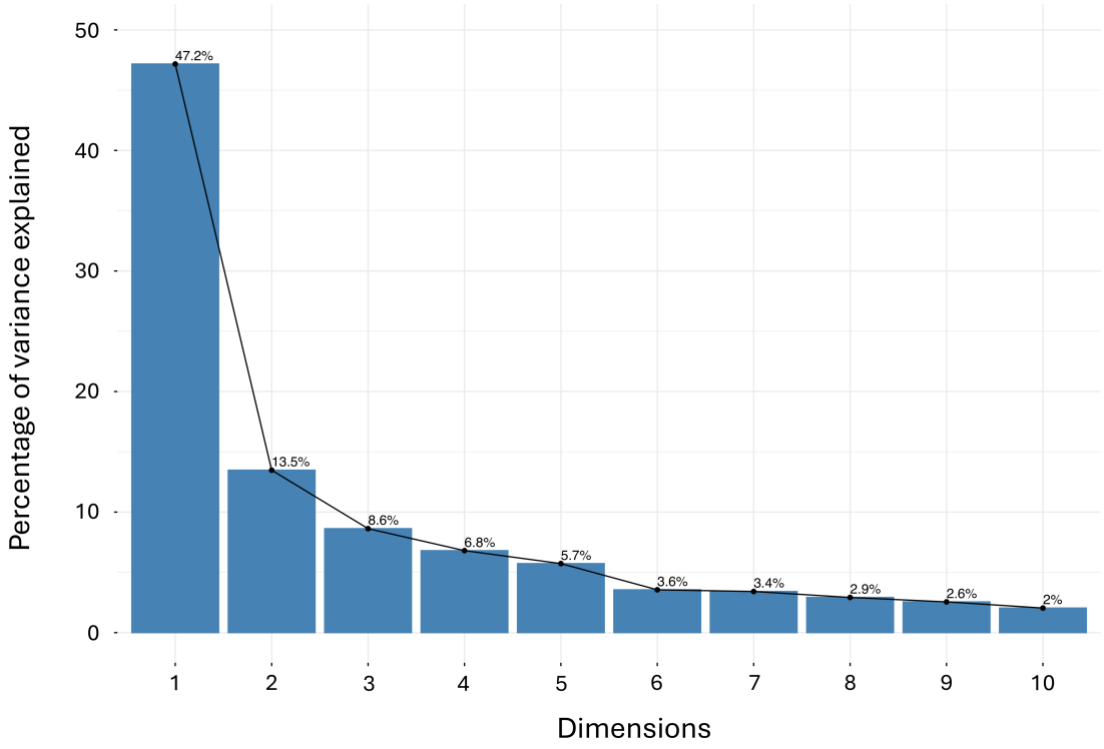

Figure S11. Scree plot displaying the percentage of variance explained by each dimension in MCA. Note the concordance with the equivalent values in Figure S8.

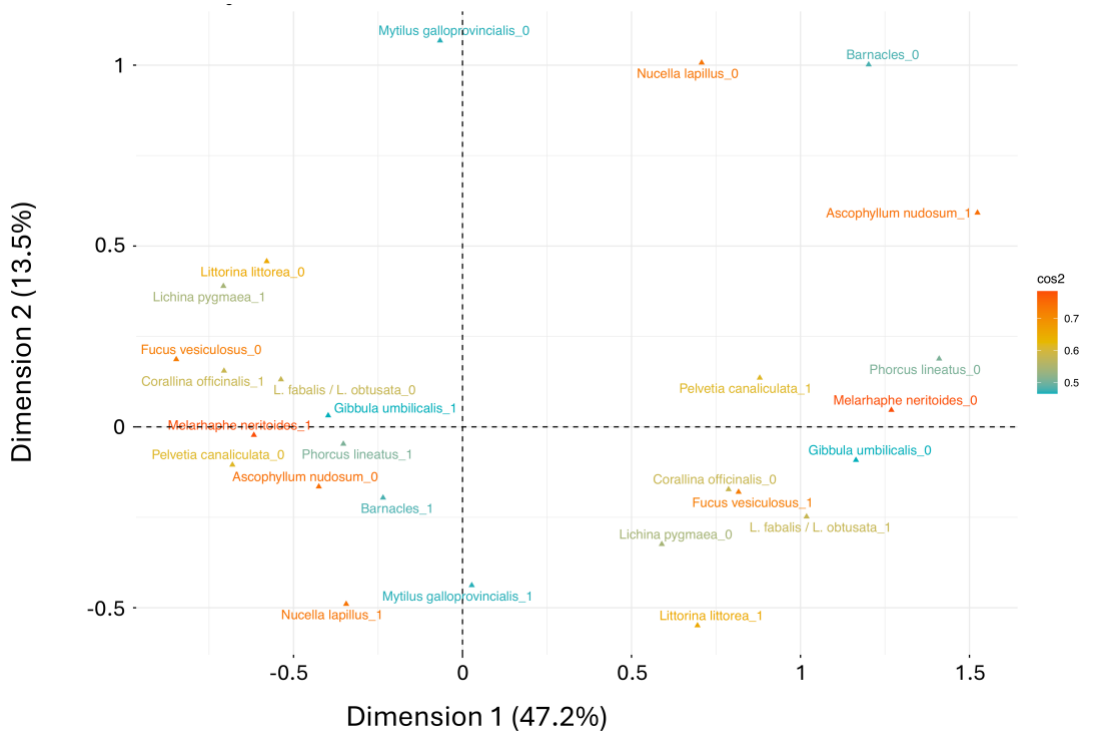

Figure S12. Plot of cases used in MCA and their positions along Dimensions 1 and 2. *Cos*^{2} ('cos2' in the plot), which measures the contribution of each variable to the MCA, is also displayed using a color gradient. Note that '_1' indicates the presence of the species, while '_0' designates its absence.

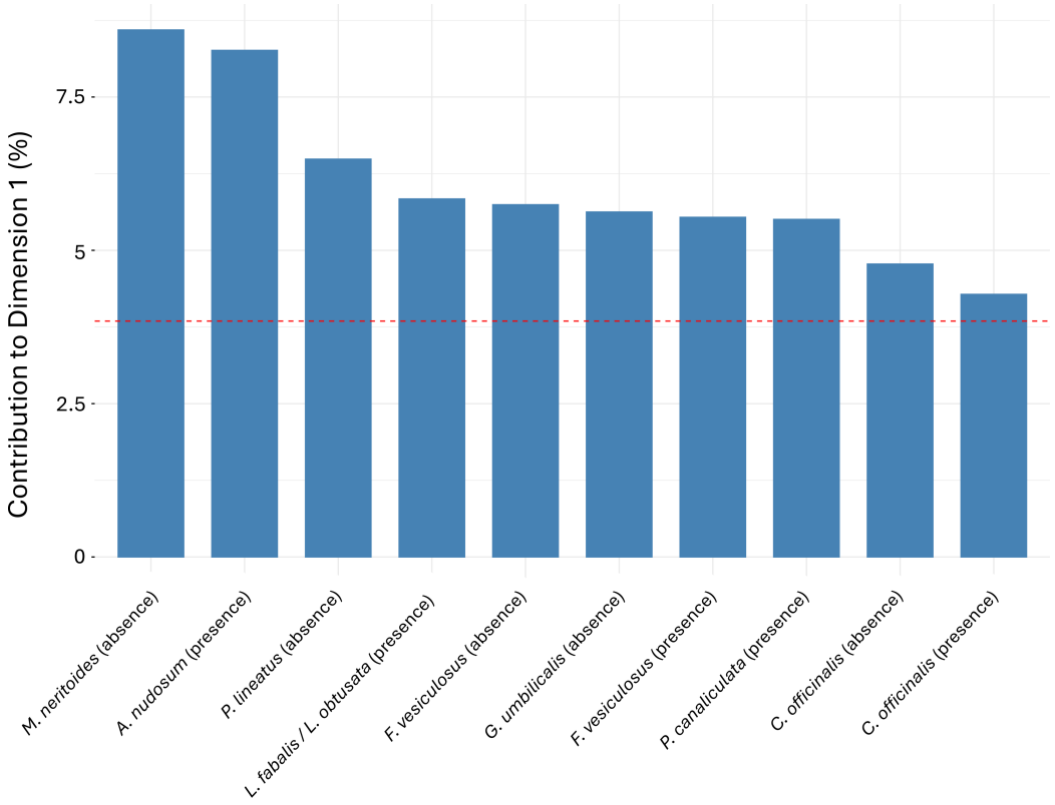

Figure S13. Contribution of each case to Dimension 1. Note that *Melarhaphe neritoides* (absence) and *Ascophyllum nudosum* (presence) make the highest contributions, much like what can be seen in PC1.

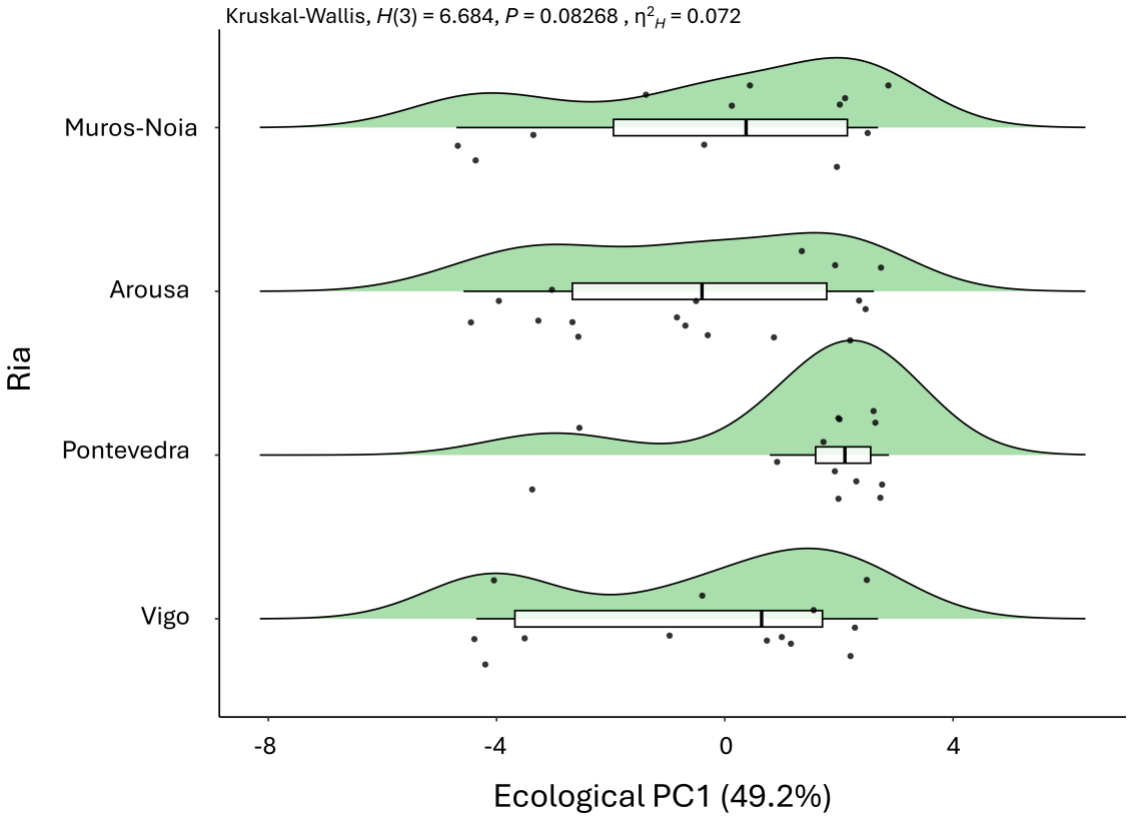

Figure S14. Boxplots and associated density plots of Ecological PC1 in each ria.

In the Sorensen Index (Figure S15), Arousa obtained the lowest value (0.56), with no instances of *Corallina officinalis*, *Lichina pygmaea*, and *Melarhaphe neritoides* found on wave-sheltered localities, and no *Ascophyllum nudosum*, *Fucus vesiculosus*, *L. fabalis* or *L. obtusata*, *L. littorea,* and *Pelvetia canaliculata* occurring in wave-exposed localities. Considering that all the rias (except for the southern coast of Pontevedra) have an approximately similar range of Ecological PC1 values (see Figure 3 in the main text), an explanation for Arousa's greatest community differences might be its relatively larger size, which might decrease connectivity between populations with different habitats. As for Pontevedra, which bears the second lowest Sorensen index value (0.59), this might have to do with the low frequency of wave-sheltered ecosystems, which might make them unrepresentative. However, the fact that the Sorensen index is obtained with presence/absence data rather than relative abundance can mask continuous –yet important– differences in community composition between types of rocky intertidal ecosystems.

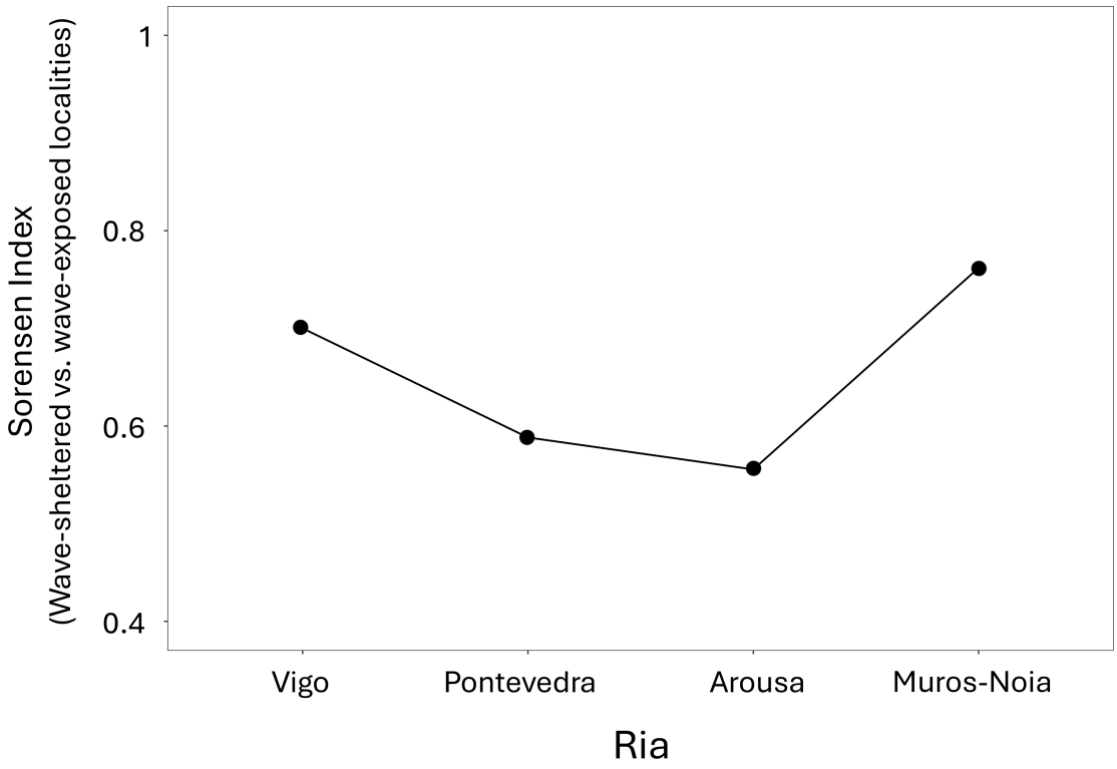

Figure S15. Sorensen similarity index between wave-sheltered and wave-exposed localities for each of the rias.

When the Moran's I values for Ecological PC1 are plotted per coast (Figure S16), an expected pattern arises: the southern coast of Pontevedra is what pushes this ria downwards, exhibiting a negative value. This indicates that the closer localities have more distinct Ecological PC1 values. This can be explained by this coast lacking a sheltered area, as already remarked. Additionally, unlike the rest of the rias, the extremest wave-exposed locality on this coast (Beluso, $P_{S\text{-}5}$) is situated between two localities with intermediate Ecological PC1 values (i.e., wave-intermediate). This effect is likely amplified by the low sample size (n = 6).

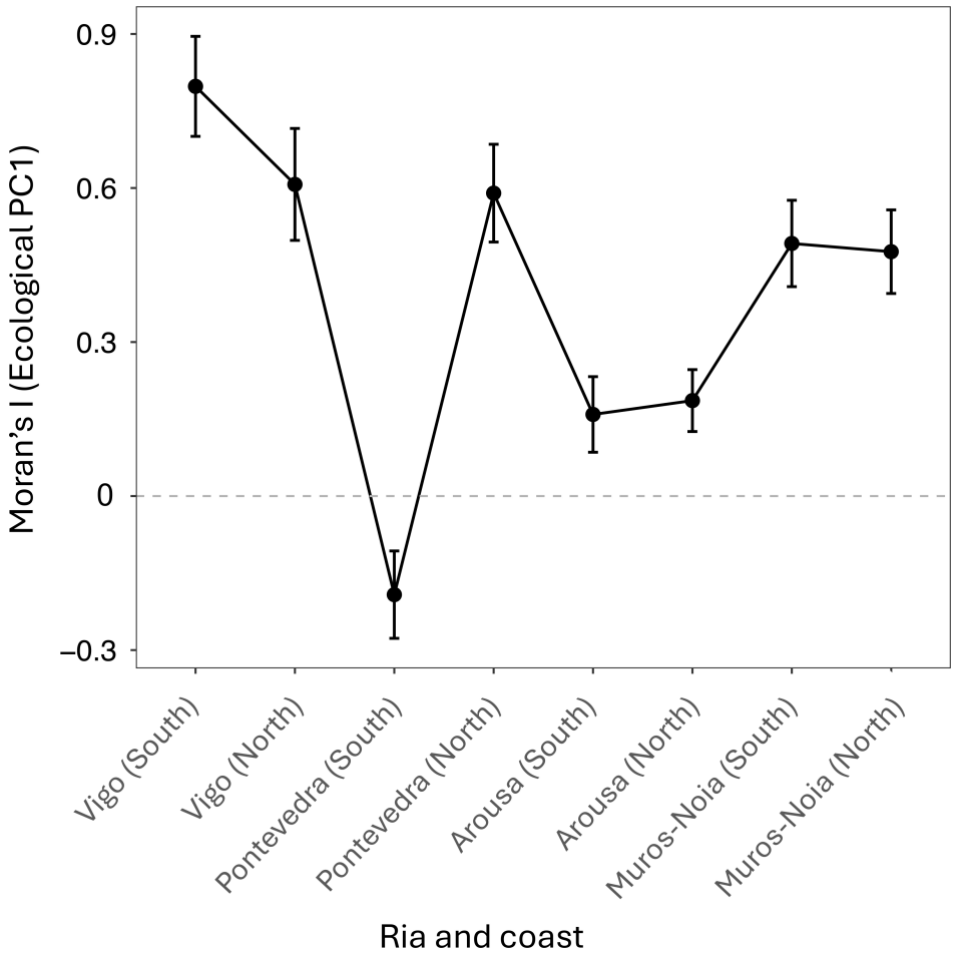

Figure S16. Moran's I Ecological PC1 values for each coast of the Rias Baixas (error bars displaying their variance). Note the discordant value in Pontevedra (South), likely explained by its geomorphology (see the text for further details).

Extended Results (3): Further analyses on shell color variation across the wave exposure gradient

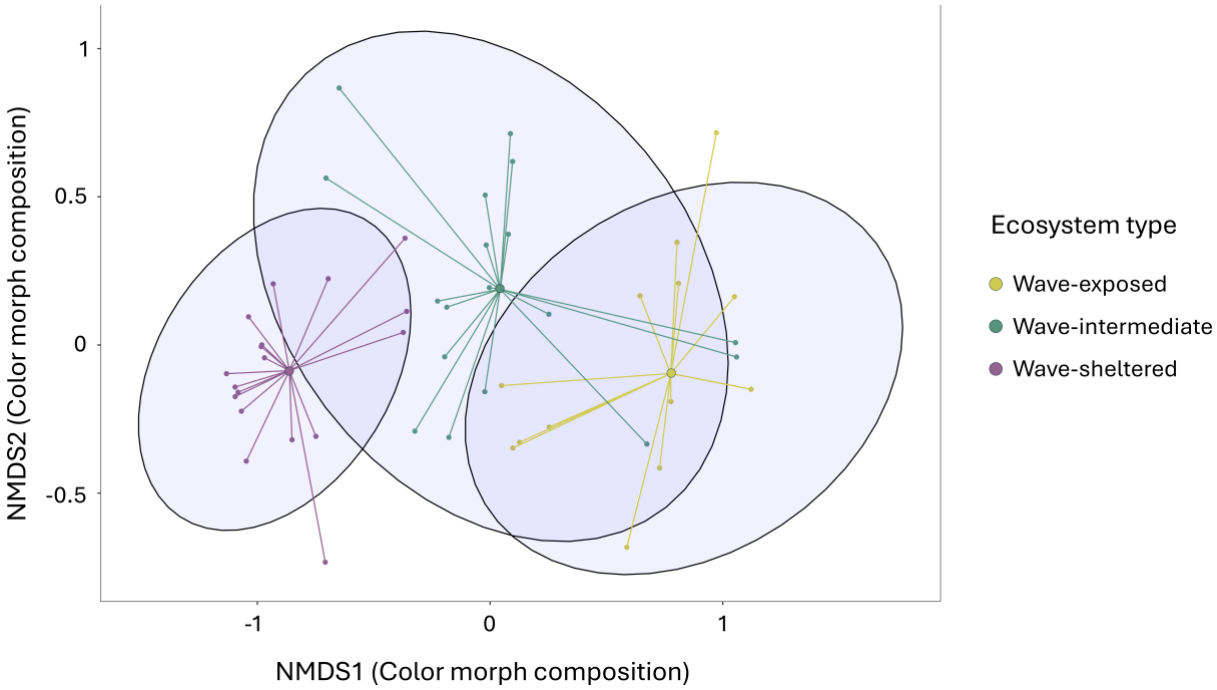

Figure S17. Non-metric Multidimensional Scaling (NMDS) plot of color morph composition across ecosystem types. The outlined dots correspond to the group centroids.

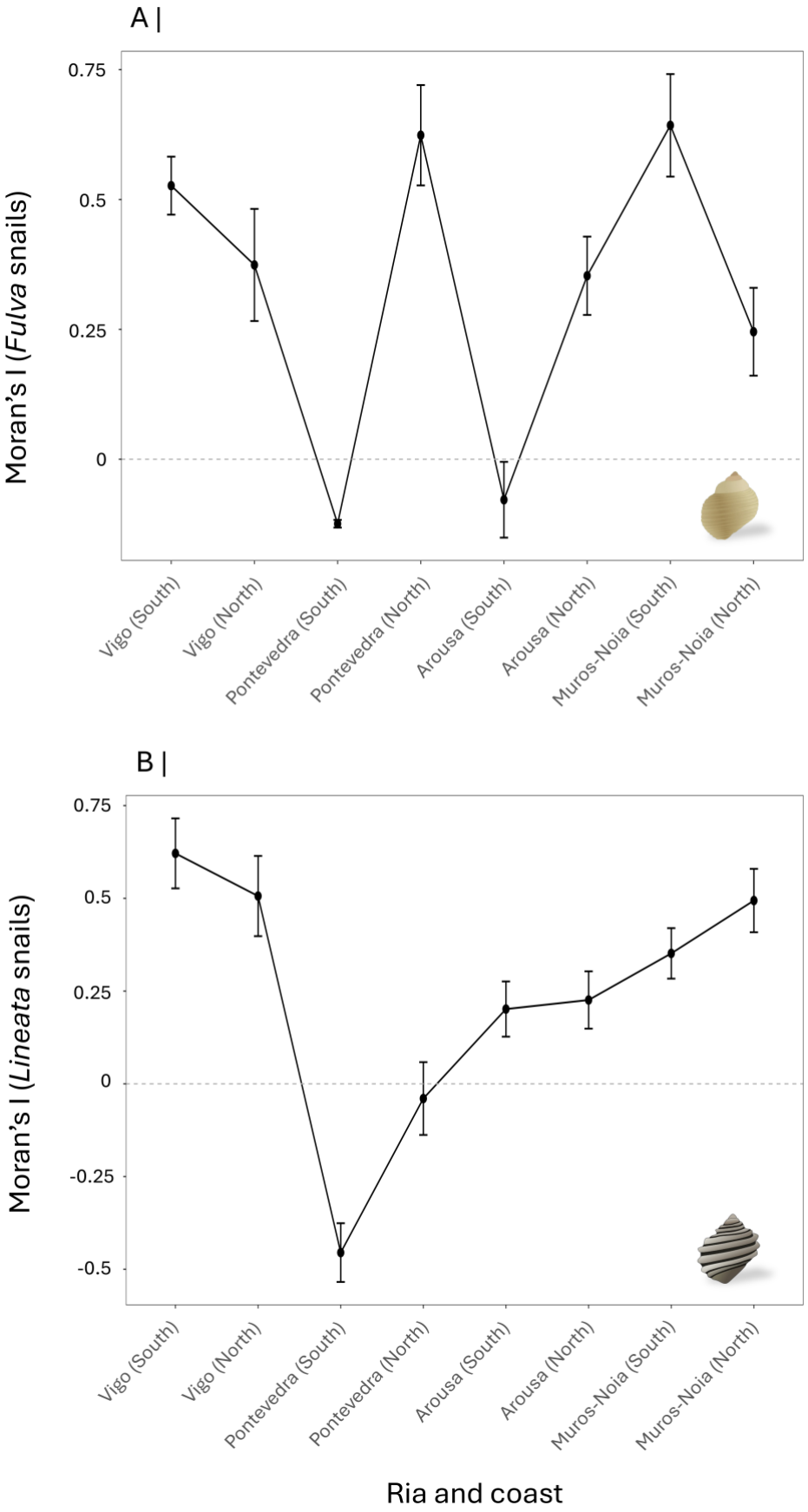

Figure S18. Moran's I of *fulva* (A) and *lineata* (B) snails per coast. Error bars are the variances.

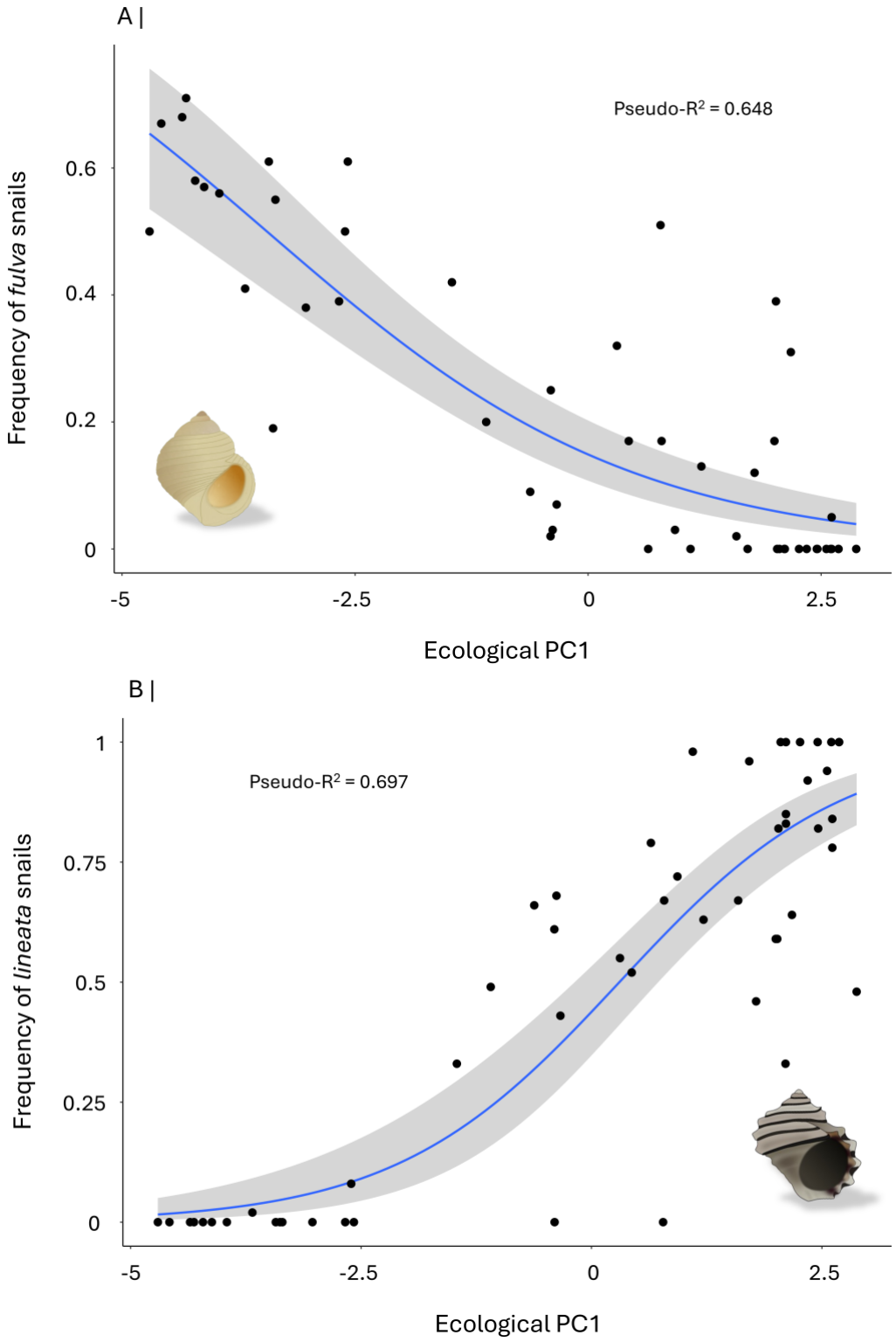

Figure S19. Quasi-binomial models fitting Ecological PC1 as a predictor of *fulva* (A) and *lineata* (B) frequencies (localities of all rias pooled). Pseudo-R^2 values correspond to McFadden's pseudo-R^2 and the shaded areas to 95% confidence intervals. Note that the x-axes of both figures are not at scale, due to *fulva* exhibiting generally lower frequencies than *lineata* in their corresponding peak localities.

Extended Results (4): Further analyses on variation in other shell traits across the wave exposure gradient

The adjustment between Ecological PC1 and shell traits is quite similar and good across rias for shell sculpture, as determined by the frequency of ribbed (*jugosa*) shells (Pseudo-R^2 values between 0.54 and 0.66, with the Ecological PC1 estimates being significant in all of them; see Figure S20). Conversely, there is substantial adjustment variation for the mean shell length of the snails (Figure S21), despite significance in all rias. Here, Vigo and Arousa cluster together (R^2 = 0.783 and 0.591, respectively), with a more noticeable cline than Pontevedra and Muros-Noia (R^2 = 0.454 and 0.396, respectively). In relative warp 1, there are also differences in adjustment cross rias (Figure S22): Vigo and Muros-Noia show good fitting (adjusted R^2 = 0.777 and 0.748, respectively), Arousa an intermediate, yet significant fitting (adjusted R^2 = 0.316, P = 0.01101), and Pontevedra no fitting at all (adjusted R^2 = 0.02435, P = 0.2785). In the first three rias, snails tend to be spikier and narrow-apertured in wave-sheltered areas than in wave-exposed zones, where they are rounder and wide-apertured (lower and higher relative warp 1 values, respectively). The adjustment between shell thickness and Ecological PC1 is, in turn, virtually nonexistent in all rias (Figure S23), in some cases yielding even negative values (i.e., Ecological PC1 being a worse predictor of shell thickness than the overall mean).

The parameters reflecting the ecological conditions also vary in adjustment quality between rias. Ecological PC1 does a good job at predicting variation in the frequency of scars in Vigo and Muros-Noia (with higher percentages of scarred snails in wave-sheltered localities) and a more modest job –yet significant– in Pontevedra and Arousa (Figure S25). Around 30% of the variance in shell wear is predicted by Ecological PC1 in all rias except Pontevedra, which performs poorly (Figure S26). However, Ecological PC1 is only significant in Arousa (β estimate = 0.20, P = 0.0116).

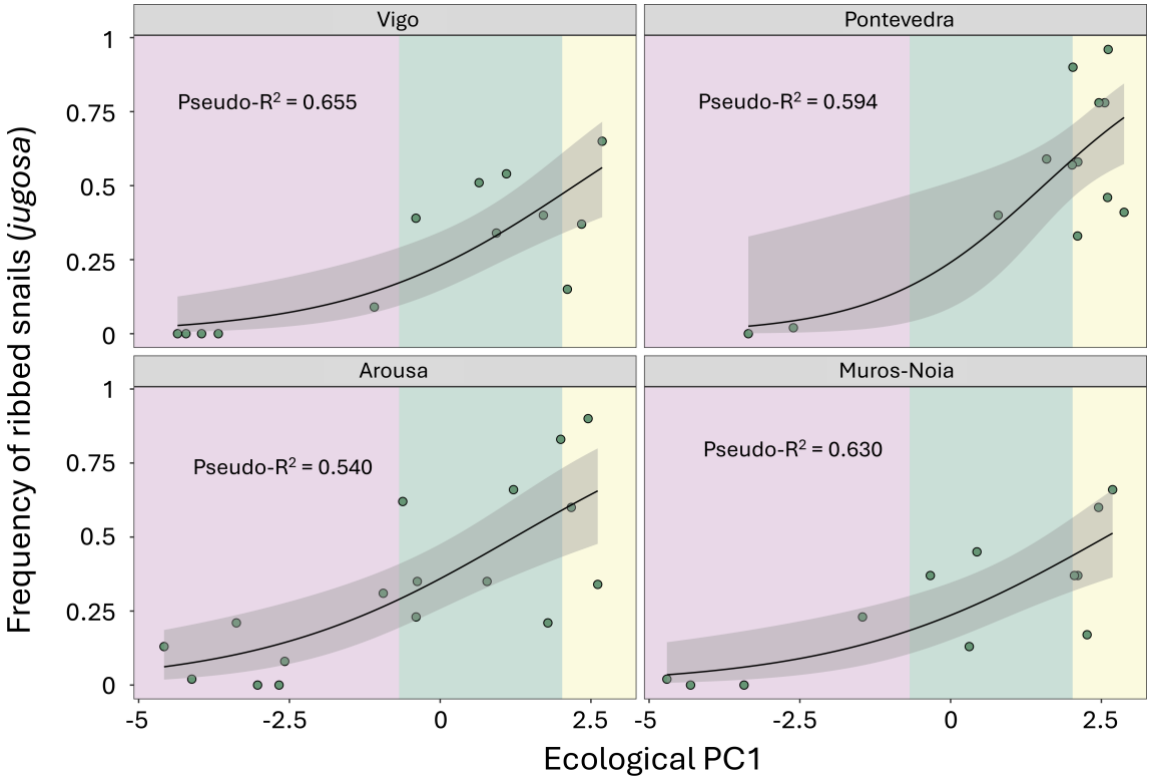

Figure S20. Quasi-binomial regressions for the frequency of ribbed snails (*jugosa*) across Ecological PC1 values per ria, with their corresponding McFadden's pseudo-R^2 values. Colors represent the Ecological PC1 limit values for each ecosystem type: pink-purple for wave-sheltered habitats, green for wave-intermediate habitats, and yellow for wave-exposed habitats.

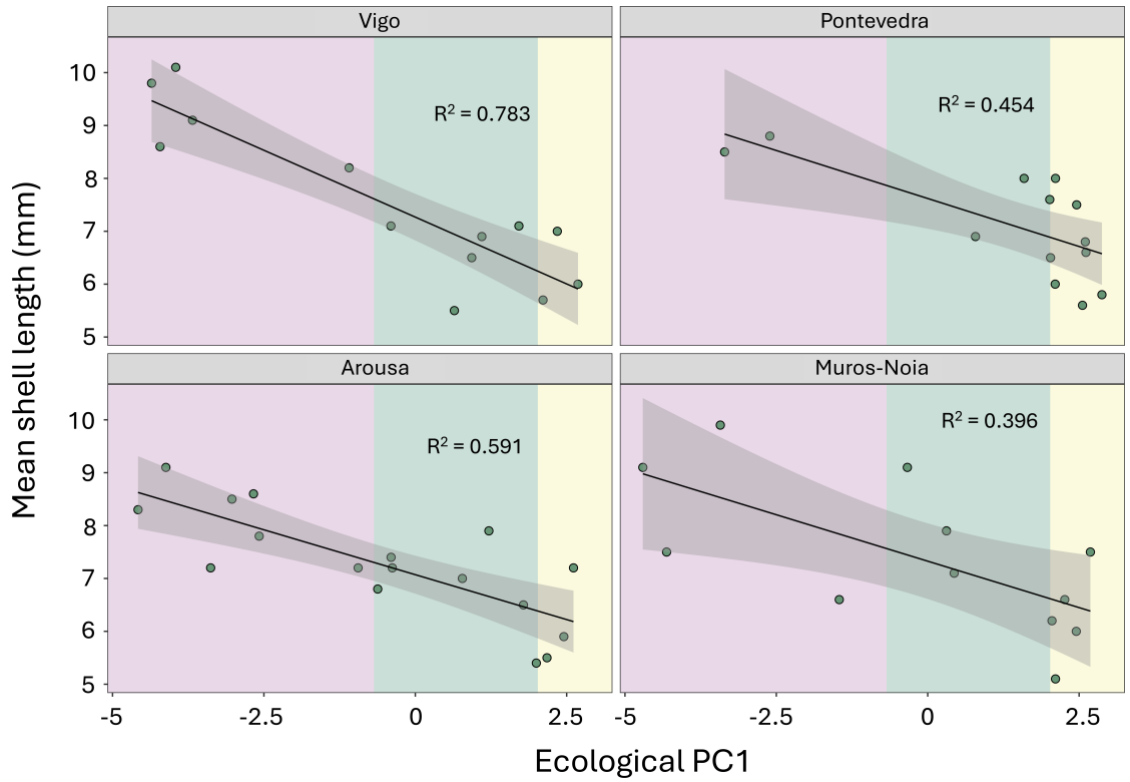

Figure S21. Linear regressions of the mean shell length (in mm) of the populations studied against Ecological PC1, with their corresponding adjusted R^2 values.

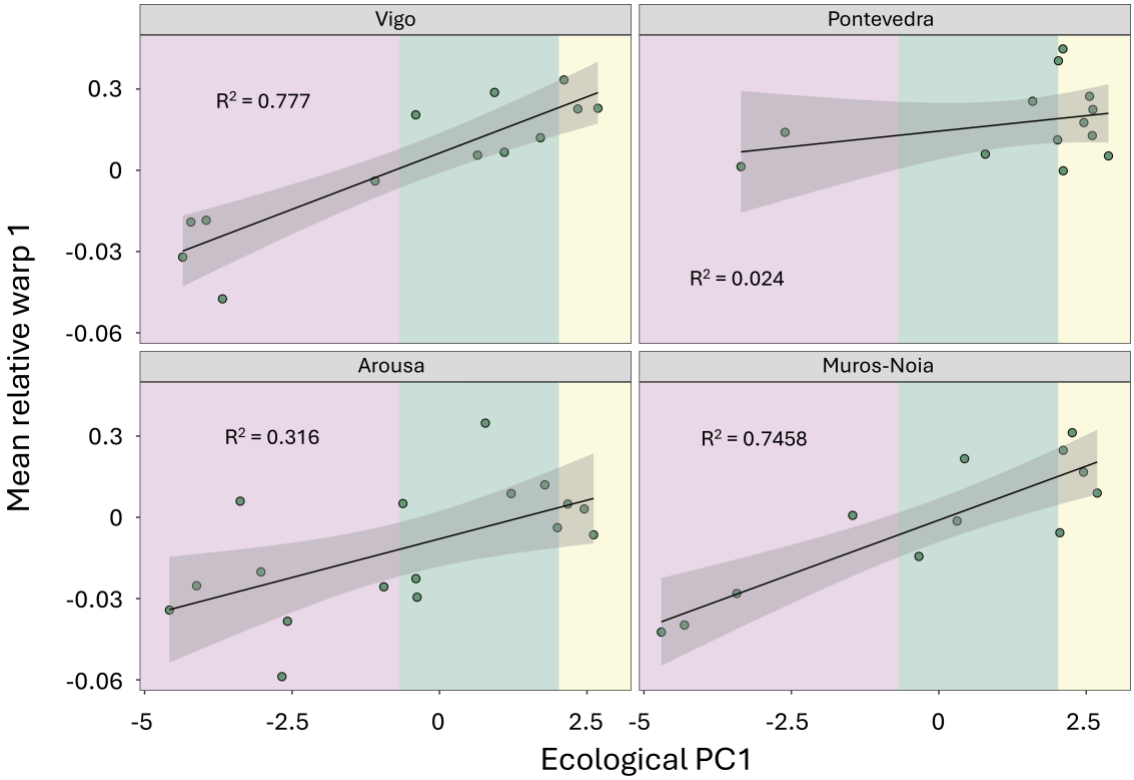

Figure S22. Relative warp 1 mean values of the populations across Ecological PC1 values, separated by ria, with their corresponding adjusted R^2 values.

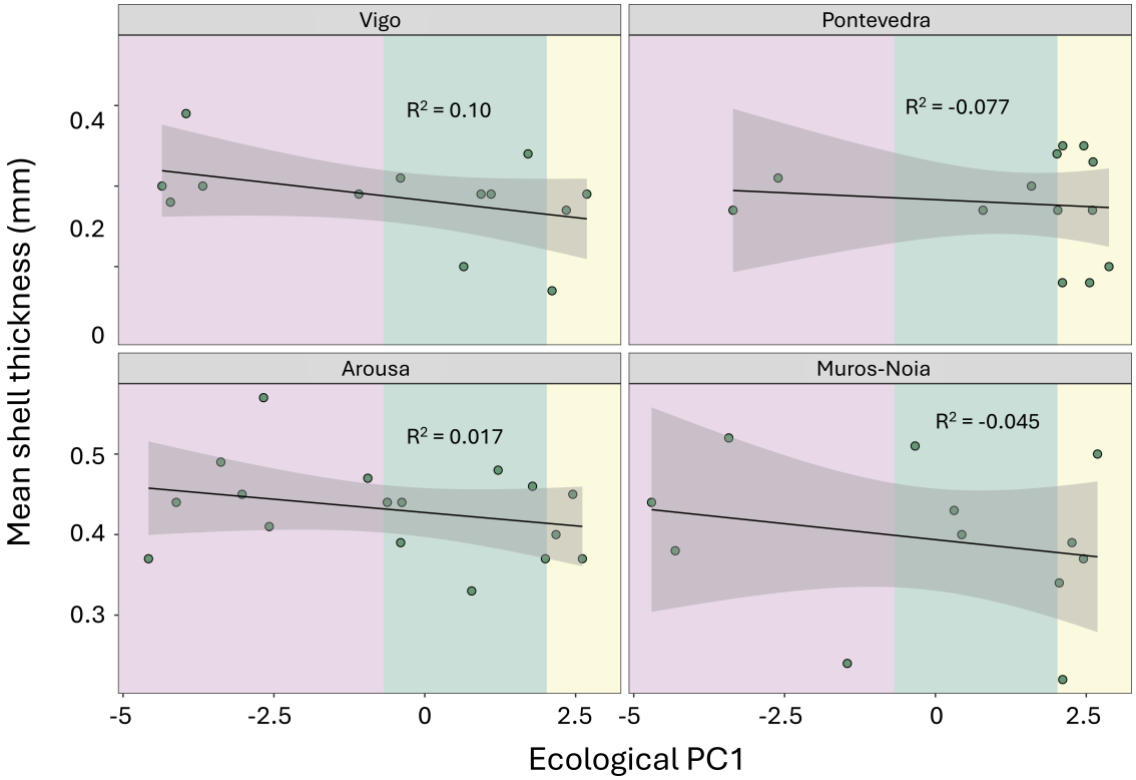

Figure S23. Mean shell thickness of the populations studied across Ecological PC1 values per ria, with their adjusted R^2 values. Note that negative adjusted R^2 values indicate a poorer predictive performance than that of the overall shell thickness mean.

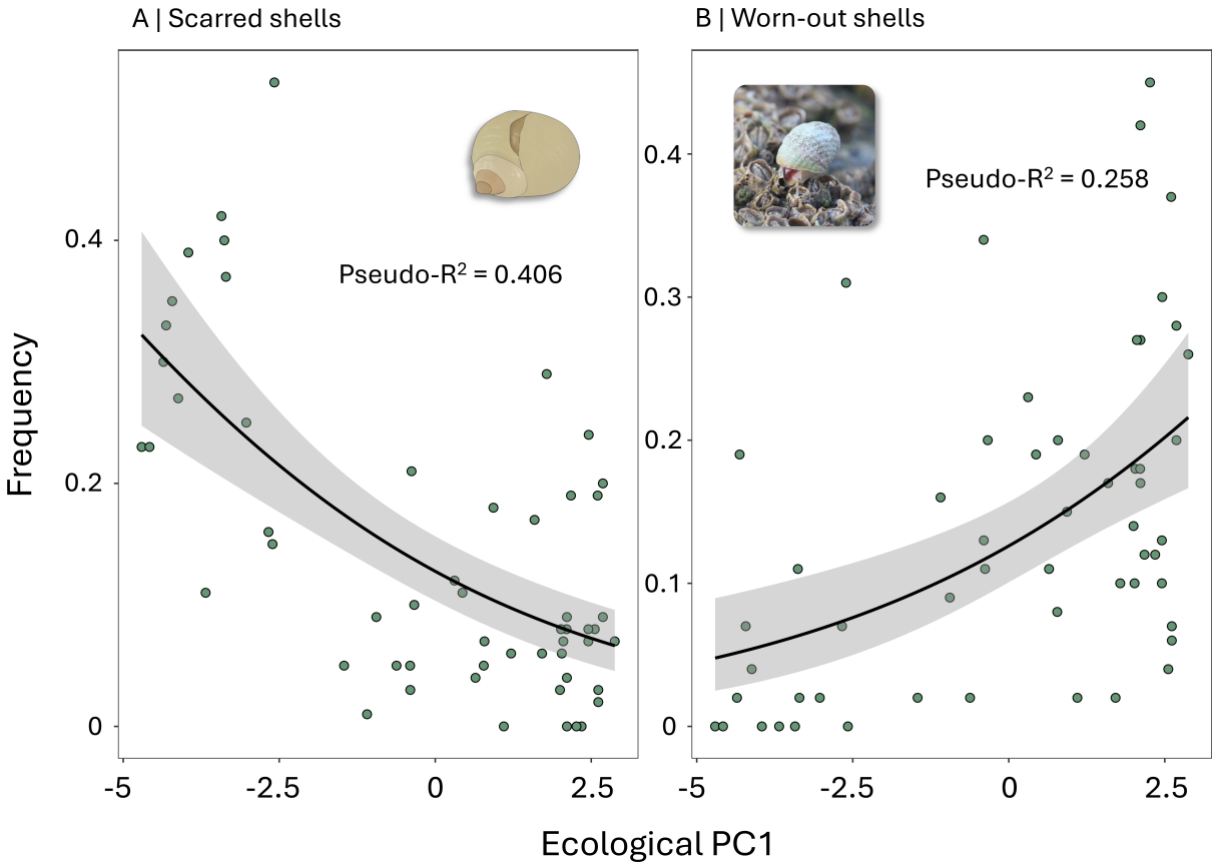

Figure S24. Quasi-binomial models of the frequency of scarred (A) or worn (B) shells across Ecological PC1 values. Pseudo-R^2 values correspond to McFadden's pseudo-R^2.

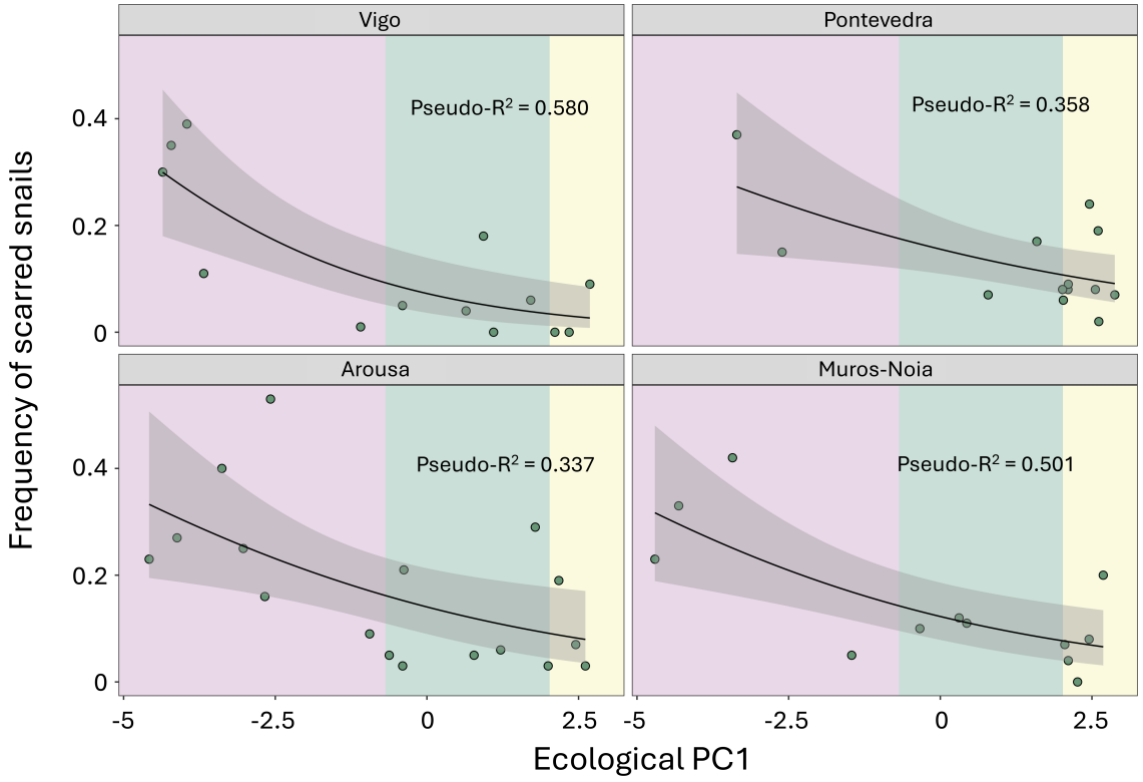

Figure S25. Frequency of scarred snails across localities, separated by ria.

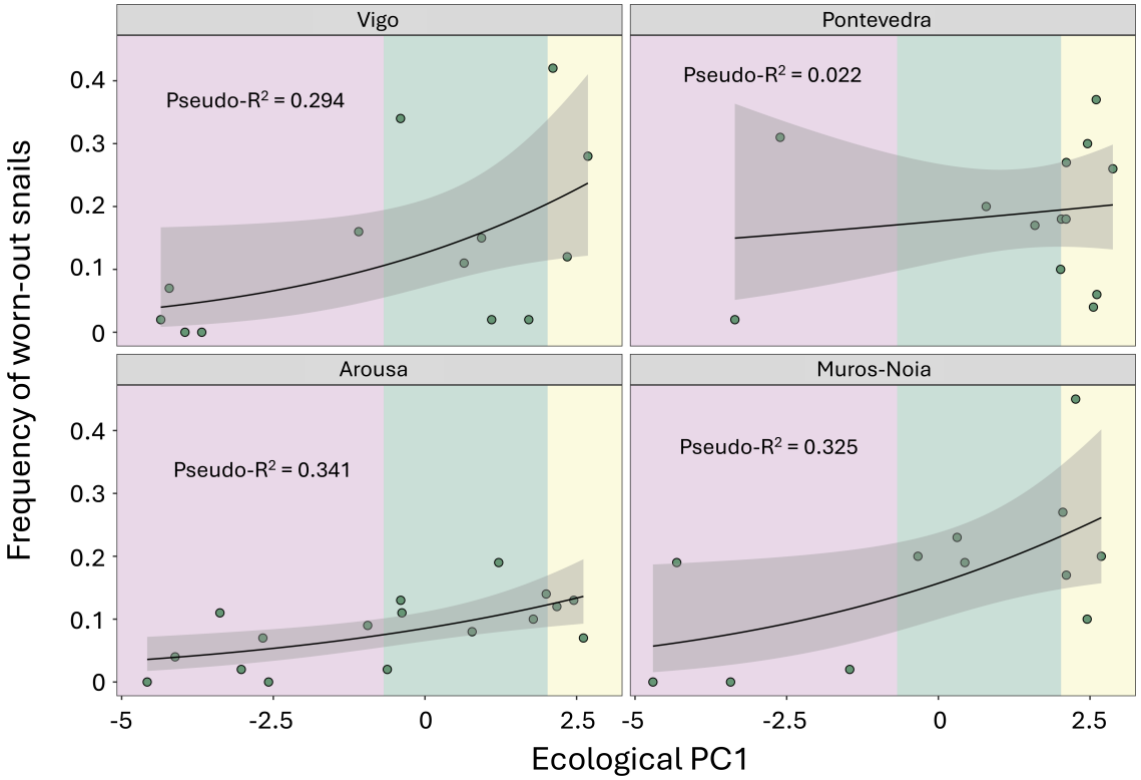

Figure S26. Frequency of worn-out shells per population along Ecological PC1 values in each ria.

Extended Results (5): Further analyses on shell trait covariation

The first step towards PCA involves assessing potential multicollinearity issues between the studied variables (shell traits with a presumably strong genetic basis), as this can compromise the analysis. Figure S27 displays a pairs plot of the considered parameters, showing that numerous statistically significant correlations exist between the variables, but not of a concerning magnitude (all with $\rho < 0.62$). Furthermore, a linear model was performed using three of the studied parameters (mean shell length, mean shell thickness, and mean relative warp 1) as predictors of the remaining one (frequency of *jugosa*) to study multicollinearity through the Variance Inflation Factor (VIF). This resulted in values < 2.06, confirming that multicollinearity is not an issue. Thus, we proceed with PCA.

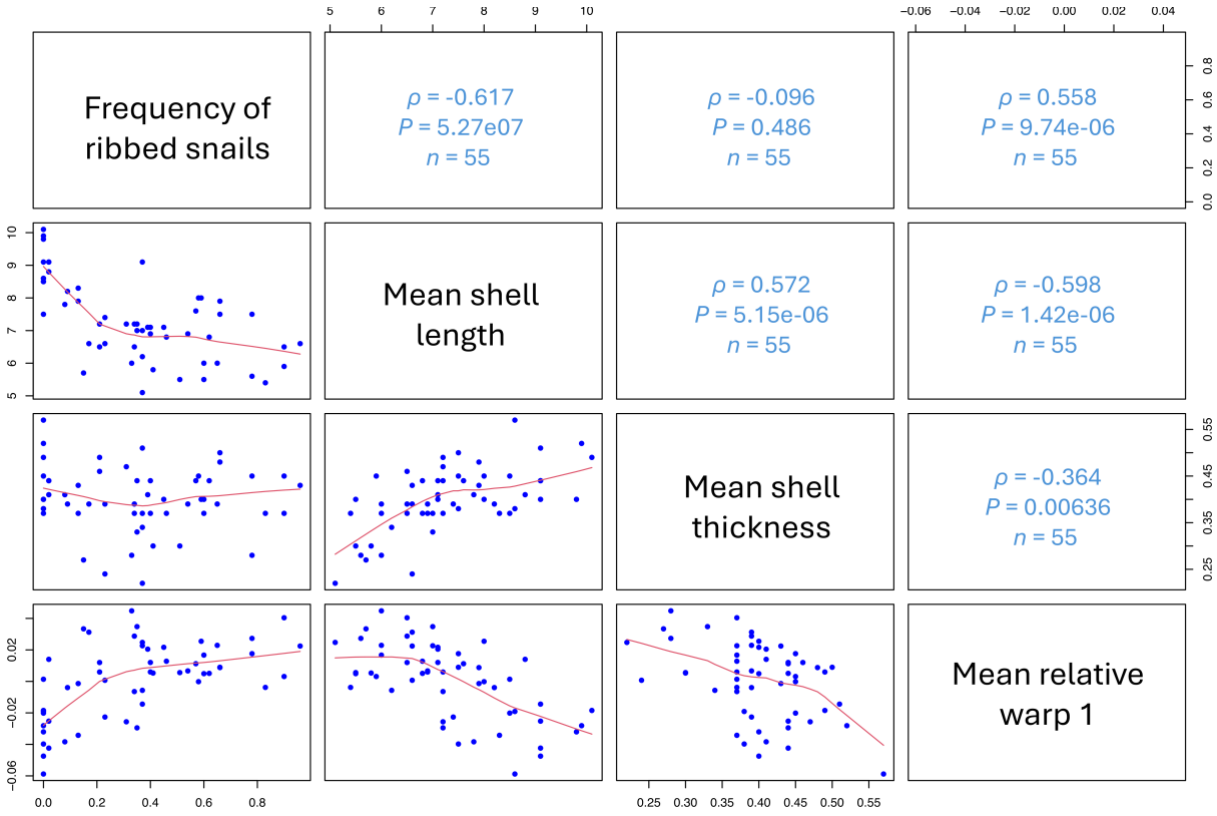

Figure S27. Pairs plot of the shell variables studied in PCA. The lower half of the plot shows the scatterplots between pairs of variables with their corresponding LOESS (locally weighted scatterplot smoothing) regression lines. The upper half shows Spearman's rank correlation coefficients (ρ), their associated probabilities (P), and the sample size (n) corresponding to the 55 studied localities.

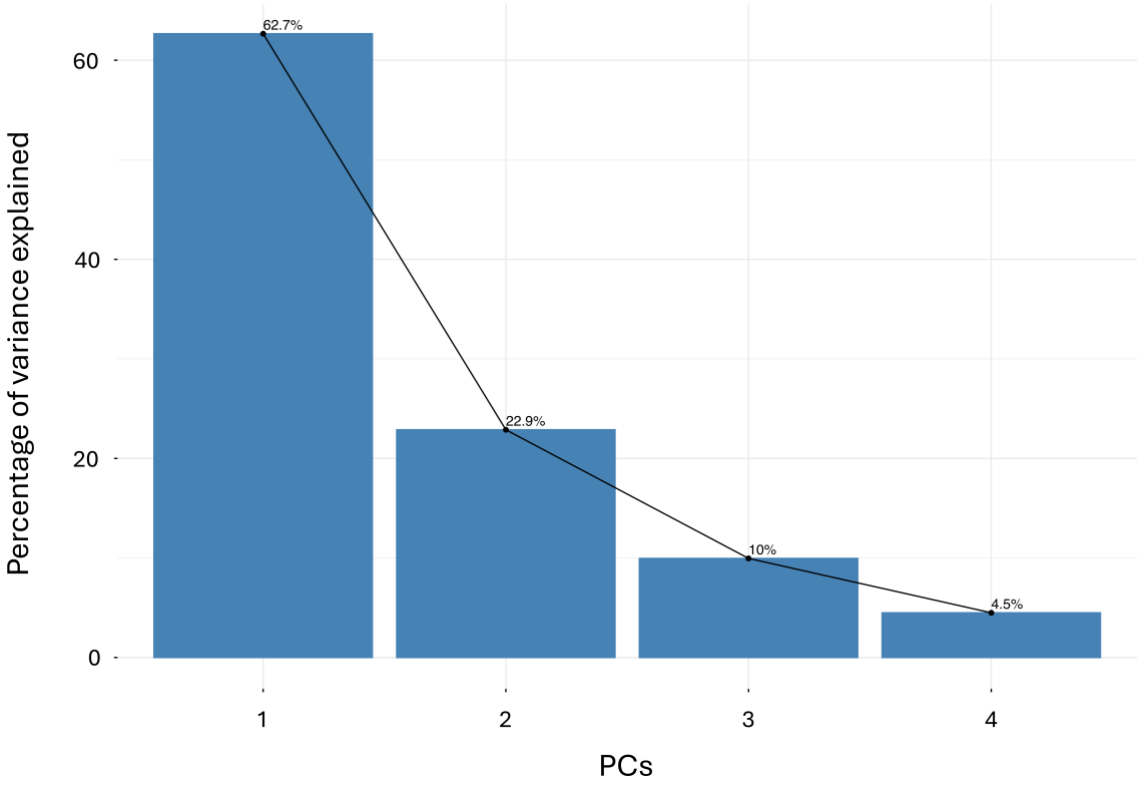

Figure S28. Scree plot of the PCA conducted on shell traits with a putatively strong genetic basis, displaying the main PCs and their percentage of variance explained.

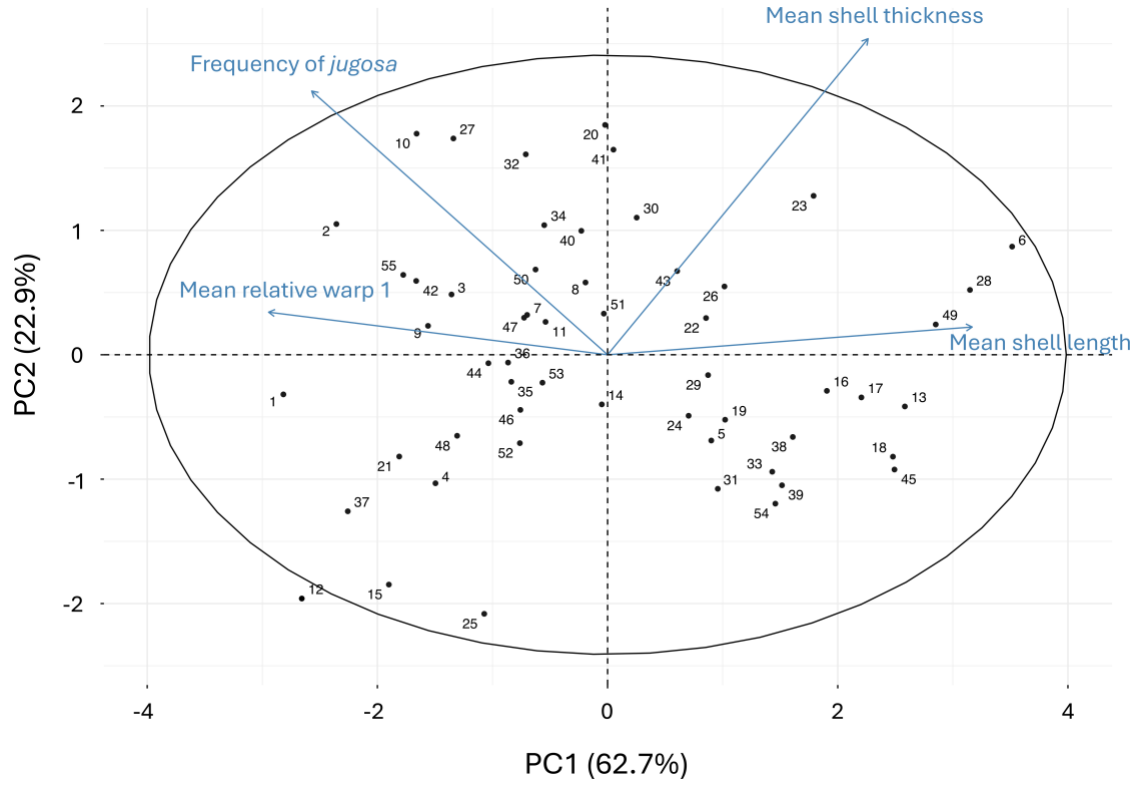

Figure S29. Biplot showing the direction of the vectors for each shell parameter studied. The dots represent the localities, along with their corresponding number in the original dataset.

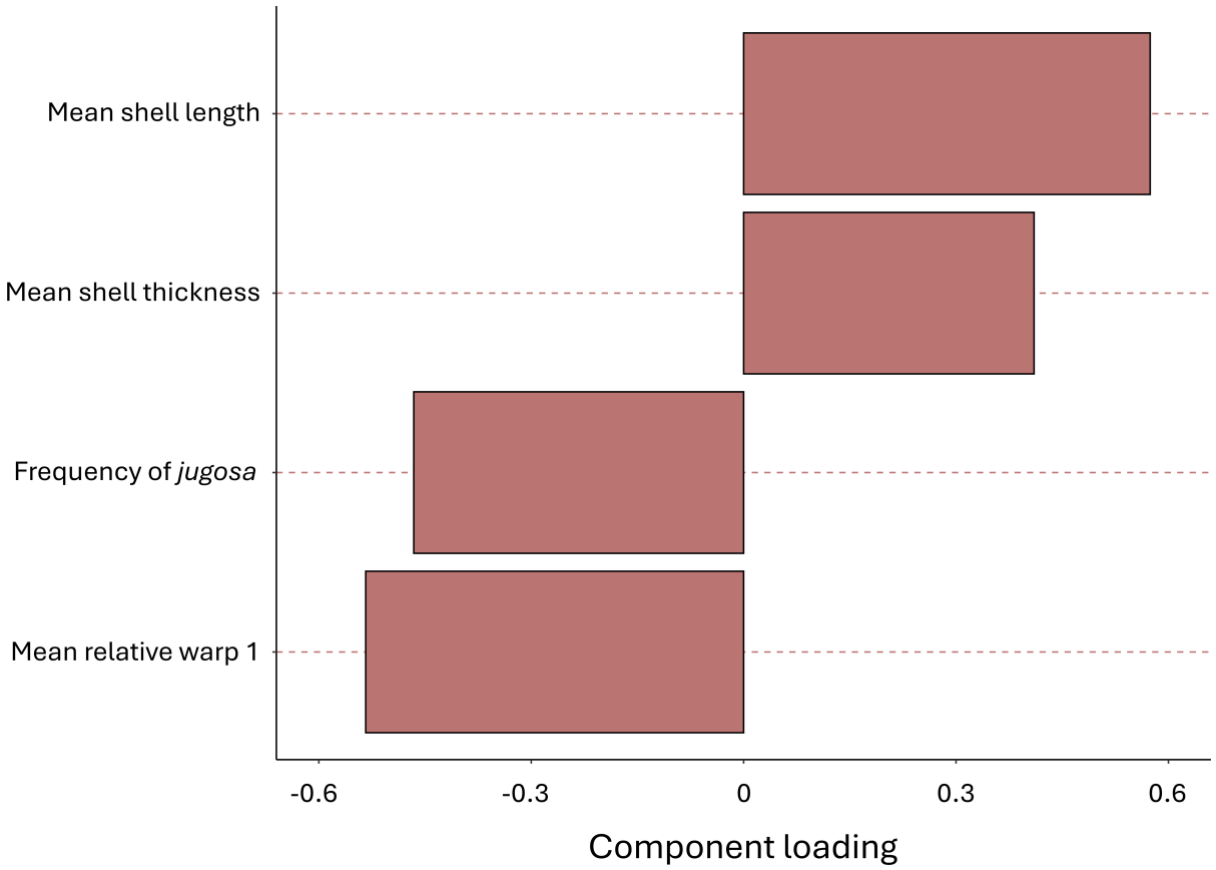

Figure S30. Plot of component loadings of the main variables studied for PC1. Two distinguishable clusters exist, with frequency of *jugosa* and mean relative warp 1 in one cluster, and mean shell length and thickness in the other.

Figure 13 from the main text might suggest that Shell traits PC1 values are similar between wave-intermediate (green) and wave-exposed (yellow) localities, which would be different from wave-sheltered localities (pink-purple). This was formally assessed with an ANOVA of Shell traits PC1 by ecosystem type (see Figure S31). The ANOVA results, along with the Tukey HSD post-hoc test, refuted this interpretation, revealing statistically significant differences between the three ecosystem types in Shell traits PC1. Wave-intermediate localities exhibit intermediate values of Shell traits PC1, indicating that variation in shell parameters appears to be continuous rather than organized into discrete clusters.

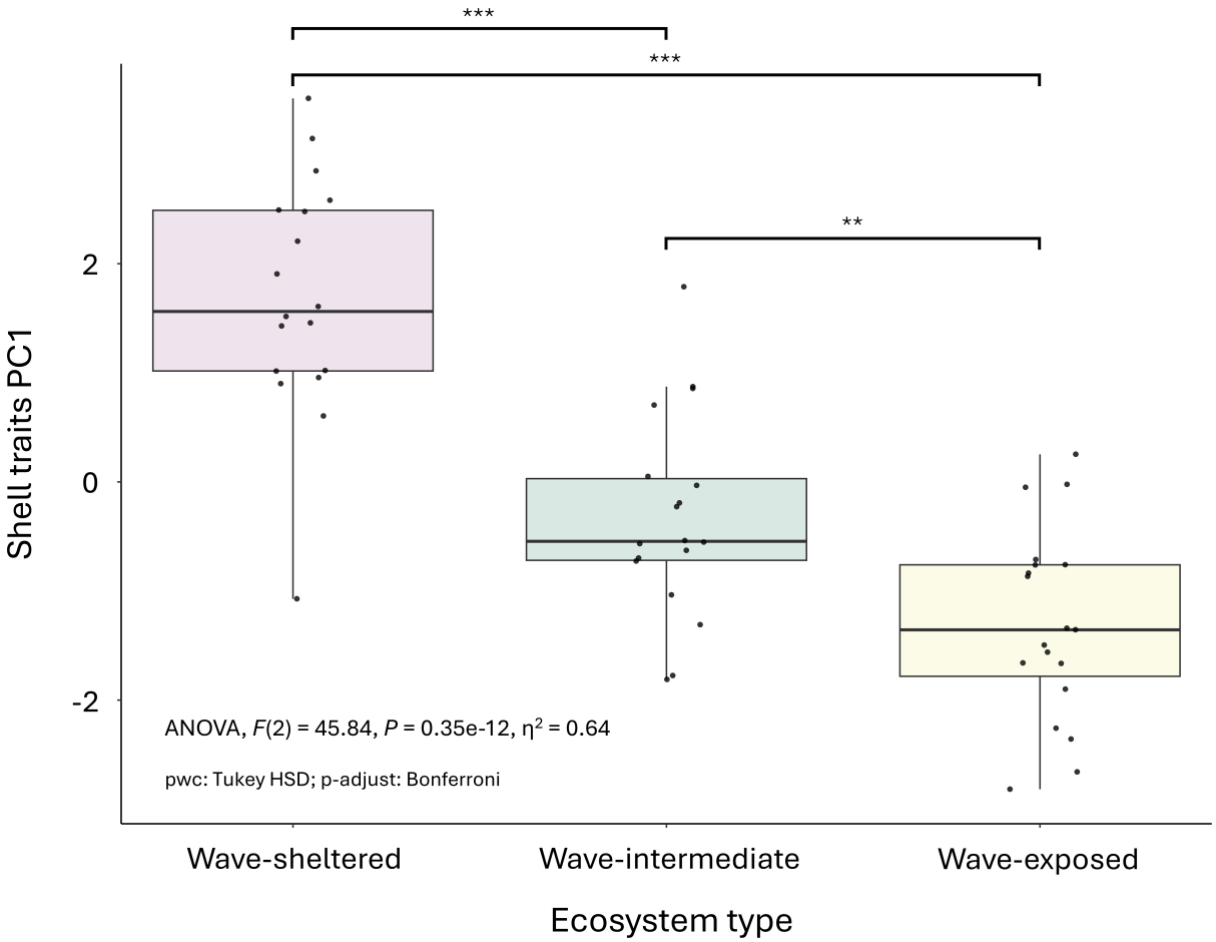

Figure S31. Boxplot of Shell traits PC1 per ecosystem type, with their corresponding ANOVA and Tukey HSD post-hoc test results. ** $P < 0.01$, *** $P < 0.001$.

Table S5. Linear models using Shell traits PC1 as a predictor of *fulva* and *lineata* frequencies, calculated per ria. * $P < 0.05$, ** $P < 0.01$, *** $P < 0.001$.

Dependent variable	Ria	Predictor	Estimate (β)	SE	*T*	*P*	Adjusted R^2
Fulva frequency	Vigo	Intercept	0.17381	0.03147	5.523	0.00018	0.8189***
		Shell traits PC1	0.14091	0.01895	7.435	1.30e-05	
	Pontevedra	Intercept	0.23694	0.0512	4.628	0.000731	0.5048**
		Shell traits PC1	0.1278	0.03513	3.637	0.003905	
	Arousa	Intercept	0.24887	0.04867	5.113	0.000127	0.1685
		Shell traits PC1	0.06803	0.03303	2.06	0.057226	
	Muros-Noia	Intercept	0.21371	0.0597	3.579	0.00502	0.4085*
		Shell traits PC1	0.1025	0.03496	2.932	0.01499	
Lineata frequency	Vigo	Intercept	0.59099	0.04956	11.925	1.24e-07	0.8152***
		Shell traits PC1	-0.21916	0.02985	-7.342	1.46e-05	
	Pontevedra	Intercept	0.521	0.1048	4.972	0.000421	0.1569
		Shell traits PC1	-0.1293	0.0719	-1.798	0.099604	
	Arousa	Intercept	0.4023	0.06996	5.75	3.83e-05	0.3532**
		Shell traits PC1	-0.14816	0.04748	-3.121	0.00701	
	Muros-Noia	Intercept	0.58859	0.08195	7.182	2.99e-05	0.5292**
		Shell traits PC1	-0.17542	0.04799	-3.655	0.00442	

Extended Results (6): Shell color and the neutral hypothesis

The neutral hypothesis predicts that higher *L. saxatilis* densities should be accompanied by a higher color diversity, irrespective of ecosystem type. Alternatively, if all (or at least most) color morphs are adaptive, color diversity might depend more on ecosystem characteristics than on *L. saxatilis* density. This was visually assessed by plotting the Shannon index values of the localities against either Ecological PC1 (Figure S32A), or average *L. saxatilis* density (Figure S32B). A LOESS (locally weighted scatterplot smoothing) regression model (span = 0.75) was fitted to each scatterplot to reveal potential trends in the data.

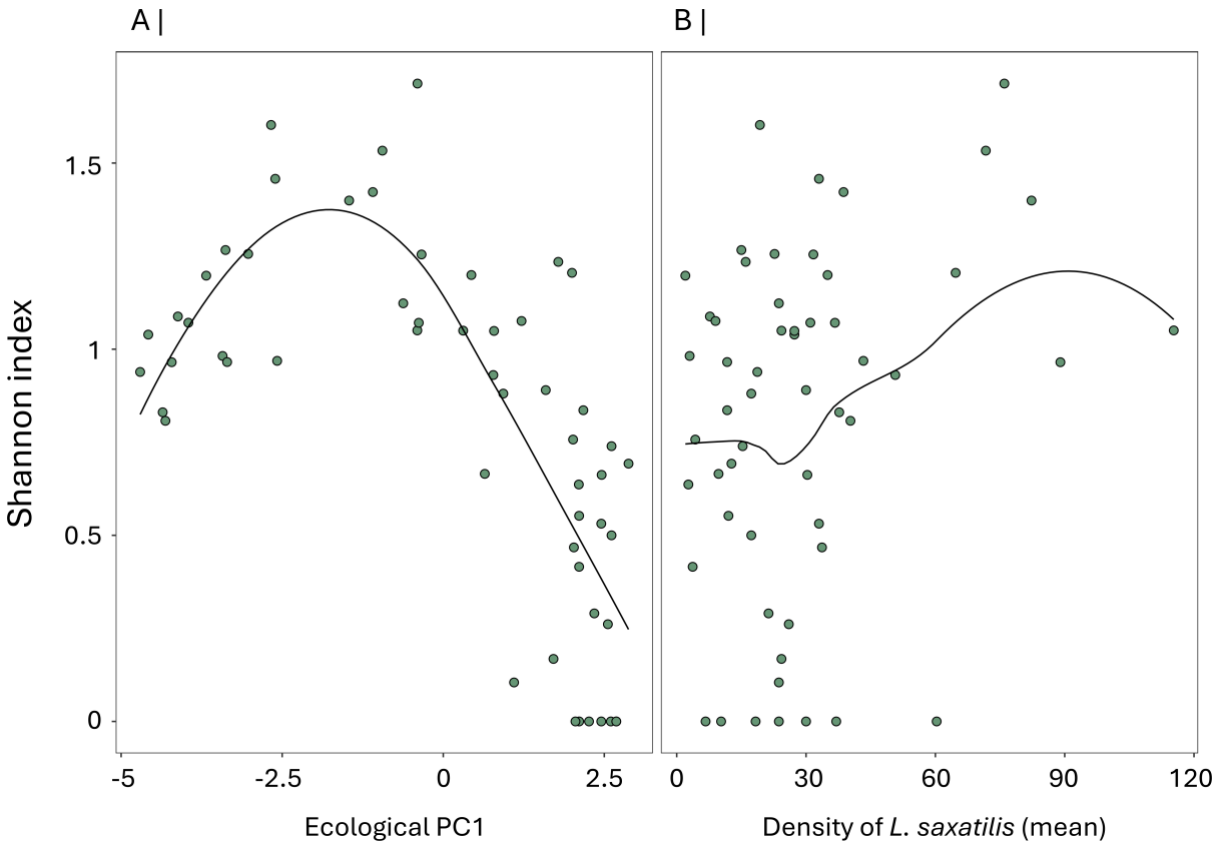

Figure S32. Scatterplots of Shannon diversity index values for each locality against Ecological PC1 (A) and *L. saxatilis* density (B). The fitted lines represent LOESS regression lines.

Figure S32A shows that the Shannon index varies across Ecological PC1 values, following an inverted 'U' shape. The Shannon index peaks at localities with mid-low Ecological PC1 values. This pattern was further examined by comparing the Shannon index between ecosystem types (Figure S33). Color diversity varied between ecosystem types (Kruskal-Wallis, $H(2) = 31.9183$, $P < 0.001$, $\eta^2_H = 0.58$), with wave-sheltered and wave-intermediate localities being more diverse than wave-exposed localities. However, no statistically significant differences were found between wave-sheltered and wave-intermediate localities, as confirmed by the Dunn post-hoc

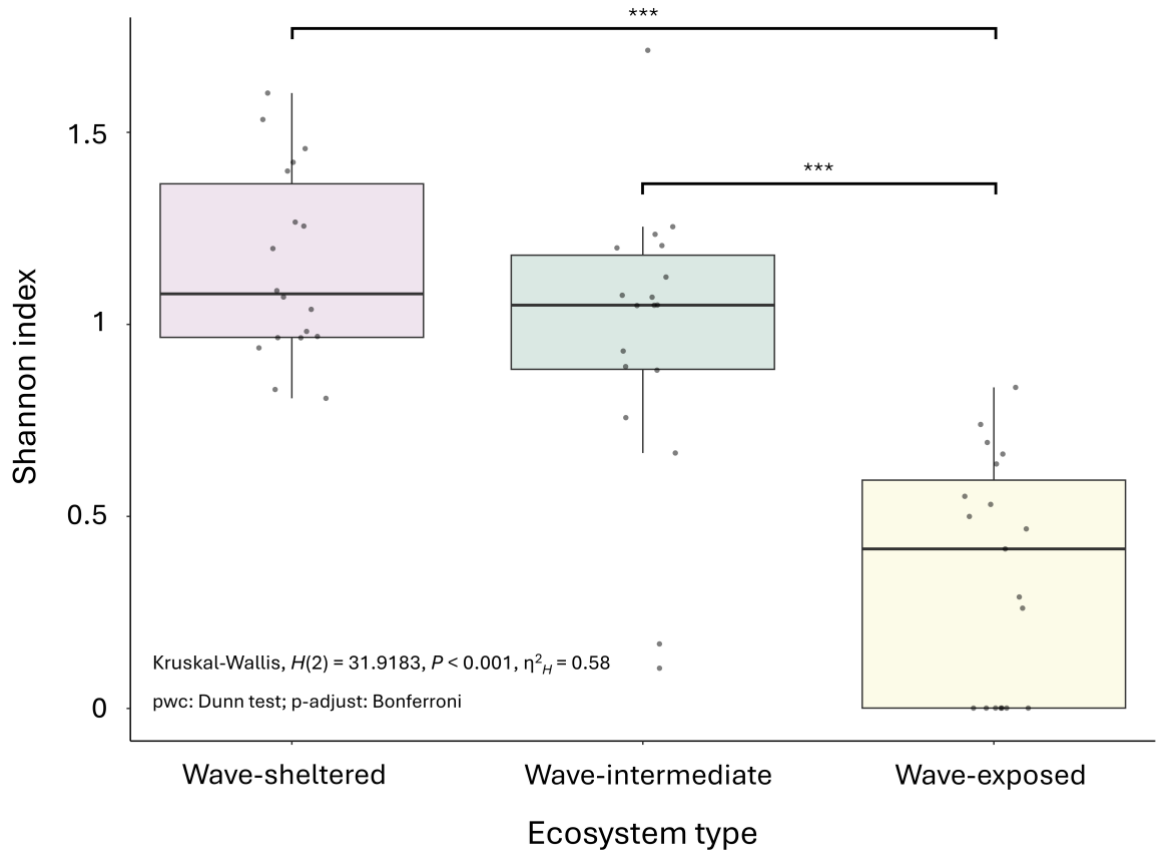

Figure S33. Boxplot of Shannon index by ecosystem type, with the corresponding Kruskal-Wallis and Dunn post-hoc tests.

test. Referring back to Figure S32, it can be concluded that the most color-diverse localities are those at the boundary between the wave-sheltered and the wave-intermediate classes.

On the other hand, localities with higher densities of *L. saxatilis* tend to have higher Shannon index values (Figure S32B), though the relationship between the two variables is not straightforward. The predictive power of Ecological PC1 and *L. saxatilis* density on the Shannon index was assessed by comparing the corresponding generalized linear models (GLMs) with a gamma error structure and log link function, using the Akaike and Bayesian Information Criteria (AIC and BIC, respectively). The gamma error structure was chosen because it is suitable for modeling skewed distributions, such as the Shannon index in our dataset. To address the issue of zero values in the Shannon index, a small constant (0.001) was added, as the gamma error structure cannot handle zero values effectively. Based on AIC and BIC values, the model using Ecological PC1 as a predictor of Shannon index values provided a significantly better fit than the model using *L. saxatilis* density (Table S6). In fact, a model incorporating both predictors yielded a worse fit than the model using only Ecological PC1. This confirms that *L. saxatilis* density is a poor predictor of the Shannon diversity index, contradicting the neutral hypothesis.

Table S6. Comparison between the 'Ecological PC1', '*L. saxatilis* density', and 'Ecological PC1 + *L. saxatilis* density' models using AIC and BIC.

	Dependent variable: Shannon index		
	Model 1: Ecological PC1 as predictor	Model 2: *L. saxatilis* density (mean) as predictor	Model 3: Ecological PC1 + *L. saxatilis* density (mean) as predictors
df	3	3	4
AIC	84.6846	89.33275	86.15085
BIC	90.7066	95.35475	94.18018

General discussion

This dissertation aimed to shed light on the patterns and mechanisms of color variation in *Littorina saxatilis* from the Rías Baixas. Intrapopulation and clinal variation were studied separately, although they tend to occur conjointly in nature. Mating trends and processes were investigated in polymorphic populations, while ecological work and a test of selection for crypsis were conducted in the case of the cline. Now, it is time to synthesize and integrate the findings. Although not many strong conclusions can be drawn from the studies presented, a few provisional inferences can be made. Additionally, some of the results obtained open the door to interesting speculations and suggest exciting future avenues of research. Here, I discuss some of these topics, concluding with a plausible unifying account of the evolution of color variation in *L. saxatilis* from the Rías Baixas. Since most chapters, except for Chapter 6, include a thorough discussion of results, comparatively more space will be devoted to the consideration of the latter chapter.

The role of natural selection in shell color variation of marine snails

Chapter 1 provided a literature review of the evolutionary causes of shell color polymorphism in marine gastropods (Gefaell *et al*., 2023). This review highlighted the primary

explanatory role of natural selection, encouraging a reflection on this process and how best to study it.

Natural selection, defined as the non-random differential reproduction of heritable phenotypic variants within a population, is the cornerstone of evolutionary biology (Futuyma & Kirkpatrick, 2017). Only this process can naturalistically explain the existence of adaptations – traits that have a functional role, match certain characteristics of an organism's ecological niche, and consequently increase its biological fitness (Dawkins, 1986; Dennet, 1996). Despite this, many of evolutionary biology's most heated debates have revolved –and continue to revolve– around the explanatory scope of natural selection (Gould, 2002; Ayala & Arp, 2010). How many traits can be considered adaptations evolved by natural selection? Do we sometimes wrongly assume that a certain trait has a functional role when in fact it does not? How should we properly study natural selection? Particularly influential in the field was Gould and Lewontin's critique of adaptationism in the 1970s (Gould & Lewontin, 1979; see Box 2 of the General introduction). These authors denounced the tendency of many biologists to assume functional explanations for characters without sufficient evidence.

Arguably, Gould and Lewontin were partly right in their criticism: both children and adult humans appear to show a tendency for teleological explanations, that is, explanations in terms of functions, goals, or aims (Kelemen & Rosset, 2009). When applied to evolutionary biology, this means that many of us are prone to hastily ascribe adaptive functions to traits, even when the evidence for this functional role is scarce. But while this tendency might be a hindrance to evolutionary biology (although see Ayala, 1970), the value of what Peter Godfrey-Smith has called 'methodological adaptationism' cannot be denied either (Godfrey-Smith, 2001). This approach involves organizing research in evolutionary biology around the search for adaptations. Natural selection and adaptation are useful concepts for studying nature. Properly applied, they can shed light on numerous puzzling traits of organisms in a way no other concepts have ever done in the history of biology (Mayr, 1983). Furthermore, there are both strong theoretical arguments and empirical evidence that support the significant role of natural selection in shaping the phenotype of organisms (Endler, 1986; Bell, 2008; Futuyma & Kirkpatrick, 2017).

Since the pioneering work of Alfred R. Wallace and Edward B. Poulton, animal coloration has attracted the attention of many evolutionary biologists due to the numerous adaptive functions that color might apparently serve (see Appendix 1 and references therein). Indeed, some of the most renowned examples of adaptation and natural selection in action involve color, from the melanism of *Biston betularia* (Majerus, 1998) to *Peromyscus* cryptic coloration (Hoekstra, 2011), and the frequency-dependent selection of *Cepaea nemoralis* morphs (Cook,

2017). Animal coloration might have three non-exclusive functions: communication with conspecifics, thermoregulation, and predator avoidance (Endler, 1978). These have been described in various organisms with varying levels of detail (see the General introduction and Appendices 1 and 2). Thus, at this point, it is beyond any reasonable doubt that natural selection plays a key role in explaining the astounding chromatic diversity of the living world.

In marine snails, numerous shreds of evidence have been gathered for the functional role of shell color, as seen in Chapter 1 (Gefaell *et al*., 2023). However, much of this evidence is indirect. As the aggregate result of numerous individual life histories across generations of a population (i.e., Walsh, 2015), natural selection is an elusive phenomenon. According to the classic account by Lewontin (1970), natural selection requires (a) phenotypic variation, (b) a hereditary basis, and (c) differential reproductive success. Each of these components is very difficult to study directly, especially the latter two. Consequently, biologists cannot help but study them indirectly through numerous indicators.

As stated by some philosophers of science, an indicator is a functional relationship between an unobservable phenomenon or process (*U*), and an observable phenomenon or process (*O*), such that *U* = f(*O*) (Bunge, 2006). This relationship can either be theoretical or empirical, depending on whether there is a robust causal explanation for the connection between the unobservable and observable phenomena. If evidence for the indicators is found, it is taken as support for the existence of the unservable phenomenon to which it connects. Indicators of natural selection include fitness estimates, F_{ST} outlier tests, functional analyses, species comparisons, or manipulative experiments. Although most of these indicators are theoretical in the sense that there are warranted causal accounts for their connection to natural selection, they vary in strength. Not all of them provide evidence of equal quality for natural selection: some of them are quite unequivocal, while others offer only hints of natural selection. In this regard, much like in other scientific disciplines such as animal behavior and psychology (i.e., Bateson & Martin, 2021), there is a hierarchy of evidence that ranks the scientific quality of the information provided by the different indicators of natural selection. The higher an indicator is in the evidence hierarchy, the fewer unknown assumptions it makes about how natural selection operates, or the less critical these assumptions are for understanding how natural selection is acting in the population.

As discussed in Chapter 1, much of the evidence for the functional role of color and color variation in marine gastropods comes from frequentist studies, in which natural selection is not directly assessed. This also applies to some investigations included in this dissertation. In frequentist studies, the frequencies of colors in various localities are compared to their

corresponding backgrounds. If there is a chromatic concordance between these (qualitatively or quantitatively assessed), it is concluded that color is likely being maintained by selection for crypsis. While the value of these studies is undeniable, their inferential power, as argued in Chapter 5, is limited. Most notably, they do not provide first-hand evidence or estimates of natural selection in action but rather assess the presumed results of such selection. Additionally, in many cases, knowledge about the basic aspects of the organisms' ecology or the hereditary basis of the trait is scarce.

We are then confronted with the fact that frequentist studies, despite their limitations, constitute an important part of the evidence supporting our belief in the role of natural selection in maintaining color variation in marine gastropods. So, what can be done to strengthen this evidence? Abandoning this methodology would be unwise, as it still provides useful information. To address this, let's examine one of the many successful research programs in evolutionary ecology: Grant and Grant's investigation into the radiation of the Galapagos finches (Grant & Grant, 2008, 2014). Although much remains to be uncovered about how these bird species have diversified, a solid picture of this process is now available. Interestingly, much of Grant and Grant's work on this system has involved numerous field studies where control of variables is usually limited and which rely on various untested assumptions, much like frequentist studies in marine gastropods. Each of these studies, on its own, provides limited information on the ecology and evolution of the Galapagos finches. However, the cumulative effect of these studies –each targeting a specific aspect of the evolutionary ecology of the studied organisms, such as their diets, their phenotypic differences in beak size, and the heritability of key traits– offers a comprehensive and well-supported account of their evolution and the role natural selection has played in the process.

The scientific soundness of Grant and Grant's approach stems, therefore, from a piecemeal method that builds a detailed understanding by combining what would otherwise be very restricted ecological and evolutionary studies. Additionally, their focus on species and populations within a specific geographical area, rather than on geographically disconnected populations of a species, has also helped them to provide a complete account of their evolution. Evolutionary biology is a historical science (Mayr, 2004), meaning it deals with phenomena that are closely connected to the region where they unfold and the specific causal chain of events that have taken place there. Thus, focusing on a specific area can also help strengthen our inferences about how a particular species has evolved.

In light of the previous considerations, I suggest that to enhance our understanding of natural selection and the maintenance of shell color variation in *L. saxatilis* from the Rías Baixas,

it would be advisable to complement frequentist studies with small, targeted studies that provide additional evidence on various aspects of the ecology and natural history of the organisms and populations involved. Fortunately, much of this knowledge already exists for some winkle species; despite this, however, a case could be made for redoing similar kinds of analyses in *L. saxatilis* from the Rías Baixas, as this can connect the evidence obtained to the particular historical trajectory under study. By doing so, we can lay the groundwork for future, more sophisticated genomic approaches to studying natural selection in these specific populations.

Negative assortative mating, negative frequency-dependent selection, and intrapopulation color polymorphism in *L. saxatilis*

Chapters 2 and 3, corresponding to Gefaell *et al*. (2021, 2024a) dealt with intrapopulation color polymorphism in *L. saxatilis* from the Rías Baixas. Specifically, they focused on color variation in the wave-sheltered localities of this region, where three morphs (*fulva*, *lutea*, and *fusca*) coexist. The main finding was that these morphs exhibit a pattern of negative assortative mating both in the field (Gefaell *et al*., 2021) and in the lab (Gefaell *et al*., 2024a). This mating pattern is likely driven by mate choice, where the choosy sex (in this case, the male) actively seeks to mate with females bearing different shell colors (Gefaell *et al*., 2024a). This suggests that negative frequency-dependent selection (NFDS) maintains color polymorphism in these environments. Although the results of both studies were not entirely consistent and some discrepancies remain, they raise interesting questions. In particular, given that *L. saxatilis* likely lacks color vision, it is puzzling how males distinguish between female colors, preferentially select dissimilar ones, and what benefits might derive from these heterotypic matings.

In addition to the explanations discussed in the papers, another alternative relates to the possible immune benefits of mating with dissimilar color morphs. If snails in the wave-sheltered areas of the Rías Baixas face a high parasite load, *L. saxatilis* individuals with more diverse immune genes might gain a fitness advantage. This diversity could be increased by engaging in heterotypic matings: since the color morphs likely differ in numerous genes, mating with a different morph could boost genetic diversity and thereby strengthen the immune response of their offspring. If this genetic diversity makes snails better equipped to deal with the increased number of potential parasites in their environment, a mating preference for dissimilar color morphs might have evolved as a way to achieve such diversity. As suggested in the papers, males

might distinguish between females of different colors through the different chemical components present in their mucus.

Although there is currently no evidence supporting this hypothesis, it remains plausible. Snails from the genus *Littorina* are the first intermediate host for many trematode parasites, particularly from the genus *Microphallus* (Elner & Raffaelli, 1980; Gorbushin & Levakin, 1999; Blakeslee *et al.*, 2015; Keogh, Sanderson, & Byers, 2016; Bojko, Grahame, & Dunn, 2017). These parasites reproduce asexually within the snails, specifically in the digestive gland and gonads, often resulting in the castration of the hosts and significantly impacting their biological fitness. In the case of *Microphallus similis*, which is present in the Rías Baixas (Sanmartín *et al.*, 2005), the parasite is released from the snails as a free-swimming cercaria and continues its life cycle by penetrating crabs such as *Carcinus maenas* (Blakeslee *et al.*, 2015), the second intermediate hosts. Crabs are then eaten by seabirds, such as gulls or cormorants, which are the definite hosts. The seabirds release *M. similis* eggs onto the rocks through their guano, from which *Littorina* snails become infected again (Keogh *et al.*, 2016).

Trematode infection rates are known to vary widely between locations and may be particularly high in wave-sheltered areas, where the studied snails live. For instance, evidence from the White Sea suggests that *L. saxatilis* populations in wave-sheltered areas have a higher prevalence of trematode infections than those in wave-exposed sites, particularly from two species, *Microphallus piriformes* and *M. pygmaeus* (Kaliberdina & Granovich, 2003). Similarly, Bojko, Grahame, & Dunn (2017) found a higher prevalence of trematodes in the British habitats of the *Crab* ecotype, which is more similar to a Galician wave-sheltered environment than to a wave-exposed one. Given the abundance of *C. maenas* crabs in the wave-sheltered localities of the Rías Baixas (see next sections), it is reasonable to assume that parasite infection prevalence might be higher in these areas than in wave-exposed localities, representing a more intense selective pressure.

Like all mollusks, *Littorina* snails only have an innate immune system (Al-Khalaifah, 2022). Although much is still unknown, evidence suggests that this immune system consists of several cellular and humoral components and is tissue-specific (Gorbushin, 2019). One of the main defense mechanisms against trematode infection is lectins. Lectins are carbohydrate-binding proteins produced by hemocytes in the hemolymph of gastropods and can either function as membrane-bound receptors or as secreted molecules (Loker, 2013; Gorbushin, 2019). These proteins bind to the carbohydrate structures of pathogen larvae, allowing their elimination or neutralization through hemocyte phagocytosis or encapsulation (Loker, 2013). Lectins can achieve significant recognition diversity through various mechanisms; for example,

they are coded by multi-gene families and can undergo alternative splicing (Loker, 2013). Although relatively few studies have analyzed genetic variation and natural selection for immunological mechanisms in mollusks –and these studies may have limitations– there is evidence from bivalves and gastropods of family-level variation in parasite resistance, including resistance to trematode infection (Seppälä *et al.*, 2021).

Considering the aforementioned factors, it seems plausible that genetic variation for immune response to *Microphallus* trematodes exists in snails from the wave-sheltered localities of the Rías Baixas. This variation might be linked to shell color, such that offspring resulting from heterotypic matings could exhibit greater immune-related allele diversity than their homozygous counterparts (i.e., those resulting from homotypic matings). Since immune compounds have also been found in snail mucus –such as achacin, an antibacterial glycoprotein present in the giant Africant land snail (Loker, 2013)– males might detect variation in these components associated with color morph variation and thus select mates with different colorations. This could have led to the evolution of their mate choice preference for dissimilar morphs, and thus to a negative assortative mating pattern and NFDS. This phenomenon would be analogous to the classic (and controversial) disassortative mating for major histocompatibility complex (MHC) genes in vertebrates (reviewed in Penn & Potts, 1999).

Although theoretically plausible, this speculation relies on several assumptions that are difficult to test. Virtually nothing is known about the genetic basis of immune function or coloration in *L. saxatilis*, and whether these are related. Additionally, no studies have been conducted on the parasitism rates of the different color morphs or the various areas of the Rías Baixas. This demands caution when asserting this hypothesis. However, it could also serve as a guide for future research and provide a reasonable explanation for a puzzling phenomenon described in this dissertation: *L. saxatilis* snails from wave-sheltered localities prefer to mate with color-dissimilar individuals.

The strength of the association between color morphs and ecological parameters

The other main set of findings in this dissertation involves the *L. saxatilis* color cline. Chapters 4 and 6 demonstrate that the cline described by Fischer-Piette & Gaillard (1960), Sacchi (1979), and Reid (1996) exists and has remained relatively stable over the last forty years. This cline is characterized by a close association between the *fulva* and *lineata* morphs and wave-sheltered and wave-exposed ecosystems, respectively. The fact that this cline is more

ecological than geographical suggests a role for spatially varying selection over isolation by distance as its primary maintenance mechanism, although both factors certainly contribute.

That said, we observe substantial variation among the rias in the strength of the association between ecosystem types, as measured by Ecological PC1, and the main color morphs of the cline (i.e., *fulva* and *lineata*; see Chapter 6, Figure 5). Specifically, this association is stronger in Vigo and Muros-Noia than in Pontevedra and Arousa, posing the question of what explains this variation in the magnitude of this association.

To provide a cogent hypothesis for this phenomenon, it is necessary to recall three things. First, as a whole, Vigo and Muros-Noia exhibit a greater spatial aggregation of ecosystem types than Pontevedra and Muros-Noia (Chapter 6, Supplementary Materials, Figure S16). Second, as seen in the General introduction (section 2.3.a), clines are the result of a tug-of-war between spatially varying selection and genetic drift on the one hand (forces leading to divergence) and gene flow on the other (a force promoting homogenization). Third, as also discussed in the General introduction (section 3.1.a), due to its reproductive system and ontogeny, which lacks a pelagic larvae stage, *L. saxatilis* is likely to migrate primarily between adjacent sites rather than distant areas.

In light of the previous observations, a likely explanation for the differences in the strength of the association between ecosystem types and color morphs is that in Vigo and Muros-Noia, the spatial clustering of ecosystem types prevents gene flow from distorting the association, allowing selection and drift to reinforce it. In these rias, adjacent localities are likely to belong to the same ecosystem type, so migrating snails are likely to be of the same morphs. In contrast, the lack of such clustering in Pontevedra and Arousa makes it easier for gene flow to disrupt the clinal pattern. In these rias (especially in Arousa, as Pontevedra may be biased due to the low number of wave-sheltered localities), wave-exposed and wave-sheltered habitats are sometimes relatively close by, which makes migrating snails between these habitats to have different colors and shell characteristics. Many of these incoming individuals are likely to survive despite not being in their optimal environments, as the selective pressures driving divergence between morphs are not expected to be so intense as to immediately wipe out all alternative phenotypes. This, in the long term, can weaken the association between morphs and ecosystem types, explaining the differences between the rias (Figure 1). If this hypothesis is true, it would predict that gene flow between ecosystem types is higher in Pontevedra and Arousa than in Vigo and Muros-Noia. Fortunately, snails for molecular analyses were collected from 25 of the 55 loc-

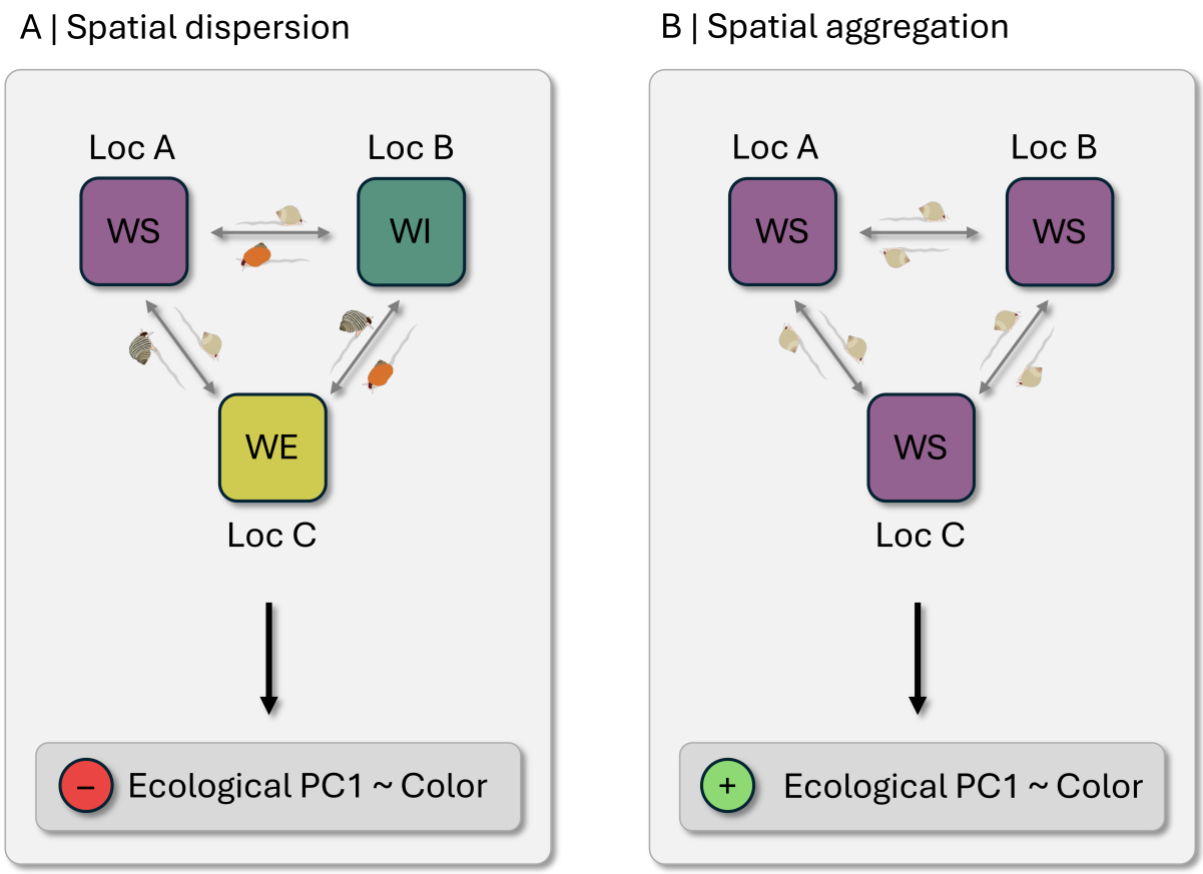

Figure 1. Two hypothetical scenarios illustrating how gene flow can either weaken or strengthen (or at least not diminish) the association between ecosystem types and color morph composition. In scenario A, there is spatial dispersion of ecosystem types. Here, three adjacent localities (Loc A, Loc B, and Loc C) represent different ecosystem types: wave-sheltered (WS, purple), wave-intermediate (WI, green-blue or turquoise), and wave-exposed (WE, yellow). As a result, migrating individuals are likely to belong to different morph types, which in turn weakens the association between ecosystem type and color morph composition (Ecological PC1 ~ Color). In scenario B, there is a spatial aggregation of ecosystem types, where all localities share the same ecosystem type (wave-sheltered). This aggregation reinforces, or at least does not erode, the association between ecosystem type and color morph composition.

alities studied in Chapter 6, so we will likely have data soon to conduct a preliminary test of this and related predictions.

Covariation between shell traits and correlational selection

Another important finding of Chapter 6 is that shell color covaries with a suite of shell traits with a putative strong genetic basis: *fulva* is associated with larger, thicker, spikier, smoother, and narrower-apertured shells, while *lineata* is linked to smaller, thinner, rounder, more ribbed and wider-apertured shells. As indicated in Chapter 1, evidence from both *L. saxatilis* and other marine gastropods suggests that shell color is largely dependent on genes (Johannesson *et al.*, 1993; Gefaell *et al.*, 2023, and references therein). The same applies to the remaining studied shell traits as well. Johannesson & Johannesson (1996) reared three *L. saxatilis* ecotypes under common laboratory conditions and found that the morphological and behavioral

differences between these ecotypes persisted, though less intensively. Additionally, Janson (1982) revealed consistent differences in growth rate between *L. saxatilis* morphs from sheltered and exposed environments, which in turn affected adult size. Analogous results have been obtained for shell shape (Newkirk & Doyle, 1975; Smith, 1981; Conde-Padín *et al.*, 2007; Chrismas *et al.*, 2014), shell thickness (Johannesson & Johannesson, 1996), and shell sculpture (Johannesson *et al.*, 1993; Kess, Brachmann, & Boulding, 2021), although phenotypic plasticity also seems to play a role in all these traits (Conde-Padín, Caballero, & Rolán-Alvarez, 2009; Hollander & Butlin, 2010; Saura *et al.*, 2012; Galindo *et al.*, 2019).

The strength of the covariation between shell color and shell traits, however, shows substantial differences between the rias. Shell traits PC1, which effectively summarizes variation in shell length, thickness, sculpture, and shape, accounts for approximately 82% of the variance in *fulva* frequency in the Ría de Vigo, but only around 17% in Arousa (see Chapter 6, Supplementary Materials, Table S5). A similar result was obtained for *lineata*, except that in this case, Pontevedra shows the lowest percentage of variance explained instead of Arousa. Additionally, we also found that a small, but significant, percentage of the variance in color morph frequencies is not explained by other shell traits, yet it is associated with ecological parameters (see Chapter 6, Table 3). What do all these data suggest about the genetic basis and selective regimes affecting the *fulva* and *lineata* morphs?

It is currently known that the *L. saxatilis* genome possesses several chromosomal inversions associated with traits under selection, such as shape, size, and color (Koch *et al.*, 2021, 2022). Many of these inversions might help maintain beneficial allele combinations through the suppression of recombination (i.e., linkage disequilibrium). Additionally, a total of 341 single nucleotide polymorphisms (SNPs) have been linked to various shell traits in this species (Kess *et al.*, 2021). There is also evidence of an association between mitochondrial haplotypes and shell shape, though their exact role is still poorly understood (Chrismas *et al.*, 2014). Finally, the available evidence suggests that the genetic basis of most shell traits under selection is polygenic, involving various loci scattered throughout the genome, as indicated by studies pinpointing quantitative trait loci (QTLs) (Koch *et al.*, 2021). However, to what extent these results apply to the *L. saxatilis* populations from the Rías Baixas remains unknown.

Given the preceding points, a plausible explanation for the obtained results is that shell color has a different genetic basis and responds to a different selective pressure than the other shell traits studied. These selective pressures, however, might occur together in Vigo and Muros-Noia, leading to the strong shell covariation observed there. This simultaneous operation of sel-

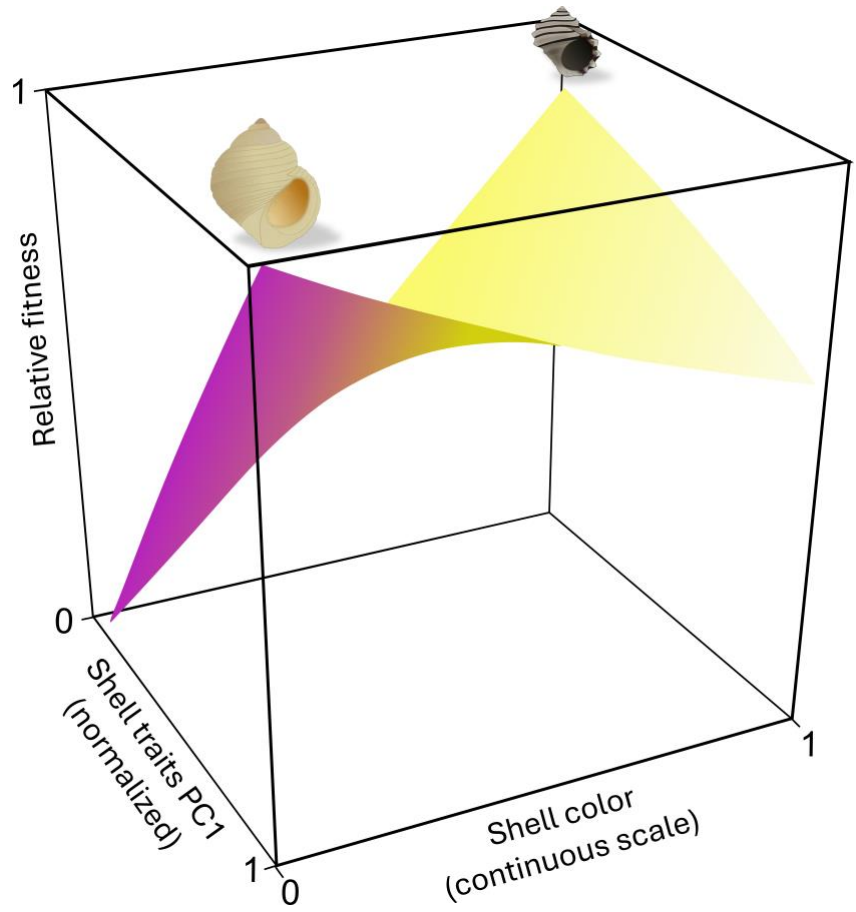

Figure 2. Hypothetical adaptive landscape of *L. saxatilis* in the Ría de Vigo and the Ría de Muros-Noia. The y-axis represents normalized values of Shell traits PC1, where low values correspond to small, ribbed, round, thin, and wide-apertured shells, and high values correspond to large, smooth, spiky, thick, and narrow-apertured shells. The x-axis represents shell color on a continuous scale, which could be either shell darkness, the intensity of lines, or any chromatic parameter that allows us to distinguish the *fulva* and *lineata* snails on a continuous scale. The z-axis corresponds to relative fitness. The resulting landscape would feature two peaks, each corresponding to a different ecosystem type (represented by the colors of the landscape; purple: wave-sheltered, yellow: wave-exposed). This figure, inspired by the figure in Box 1 of Svensson *et al*. (2021), is intended for illustrative purposes, so the fitness values are purely conjectural.

ective pressures for shell color and other shell traits in these rias might be due to their greater spatial clustering of ecosystem types, as seen in the previous section. In practice, this may have reduced the adaptive landscape of their *L. saxatilis* populations to just two optimal phenotypic combinations (Figure 2), in a putative case of correlational selection (see General introduction, section 1.2.c): either a *fulva* color with a large, thick, spiky, smooth, and narrow-apertured shell (in wave-sheltered ecosystems), or a *lineata* color with a small, thin, round, ribbed, and wide-apertured shell (in wave-exposed ecosystems). Correlational selection may have led to linkage disequilibrium between shell trait loci, but the extent to which chromosomal inversions are involved remains to be studied. Although correlational selection might also be operating in the

remaining rias, their less pronounced clustering of ecosystem types may result in a weaker effect, thus explaining why covariation between shell traits is much less pronounced there.

The selective forces behind the *fulva* and *lineata* morphs

Up to this point, there is suggestive evidence that the cline is maintained by spatially varying selection, that the association between shell colors and ecological parameters is sometimes weakened by migration among ecosystem types, and that shell color and other shell traits may respond to different selective pressures. But what could these selective pressures be? Let's start with color.

Selective pressure for colors

As noted earlier, shell color might serve three different functions: communication with conspecifics, thermoregulation, and predator avoidance (Endler, 1978). In a species without chromatic vision, such as *L. saxatilis*, the first function is the least plausible (though see the section on negative assortative mating). Thermoregulation, in turn, has not been assessed in this dissertation, but preliminary studies conducted by our group suggest that thermal selection is unlikely to operate in the *fulva* and *lineata* morphs, despite an overall temperature cline in the Rías Baixas (Vigo *et al.*, unpublished data). The remaining candidate, therefore, is predator avoidance, particularly through crypsis. Since most *L. saxatilis* predators are visual organisms (Pettitt, 1975), this hypothesis seems plausible. In fact, informal field observations of *L. saxatilis* reveal a high degree of crypsis in their natural environments, both in the Rías Baixas and beyond (see Figure 3).

If selection for crypsis is assumed as a working hypothesis, two questions become particularly important: (1) which predator plays the primary role in promoting crypsis in *L. saxatilis*?; and (2) against which specific backgrounds are the *fulva* and *lineata* morphs cryptic?

Regarding question (1), three predators stand out as the most plausible contenders for selection for crypsis: shorebirds, such as turnstones (*Arenaria* sp.) and sandpipers (Scolopacidae); percomorph fishes, such as blennies (Blenniidae) and gobies (Gobiidae); and crabs, such as the European green crab (*Carcinus maenas*) or the marbled rock crab (*Pachygrapsus marmoratus*). Of these, only the latter has been tested in this dissertation, yielding inconclusive results (see Chapter 5). Although we cannot rule out that this outcome was due to low statistical power, other studies testing differential predation of snail morphs by crabs have also produced negative or mixed results (e.g., Tucker, 1988; Ekendahl, 1998). This makes crabs the least plausible contenders of the three groups. In contrast, shorebirds and percomorph

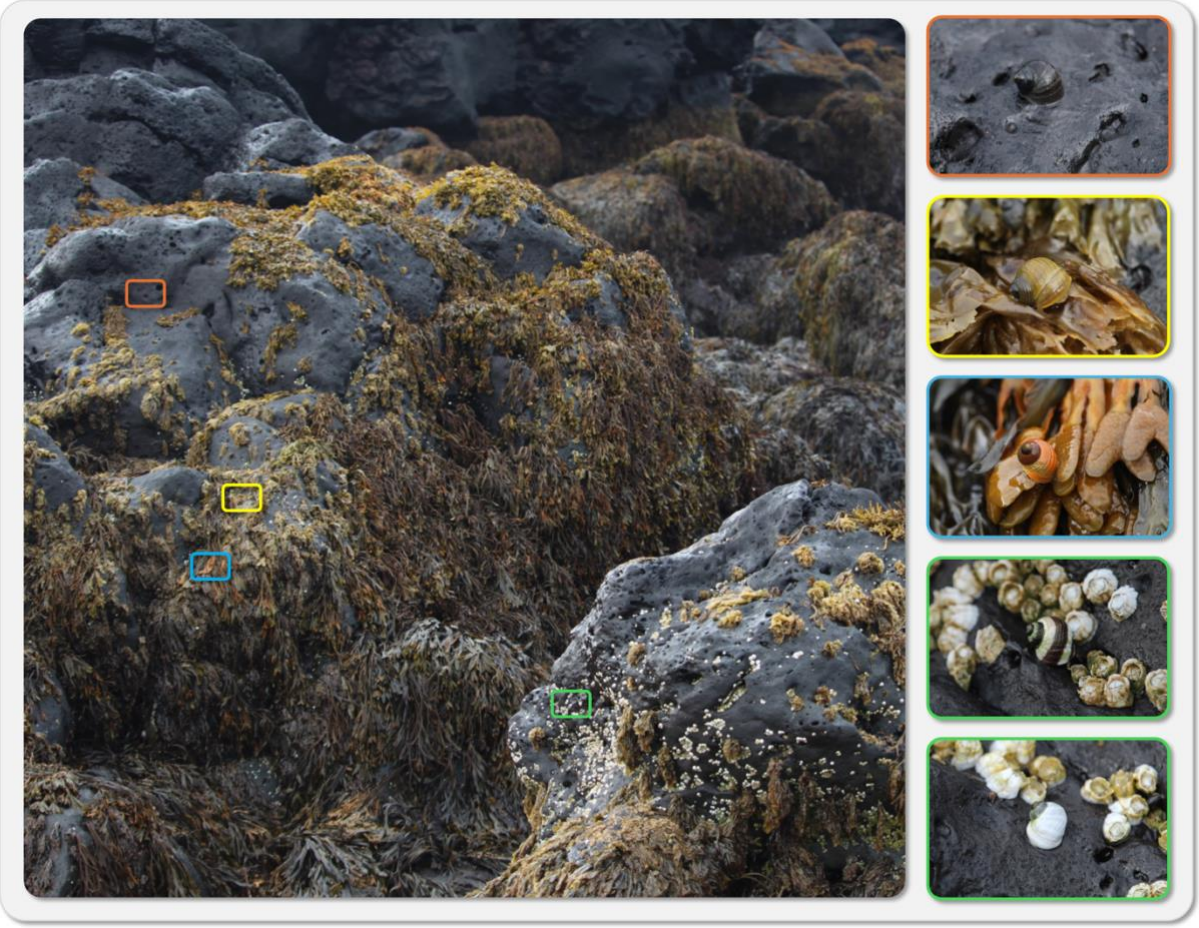

Figure 3. An Icelandic intertidal ecosystem (Skarðsvík Beach, August 2023) composed of various microhabitats (in different colors), each of which has at least one cryptic snail color morph. For the photographs, the snails were relocated from the specific spots where they were originally found –which did not always correspond to the microhabitats where they are most cryptic– and placed in the microhabitats where they exhibit the highest degree of crypsis. This adjustment was made for illustration purposes only.

fishes, which have not been tested in this dissertation, seem more plausible options, as previous research suggests that they can exert substantial predation pressure on *Littorina* populations. For instance, during spring stopovers in Iceland, migrating red knots (*Calidris canutus*) largely rely on *Littorina* sp. as an energy source (Alerstam, Gudmundsson, & Johannesson, 1992). Similarly, snails from this genus are a key component of the diet of the percomorph fish *Gobius niger* in San Simón Bay, Ría de Vigo (Filgueira & Castro, 2011).

Informal sightings along the Rías Baixas suggest that the ruddy turnstone (*Arenaria interpres*) preys upon *L. saxatilis* snails (Vigo, pers. obs.), although these observations should be formally assessed. However, in a manipulative field experiment conducted in Iceland, Johannesson & Ekendahl (2002) did not find evidence of differential morph predation by birds, despite confirming them as significant predators of *L. saxatilis* (removing around 16% of experimental snails). Instead, these researchers concluded that intertidal fishes were the most likely agents of differential morph predation. Johannesson & Butlin (2017) also suggest a role for

both fish and shorebirds, although they do not provide direct evidence for this claim. Finally, in a comprehensive study that included field and lab observations, as well as a predation experiment, Reimchen (1979) found evidence of selection for crypsis by intertidal percomorph fishes on *L. fabalis* (then *L. mariae*). The combined results of these studies therefore tip the balance in favor of fishes as the most likely predators driving the evolution and maintenance of shell color in the *fulva* and *lineata* morphs. Although we did not find evidence of differential predation by percomorph fishes in our tethering experiment, described in Chapter 5 (as determined by the 'Empty Shell' category, since these fishes prey by swallowing the whole snail and regurgitating the empty shell; Reimchen, 1979, 1982), this experimental protocol might not be the most appropriate for assessing cryptic predation by these animals.

Regarding question (2) ('*Against which specific backgrounds are the fulva and lineata morphs cryptic?*'), it is worth noting that Chapter 5 demonstrated that the color spectrum of the *lineata* morph closely resembles that of rocks from a wave-exposed locality. Given this, a plausible hypothesis is that the striped pattern exhibited by *lineata* serves as a cryptic adaptation to the barnacle-covered or bare alkaline granitoid rocks of wave-exposed areas. Specifically, barnacle plate boundaries may create a stripe-like pattern against which *lineata* could be disruptive to many predators, including fish.

Fulva, for its part, inhabits a chromatically distinct environment from *lineata*, needing to blend in a brown algae-dominated setting. Thus, a sensible conjecture is that *fulva* is cryptic against the brown algae species prevalent in the wave-sheltered ecosystems of the Rías Baixas: *Ascophyllum nudosum*, *Fucus vesiculosus*, and *Pelvetia canaliculata*. This hypothesis is supported by the fact that two of the three studied brown algae (*A. nudosum* and *F. vesiculosus*) are absent from wave-exposed habitats and are found exclusively in wave-sheltered, and possibly some wave-intermediate, ecosystems. As such, these algae are a major distinguishing feature of the latter areas, likely generating specific selective pressures. The brown algae, which are scattered on the rocks at low tide, may form a dense forest at high tide, where cryptic snails could have a selective advantage over more conspicuous individuals.

Selective pressures for covarying shell traits

What about the remaining shell traits? The existing literature on *L. saxatilis* strongly suggests that shell length, shape, thickness, sculpture, and aperture size have functional roles (Reid, 1996; Johannesson, 2003, 2016; Rolán-Alvarez, 2007; Butlin *et al.*, 2014), with different phenotypic variants being adaptive to contrasting selective pressures. Specifically, many shell traits covarying with the *fulva* morph (such as larger, thicker, and narrower-apertured shells) have

traditionally been considered adaptations against crab predation, while many traits associated with the *lineata* morph (particularly smaller, rounder, and wider-apertured shells) have been interpreted as adaptations against wave-action. Let's explore these further.

Intense crab predation, inferred from the high frequency of shell scars (Raffaelli, 1978; Stafford, Tyler, & Leighton, 2015), predicts the presence of larger, thicker, and narrower-apertured shells, which may result from either genetic adaptation or plastic responses to this selective pressure (Heller, 1976; Seeley, 1986; Johannesson, 1986; Trussell, 2000; Rolán-Alvarez, 2007; Brookes & Rochette, 2007; Bourdeau, 2009; Hollander & Butlin, 2010). Larger individuals are less frequently preyed upon by crabs (Elner & Raffaelli, 1980; Reimchen, 1982), thicker shells increase resistance to crushing (Johannesson, 1986; Quensen Iii & S Woodruff, 1997), and narrower apertures may complicate the manipulation of the snails (Heller, 1976; Johannesson, 1986). This pattern is precisely what we observe in wave-sheltered ecosystems across the four rias, where the *fulva* morph predominates.

One crab species, *Carcinus maenas*, is notably abundant in wave-sheltered shores and estuaries, where it can tolerate low salinity concentrations (Crothers, 1968). In the Rías Baixas, a significant portion of *L. saxatilis* from these habitats exhibits shell scars, indicating a high abundance of *C. maenas* and possibly other crab species (though *P. marmoratus*, typically found in wave-exposed localities, may be less prevalent) (Reimchen, 1982; Stafford *et al.*, 2015). This crab species likely preys on *L. saxatilis* snails during high tides, as *C. maenas* remains below the tide mark during low tides (Crothers, 1968; Heller, 1976). Consequently, many of the shell traits associated with *fulva* might be a result of adaptation specifically against this crab species.

The habitats of the *lineata* morph differ significantly from those of the *fulva* morph. While crab predation, likely by *P. marmoratus* (see Chapter 5), does occur and can be significant in some localities, it is unlikely to be the primary driver of most shell traits associated with the *lineata* morph. First, predation by these crabs, inferred from shell scars, may be comparatively low in most wave-exposed ecosystems. Second, many traits associated with the *lineata* morph –like larger apertures or thinner shells– actually facilitate crab attacks rather than deter them (Elner & Raffaelli, 1980; Reimchen, 1982; Johannesson, 1986; Quensen Iii & S Woodruff, 1997). A more plausible explanation, therefore, is that these traits are adaptations to the primary selective pressure in wave-exposed areas: wave action. In these environments, wave action exerts substantial mechanical force that can easily dislodge snails from rocks (Heller, 1976; Le Pennec *et al.*, 2017). Consequently, many snail adaptations in these areas aim to resist or minimize the impact of waves. For example, smaller shells allow snails to seek shelter in small crevices (Heller, 1976; Rolán-Alvarez, 2007), while wide apertures and round shapes accommodate a

relatively larger foot, increasing the snail's attachment surface and making dislodgement more difficult (Newkirk & Doyle, 1975; Heller, 1976; Le Pennec *et al.*, 2017).

To sum up, many shell traits covarying with the *fulva* and *lineata* morphs seem either adaptations to crab predation (in the case of *fulva*) or wave action (in the case of *lineata*). There are, however, other traits we haven't yet discussed, particularly shell sculpture and, specifically for the *fulva* morph, shell shape (spiky sculptures). Can these traits also have adaptive value? Consider shell sculpture, with its smooth (*fulva*) or ribbed (*lineata*) variants. For the *fulva* morph, smoothness might serve a dual function: enhancing camouflage and providing protection against crab predation. Regarding the first function, smooth shells could blend with the smooth surfaces of brown algae, increasing camouflage. Regarding the second function, structurally speaking, ribs in *L. saxatilis* are apparently formed by decreasing shell growth along grooves (see blue arrows in Chapter 6, Supplementary Materials, Figure 1A). Consequently, the strengthening effect of ribs observed in other gastropod species (e.g., Quensen Iii & S Woodruff, 1997) may not apply to *L. saxatilis*. In contrast, in the rough periwinkle, smooth shells might be associated with thicker and stronger shells, thereby increasing resistance to crab predation. This conjecture also offers insights into why *lineata* shells may be ribbed. In the near absence of crab predation, ribbed shells might represent a structurally efficient way to produce a thin shell, given the development of *L. saxatilis* shells. Alternatively, the ribs might enhance disruptive coloration in barnacle-covered rocks by further breaking up the outlines of the animal.

The remaining trait –spikier shapes, typical of *fulva* snails– may be more challenging to interpret adaptively. Contrary to the crab predation hypothesis, gastropod species with spikier shells, such as those seen in the *fulva* morph, are crushed faster than those with flatter shells (Cotton, Rundle, & Smith, 2004). Future studies should determine whether this pattern holds true for *L. saxatilis* in the Rías Baixas as well.

In conclusion, a plausible hypothesis is that the colors of the *fulva* and *lineata* morphs have evolved as cryptic adaptations against predation by percomorph fishes, while their covarying traits might have evolved as adaptations against predation by *C. maenas* crabs (for *fulva*) or wave action (for *lineata*) (Figure 4). If true, this hypothesis would support the view put forward by Galindo *et al.* (2019) and Tirado *et al.* (2016) that rough periwinkles from intermediate-to-inner ria regions (i.e., *fulva*) represent an exaggerated *Crab* ecotype. Our hypothesis, however, requires thorough empirical testing. In compliance with our recommendations from the first section, a piecemeal approach could yield promising results. A first step, for example, might include assessing the crypsis of the *fulva* morph against various brown algae species using reflectance spectrometry. A subsequent step could involve a field or mesocosms experiment to

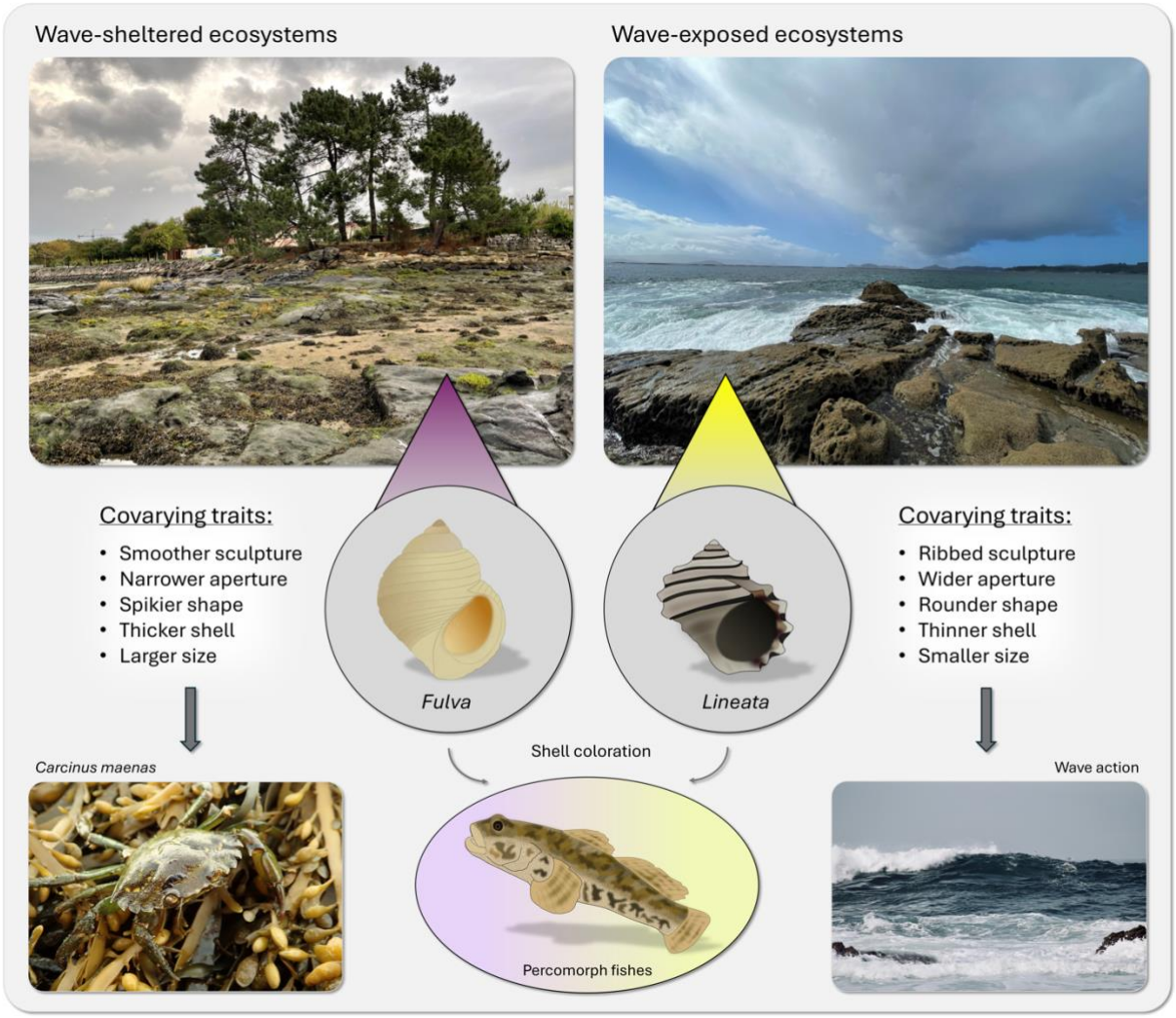

Figure 4. Graphical summary of the selective pressures driving shell trait evolution in the two main *L. saxatilis* morphs from the Rías Baixas, *fulva* and *lineata*, each associated with a different ecosystem type. The coloration of both morphs may have evolved through selection for crypsis, primarily driven by percomorph fishes. In contrast, many traits covarying with each morph may have evolved as adaptations to either crab predation (in the case of *fulva*) or wave action (in the case of *lineata*).

test differential fish predation of *L. saxatilis* morphs, along with additional studies to evaluate the various assumptions of the model (such as if ribbed shells are really constructed by decreasing shell growth along grooves).

A further question concerning the *fulva* and *lineata* selective pressures

Before moving on to the last topic, there is yet another aspect regarding the selective pressures maintaining the *fulva* and *lineata* morphs that deserves brief commentary. As shown in Figure 6 from Chapter 6, most wave-exposed populations of *L. saxatilis* are monomorphic or nearly monomorphic for *lineata*, while no wave-sheltered population is monomorphic for *fulva*.

This morph typically co-occurs with the *lutea* and *fusca* morphs, which exhibit substantial variation in frequency from locality to locality. In addition to the immune-related mate choice mechanism discussed above, this phenomenon may also relate to differences in species composition between ecosystem types: as already discussed, most rock substrate in wave-exposed ecosystems is covered by a single type of animal –barnacles–, while three algae species (*A. nudosum*, *F. vesiculosus*, and *P. canaliculata*) dominate the wave-sheltered intertidal ecosystems. This likely creates multiple microhabitats to which each of the morphs (*fulva*, *lutea*, and *fusca*) may be chromatically adapted. Furthermore, species composition is more varied in wave-sheltered ecosystems than in wave-exposed habitats (see dispersion from centroids in Chapter 6, Figure 4). This variability might explain why some localities have more *lutea* than *fusca* snails, and vice versa.

Colonization, cline formation, and parallel evolution

A final topic that deserves discussion is the formation of the analogous *L. saxatilis* color clines observed in the Rías Baixas. The Rías Baixas have a geologically recent origin, formed by the inundation of their corresponding river valleys in a process that began approximately 12,000 years ago and concluded around 6,000 to 3,000 years ago (General introduction, Figure 8C and section 3.2.c). During this period, several climatic phases occurred, leading to fluctuations in temperature, rainfall, and other environmental factors. It is also important to note that, according to theoretical models, clines can form through either primary differentiation or secondary contact (Endler, 1977; see General introduction, section 2.3.b). In primary differentiation, environmental heterogeneity along a species' range leads to its differentiation. Alternatively, primary differentiation might follow a range expansion, where colonizing populations diverge from their parental populations if the environmental conditions of the new habitats differ significantly from their original ones. In secondary contact, a cline forms through the reconnection of previously isolated populations, likely resulting in a hybrid zone.

Based on this information, can we construct a historical narrative (sensu Mayr, 2004) of how the analogous color clines came into existence? A plausible scenario is as follows (see Figure 5): before the rias were flooded and the coastline extended well beyond the current islands at the mouths of the rias, there were interspersed populations of *L. saxatilis* along the shoreline. At that time, most of the coast was composed of aeolian sands (i.e., wind-blown sands; Arce-Chamorro, Vidal-Romaní, & Sanjurjo-Sánchez, 2022), so *L. saxatilis* populations might have been concentrated along the patchy outcrops of granitic rock emerging from the granitic basement. Given *L. saxatilis*' high phenotypic and genetic diversity and its capacity to evolve local adaptati-

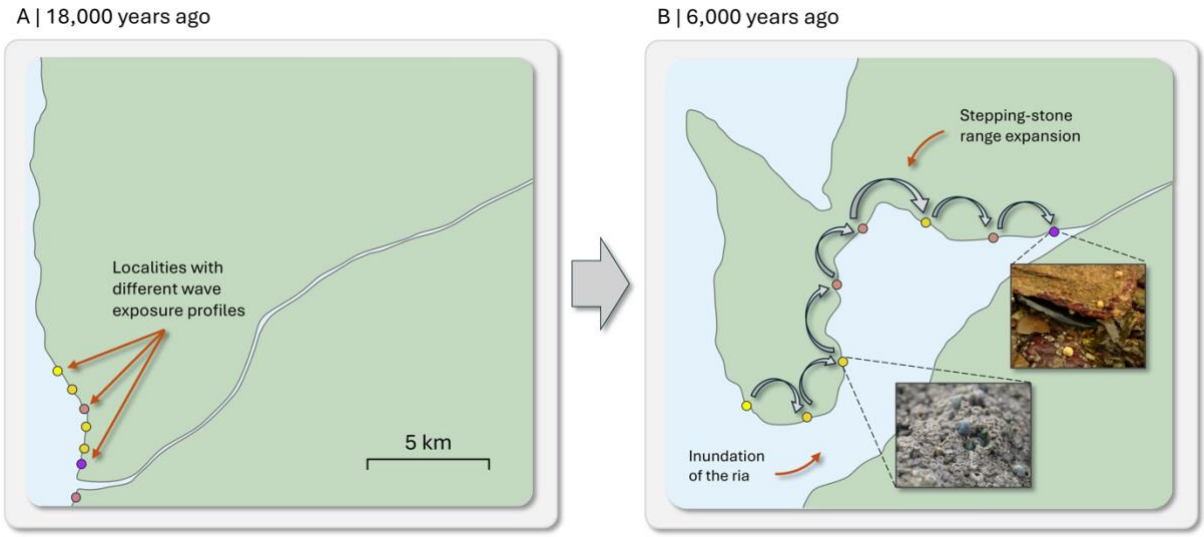

Figure 5. Scenario for the colonization of the Rías Baixas by *L. saxatilis*, using the Ría de Vigo as a model. A: Situation before the formation of the rias, approximately 18,000 years ago. At that time, numerous *L. saxatilis* populations coexisted along the coastline, likely in localities with different wave exposure profiles (dots on a color gradient, with yellow representing extreme wave-exposed localities and purple representing extreme wave-sheltered localities). Many of these populations were likely polymorphic for most shell traits, including color. B: The Ría de Vigo around 6,000 years ago, when colonization by *L. saxatilis* might have already been advanced. By this time, the ria was almost halfway formed, and ecosystem types had probably begun to cluster spatially. Many populations of *L. saxatilis* were likely similar to those observed today: monomorphic populations of *lineata* at wave-exposed shores and populations dominated by *fulva* in wave-sheltered areas (see inset photographs). The peninsula is what nowadays are the Cíes Islands. This scenario, although plausible, is purely conjectural and should be properly tested in the future.

ons (Behrens Yamada, 1987; Reid, 1996; Johannesson, 2016), it is reasonable to assume that many of the genetic variants responsible for shell color and other trait variations were already present in these populations. Ultimately, these variations were likely due to the wave exposure profiles of the localities in which the snails settled (Figure 5A).

As the rias began to flood, individuals from the source populations probably started to colonize the newly formed habitats, following a stepping-stone model of gradual range expansion (Rafajlović *et al.*, 2013; Figure 5B). This colonization could have accelerated during periods of increased temperature and decreased rainfall, as extreme rainfalls are known to cause high mortality rates in the populations of *L. saxatilis* from the Rías Baixas (Vigo and Rolán-Alvarez, unpublished data). Since the wave exposure profile of these newly colonized habitats probably varied gradually, this may have led to the progressive evolution of the color cline through primary differentiation. The formation of the cline could have occurred independently in each ria or even on each coast within a ria, as rafting from coast to coast likely is a rare phenomenon. However, preexisting standing genetic variation was likely the source of all the *fulva* and *lineata* variants, since it is known that due to multiple paternity, all founder groups in this species contain considerable genetic variation (Rafajlović *et al.*, 2013; Johannesson, 2016). In this sense, what evolved in parallel was probably the cline itself –the gradual progression in color frequencies– rather than the specific color morphs, although the various possibilities

remain open (Johannesson *et al.*, 2010; Westram *et al.*, 2014; Kess *et al.*, 2018). Finally, the ecological peculiarities of each ria might have led to the uneven evolution of trait combinations, with only the ones exhibiting isolated wave-sheltered and wave-exposed areas (i.e., Vigo and Muros-Noia) giving rise to the most tightly coupled covariation between shell colors and other shell traits.

If our historical narrative is accurate, we would expect the different morphs within each ria to be more genetically similar at neutral markers than the same morphs from different rias. The available evidence seems to support this account. A phylogram of various Galician *L. saxatilis* populations based on mitochondrial DNA data suggests that, despite differences in shell characteristics and habitat, snails from the same ria in the Rías Baixas are genetically closer than snails from different rias (Tirado *et al.*, 2016). However, the populations used to construct this phylogram may not be the most suitable for evaluating the present historical narrative.

A unified account of color variation in *L. saxatilis* from the Rías Baixas

The results of this dissertation are limited but provide a unified account of color variation in *L. saxatilis* from the Rías Baixas that can serve as a foundation for further, more detailed studies. This account begins with the colonization of the Rías Baixas: as the rias started forming, *L. saxatilis* snails from nearby polymorphic populations may have gradually begun to colonize them. Driven by spatially varying selection and isolation by distance, a parallel color cline may have developed, with *fulva* snails predominating in the wave-sheltered areas and the *lineata* morph dominating the wave-exposed zones. Since its establishment, this color cline might have remained more or less stable, without much directional change in the overall shape of the cline.

The colorations of *fulva* and *lineata* might have evolved through selection for crypsis. The *fulva* morph (and maybe the *lutea* and *fusca* morphs) is probably cryptic against brown algae, while the *lineata* morph is cryptic against alkaline granitoid rocks, whether fully covered with barnacles or not. The primary agent of selection for crypsis might have been a percomorph fish, with birds and crabs perhaps playing a secondary role. During the colonization of the rias, covariations between shell color and numerous shell traits with independent genetic bases evolved. Specifically, the *fulva* morph became associated with larger, smoother, spikier, thicker, and narrower-apertured shells, whereas the *lineata* morph became linked to smaller, more ribbed, rounder, thinner, and wider-apertured shells. These covariations might have been driven by predation by *C. maenas* crabs in the case of *fulva* and by wave action in the case of *lineata*.

However, the covariation between shell color and other shell traits did not develop equally in the different rias, as didn't the association between color morphs and ecological parameters. Due to Pontevedra and Arousa showing much less spatial aggregation of ecosystem types compared to Vigo and Muros-Noia, this led to disruptions in the magnitude of the connection between shell color and ecological parameters caused by gene flow. Additionally, populations in wave-sheltered habitats, which evolved various color morphs, maintained their polymorphic condition through divergent selection in heterogeneous microhabitats of brown algae, as well as through mate choice-driven NFDS ultimately caused by the potential fitness benefits of increased immune diversity against trematode parasites.

This is how color variation in *Littorina saxatilis* from the Rías Baixas might have evolved. Future work should put this historical narrative to the test.

Conclusions

1. Shell color polymorphism in marine gastropods is primarily driven by natural selection, which has been the most extensively studied mechanism. However, much of the evidence supporting natural selection is indirect. The advent of new techniques and approaches, alongside classic, ecologically-targeted studies, can enhance our understanding of the patterns and maintenance mechanisms of shell color polymorphism in this group of animals.
2. Negative assortative mating occurs in the color polymorphic populations of *Littorina saxatilis* from the wave-sheltered areas of the Rías Baixas. This pattern is likely ancestral, as it is also observed in other marine snails from the subgenus *Neritrema*.
3. Negative assortative mating in *Littorina saxatilis* is most likely driven by mate choice, with males selecting color-dissimilar sexual partners. This suggests that color polymorphism in the wave-sheltered localities of the Rías Baixas is maintained through negative frequency-dependent selection. However, how snails distinguish between colors and what benefits derive from heterotypic matings remain unknown.
4. A color cline exists in the Ría de Vigo, with a fawn morph (*fulva*) being prevalent in the wave-sheltered localities and a lineated morph (*lineata*) predominating in the wave-exposed zones of the ria. This cline has remained relatively stable over the last four decades and shows a clear association with environmental variables, thus suggesting selection.

5. The *lineata* morph is cryptic against the rock substrates of wave-exposed localities, but experimental evidence for selection for crypsis by marbled rock crabs (*Pachygrapsus marmoratus*) is inconclusive.
6. The four Rías Baixas (from north to south: Vigo, Pontevedra, Arousa, and Muros-Noia) exhibit analogous color clines along the wave-exposure continuum. The two main color morphs involved in the cline (*fulva* and *lineata*) are associated with both environmental parameters and other adaptive shell traits. However, the strength of these associations varies between rias. This cline may have evolved in parallel in the four rias, with the *fulva* and *lineata* colorations resulting from selection for crypsis by percomorph fishes. Other covarying shell traits might constitute adaptations to either crab predation (in the case of *fulva*) or wave action (in the case of *lineata*).

Appendix 1.- Color polymorphism: Supplementary topics

In this and the following appendix, which serve as supplementary material to the General introduction, a series of subsidiary topics not directly related to the dissertation's core content but of high interest to the evolutionary study of color variation are explored. Appendix 1 covers supplementary topics related to color polymorphism, while Appendix 2 addresses similar aspects for color clines. Both appendices show a comparable structure: firstly, they provide a brief historical account of major breakthroughs in the study of color polymorphism and clines, respectively. Modern science curricula, including PhD programs, rarely cover the history of science, leading to the oblivion of the contributions of many great scientists, which is certainly unfair. Secondly, each appendix includes a preliminary assessment of the prevalence of color polymorphism and clines across major animal taxa, alongside discussions on their correlates, typology, or consequences. These sections help us understand the relative importance of color variation in the Animal Kingdom and gain insights into the factors driving its evolution. Lastly, the appendices offer extended discussions on the maintenance mechanisms of color variation, incorporating additional examples of both color polymorphism and clines and discussing some of their nuances.

A1.1.- History of the study of color polymorphism

a) Research in coloration before Darwin

Since the inception of biology as a scientific discipline, animal coloration has captivated researchers of the natural world. As early as the fourth century BCE, Aristotle (384–322 BCE), the

founder of biological science, documented various observations about the colors of animals in his magnum opus *A History of Animals* (Aristotle, 1910). There he noticed, for instance, that humans exhibit eye color polymorphism (I. 492a), that some birds change their plumage color seasonally (III. 519a), or certain animals, like octopuses and cuttlefish, use color to evade both predator and prey (IX. 621b–622a).

However, the interest in intrapopulation color variation per se (i.e., color polymorphism) likely emerged much later in the history of biology, probably because it requires the explicit adoption of what Ernst Mayr called *population thinking*, that is, the recognition that individual differences, far from being mere errors in actualizing an ideal form, are a real phenomenon, and therefore a topic worthy of study (Mayr, 2004). According to Mayr, population thinking was introduced into biology by Charles Darwin (1809–1882), who considered intrapopulation variation as the raw material for natural selection. Consequently, it is reasonable to assume that, if not founded, the study of color polymorphism entered a new phase after Darwin consolidated evolutionary biology as a respectable scientific discipline.

It is true that, before Darwin, several researchers probably noticed that animals and plants of a given species sometimes exhibit various discrete color morphs. Notable among these early observers were the great taxonomists and naturalists of the Enlightenment, such as Carl Linnaeus (1707–1778), or Georges-Louis Leclerc, Comte de Buffon (1707–1788) (Morange, 2021). However, it is also highly likely that the relative importance granted to color polymorphism by these researchers was certainly minor compared to researchers in the post-Darwin era. Indeed, many post-Darwin scientists devoted significant portions of their careers to studying how color varies among individuals of a given species, and how such variations can evolve and be maintained in a population. Several early evolutionists are worth mentioning in this regard.

b) Pioneers in the study of color polymorphism

Alfred R. Wallace (1823–1913), the co-discoverer of natural selection alongside Darwin, was the first researcher to publish extensively on coloration from an evolutionary perspective (Wallace, 1877; for an overview, see Caro, 2017). Wallace conducted a detailed classification of the different possible adaptive roles of color in nature, arguing for the functional role of this trait:

> An extensive survey of the organic world thus leads us to the conclusion that colour is by no means so unimportant or inconstant a character as at first sight it appears to be; and the more we examine it the more convinced we shall become that it must serve some purpose in nature, and that besides charming us by its diversity and beauty it must be well worthy of our attentive study, and have many secrets to unfold to us. (Wallace, 1877: 643)

Another prominent early figure in the study of color was Edward B. Poulton (1856–1943), a colleague of Wallace, who published an entire book on the subject, entitled *The Colours of Animals* (1890). Like many of Wallace's works, *The Colours of Animals* described the various functions of color in nature, from camouflage to mimicry, defending as well that these should have an adaptive value (Bowler, 2003). Poulton's work was a significant innovation in the study of color polymorphism, introducing important concepts that are still common currency in contemporary evolutionary biology, notably negative frequency-dependent selection (Poulton, 1884; Allen & Clarke, 1984). Poulton used this concept to explain how color dimorphism could prevail in the larvae and pupae of the Geometrid moths *Cyclophora albipunctata* and *C. annularia* (then known as *Ephyris pendularia* and *E. omicronaria*, respectively).

A third early researcher who made significant contributions to the study of coloration and color polymorphism was William Bateson (1861–1926). In his *Materials for the Study of Variation* (1984), Bateson described several examples of discrete color morphs coexisting within a single species. In that respect, he stated: 'Nature abounds with examples of colour-polymorphism, and in numerous instances such Variation is discontinuous' (Bateson, 1894: 42). Bateson's book was a major milestone in the advent of mutationism, so his emphasis on discontinuous variation should be read as an argument against the biometric school, whose advocates defended that natural selection acted primarily on continuous variation (Bowler, 2003).

c) Classical research lines on the evolution of color polymorphism

Once the controversies surrounding the nature of heredity and its relationship to natural selection were finally worked out with the emergence of population genetics and the Modern Synthesis (Smocovitis, 1992; Provine, 2001), color polymorphism became an active area of study for many evolutionary biologists. Being conspicuous and present in many taxa, color polymorphism provided an ideal model phenomenon to test many of the fundamental tenets of theoretical population genetics (Svensson, 2017; Charlesworth & Charlesworth, 2017). These researchers were interested in the evolutionary processes maintaining color polymorphism rather than the polymorphism itself or the organisms exhibiting it. In this sense, they were 'process-oriented' rather than 'organism-oriented' natural historians (Svensson, 2023). This focus led to the establishment of many research lines aimed at understanding why genetic (and particularly color) polymorphisms were a thing in many natural populations (Ford, 1964). For brevity, I will only cover two of the most influential ones.

One particularly influential line of research in the study of color polymorphism –one that has ever since become a milestone in the history of evolutionary biology– is the case of industrial

melanism in the peppered moth, *Biston betularia* (Majerus, 1998). The peppered moth case has been recounted extensively, so I will briefly mention the main architects of this groundbreaking investigation. Empirical research on the peppered moth began in the 1950s by Bernard Kettlewell (1907–1979), a student of the prominent geneticist E. B. Ford (1901–1988), although J. B. S. Haldane (1892–1964) had previously published theoretical works on this system in the 1920s (e.g., Haldane, 1927). Kettlewell, a key member of what is sometimes known as the Oxford School of Ecological Genetics (Ruse & Travis, 2009), provided experimental evidence for the differential selective advantage of the wild and melanic morphs in clean and polluted environments, respectively (Kettlewell, 1973). However, subsequent challenges in replicating his findings raised doubts about his main conclusions (Ruse & Travis, 2009). It wasn't until decades later that further work on this system, largely conducted by Michael Majerus (1954–2009), corroborated Kettlewell's initial results, thereby establishing the industrial melanism of *B. betularia* as a paradigmatic example of natural selection in action (Ridley, 2004; Cook *et al.*, 2012).

Another significant research line on the evolutionary causes of color polymorphism was also led by two members of the Oxford School of Ecological Genetics, Arthur J. Cain (1921–1999), and Philip M. Sheppard (1921–1976), and focused on the shell color polymorphism of snails in the genus *Cepaea* (Cain & Sheppard, 1950, 1954; Cain, 1954). The two species within this group, *C. nemoralis* and *C. hortensis*, exhibit multiple color morphs determined by variations in background color and banding patterns (see Figure 1A of the General introduction). At a time when Oxonian zoology was highly anti-selectionist, Cain and Sheppard utilized Fisher's mathematical tools and Ford's ecological approach to provide evidence for the role of natural selection in maintaining *Cepaea*'s shell color polymorphism (Ruse & Travis, 2009). Their studies revealed, among other things, that snails of different colors faced differential predation by birds, particularly song thrushes (*Turdus philomenos*) depending on the habitat's background color. Besides that, Cain also pioneered the use of sub-fossil material to track spatiotemporal changes in morph frequencies (Currey & Cain, 1968).

Also notable in the context of *Cepaea*'s shell color polymorphism is Bryan Clarke (1932–2014). Building upon Cain and Sheppard's pioneering work, Clarke revitalized Poulton's concept of negative frequency-dependent selection by proposing that certain instances of shell polymorphism in *Cepaea* might be maintained by what he termed 'apostatic selection' (Clarke, 1962; for a discussion see section 1.2a of the General introduction, and A1.2d of the present Appendix). Although the status of apostatic selection in *Cepaea* remains debated (see Cook, 2017), the species within this genus continue to play a key role in the contemporary study of color polymorphism (e.g., Kerkvliet *et al.*, 2017; Affenzeller *et al.*, 2020; Johansen *et al.*, 2023).

A1.2.- Prevalence, correlates, consequences, and further mechanisms of color polymorphism

a) Prevalence of color polymorphism

A relevant question regarding color polymorphism is how widespread it is. Instances of this phenomenon have been reported in various kinds of organisms, encompassing not only animals but also gymnosperm and angiosperm plants (Malerba & Nattero, 2012; White & Kemp, 2016; Rudall, 2020). From now on I will focus on Kingdom Animalia, as it is the group of organisms to which the species under study in this dissertation belongs. Broadly construed, color polymorphism has been described in several animal phyla, including Cnidaria (Nevárez-López *et al.*, 2020; Clarke *et al.*, 2024), Annelida (Nygren *et al.*, 2011), Mollusca (Cook, 2017; Gefaell *et al.*, 2023), Chordata (e.g., Roulin, 2004; Hofreiter & Schöneberg, 2010; Abalos, 2021), Arthropoda (Caillaud & Losey, 2010; Nasir & Faulkes, 2011; Sánchez-Guillén *et al.*, 2020), and Echinodermata (Growns & Ritz, 1994; Harley *et al.*, 2006). While these constitute only a fraction of the total metazoan phyla that exist (estimated between 35 and 40 according to modern-day zoologists, although this remains a matter of debate; e.g., Valentine, 2004; Zhang, 2013; Levin *et al.*, 2016), they comprise most animal species, considering that arthropods, the most diverse phylum by far, account for 78.5% of species, and Mollusca and Chordata constitute the second and third most populous phyla, respectively (Zhang, 2013).

If we examine these three phyla, we find that their principal taxa include color polymorphic species. Among chordates, color polymorphism has been described in ray-finned fishes (Olendorf *et al.*, 2006; Rodriguez-Silva *et al.*, 2023), amphibians (Hoffman & Blouin, 2000; Beukema *et al.*, 2016), turtles (Garbin & Caramaschi, 2015), squamates (Abalos, 2021; Stuart-Fox *et al.*, 2021; Brock, McTavish, & Edwards, 2022; Madsen *et al.*, 2022), mammals (Caro, 2005; Römpler *et al.*, 2006; Hofreiter & Schöneberg, 2010), and birds (Galeotti *et al.*, 2003; Roulin, 2004). Among mollusks, the three most diverse classes –gastropods, bivalves, and polyplacophorans– also include color polymorphic species (Gonçalves Rodrigues & Silva Absalão, 2005; Luttikhuizen & Drent, 2008; Cook, 2017; Mendonça *et al.*, 2015, 2016; Williams, 2017; Gefaell *et al.*, 2023). Finally, among arthropods, color polymorphic species have been reported in chelicerates (particularly arachnids; Oxford & Gillespie, 1998; Cotoras *et al.*, 2017; Ximenes & Gawryszewski, 2019), crustaceans (Bauer, 1982; Nasir & Faulkes, 2011; Nokelainen, Stevens, & Caro, 2018), and hexapods (e.g., Caillaud & Losey, 2010; Sánchez-Guillén *et al.*, 2020; Winter *et al.*, 2021).

We can even go further downwards in the taxonomic hierarchy. For instance, color polymorphism has been documented at least once in the main groups of insects (which, in turn,

constitutes the most important group of hexapods): Coleoptera (Roggero *et al.*, 2024), Lepidoptera (Kettlewell, 1973; Woronik & Wheat, 2017), Hymenoptera (Naito, 1983), Diptera (Parkash, Singh, & Ramniwas, 2009), Orthoptera (Karpestam, Merilaita, & Forsman, 2016; Schielzeth, 2020; Winter *et al.*, 2021), Odonata (Fincke *et al.*, 2005; Willink, Duryea, & Svensson, 2019; Sánchez-Guillén *et al.*, 2020; Okude & Futahashi, 2021), Homoptera (Thompson & Halkka, 1973; Stewart & Lees, 1997), and Hemiptera (McLennan *et al.*, 1994; Caillaud & Losey, 2010; Watanabe *et al.*, 2016).

So, from the present review, we can conclude that color polymorphism does exist across the main groups of animals. However, this short review doesn't provide information about its prevalence (i.e., how common it is within a given taxa), which is crucial for understanding the significance of a biological phenomenon. To date, few studies have systematically analyzed the pervasiveness of color polymorphism within the different animal groups. An exception to this is Galeotti *et al*. (2003; see also Roulin, 2004), who investigated the prevalence of color polymorphism in birds. In this taxon, color polymorphism occurs in 3.5% of species, spanning 61% of orders and 37% of families. Within these orders, an average of 7% of species exhibit color polymorphism, with prevalence rates ranging between 0.6 and 33.5%. Notably, the order Strigiformed (owls and nightjars), primarily nocturnal species, shows the highest prevalence of color polymorphism at 33.5%. In contrast, Piciformes (which includes woodpeckers and toucans), exhibit the lowest incidence at 0.6%, followed by Passeriformes (perching birds, the largest order of birds) at 0.9%.

Another notable exception is Schielzeth (2020), who analyzed the prevalence and distribution of the green-brown color polymorphism observed in many European orthopterans (the insect order that includes grasshoppers, locusts, and crickets). This researcher concluded, among other things, that 30% of them exhibit this polymorphism. Further studies exploring the prevalence of color polymorphism include Gefaell *et al*. (2023; corresponding to Chapter 1 of this dissertation) for marine gastropods, Fincke *et al*. (2005) for Odonata, and Chapple *et al*. (2008) for the lizards of the genus *Liopholis*.

b) Correlates of color polymorphism

The scarcity of studies on the prevalence of color polymorphism complicates drawing definitive conclusions about the phylogenetic, geographic, and ecological factors correlates of this phenomenon. From a very general perspective, color polymorphism appears to be predominantly present in relatively complex animals. If, following Dawkins (1992), we understand complexity as the amount of information needed to describe an organism, then it becomes

evident that color polymorphism is mostly present in taxa containing complex animals not only in terms of their anatomies and physiologies but also regarding their life history traits, mating systems, or trophic relationships. Therefore, it may not be reasonable to expect a color polymorphism to be discovered in a kinorhynch or a rotifer. Only complex animals appear to display color polymorphism.

But it is hard to go beyond that truism due to the lack of sufficient evidence. Galeotti *et al*. (2003; Figure 5 and Table 2) found that when accounting for phylogeny, color polymorphism in birds was associated with factors such as a daily activity rhythm extended to day and night. This prompts these authors to link this phenomenon to selective pressures related to individual detectability by predators, prey, and conspecifics. As for orthopterans, Schielzeth (2020) found that polymorphic species were particularly prevalent in moist to mesic grasslands, alpine environments, and arboreal habitats, with higher abundance in northern latitudes. In contrast, monomorphic species tended to inhabit rocky and cave environments or to exhibit nocturnal habits. This provides further support to the detectability hypothesis, which is also in line with what is found in plants: angiosperms, which rely on animals for pollination, seem to be much more color polymorphic than gymnosperms (Rudall, 2020). Additionally, closely related species often display similar types of color polymorphism, emphasizing the role of phylogeny in shaping this trait (Jamie & Meier, 2020). Nonetheless, more work is needed to reveal the macroscale patterns and factors that correlate with color polymorphism.

c) Consequences of color polymorphism

Another intriguing aspect of color polymorphism is its potential consequences. Does being color polymorphic increases the likelihood of certain evolutionary outcomes? Traditionally, polymorphism, whether in color or other traits, has been thought to promote speciation under certain conditions. If morphs become isolated and start to diverge over time, or if assortative mating emerges in sympatry, preventing the breakdown of optimal trait combinations, this can lead to sexual isolation between morphs and potentially drive speciation (Gray & McKinnon, 2007; Stuart-Fox *et al*., 2021; Brock et al., 2022). Indeed, some studies have suggested a link between color polymorphism and higher rates of speciation (Forsman *et al.*, 2008; Corl *et al.*, 2010; Hugall & Stuart-Fox, 2012; McLean & Stuart-Fox, 2014; Brock *et al.*, 2022). For instance, Brock *et al*. (2022) found that the evolutionary transition to color polymorphism in the lizard family Lacertidae was associated with an increased net diversification rate. However, whether this pattern applies universally across taxa remains uncertain.

Connected to this, a recent debate has centered on whether color polymorphism increases or decreases vulnerability to extinction (Bolton *et al.*, 2015, 2016; Forsman, 2016; see also Svensson, 2017). Interestingly, both sides of the debate rely on the same argument: color polymorphism is often correlated with various sexual, ecological, or life history traits (see section 1.2c of the General introduction). This is seen by some researchers as a potential disadvantage in the face of disruptive environmental change (Bolton *et al.*, 2015), and by others as an opportunity for colonization success, range expansions, or utilization of environmental resources (Forsman *et al.*, 2008; Forsman, 2016). Although the debate is far from settled, recent evidence from a study on birds supports the latter viewpoint (Ducatez *et al.*, 2017).

In sum, it seems that color polymorphism tends to speed up the speciation process, although not much is known about this and other potential outcomes of this phenomenon. To close this section, it is worth mentioning that while the evolutionary consequences of color polymorphism may seem at first sight disconnected from discussions on its definition and prevalence, there are potential interconnections between these areas. For instance, part of the controversy on whether color polymorphism spurs or hinders extinctions hinged around how to define this phenomenon (Bolton *et al.*, 2015, 2016; Forsman, 2016; Svensson, 2017). As for the link between the prevalence of color polymorphism and its consequences, Hugall and Stuart-Fox (2012) found that the fact that color polymorphism in birds is rare and phylogenetically scattered may be partly attributed to the higher speciation rate it promotes, as well as to the observed tendency for transitions from polymorphism to monomorphism to be more common than the reverse. So future studies should delve deeper into the relationships between these three potentially interconnected topics.

d) Maintenance mechanisms: Further ecological contexts that can result in color polymorphism through apostatic negative frequency-dependent selection (NFDS)

In the General Introduction, apostatic selection was defined as the process whereby predators develop a search image for their most common prey morph, leading to reduced predation pressure on less frequent morphs and thus an increase in their fitness. However, apostatic selection encompasses various forms, all rooted in the same principle of search image formation. For instance, in cases involving trophic relationships, apostatic negative frequency-dependent selection (NFDS) can affect predators instead of prey. In situations where prey possess sophisticated perceptual or cognitive abilities, apostatic selection can impact the color pattern of the predator species themselves. Preys may implement anti-predator behavioral strategies only when they encounter predator individuals with common color patterns, ignoring

those with rarer colors. This is so because presumably, such prey would form avoidance images of their commonest predators (Galeotti *et al.*, 2003). This form of prey-induced apostatic NFDS has been invoked to explain color polymorphism in certain avian predators, such as hawks and skuas, which prey on highly astute and visually advanced mammals and birds (Paulson, 1973; Rohwer & Paulson, 1987). The empirical status of this hypothesis, though, is still controversial (Galeotti *et al.*, 2003).

Apostatic-NFDS can also involve parasite-host interactions (Ajuria Ibarra & Reader, 2013), including parasitoid and brood or klepto-parasitic species (Arnason, 1978; Galeotti *et al.*, 2003). As said, the underlying principle remains the same: if parasites detect their hosts by forming a search image, the rarest host color morphs may go unnoticed. For example, the parasitoid wasp *Aphidius ervi*, which targets the color-dimorphic pea aphid (*Acyrthosiphon pisum*, with red and green morphs; see Figure 1B and section 1.2b of the General introduction), tends to preferentially lay its eggs inside the green aphid morph. However, this preference can be influenced by prior experience, such that encounters with red aphids increase the likelihood of this morph being attacked (Langley *et al.*, 2006). This would be so because the parasitoid wasp forms a search image based on its previous experience. A similar putative example of parasitoid apostatic NFDS maintaining a color polymorphism has been suggested in marine gastropods (McKillup & McKillup, 2008), a case discussed in more detail in Chapter 1 of this dissertation (Gefaell *et al.*, 2023).

Finally, apostatic NFDS can involve interactions between the different sexes in a male-female mating context (Wellenreuther, Svensson, & Hansson, 2014). This could be driven by sexual conflict, where there are antagonistic optimal fitness strategies between the sexes, or a discrepancy between the genetic interests of males and females (Chapman *et al.*, 2003). A potential example of this might occur in the blue-tailed damselfly *Ischnura elegans* (Svensson, Abbott, & Härdling, 2005; Wellenreuther *et al.*, 2014; see Figure 1D of the General introduction). In *I. elegans*, males harass females in mating contexts, reducing female fitness by diminishing both fecundity and food intake. It has been suggested that a male-driven apostatic NFDS might drive the female polymorphism shown by this species. Males might form a search image of the most common female morph, thereby ignoring –and not harassing– the less frequent morphs (Svensson *et al.*, 2005; Svensson, 2017).

In summary, while the most well-known cases of apostatic selection involve image-forming predators, this mechanism can operate across various ecological and sexual backgrounds, ultimately maintaining color polymorphism through NFDS.

Appendix 2.- Color clines: Supplementary topics

A2.1.- History of the study of color clines

a) Pioneers in the study of color clines

Much of what can be said about the early study of color clines overlaps with the history of research in color polymorphism. Although descriptions of color clines likely existed long before the advent of modern evolutionary biology, they were probably not analyzed in detail or given much attention. Therefore, I will skip the ancients and jump straight into the nineteenth century and later, where the first modern descriptions of color clines show up.

Interestingly, the study of color clines –broadly construed– predates the very coining of the term *cline* itself. For instance, Constantin Wilhelm Lambert Gloger (1803–1863) described what we now refer to as Gloger's rule (see section A2.2b below) as early as 1833, although some have traced the origins of this ecogeographical rule even further back, specifically to the work of the Prussian naturalist Peter Simon Pallas (1741–1811) at the beginning of the nineteenth century (Stresemann, 1975). While Gloger formulated his rule primarily in terms of temperature's influence on bird coloration, the prominent German ornithologist Bernhard Rensch (1900–1990) later incorporated the effect of humidity (Rensch, 1929; see also Goldenberg *et al.*, 2022).

Causes of color variation

Darwin himself noted the existence of continuous geographic intergradation of characters within certain species, considering this phenomenon of uttermost importance to the study of the natural world:

> Those forms which possess in some considerable degree the character of species, but which are so closely similar to some other forms, or are so closely linked to them by intermediate gradations, that naturalists do not like to rank them as distinct species, are in several respects the most important to us. (Darwin, 1859: 48)

Already in the twentieth century and under a fully Darwinian framework, Francis B. Sumner (1874–1945) made significant contributions to the study of color clines. Sumner investigated the geographic variation in fur coloration shown by *Peromyscus* mice across the western United States (e.g., Sumner, 1926; for a summary, see Mullen & Hoekstra, 2008). His work aimed at understanding how variation was maintained in nature and was truly innovative. By examining numerous samples of mice across a large geographic area, he established that color variation in *Peromyscus* was likely maintained through selection for crypsis and that coat color in this genus had a genetic basis (Hoekstra, 2011). Sumner's original samples are now stored at UC Berkeley's Museum of Vertebrate Zoology and continue to be used in contemporary research on color clines (Mullen & Hoekstra, 2008). His work was widely regarded as top-notch science by his contemporaries and was discussed in many classic works on evolutionary biology from the time of the Modern Synthesis (Dobzhansky, 1937; Haldane, 1948; Huxley, 1942; Mayr, 1963; see Mullen & Hoekstra, 2008).

b) The coining of the concept of a cline: Julian Huxley

Despite the contributions of earlier pioneers, it was not until 1938 that the concept of a cline was coined by the distinguished evolutionary biologist Julian Huxley (1887–1975) (Huxley, 1938; see also Huxley, 1939). The main argument for the invention of this new term was straightforward: until then, geographic variation in color or other traits was understood under a subspecies framework, that is, as if such variation suggested the existence of various geographic races within a species. This approach, however, failed to capture the full spectrum of gradation in some traits and led to quite convoluted descriptions of within-species variation. To grasp such character gradients more parsimoniously, Huxley proposed the concept of a cline. In his own words:

> Phenomena of this type [character gradients] are so widespread and so inadequately dealt with by the usual taxonomic practice of merely assigning subspecific or specific names to forms which replace each other geographically, that an addition to current terminology seems called for. I have therefore proposed the term cline for a geographical gradient in phenotypic characters. (Huxley, 1939: 492–493)

In his foundational papers, Huxley not only introduced the term *cline*, but he also offered a thorough classification of the different types of clines based on their degree of interconnectedness (Huxley, 1939; see section A2.2b below). This classification is still discussed in contemporary literature (e.g., Salomon, 2002). Additionally, in his influential book *Evolution: The Modern Synthesis*, Huxley conducted a comprehensive review of most clines described up to that point (Huxley, 1942: 206–221). Huxley's introduction of the concept of cline exemplifies Mayr's thesis that progress in biology is sometimes achieved through the invention of new terms rather than the discovery of new facts, as these terms can help make sense of already-known natural phenomena more easily (Mayr, 1998). Indeed, Huxley's term led to the blossoming of research in clines, as geographic variation in traits, like color polymorphism, provided an intriguing model phenomenon for studying the interaction of evolutionary forces in natural populations.

c) Population genetics and clines

Part of these research efforts involved quantifying evolutionary processes using the tools of population genetics. These mathematical models could then be tested against real cases studied by naturalists and field biologists to determine whether they provided accurate predictions of biological phenomena, much like theoretical physicists do with data from experimentalists. Clines, in particular, offered a fascinating case for modeling the interactions between selection, gene flow, and drift. J. B. S. Haldane (1892–1964), Ronald A. Fisher (1890–1962), and Sewall Wright (1889–1988), considered the fathers of modern population genetics theory, all made contributions to this topic (Wright, 1943; Haldane, 1948; Fisher, 1950). For brevity, I will only say a few words about Haldane's work.

In a now-classical paper, Haldane set himself to show that '(…) in certain cases a cline can give quantitative information regarding the intensity of natural selection' (Haldane, 1948: 277). Using a series of differential equations and data from Sumner's investigations on the dynamics of coat color in *Peromyscus* (specifically the check patch), Haldane provided tentative estimates of the intensity of natural selection. Still, Haldane himself seemed somewhat skeptical of his calculations, as they relied on several untested assumptions that would make them 'at best approximate', and at worst, 'wholly fallacious' (Haldane, 1948: 283). Modern estimations of selection coefficients appear to confirm Haldane's calculations, but only under the assumption that the dispersal distance of the mice is equal to –or even larger than– the width of the cline (Mullen & Hoekstra, 2008).

d) John A. Endler and beyond

The next major figure in the study of clines is most likely John A. Endler (born 1947). Endler, a pioneer in the study of animal coloration (Endler, 1978) and natural selection (Endler, 1986), devoted an entire book to the study of geographic variation and clines (Endler, 1977). Endler's book on clines, much like his other works from that period (e.g., Endler, 1986), offers a comprehensive review of existing knowledge on the subject, along with an insightful discussion on various related topics. In his book on clines, Endler examined in detail the different mechanisms by which this phenomenon can arise, among other things. Additionally, he vindicated the importance of spatially continuous variation for differentiation and speciation, challenging Mayr's popular hypothesis that no speciation could occur without geographic barriers (Mayr, 1963). In this regard, it is hard to overestimate the breadth and significance of Endler's contributions to the study of clines, geographic variation, and evolutionary biology as a whole.

After Endler's seminal work, other important contributions ensued, among which Barton and Hewitt's review on hybrid zones probably stands out (Barton & Hewitt, 1985). Moreover, many empirical examples of color clines began to accumulate. More recently, the introduction of various molecular techniques (e.g., Takahashi, 2015) has provided a valuable addition to the more classical ecologically oriented approaches to the study of clines, which nonetheless remain as essential as ever in evolutionary biology.

A2.2.- Prevalence, diversity, consequences, and further mechanisms of color clines

a) Prevalence of color clines

Color clines are a ubiquitous phenomenon in nature. Like color polymorphism, they have been described in both gymnosperm and angiosperm plants (Sturgeon & Mitton, 1980; Streisfeld & Kohn, 2005; Schemske & Bierzychudek, 2007), as well as in animals. Following the previous appendix, in this section I will focus on this last group, where color clines have been reported in the most prominent phyla: Cnidaria (Stoletzki & Schierwater, 2005), Mollusca (Cook, 2017; Gefaell *et al.*, 2023), Chordata (Wunderle, 1981; Gray, 1983; Galeotti *et al.*, 2003; Corl *et al.*, 2010; Hoffman *et al.*, 2018), Arthropoda (de Jong & Brakefield, 1998; Oxford, 2005; Takahashi *et al.*, 2011; Brakefield & de Jong, 2011), and Echinodermata (Beck & Styan, 2010). Compared to the equivalent list in Appendix 1, section A1.2a, only annelids appear to drop off. However, given the variety of concepts often used to refer to clines in the literature, it wouldn't be surprising to find

descriptions of clines in papers using different terminology than that used in most evolutionary biology articles.

Anyway, similarly to color polymorphism, the three most populous phyla (Arthropoda, Mollusca, and Chordata) include species with populations showing gradual geographic variation in coloration. Delving into these taxonomic groups, we find that in chordates, color clines in a broad sense –that is, not always meeting all the criteria outlined in section 2.1 of the General introduction– have been reported in Actninopterygii (i.e., ray-finned fishes; Planes & Doherty, 1997; French *et al.*, 2018; Domínguez-Castanedo *et al.*, 2021), Amphibia (Gray, 1983), Squamata (Corl *et al.*, 2010), Mammalia (e.g., Mullen & Hoekstra, 2008; Hoffman *et al.*, 2018; Cosentino *et al.*, 2023), and Aves (Galeotti *et al.*, 2003; Antoniazza *et al.*, 2010). Among mollusks, they have been described in Gastropoda (Gefaell *et al.*, 2023), and Bivalvia (e.g., Beukema & Meehan, 1985). Finally, within arthropods, instances have been documented in Chelicerata (e.g., Reillo & Wise, 1988; Oxford, 2005), and Hexapoda (e.g., Takahashi *et al.*, 2011; Brakefield & de Jong, 2011). In this sense, it seems that in Polyplacophora among mollusks, and in Crustacea among arthropods, no instances of color clines have been reported so far.

Once again, insects appear to stand out in terms of color clines, possibly due to their status as the most biodiverse taxon (Zhang, 2013), their extensive study, or because they really surpass other groups in terms of gradual geographic variation in color. Instances of this phenomenon have been described in Coleoptera (Brakefield & de Jong, 2011), Lepidoptera (Mallet *et al.*, 1990; Grant *et al.*, 1998; Saccheri *et al.*, 2008), Hymenoptera (De Souza *et al.*, 2017), Diptera (Hollocher, Hatcher, & Dyreson, 2000; Bastide *et al.*, 2014), Orthoptera (Dearn, 1981, 1984; Köhler, Samietz, & Schielzeth, 2017), Odonata (Takahashi *et al.*, 2011), Homoptera (Brakefield, 1990; Stewart & Lees, 1997), and Hemiptera (Stewart, 1986).

Upon examining the previous list, we once again encounter the challenge that, despite its interest, it does not offer a direct estimation of the relative significance of color clines. Such assessments appear to be relatively scarce in the literature. Once more, Galeotti *et al*. (2003) stand out as an exception; these researchers found that 67 polymorphic bird species exhibit a clinal pattern in the frequency of color morphs. Beyond this, other authors have provided guesses of the prevalence of certain types of clines, such as thermal melanic clines in ectothermic animals by Clusella-Trullas *et al*. (2007), but without the quantitative detail provided by Galeotti *et al*. (2003). Therefore, future review studies should aim to provide precise estimates of the prevalence of color clines across different animal taxa.

b) Further properties that can help understand clines

In the General introduction, I introduced the shape and temporal dynamics of color clines, which provide valuable insights into their maintenance mechanisms. However, other properties such as geographic scale and degree of connectedness (Figure A2.1) are also crucial for distinguishing between different types of color clines.

Geographic scale helps differentiate between ecogeographical rules and small-scaled (or even microhabitat) clines (Figure A2.1A). Ecogeographical rules describe broad patterns of variation in one or more traits across the geographic ranges of multiple species (usually entire taxa) (Goldenberg *et al.*, 2022). Examples of ecogeographical rules involving color include Gloger's rule (see section A2.1a of the present Appendix), which posits that endotherms, particularly birds, exhibit darker colorations in warm and humid areas (Goldenberg *et al.*, 2022), and the thermal melanism hypothesis (also known as Borgert's rule), where at least arthropods and squamates tend to be darker in cooler environments (Clusella-Trullas *et al.*, 2007; Goldenberg *et al.*, 2022). Strictly speaking, ecogeographical rules are not clines proper, as they typically involve variation across species within a taxon (as in Bishop *et al.*, 2016) rather than populations within a species. Nevertheless, clines and ecogeographical rules are often discussed together (e.g., Futuyma & Kirkpatrick, 2017), as both usually involve selection as a primary maintenance mechanism and exhibit a gradual geographic progression in the trait of interest. On other occasions, though, clines occur over much shorter distances. For example, they might take place between microhabitats within the same ecosystem, as observed with the *Crab* and *Wave* ecotypes of *L. saxatilis* (e.g., Rolán-Alvarez, 2007). When categorizing clines based on geographic scales, it is crucial to consider the dispersal capabilities of the species, which significantly influence their distribution patterns.

Actually, gene flow itself provides another way of sorting out clines (Figure A2.1B). A key distinction lies in whether the populations involved in a cline are connected or disconnected – that is, whether there is gene flow between them (*continuous clines*) or whether they are allopatric (*discontinuous clines*). This classification method was originally proposed by Julian Huxley (1939), as discussed in section A2.1b. Discontinuous clines are more susceptible to breaks in trait values compared to continuous clines, as gene flow tends to homogenize the populations within the cline.

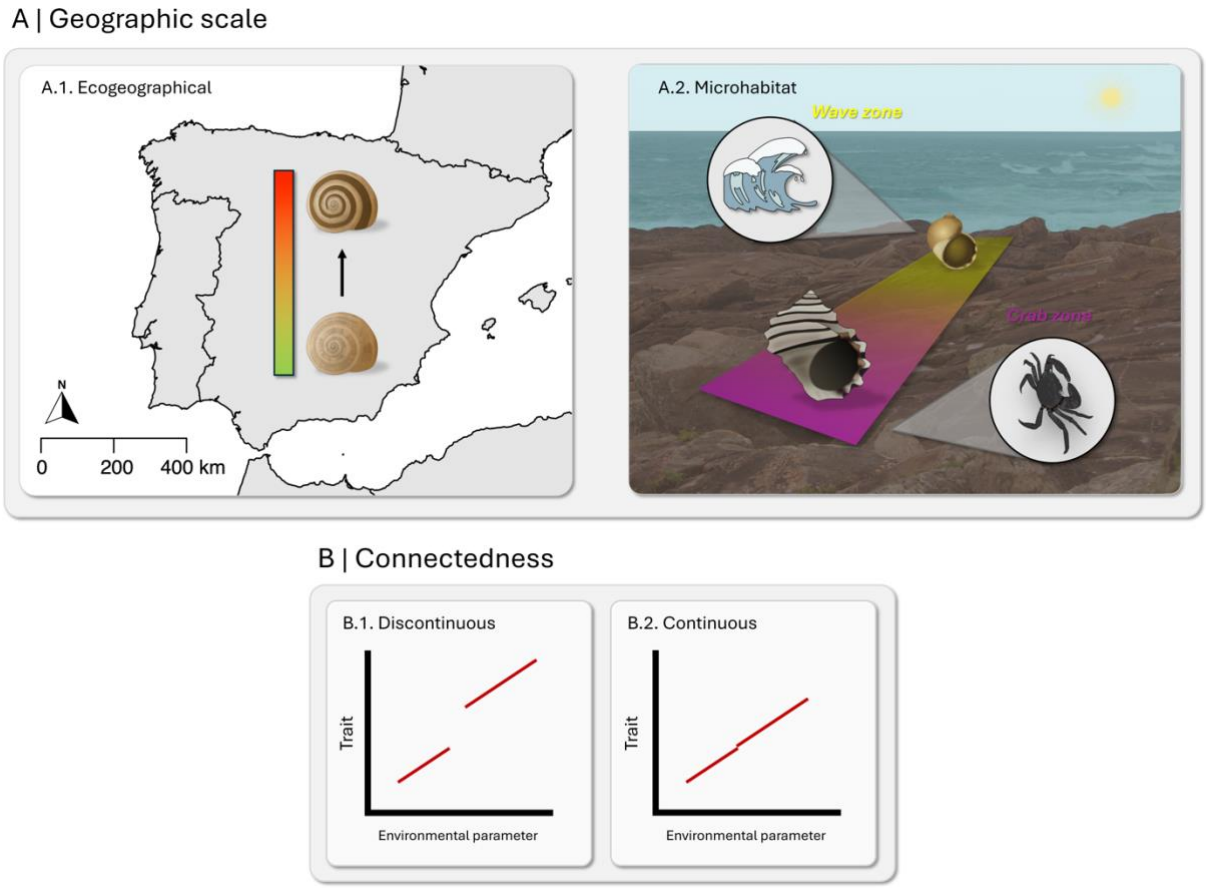

Figure A2.1. Two properties that can help distinguish between different types of clines. A: Geographic scale; A.1. shows a hypothetical ecogeographical cline (i.e., rule) in a terrestrial gastropod, with darker colorations (caused by bands) at higher latitudes and lighter colorations at lower latitudes. This could represent a case of Bogert's rule in a gastropod, provided that lower temperatures accompany the higher latitudes. However, this example is purely hypothetical and for illustrative purposes only. A.2. displays the *Crab* and *Wave* ecotypes of *L. saxatilis* as an example of microhabitat cline. B: Degree of connectedness of the populations involved in a cline, with discontinuous (B.1) and continuous (B.2) clines.

c) Consequences of color clines

At this juncture, one might ask whether color clines have similar or different evolutionary consequences than color polymorphism. As we saw in Appendix 1, section A1.2c, color polymorphism has been associated with speciation among other outcomes, although the debate remains unresolved. Similarly, color clines have also been linked to the process of species formation, yet uncertainties abound too, largely due to the challenges in studying the speciation itself (e.g., Via, 2009). Interestingly, the pathway from clines to species formation involves two contrasting scenarios, both contingent on the degree of gene flow between the populations taking part in a cline. Let's see.

The first pathway through which a color cline may lead to the generation of new species involves maintaining its polymorphic condition (McLean & Stuart-Fox, 2014). Consider a morph ratio cline with two distinct color morphs, *A* and *B*. Suppose that the distribution of these morphs along an environmental gradient is the following: at one extreme morph *A* predominates at 75%

while morph *B* is at 25%, with the reverse distribution at the opposed extreme. Despite morphs *A* and *B* being maintained at their respective extremes by a relatively strong spatially varying selection, a sufficiently high and symmetric gene flow along the cline can prevent their extinction at the less favorable edges. This preservation of polymorphism can lead to various evolutionary outcomes attributed to polymorphic populations, as discussed in Appendix 1, section A1.2c. Therefore, in this scenario, all potential evolutionary consequences of color clines stem from being a special case of color polymorphism.

The second pathway through which color clines could be conducive to speciation involves the relative isolation of their populations, ultimately driven by spatially varying selection. Let's take our previous hypothetical example with morphs *A* and *B*, but now assume that the gene flow between the populations at their edges is much lower than in the previous scenario. Under these conditions, the polymorphic condition may be lost, giving rise instead to two monomorphic populations. Through local adaptation and assortative mating, these populations could speed up the speciation process (McLean & Stuart-Fox, 2014). This scenario becomes particularly plausible if reinforcement –where hybrids of morphs *A* and *B* have lower fitness– or genetic incompatibilities arise. Such events allegedly happen in populations that were previously isolated but later come into contact again (Pryke & Griffith, 2009; McLean & Stuart-Fox, 2014), or in cases where there are significant fitness costs associated with hybridization (Nosil, Crespi, & Sandoval, 2003).

This last pathway is tightly linked to parapatric speciation, where populations are separated by a partial extrinsic barrier (Butlin, Galindo, & Grahame, 2008). The significance of parapatric speciation has been extensively discussed at least since Ronald Fisher addressed it in his classic *The Genetical Theory of Natural Selection* (1930), and while some putative examples have been identified, its extent and relative significance are still a matter of debate (Coyne & Orr, 2004). Today, it is assumed that spatially varying selection in the face of gene flow can –but not necessarily will always– lead to species formation (Stankowski & Ravinet, 2021). For this to happen, selection must be strong enough to counteract the homogenizing effects of gene flow (Via, 2009). However, these debates are beyond the scope of this dissertation. What concerns us here is that clines fueled by spatially varying selection can also, provided the right conditions, result in speciation.

d) Maintenance mechanisms: Examples of environmental factors that can result in the formation of color clines

In the General introduction (section 2.3a), I numbered a series of environmental factors – temperature, substrate, humidity, urbanization, pollution, and ecological communities– that can potentially drive the formation of a color cline. Numerous instances of color clines resulting from these factors have been documented in the literature. Examples of climatic or temperature gradients maintaining color clines have been found in several animal groups, some of which have already been discussed in the General introduction when examining the impact of climate change on clines (section 2.2b). Additional cases include *C. nemoralis* (Ożgo, 2011) and other helicid snails like *Arianta arbustorum* and *Theba pisana* (Burla & Gosteli, 1993; Johnson, 2011), *Polistes* paper wasps (De Souza *et al.*, 2017), or the meadow grasshopper (*Pseudochorthippus parallelus*; Köhler *et al.*, 2017).

Other color clines are influenced by spatially varying substrates, where the most cryptic morph tends to predominate. Francis B. Sumner's investigations on *Peromyscus* (see section A2.1a), taken over by Hoekstra and colleagues (Mullen & Hoekstra, 2008; Hoekstra, 2011), exemplify this type of color cline. Substrate-driven color clines are also abundant in marine gastropods, as can be seen in Gefaell *et al*. (2023), corresponding to Chapter 1 of this dissertation, although it remains unclear whether it affects *L. saxatilis* from the Rías Baixas (see Chapter 5). Humidity also influences pigmentation clines along hygric gradients in several *Drosophila* species, with darker colorations predominating in drier environments, except for *D. americana*, which shows the opposite trend (Wittkopp *et al.*, 2011 and references therein). However, experiments on desiccation resistance in the latter species have not shown a direct link between this physiological trait and body color, so the exact link between humidity and pigmentation is still an open question (Wittkopp *et al.*, 2011).

Urbanization has also been shown to induce color clines. For instance, the spittlebug *Philaenus spumarius*, which includes several color morphs that can be categorized as either melanic or non-melanic forms, shows higher melanic frequencies in urban compared to rural areas across England and Wales (Stewart & Lees, 1997). Similarly, eastern grey squirrels (*Sciurus carolinensis*) from Syracuse (New York, United States) exhibit increasing melanic morph frequencies along an urbanization gradient, ranging from 48% in the city center to less than 5% in rural woodlands. A transplantation experiment suggests that the melanic morph faces lower fitness than its grey counterpart in rural environments (Cosentino *et al.*, 2023), supporting the hypothesis that the relatively high frequencies of the melanic morph in urban areas might be due

to the relaxed predation pressures against squirrels in those settings. Color clines along urbanization continuums may ultimately be driven by pollution gradients, which have also been implicated in the evolution of color clines, as demonstrated in the peppered moth *Biston betularia* (Majerus, 1998; Grant *et al.*, 1998; Saccheri *et al.*, 2008).

Ecological factors, such as habitat type and vegetation cover, have also been linked to color clines. For example, McLean *et al*. (2015) found strong statistical associations between color morph frequencies in the Australian tawny dragon lizard, *Ctenophorus decresii*, and vegetation cover, such that the frequency of orange and yellow morphs tends to increase in more vegetated localities, while the grey morph exhibits the opposite trend. Although uncertainties remain, the authors suggest that this pattern may relate to increased detectability by conspecifics, as the orange and yellow morphs are more conspicuous in vegetated habitats compared to the grey morph (McLean *et al.*, 2015). Lastly, differences in community composition and ecological relations can also contribute to the formation of color clines. Several anti-predator traits exhibit clinal variations that ultimately depend on predator density (Laurila, Lindgren, & Laugen, 2008; Matsumura & Miyatake, 2023). This might affect color as well. For instance, predation intensity has been identified as the main factor explaining color morph frequency differences across latitudes in the lizard *Liopholis whitti* (Matthews *et al.*, 2018). However, caution is warranted due to some contradictory findings in this case this case[1].

[1] Even though differences in community composition can sometimes lead to the formation of clines, these processes often result in non-clinal selective patterns. The main reason for this is that predation strategies tend to vary depending on prey density and types, parameters that do not always follow clinal trends (McLean & Stuart-Fox, 2014 and references therein).

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